

## 24. Multi-Function Timer Pulse Unit 3 (MTU3a)

### 24.1 Overview

This MCU has an on-chip multi-function timer pulse unit 3 (MTU3a), consisting of eight 16-bit timer channels and one 32-bit timer channel.

Table 24.1 shows the specifications of the MTU and Table 24.2 lists the functions of the MTU. Figure 24.1 and Figure 24.2 show block diagrams of the MTU.

**Table 24.1 MTU Specifications**

| Item                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pulse input/output             | 28 lines max.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Pulse input                    | 3 lines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Count clock                    | 11 clocks for each channel (14 clocks for MTU0, 12 clocks for MTU2, 10 clocks for MTU5, and four clocks for MTU1 & MTU2 (LWA = 1))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Available operations           | <div> [MTU0 to MTU4, MTU6, MTU7, MTU8] <ul style="list-style-type: none"> <li>• Waveform output on compare match</li> <li>• Input capture function (noise filter setting available)</li> <li>• Counter-clearing operation</li> <li>• Simultaneous writing to multiple timer counters (TCNT) (excluding MTU8)</li> <li>• Simultaneous clearing on compare match or input capture (excluding MTU8)</li> <li>• Simultaneous input and output to registers in synchronization with counter operations (excluding MTU8)</li> <li>• Up to 12-phase PWM output in combination with synchronous operation (excluding MTU8)</li> </ul> </div> <hr/> <div> [MTU0, MTU3, MTU4, MTU6, MTU7, MTU8] <ul style="list-style-type: none"> <li>• Buffer operation specifiable</li> </ul> </div> <hr/> <div> [MTU1, MTU2] <ul style="list-style-type: none"> <li>• Phase counting mode can be specified independently</li> <li>• 32-bit phase counting mode can be specified for interlocked operation of MTU1 and MTU2 (when TMDR3.LWA = 1)</li> <li>• Cascade connection operation available</li> </ul> </div> <hr/> <div> [MTU3, MTU4, MTU6, MTU7] <ul style="list-style-type: none"> <li>• Through interlocked operation of MTU3/4 and MTU6/7, the positive and negative signals in six phases (12 phases in total) can be output in complementary PWM and reset-synchronized PWM operation.</li> <li>• In complementary PWM mode, transfer of values from buffer registers to temporary registers on crests or troughs of the timer-counter values or writing to the buffer registers (MTU4.TGRD and MTU7.TGRD)</li> <li>• Double-buffering selectable in complementary PWM mode</li> </ul> </div> <hr/> <div> [MTU3, MTU4] <ul style="list-style-type: none"> <li>• Through interlocking with MTU0, a mode for driving AC synchronous motors (brushless DC motors) by using complementary PWM output and reset-synchronized PWM output is settable and allows the selection of two types of waveform output (chopping or level)</li> </ul> </div> <hr/> <div> [MTU5] <ul style="list-style-type: none"> <li>• Capable of operation as a dead-time compensation counter</li> </ul> </div> <hr/> <div> [MTU0/MTU5, MTU1, MTU2, MTU8] <p>32-bit phase counting mode specifiable by combining MTU1 and MTU2 and through interlocked operation with MTU0/MTU5 and MTU8</p> </div> |
| Interrupt skipping function    | <ul style="list-style-type: none"> <li>• In complementary PWM mode, interrupts on crests and troughs of counter values and triggers to start conversion by the A/D converter can be skipped</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Interrupt sources              | 43 sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Buffer operation               | Automatic transfer of register data (transfer from the buffer register to the timer register)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Trigger generation             | <div>A/D conversion start triggers can be generated</div> <hr/> <div>A/D conversion start request delaying function enables A/D converter to be started with any desired timing and to be synchronized with PWM output</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Low power consumption function | Module stop mode can be set                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Table 24.2 MTU Functions (1/2)

| Item                                    | MTU0                                                                                                                                                       | MTU1                                                                                                                                                       | MTU2                                                                                                                                                       | MTU1 &<br>MTU2<br>(LWA = 1)                      | MTU3                                                                                                                        | MTU4                                                                                                                        | MTU5                                                                                                                                   | MTU6                                                                                                                        | MTU7                                                                                                                        | MTU8                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Count clock                             | PCLKA/1<br>PCLKA/2<br>PCLKA/4<br>PCLKA/8<br>PCLKA/16<br>PCLKA/32<br>PCLKA/64<br>PCLKA/256<br>PCLKA/1024<br>MTCLKA<br>MTCLKB<br>MTCLKC<br>MTCLKD<br>MTIOC1A | PCLKA/1<br>PCLKA/2<br>PCLKA/4<br>PCLKA/8<br>PCLKA/16<br>PCLKA/32<br>PCLKA/64<br>PCLKA/256<br>PCLKA/1024<br>MTCLKA<br>MTCLKB<br>MTCLKC<br>MTCLKD<br>MTIOC1A | PCLKA/1<br>PCLKA/2<br>PCLKA/4<br>PCLKA/8<br>PCLKA/16<br>PCLKA/32<br>PCLKA/64<br>PCLKA/256<br>PCLKA/1024<br>MTCLKA<br>MTCLKB<br>MTCLKC<br>MTCLKD<br>MTIOC1A | MTCLKA<br>MTCLKB<br>MTCLKC<br>MTCLKD             | PCLKA/1<br>PCLKA/2<br>PCLKA/4<br>PCLKA/8<br>PCLKA/16<br>PCLKA/32<br>PCLKA/64<br>PCLKA/256<br>PCLKA/1024<br>MTCLKA<br>MTCLKB | PCLKA/1<br>PCLKA/2<br>PCLKA/4<br>PCLKA/8<br>PCLKA/16<br>PCLKA/32<br>PCLKA/64<br>PCLKA/256<br>PCLKA/1024<br>MTCLKA<br>MTCLKB | PCLKA/1<br>PCLKA/2<br>PCLKA/4<br>PCLKA/8<br>PCLKA/16<br>PCLKA/32<br>PCLKA/64<br>PCLKA/256<br>PCLKA/1024<br>MTIOC1A<br>MTCLKA<br>MTCLKB | PCLKA/1<br>PCLKA/2<br>PCLKA/4<br>PCLKA/8<br>PCLKA/16<br>PCLKA/32<br>PCLKA/64<br>PCLKA/256<br>PCLKA/1024<br>MTCLKA<br>MTCLKB | PCLKA/1<br>PCLKA/2<br>PCLKA/4<br>PCLKA/8<br>PCLKA/16<br>PCLKA/32<br>PCLKA/64<br>PCLKA/256<br>PCLKA/1024<br>MTCLKA<br>MTCLKB | PCLKA/1<br>PCLKA/2<br>PCLKA/4<br>PCLKA/8<br>PCLKA/16<br>PCLKA/32<br>PCLKA/64<br>PCLKA/256<br>PCLKA/1024<br>MTCLKA<br>MTCLKB |
| External clock for phase counting mode  | —                                                                                                                                                          | MTCLKA<br>MTCLKB                                                                                                                                           | MTCLKA<br>MTCLKB<br>MTCLKC<br>MTCLKD                                                                                                                       | MTCLKA<br>MTCLKB<br>MTCLKC<br>MTCLKD             | —                                                                                                                           | —                                                                                                                           | —                                                                                                                                      | —                                                                                                                           | —                                                                                                                           | —                                                                                                                           |
| General registers (TGR)                 | TGRA<br>TGRB<br>TGRE                                                                                                                                       | TGRA<br>TGRB                                                                                                                                               | TGRA<br>TGRB                                                                                                                                               | TGRALW<br>TGRBLW                                 | TGRA<br>TGRB                                                                                                                | TGRA<br>TGRB                                                                                                                | TGRU<br>TGRV<br>TGRW                                                                                                                   | TGRA<br>TGRB                                                                                                                | TGRA<br>TGRB                                                                                                                | TGRA<br>TGRB                                                                                                                |
| General registers/<br>buffer registers  | TGRC<br>TGRD<br>TGRF                                                                                                                                       | —                                                                                                                                                          | —                                                                                                                                                          | —                                                | TGRC<br>TGRD<br>TGRE                                                                                                        | TGRC<br>TGRD<br>TGRE<br>TGRF                                                                                                | —                                                                                                                                      | TGRC<br>TGRD<br>TGRE                                                                                                        | TGRC<br>TGRD<br>TGRE<br>TGRF                                                                                                | TGRC<br>TGRD                                                                                                                |
| I/O pins                                | MTIOC0A<br>MTIOC0B<br>MTIOC0C<br>MTIOC0D                                                                                                                   | MTIOC1A<br>MTIOC1B                                                                                                                                         | MTIOC2A<br>MTIOC2B                                                                                                                                         | MTIOC1A<br>MTIOC1B                               | MTIOC3A<br>MTIOC3B<br>MTIOC3C<br>MTIOC3D                                                                                    | MTIOC4A<br>MTIOC4B<br>MTIOC4C<br>MTIOC4D                                                                                    | MTIC5U<br>MTIC5V<br>MTIC5W                                                                                                             | MTIOC6A<br>MTIOC6B<br>MTIOC6C<br>MTIOC6D                                                                                    | MTIOC7A<br>MTIOC7B<br>MTIOC7C<br>MTIOC7D                                                                                    | MTIOC8A<br>MTIOC8B<br>MTIOC8C<br>MTIOC8D                                                                                    |
| Counter clear function                  | TGR compare match or input capture                                                                                                                         | TGR compare match or input capture                                                                                                                         | TGR compare match or input capture                                                                                                                         | TGRALW/<br>TGRBLW compare match or input capture | TGR compare match or input capture                                                                                          | TGR compare match or input capture                                                                                          | TGR compare match or input capture                                                                                                     | TGR compare match or input capture                                                                                          | TGR compare match or input capture                                                                                          | TGR compare match or input capture                                                                                          |
| Compare match output                    | 0 output<br>1 output<br>Toggle output                                                                                                                      | ✓<br>✓<br>✓                                                                                                                                                | ✓<br>✓<br>✓                                                                                                                                                | ✓<br>✓<br>✓                                      | ✓<br>✓<br>✓                                                                                                                 | ✓<br>✓<br>✓                                                                                                                 | —<br>—<br>—                                                                                                                            | ✓<br>✓<br>✓                                                                                                                 | ✓<br>✓<br>✓                                                                                                                 | ✓<br>✓<br>✓                                                                                                                 |
| Input capture function                  | ✓                                                                                                                                                          | ✓                                                                                                                                                          | ✓                                                                                                                                                          | ✓*1                                              | ✓                                                                                                                           | ✓                                                                                                                           | ✓                                                                                                                                      | ✓                                                                                                                           | ✓                                                                                                                           | ✓*2                                                                                                                         |
| Synchronous operation                   | ✓                                                                                                                                                          | ✓                                                                                                                                                          | ✓                                                                                                                                                          | —                                                | ✓                                                                                                                           | ✓                                                                                                                           | —                                                                                                                                      | ✓                                                                                                                           | ✓                                                                                                                           | —                                                                                                                           |
| PWM mode 1                              | ✓                                                                                                                                                          | ✓                                                                                                                                                          | ✓                                                                                                                                                          | —                                                | ✓                                                                                                                           | ✓                                                                                                                           | —                                                                                                                                      | ✓                                                                                                                           | ✓                                                                                                                           | —                                                                                                                           |
| PWM mode 2                              | ✓                                                                                                                                                          | ✓                                                                                                                                                          | ✓                                                                                                                                                          | —                                                | —                                                                                                                           | —                                                                                                                           | —                                                                                                                                      | —                                                                                                                           | —                                                                                                                           | —                                                                                                                           |
| Complementary PWM mode                  | —                                                                                                                                                          | —                                                                                                                                                          | —                                                                                                                                                          | —                                                | ✓                                                                                                                           | ✓                                                                                                                           | —                                                                                                                                      | ✓                                                                                                                           | ✓                                                                                                                           | —                                                                                                                           |
| Reset-synchronized PWM mode             | —                                                                                                                                                          | —                                                                                                                                                          | —                                                                                                                                                          | —                                                | ✓                                                                                                                           | ✓                                                                                                                           | —                                                                                                                                      | ✓                                                                                                                           | ✓                                                                                                                           | —                                                                                                                           |
| AC synchronous motor drive mode         | ✓                                                                                                                                                          | —                                                                                                                                                          | —                                                                                                                                                          | —                                                | ✓                                                                                                                           | ✓                                                                                                                           | —                                                                                                                                      | —                                                                                                                           | —                                                                                                                           | —                                                                                                                           |
| Phase counting mode                     | —                                                                                                                                                          | ✓                                                                                                                                                          | ✓                                                                                                                                                          | ✓                                                | —                                                                                                                           | —                                                                                                                           | —                                                                                                                                      | —                                                                                                                           | —                                                                                                                           | —                                                                                                                           |
| Buffer operation                        | ✓                                                                                                                                                          | —                                                                                                                                                          | —                                                                                                                                                          | —                                                | ✓                                                                                                                           | ✓                                                                                                                           | —                                                                                                                                      | ✓                                                                                                                           | ✓                                                                                                                           | ✓                                                                                                                           |
| Dead time compensation counter function | —                                                                                                                                                          | —                                                                                                                                                          | —                                                                                                                                                          | —                                                | —                                                                                                                           | —                                                                                                                           | ✓                                                                                                                                      | —                                                                                                                           | —                                                                                                                           | —                                                                                                                           |
| DMAC/DTC trigger sources                | TGR compare match or input capture                                                                                                                         | TGR compare match or input capture                                                                                                                         | TGR compare match or input capture                                                                                                                         | TGRALW/<br>TGRBLW compare match or input capture | TGR compare match or input capture                                                                                          | TGR compare match or input capture and TCNT overflow/underflow*3                                                            | TGR compare match or input capture                                                                                                     | TGR compare match or input capture                                                                                          | TGR compare match or input capture and TCNT overflow/underflow*3                                                            | TGR compare match or input capture                                                                                          |
| A/D conversion start trigger            | TGRA compare match or input capture<br>TGRE compare match                                                                                                  | TGRA compare match or input capture                                                                                                                        | TGRA compare match or input capture                                                                                                                        | TGRALW compare match or input capture            | TGRA compare match or input capture                                                                                         | TGRA compare match or input capture, or TCNT underflow (trough) in complementary PWM mode                                   | —                                                                                                                                      | TGRA compare match or input capture                                                                                         | TGRA compare match or input capture, or TCNT underflow (trough) in complementary PWM mode                                   | —                                                                                                                           |

**Table 24.2 MTU Functions (2/2)**

| Item                                           | MTU0                                                                                                                                                                                                                                | MTU1                                                                                                                    | MTU2                                                                                                                    | MTU1 & MTU2 (LWA = 1)                                                                                                   | MTU3                                                                                                                                                                                   | MTU4                                                                                                                                                                                                  | MTU5                                                                                                                               | MTU6                                                                                                                                                                                   | MTU7                                                                                                                                                                                                  | MTU8                                                                                                                                                                                   |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interrupt sources                              | Seven sources<br>• Compare match or input capture 0A<br>• Compare match or input capture 0B<br>• Compare match or input capture 0C<br>• Compare match or input capture 0D<br>• Compare match 0E<br>• Compare match 0F<br>• Overflow | Four sources<br>• Compare match or input capture 1A<br>• Compare match or input capture 1B<br>• Overflow<br>• Underflow | Four sources<br>• Compare match or input capture 2A<br>• Compare match or input capture 2B<br>• Overflow<br>• Underflow | Four sources<br>• Compare match or input capture 1A<br>• Compare match or input capture 1B<br>• Overflow<br>• Underflow | Five sources<br>• Compare match or input capture 3A<br>• Compare match or input capture 3B<br>• Compare match or input capture 3C<br>• Compare match or input capture 3D<br>• Overflow | Five sources<br>• Compare match or input capture 4A<br>• Compare match or input capture 4B<br>• Compare match or input capture 4C<br>• Compare match or input capture 4D<br>• Overflow or underflow*3 | Three sources<br>• Compare match or input capture 5U<br>• Compare match or input capture 5V<br>• Compare match or input capture 5W | Five sources<br>• Compare match or input capture 6A<br>• Compare match or input capture 6B<br>• Compare match or input capture 6C<br>• Compare match or input capture 6D<br>• Overflow | Five sources<br>• Compare match or input capture 7A<br>• Compare match or input capture 7B<br>• Compare match or input capture 7C<br>• Compare match or input capture 7D<br>• Overflow or underflow*3 | Five sources<br>• Compare match or input capture 8A<br>• Compare match or input capture 8B<br>• Compare match or input capture 8C<br>• Compare match or input capture 8D<br>• Overflow |
| Event link function (output)                   | Seven sources<br>• Compare match 0A<br>• Compare match 0B<br>• Compare match 0C<br>• Compare match 0D<br>• Compare match 0E<br>• Compare match 0F<br>• Overflow                                                                     | —                                                                                                                       | —                                                                                                                       | —                                                                                                                       | Five sources<br>• Compare match 3A<br>• Compare match 3B<br>• Compare match 3C<br>• Compare match 3D<br>• Overflow                                                                     | Six sources<br>• Compare match 4A<br>• Compare match 4B<br>• Compare match 4C<br>• Compare match 4D<br>• Overflow<br>• Underflow*3                                                                    | —                                                                                                                                  | —                                                                                                                                                                                      | —                                                                                                                                                                                                     | —                                                                                                                                                                                      |
| Event link function (input)                    | • Start counting<br>• Input capture (to be captured in the TGRA)<br>• Restart counting (clear counter)                                                                                                                              | —                                                                                                                       | —                                                                                                                       | —                                                                                                                       | • Start counting<br>• Input capture (to be captured in the TGRA)<br>• Restart counting (clear counter)                                                                                 | • Start counting<br>• Input capture (to be captured in the TGRA)<br>• Restart counting (clear counter)                                                                                                | —                                                                                                                                  | —                                                                                                                                                                                      | —                                                                                                                                                                                                     | —                                                                                                                                                                                      |
| A/D conversion start request delaying function | —                                                                                                                                                                                                                                   | —                                                                                                                       | —                                                                                                                       | —                                                                                                                       | —                                                                                                                                                                                      | A/D conversion start request at a match between TADCORA and TCNT or A/D conversion start request at a match between TADCORB and TCNT                                                                  | —                                                                                                                                  | —                                                                                                                                                                                      | A/D conversion start request at a match between TADCORA and TCNT or A/D conversion start request at a match between TADCORB and TCNT                                                                  | —                                                                                                                                                                                      |
| Interrupt skipping 1                           | —                                                                                                                                                                                                                                   | —                                                                                                                       | —                                                                                                                       | —                                                                                                                       | Skips TGRA compare match interrupts                                                                                                                                                    | Skips TCIV interrupts                                                                                                                                                                                 | —                                                                                                                                  | Skips TGRA compare match interrupts                                                                                                                                                    | Skips TCIV interrupts                                                                                                                                                                                 | —                                                                                                                                                                                      |
| Interrupt skipping 2                           | —                                                                                                                                                                                                                                   | —                                                                                                                       | —                                                                                                                       | —                                                                                                                       | —                                                                                                                                                                                      | Skipping in compare count between TADCORA and TCNT, and TADCORB and TCNT                                                                                                                              | —                                                                                                                                  | —                                                                                                                                                                                      | Skipping in compare count between TADCORA and TCNT, and TADCORB and TCNT                                                                                                                              | —                                                                                                                                                                                      |
| Module stop function                           | MSTPCRA.MSTPA9*4                                                                                                                                                                                                                    |                                                                                                                         |                                                                                                                         |                                                                                                                         |                                                                                                                                                                                        |                                                                                                                                                                                                       |                                                                                                                                    |                                                                                                                                                                                        |                                                                                                                                                                                                       |                                                                                                                                                                                        |

✓: Possible —: Not possible

Note 1. When LWA is 1, the TGRALW capture source can be selected from either of the following: an input from MTIOC1A or MTU0.TGRA compare match/input capture event.

The TGRBLW capture source can be selected from any of the following: an input from MTIOC1B, MTU0.TGRC compare match/input capture event, or MTU8.TGRC compare match event.

Note 2. Capture in MTU8 is supported only in normal mode.

Note 3. Underflow is available only in complementary PWM mode.

Note 4. For details on the module stop function, refer to section 11, Low Power Consumption.

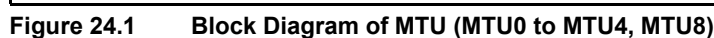




Table 24.3 shows the configuration of pins for the MTU.

**Table 24.3 Pin Configuration of the MTU**

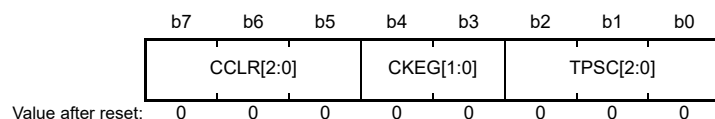
| Channel | Pin Name | I/O   | Function                                                                     |
|---------|----------|-------|------------------------------------------------------------------------------|
| MTU     | MTCLKA   | Input | External clock A input pin (MTU1 and MTU2 phase counting mode A phase input) |
|         | MTCLKB   | Input | External clock B input pin (MTU1 and MTU2 phase counting mode B phase input) |
|         | MTCLKC   | Input | External clock C input pin (MTU2 phase counting mode A phase input)          |
|         | MTCLKD   | Input | External clock D input pin (MTU2 phase counting mode B phase input)          |
| MTU0    | MTIOC0A  | I/O   | MTU0 TGRA input capture input/output compare output/PWM output pin           |
|         | MTIOC0B  | I/O   | MTU0 TGRB input capture input/output compare output/PWM output pin           |
|         | MTIOC0C  | I/O   | MTU0 TGRC input capture input/output compare output/PWM output pin           |
|         | MTIOC0D  | I/O   | MTU0 TGRD input capture input/output compare output/PWM output pin           |
| MTU1    | MTIOC1A  | I/O   | MTU1 TGRA input capture input/output compare output/PWM output pin           |
|         | MTIOC1B  | I/O   | MTU1 TGRB input capture input/output compare output/PWM output pin           |
| MTU2    | MTIOC2A  | I/O   | MTU2 TGRA input capture input/output compare output/PWM output pin           |
|         | MTIOC2B  | I/O   | MTU2 TGRB input capture input/output compare output/PWM output pin           |
| MTU3    | MTIOC3A  | I/O   | MTU3 TGRA input capture input/output compare output/PWM output pin           |
|         | MTIOC3B  | I/O   | MTU3 TGRB input capture input/output compare output/PWM output pin           |
|         | MTIOC3C  | I/O   | MTU3 TGRC input capture input/output compare output/PWM output pin           |
|         | MTIOC3D  | I/O   | MTU3 TGRD input capture input/output compare output/PWM output pin           |
| MTU4    | MTIOC4A  | I/O   | MTU4 TGRA input capture input/output compare output/PWM output pin           |
|         | MTIOC4B  | I/O   | MTU4 TGRB input capture input/output compare output/PWM output pin           |
|         | MTIOC4C  | I/O   | MTU4 TGRC input capture input/output compare output/PWM output pin           |
|         | MTIOC4D  | I/O   | MTU4 TGRD input capture input/output compare output/PWM output pin           |
| MTU5    | MTIC5U   | Input | MTU5 TGRU input capture input/external pulse input pin                       |
|         | MTIC5V   | Input | MTU5 TGRV input capture input/external pulse input pin                       |
|         | MTIC5W   | Input | MTU5 TGRW input capture input/external pulse input pin                       |
| MTU6    | MTIOC6A  | I/O   | MTU6 TGRA input capture input/output compare output/PWM output pin           |
|         | MTIOC6B  | I/O   | MTU6 TGRB input capture input/output compare output/PWM output pin           |
|         | MTIOC6C  | I/O   | MTU6 TGRC input capture input/output compare output/PWM output pin           |
|         | MTIOC6D  | I/O   | MTU6 TGRD input capture input/output compare output/PWM output pin           |
| MTU7    | MTIOC7A  | I/O   | MTU7 TGRA input capture input/output compare output/PWM output pin           |
|         | MTIOC7B  | I/O   | MTU7 TGRB input capture input/output compare output/PWM output pin           |
|         | MTIOC7C  | I/O   | MTU7 TGRC input capture input/output compare output/PWM output pin           |
|         | MTIOC7D  | I/O   | MTU7 TGRD input capture input/output compare output/PWM output pin           |
| MTU8    | MTIOC8A  | I/O   | MTU8.TGRA input capture input/output compare output pin                      |
|         | MTIOC8B  | I/O   | MTU8.TGRB input capture input/output compare output pin                      |
|         | MTIOC8C  | I/O   | MTU8.TGRC input capture input/output compare output pin                      |
|         | MTIOC8D  | I/O   | MTU8.TGRD input capture input/output compare output pin                      |

## 24.2 Register Descriptions

### 24.2.1 Timer Control Register (TCR)

- MTU0.TCR, MTU1.TCR, MTU2.TCR, MTU3.TCR, MTU4.TCR, MTU6.TCR, MTU7.TCR, MTU8.TCR

Address(es): MTU0.TCR 000C 1300h, MTU1.TCR 000C 1380h, MTU2.TCR 000C 1400h, MTU3.TCR 000C 1200h, MTU4.TCR 000C 1201h, MTU6.TCR 000C 1A00h, MTU7.TCR 000C 1A01h, MTU8.TCR 000C 1600h



| Bit      | Symbol    | Bit Name                    | Description                                                                                  | R/W |
|----------|-----------|-----------------------------|----------------------------------------------------------------------------------------------|-----|
| b2 to b0 | TPSC[2:0] | Time Prescaler Select       | Refer to Table 24.6 to Table 24.9.                                                           | R/W |
| b4, b3   | CKEG[1:0] | Clock Edge Select           | b4 b3<br>0 0: Count at rising edge<br>0 1: Count at falling edge<br>1 x: Count at both edges | R/W |
| b7 to b5 | CCLR[2:0] | Counter Clear Source Select | Refer to Table 24.4 and Table 24.5.                                                          | R/W |

x: Don't care

The TCR register controls the TCNT operation for each channel in combination with the TCR2 register. The MTU has a total of 11 TCR registers, one each for MTU0 to MTU4, MTU6, MTU7, and MTU8 and three (TCRU, TCRV, and TCRW) for MTU5. TCR values should be specified only while TCNT operation is stopped.

#### TPSC[2:0] Bits (Time Prescaler Select)

These bits select the TCNT count clock source. The count clock source can be selected independently for each channel. Refer to Table 24.6 to Table 24.9 for details.

#### CKEG[1:0] Bits (Clock Edge Select)

These bits select the clock edge, including the MTIOC1A pin. When the internal clock is counted at both edges, the count clock period is halved (e.g. PCLKA/4 at both edges = PCLKA/2 at rising edge). If phase counting mode is used on MTU1 and MTU2, the setting of these bits is ignored and the phase counting mode setting has priority. Internal clock edge selection is valid when the count clock source is PCLKA/2 or slower. When PCLKA/1 or the overflow/underflow in another channel is selected for the count clock source, a value can be written to these bits but counter operation compiles with the initial value.

#### CCLR[2:0] Bits (Counter Clear Source Select)

These bits select the TCNT counter clearing source. Refer to Table 24.4 and Table 24.5 for details.

**Table 24.4 CCLR[2:0] (MTU0, MTU3, MTU4, MTU6, MTU7, MTU8)**

|         | Bit 7   | Bit 6   | Bit 5   |                                                                                                             |
|---------|---------|---------|---------|-------------------------------------------------------------------------------------------------------------|
| Channel | CCLR[2] | CCLR[1] | CCLR[0] | Description                                                                                                 |
| MTU0    | 0       | 0       | 0       | TCNT clearing disabled                                                                                      |
| MTU3    | 0       | 0       | 1       | TCNT cleared by TGRA compare match/input capture                                                            |
| MTU4    | 0       | 1       | 0       | TCNT cleared by TGRB compare match/input capture                                                            |
| MTU6    | 0       | 1       | 1       | TCNT cleared by counter clearing in another channel performing synchronous clearing/synchronous operation*1 |
| MTU7    | 1       | 0       | 0       | TCNT clearing disabled                                                                                      |
| MTU8    | 1       | 0       | 1       | TCNT cleared by TGRC compare match/input capture*2                                                          |
|         | 1       | 1       | 0       | TCNT cleared by TGRD compare match/input capture*2                                                          |
|         | 1       | 1       | 1       | TCNT cleared by counter clearing in another channel performing synchronous clearing/synchronous operation*1 |

Note 1. Synchronous operation is selected by setting the TSYRA.SYNCn or TSYRB.SYNCn bit to 1 except for MTU8.

Note 2. When TGRC or TGRD is used as a buffer register, TCNT is not cleared because the buffer register setting has priority and compare match/input capture does not occur.

**Table 24.5 CCLR[2:0] (MTU1 and MTU2)**

|         | Bit 7      | Bit 6   | Bit 5   |                                                                                                                                        |
|---------|------------|---------|---------|----------------------------------------------------------------------------------------------------------------------------------------|
| Channel | Reserved*2 | CCLR[1] | CCLR[0] | Description                                                                                                                            |
| MTU1    | 0          | 0       | 0       | TCNT clearing disabled                                                                                                                 |
| MTU2    | 0          | 0       | 1       | TCNT cleared by TGRA compare match/input capture (when LWA = 0)<br>TCNTLW cleared by TGRALW compare match/input capture (when LWA = 1) |
|         | 0          | 1       | 0       | TCNT cleared by TGRB compare match/input capture (when LWA = 0)<br>TCNTLW cleared by TGRBLW compare match/input capture (when LWA = 1) |
|         | 0          | 1       | 1       | TCNT cleared by counter clearing in another channel performing synchronous clearing/synchronous operation*1                            |

Note 1. Synchronous operation is selected by setting the TSYRA.SYNCn and TSYRB.SYNCn bits to 1.

Note 2. Bit 7 is reserved in MTU1 and MTU2. It is read as 0. The write value is ignored.

- MTU5.TCRU, MTU5.TCRV, MTU5.TCRW

Address(es): MTU5.TCRU 000C 1C84h, MTU5.TCRV 000C 1C94h, MTU5.TCRW 000C 1CA4h

|                    | b7 | b6 | b5 | b4 | b3 | b2 | b1        | b0 |
|--------------------|----|----|----|----|----|----|-----------|----|
|                    | —  | —  | —  | —  | —  | —  | TPSC[1:0] | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0         | 0  |

| Bit      | Symbol    | Bit Name              | Description                                            | R/W |
|----------|-----------|-----------------------|--------------------------------------------------------|-----|
| b1, b0   | TPSC[1:0] | Time Prescaler Select | Refer to Table 24.10.                                  | R/W |
| b7 to b2 | —         | Reserved              | These bits are read as 0. The write value should be 0. | R/W |

**TPSC[1:0] Bits (Time Prescaler Select)**

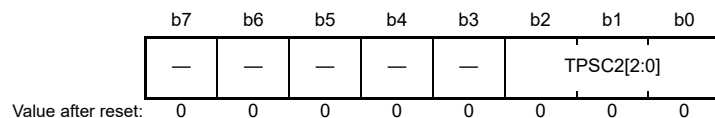
These bits select the TCNT count clock source. Refer to Table 24.10 for details.



### 24.2.2 Timer Control Register 2 (TCR2)

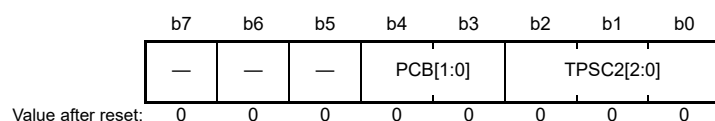
- MTU0.TCR2, MTU3.TCR2, MTU4.TCR2, MTU6.TCR2, MTU7.TCR2, MTU8.TCR2

Address(es): MTU0.TCR2 000C 1328h, MTU3.TCR2 000C 124Ch, MTU4.TCR2 000C 124Dh, MTU6.TCR2 000C 1A4Ch, MTU7.TCR2 000C 1A4Dh, MTU8.TCR2 000C 1606h



- MTU1.TCR2, MTU2.TCR2

Address(es): MTU1.TCR2 000C 1394h, MTU2.TCR2 000C 140Ch



| Bit      | Symbol     | Bit Name                                       | Description                                                       | R/W |
|----------|------------|------------------------------------------------|-------------------------------------------------------------------|-----|
| b2 to b0 | TPSC2[2:0] | Time Prescaler Select                          | Refer to Table 24.6 to Table 24.9.                                | R/W |
| b4, b3   | PCB[1:0]   | Phase Counting Mode Function Expansion Control | Functional Expansion Control for Phase Counting Modes 2, 3, and 5 | R/W |
| b7 to b5 | —          | Reserved                                       | These bits are read as 0. The write value should be 0.            | R/W |

The TCR2 register controls the TCNT operation for each channel in combination with the TCR register. The MTU has a total of 11 TCR2 registers, one each for MTU0 to MTU4, MTU6, MTU7, and MTU8 and three (TCR2U, TCR2V, and TCR2W) for MTU5. TCR2 values should be specified only while TCNT operation is stopped.

#### TPSC2[2:0] Bits (Time Prescaler Select)

These bits select the TCNT count clock source. The count clock source can be selected independently for each channel. Refer to Table 24.6 to Table 24.9 for details.

#### PCB[1:0] Bits (Phase Counting Mode Function Expansion Control)

These bits control extended functions for phase counting mode 2, 3, and 5 in MTU1 and MTU2. Refer to section 24.3.6, Phase Counting Mode.

- MTU5.TCR2U, MTU5.TCR2V, MTU5.TCR2W

Address(es): MTU5.TCR2U 000C 1C85h, MTU5.TCR2V 000C 1C95h, MTU5.TCR2W 000C 1CA5h

| b7 | b6 | b5 | b4        | b3 | b2         | b1 | b0 |
|----|----|----|-----------|----|------------|----|----|
| —  | —  | —  | CKEG[1:0] |    | TPSC2[2:0] |    |    |
| 0  | 0  | 0  | 0         | 0  | 0          | 0  | 0  |

Value after reset:

| Bit      | Symbol     | Bit Name              | Description                                                                                                | R/W |
|----------|------------|-----------------------|------------------------------------------------------------------------------------------------------------|-----|
| b2 to b0 | TPSC2[2:0] | Time Prescaler Select | Refer to Table 24.10.                                                                                      | R/W |
| b4, b3   | CKEG[1:0]  | Clock Edge Select     | b4 b3<br>0 0: Counts at the rising edge.<br>0 1: Counts at the falling edge.<br>1 x: Counts at both edges. | R/W |
| b7 to b5 | —          | Reserved              | These bits are read as 0. The write value should be 0.                                                     | R/W |

x: Don't care

### TPSC2[2:0] Bits (Time Prescaler Select)

These bits select the TCNT count clock source. Refer to Table 24.10 for details.

### CKEG[1:0] Bits (Clock Edge Select)

These bits select the edge of the count clock signal input from the MTIOC1A pin.

**Table 24.6 TPSC[2:0], TPSC2[2:0] (MTU0)**

| Channel | TCR2 register |          |          | TCR register |         |         | Description                                 |
|---------|---------------|----------|----------|--------------|---------|---------|---------------------------------------------|
|         | Bit 2         | Bit 1    | Bit 0    | Bit 2        | Bit 1   | Bit 0   |                                             |
|         | TPSC2[2]      | TPSC2[1] | TPSC2[0] | TPSC[2]      | TPSC[1] | TPSC[0] |                                             |
| MTU0    | 0             | 0        | 0        | 0            | 0       | 0       | Internal clock: counts on PCLKA/1           |
|         | 0             | 0        | 0        | 0            | 0       | 1       | Internal clock: counts on PCLKA/4           |
|         | 0             | 0        | 0        | 0            | 1       | 0       | Internal clock: counts on PCLKA/16          |
|         | 0             | 0        | 0        | 0            | 1       | 1       | Internal clock: counts on PCLKA/64          |
|         | 0             | 0        | 0        | 1            | 0       | 0       | External clock: counts on MTCLKA pin input  |
|         | 0             | 0        | 0        | 1            | 0       | 1       | External clock: counts on MTCLKB pin input  |
|         | 0             | 0        | 0        | 1            | 1       | 0       | External clock: counts on MTCLKC pin input  |
|         | 0             | 0        | 0        | 1            | 1       | 1       | External clock: counts on MTCLKD pin input  |
|         | 0             | 0        | 1        | x            | x       | x       | Internal clock: counts on PCLKA/2           |
|         | 0             | 1        | 0        | x            | x       | x       | Internal clock: counts on PCLKA/8           |
|         | 0             | 1        | 1        | x            | x       | x       | Internal clock: counts on PCLKA/32          |
|         | 1             | 0        | 0        | x            | x       | x       | Internal clock: counts on PCLKA/256         |
|         | 1             | 0        | 1        | x            | x       | x       | Internal clock: counts on PCLKA/1024        |
|         | 1             | 1        | 0        | x            | x       | x       | Setting prohibited                          |
|         | 1             | 1        | 1        | x            | x       | x       | External clock: counts on MTIOC1A pin input |

x: Don't care

**Table 24.7 TPSC[2:0], TPSC2[2:0] (MTU1)**

| Channel | TCR2 register |          |          | TCR register |         |         | Description                                |
|---------|---------------|----------|----------|--------------|---------|---------|--------------------------------------------|
|         | Bit 2         | Bit 1    | Bit 0    | Bit 2        | Bit 1   | Bit 0   |                                            |
| Channel | TPSC2[2]      | TPSC2[1] | TPSC2[0] | TPSC[2]      | TPSC[1] | TPSC[0] | Description                                |
| MTU1    | 0             | 0        | 0        | 0            | 0       | 0       | Internal clock: counts on PCLKA/1          |
|         | 0             | 0        | 0        | 0            | 0       | 1       | Internal clock: counts on PCLKA/4          |
|         | 0             | 0        | 0        | 0            | 1       | 0       | Internal clock: counts on PCLKA/16         |
|         | 0             | 0        | 0        | 0            | 1       | 1       | Internal clock: counts on PCLKA/64         |
|         | 0             | 0        | 0        | 1            | 0       | 0       | External clock: counts on MTCLKA pin input |
|         | 0             | 0        | 0        | 1            | 0       | 1       | External clock: counts on MTCLKB pin input |
|         | 0             | 0        | 0        | 1            | 1       | 0       | Internal clock: counts on PCLKA/256        |
|         | 0             | 0        | 0        | 1            | 1       | 1       | Overflow/underflow of MTU2.TCNT            |
|         | 0             | 0        | 1        | x            | x       | x       | Internal clock: counts on PCLKA/2          |
|         | 0             | 1        | 0        | x            | x       | x       | Internal clock: counts on PCLKA/8          |
|         | 0             | 1        | 1        | x            | x       | x       | Internal clock: counts on PCLKA/32         |
|         | 1             | 0        | 0        | x            | x       | x       | Internal clock: counts on PCLKA/1024       |
|         | 1             | 0        | 1        | x            | x       | x       | Setting prohibited                         |
|         | 1             | 1        | 0        | x            | x       | x       | Setting prohibited                         |
|         | 1             | 1        | 1        | x            | x       | x       | Setting prohibited                         |

x: Don't care

Note: This setting has no effect when MTU1 is in phase counting mode.

**Table 24.8 TPSC[2:0], TPSC2[2:0] (MTU2)**

| Channel | TCR2 register |          |          | TCR register |         |         | Description                                |
|---------|---------------|----------|----------|--------------|---------|---------|--------------------------------------------|
|         | Bit 2         | Bit 1    | Bit 0    | Bit 2        | Bit 1   | Bit 0   |                                            |
| Channel | TPSC2[2]      | TPSC2[1] | TPSC2[0] | TPSC[2]      | TPSC[1] | TPSC[0] | Description                                |
| MTU2    | 0             | 0        | 0        | 0            | 0       | 0       | Internal clock: counts on PCLKA/1          |
|         | 0             | 0        | 0        | 0            | 0       | 1       | Internal clock: counts on PCLKA/4          |
|         | 0             | 0        | 0        | 0            | 1       | 0       | Internal clock: counts on PCLKA/16         |
|         | 0             | 0        | 0        | 0            | 1       | 1       | Internal clock: counts on PCLKA/64         |
|         | 0             | 0        | 0        | 1            | 0       | 0       | External clock: counts on MTCLKA pin input |
|         | 0             | 0        | 0        | 1            | 0       | 1       | External clock: counts on MTCLKB pin input |
|         | 0             | 0        | 0        | 1            | 1       | 0       | External clock: counts on MTCLKC pin input |
|         | 0             | 0        | 0        | 1            | 1       | 1       | Internal clock: counts on PCLKA/1024       |
|         | 0             | 0        | 1        | x            | x       | x       | Internal clock: counts on PCLKA/2          |
|         | 0             | 1        | 0        | x            | x       | x       | Internal clock: counts on PCLKA/8          |
|         | 0             | 1        | 1        | x            | x       | x       | Internal clock: counts on PCLKA/32         |
|         | 1             | 0        | 0        | x            | x       | x       | Internal clock: counts on PCLKA/256        |
|         | 1             | 0        | 1        | x            | x       | x       | Setting prohibited                         |
|         | 1             | 1        | 0        | x            | x       | x       | Setting prohibited                         |
|         | 1             | 1        | 1        | x            | x       | x       | Setting prohibited                         |

x: Don't care

Note: This setting has no effect when the MTU2 is in phase counting mode.

**Table 24.9 TPSC[2:0], TPSC2[2:0] (MTU3, MTU4, MTU6, MTU7, MTU8)**

| Channel | TCR2 register |          |          | TCR register |         |         | Description                                |
|---------|---------------|----------|----------|--------------|---------|---------|--------------------------------------------|
|         | Bit 2         | Bit 1    | Bit 0    | Bit 2        | Bit 1   | Bit 0   |                                            |
|         | TPSC2[2]      | TPSC2[1] | TPSC2[0] | TPSC[2]      | TPSC[1] | TPSC[0] |                                            |
| MTU3    | 0             | 0        | 0        | 0            | 0       | 0       | Internal clock: counts on PCLKA/1          |
| MTU4    | 0             | 0        | 0        | 0            | 0       | 1       | Internal clock: counts on PCLKA/4          |
| MTU6    | 0             | 0        | 0        | 0            | 1       | 0       | Internal clock: counts on PCLKA/16         |
| MTU7    | 0             | 0        | 0        | 0            | 1       | 1       | Internal clock: counts on PCLKA/64         |
| MTU8    | 0             | 0        | 0        | 1            | 0       | 0       | Internal clock: counts on PCLKA/256        |
|         | 0             | 0        | 0        | 1            | 0       | 1       | Internal clock: counts on PCLKA/1024       |
|         | 0             | 0        | 0        | 1            | 1       | 0       | External clock: counts on MTCLKA pin input |
|         | 0             | 0        | 0        | 1            | 1       | 1       | External clock: counts on MTCLKB pin input |
|         | 0             | 0        | 1        | x            | x       | x       | Internal clock: counts on PCLKA/2          |
|         | 0             | 1        | 0        | x            | x       | x       | Internal clock: counts on PCLKA/8          |
|         | 0             | 1        | 1        | x            | x       | x       | Internal clock: counts on PCLKA/32         |
|         | 1             | 0        | 0        | x            | x       | x       | Setting prohibited                         |
|         | 1             | 0        | 1        | x            | x       | x       | Setting prohibited                         |
|         | 1             | 1        | 0        | x            | x       | x       | Setting prohibited                         |
|         | 1             | 1        | 1        | x            | x       | x       | Setting prohibited                         |

x: Don't care

**Table 24.10 TPSC[1:0], TPSC2[2:0] (MTU5)**

| Channel | TCR2 register |          |          | TCR register |         | Description                                 |
|---------|---------------|----------|----------|--------------|---------|---------------------------------------------|
|         | Bit 2         | Bit 1    | Bit 0    | Bit 1        | Bit 0   |                                             |
|         | TPSC2[2]      | TPSC2[1] | TPSC2[0] | TPSC[1]      | TPSC[0] |                                             |
| MTU5    | 0             | 0        | 0        | 0            | 0       | Internal clock: counts on PCLKA/1           |
|         | 0             | 0        | 0        | 0            | 1       | Internal clock: counts on PCLKA/4           |
|         | 0             | 0        | 0        | 1            | 0       | Internal clock: counts on PCLKA/16          |
|         | 0             | 0        | 0        | 1            | 1       | Internal clock: counts on PCLKA/64          |
|         | 0             | 0        | 1        | x            | x       | Internal clock: counts on PCLKA/2           |
|         | 0             | 1        | 0        | x            | x       | Internal clock: counts on PCLKA/8           |
|         | 0             | 1        | 1        | x            | x       | Internal clock: counts on PCLKA/32          |
|         | 1             | 0        | 0        | x            | x       | Internal clock: counts on PCLKA/256         |
|         | 1             | 0        | 1        | x            | x       | Internal clock: counts on PCLKA/1024        |
|         | 1             | 1        | 0        | x            | x       | Setting prohibited                          |
|         | 1             | 1        | 1        | x            | x       | External clock: counts on MTIOC1A pin input |

x: Don't care

### 24.2.3 Timer Mode Register 1 (TMDR1)

- MTU0.TMDR1

Address(es): MTU0.TMDR1 000C 1301h

| b7                 | b6  | b5  | b4  | b3      | b2 | b1 | b0 |
|--------------------|-----|-----|-----|---------|----|----|----|
| —                  | BFE | BFB | BFA | MD[3:0] |    |    |    |
| Value after reset: | 0   | 0   | 0   | 0       | 0  | 0  | 0  |

- MTU1.TMDR1, MTU2.TMDR1

Address(es): MTU1.TMDR1 000C 1381h, MTU2.TMDR1 000C 1401h

| b7                 | b6 | b5 | b4 | b3      | b2 | b1 | b0 |
|--------------------|----|----|----|---------|----|----|----|
| —                  | —  | —  | —  | MD[3:0] |    |    |    |
| Value after reset: | 0  | 0  | 0  | 0       | 0  | 0  | 0  |

- MTU3.TMDR1, MTU4.TMDR1, MTU6.TMDR1, MTU7.TMDR1, MTU8.TMDR1

Address(es): MTU3.TMDR1 000C 1202h, MTU4.TMDR1 000C 1203h, MTU6.TMDR1 000C 1A02h, MTU7.TMDR1 000C 1A03h, MTU8.TMDR1 000C 1601h

| b7                 | b6 | b5  | b4  | b3      | b2 | b1 | b0 |
|--------------------|----|-----|-----|---------|----|----|----|
| —                  | —  | BFB | BFA | MD[3:0] |    |    |    |
| Value after reset: | 0  | 0   | 0   | 0       | 0  | 0  | 0  |

| Bit      | Symbol  | Bit Name           | Description                                                                                                  | R/W |
|----------|---------|--------------------|--------------------------------------------------------------------------------------------------------------|-----|
| b3 to b0 | MD[3:0] | Mode Select        | These bits specify the timer operating mode. Refer to Table 24.11 for details.                               | R/W |
| b4       | BFA     | Buffer Operation A | 0: TGRA and TGRC operate normally<br>1: TGRA and TGRC used together for buffer operation                     | R/W |
| b5       | BFB     | Buffer Operation B | 0: TGRB and TGRD operate normally<br>1: TGRB and TGRD used together for buffer operation                     | R/W |
| b6       | BFE     | Buffer Operation E | 0: MTU0.TGRE and MTU0.TGRF operate normally<br>1: MTU0.TGRE and MTU0.TGRF used together for buffer operation | R/W |
| b7       | —       | Reserved           | This bit is read as 0. The write value should be 0.                                                          | R/W |

The TMDR1 register specifies the operating mode of each channel. The MTU has a total of eight TMDR1 registers, one each for MTU0 to MTU4, MTU6, MTU7, and MTU8. TMDR1 register values should be specified only while TCNT operation is stopped.

**Table 24.11 Operating Mode Setting by MD[3:0] Bits (MTU0 to MTU4 and MTU6 to MTU8)**

| Bit 3 | Bit 2 | Bit 1 | Bit 0 |                                                           | MTU0 | MTU1 | MTU2 | MTU1 & MTU2<br>(LWA = 1) | MTU3 | MTU4 | MTU6 | MTU7 | MTU8 |
|-------|-------|-------|-------|-----------------------------------------------------------|------|------|------|--------------------------|------|------|------|------|------|
| MD[3] | MD[2] | MD[1] | MD[0] | Description                                               |      |      |      |                          |      |      |      |      |      |
| 0     | 0     | 0     | 0     | Normal mode                                               | ✓    | ✓    | ✓    |                          | ✓    | ✓    | ✓    | ✓    | ✓    |
| 0     | 0     | 0     | 1     | Setting prohibited                                        |      |      |      |                          |      |      |      |      |      |
| 0     | 0     | 1     | 0     | PWM mode 1                                                | ✓    | ✓    | ✓    |                          | ✓    | ✓    | ✓    | ✓    |      |
| 0     | 0     | 1     | 1     | PWM mode 2                                                | ✓    | ✓    | ✓    |                          |      |      |      |      |      |
| 0     | 1     | 0     | 0     | Phase counting mode 1                                     |      | ✓    | ✓    | ✓                        |      |      |      |      |      |
| 0     | 1     | 0     | 1     | Phase counting mode 2                                     |      | ✓    | ✓    | ✓                        |      |      |      |      |      |
| 0     | 1     | 1     | 0     | Phase counting mode 3                                     |      | ✓    | ✓    | ✓                        |      |      |      |      |      |
| 0     | 1     | 1     | 1     | Phase counting mode 4                                     |      | ✓    | ✓    | ✓                        |      |      |      |      |      |
| 1     | 0     | 0     | 0     | Reset-synchronized PWM mode*1                             |      |      |      |                          | ✓    |      | ✓    |      |      |
| 1     | 0     | 0     | 1     | Phase counting mode 5                                     |      | ✓    | ✓    | ✓                        |      |      |      |      |      |
| 1     | 0     | 1     | x     | Setting prohibited                                        |      |      |      |                          |      |      |      |      |      |
| 1     | 1     | 0     | 0     | Setting prohibited                                        |      |      |      |                          |      |      |      |      |      |
| 1     | 1     | 0     | 1     | Complementary PWM mode 1 (transfer at crest)*1            |      |      |      |                          | ✓    |      | ✓    |      |      |
| 1     | 1     | 1     | 0     | Complementary PWM mode 2 (transfer at trough)*1           |      |      |      |                          | ✓    |      | ✓    |      |      |
| 1     | 1     | 1     | 1     | Complementary PWM mode 3 (transfer at crest and trough)*1 |      |      |      |                          | ✓    |      | ✓    |      |      |

x: Don't care

Note: Only set the corresponding operating mode listed above for each channel.

Note 1. Reset-synchronized PWM mode and complementary PWM mode can only be set for MTU3 and MTU6.

When MTU3 or MTU6 is set to reset-synchronized PWM mode or complementary PWM mode, the MTU4 or MTU7 settings become ineffective and automatically conform to the MTU3 or MTU6 setting, respectively. MTU4 and MTU7 should be set to the initial values (normal mode).

### BFA Bit (Buffer Operation A)

This bit specifies whether to operate TGRA in the normal way or to use TGRA and TGRC together for buffer operation. When TGRC is used as a buffer register, TGRC input capture/output compare does not take place in modes other than complementary PWM mode, but compare match with TGRC occurs in complementary PWM mode.

If a compare match occurs on MTU4 in the Tb interval in complementary PWM mode, the TGIEC bit in timer interrupt enable register (MTU4.TIER) should be set to 0.

In reset-synchronized PWM mode or complementary PWM mode, buffer operation in MTU3 and MTU4 (or MTU6 and MTU7) depends on the settings in the BFA bit of MTU3.TMDR1 (MTU6.TMDR1). The BFA bit of MTU4.TMDR1 (MTU7.TMDR1) should be set to 0.

In MTU1 and MTU2, which have no TGRC, this bit is reserved. It is read as 0. The write value should be 0.

Refer to Figure 24.50 for an illustration of the Tb interval in complementary PWM mode.

### BFB Bit (Buffer Operation B)

This bit specifies whether to operate TGRB in the normal way or to use TGRB and TGRD together for buffer operation. When TGRD is used as a buffer register, TGRD input capture/output compare does not take place in modes other than complementary PWM mode, but compare match with TGRD occurs in complementary PWM mode.

If a compare match occurs on MTU4 in the Tb interval in complementary PWM mode, the TGIED bit in timer interrupt enable register (MTU4.TIER) should be set to 0.

In reset-synchronized PWM mode or complementary PWM mode, buffer operation in MTU3 and MTU4 (or MTU6 and MTU7) depends on the settings in the BFB bit of MTU3.TMDR1 (MTU6.TMDR1). The BFB bit of MTU4.TMDR1

(MTU7.TMDR1) should be set to 0.

In MTU1 and MTU2, which have no TGRD, this bit is reserved. It is read as 0. The write value should be 0.

Refer to Figure 24.50 for an illustration of the Tb interval in complementary PWM mode.

### BFE Bit (Buffer Operation E)

This bit specifies whether to operate MTU0.TGRE and MTU0.TGRF in the normal way or to use them together for buffer operation. Compare match with TGRF occurs even when TGRF is used as a buffer register.

In MTU0 to MTU4, MTU6, MTU7, and MTU8, this bit is reserved. It is read as 0. The write value should be 0.

## 24.2.4 Timer Mode Register 2m (TMDR2m) (m = A, B)

Address(es): MTU.TMDR2A 000C 1270h, MTU.TMDR2B 000C 1A70h

| b7                 | b6 | b5 | b4 | b3 | b2 | b1 | b0  |
|--------------------|----|----|----|----|----|----|-----|
| —                  | —  | —  | —  | —  | —  | —  | DRS |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0   |

| Bit      | Symbol | Bit Name             | Description                                                                   | R/W |
|----------|--------|----------------------|-------------------------------------------------------------------------------|-----|
| b0       | DRS    | Double Buffer Select | 0: Double buffer function is disabled<br>1: Double buffer function is enabled | R/W |
| b7 to b1 | —      | Reserved             | These bits are read as 0. The write value should be 0.                        | R/W |

TMDR2A and TMDR2B specify the double buffer function in complementary PWM mode 3 (transfer at the crest and trough of the counter value). The MTU has two TMDR2m registers: one each for MTU3 (TMDR2A) and MTU6 (TMDR2B). TMDR2A and TMDR2B values should be specified only while TCNT operation is stopped.

### DRS Bit (Double Buffer Select)

This bit enables or disables the double buffer function in complementary PWM mode.

### 24.2.5 Timer Mode Register 3 (TMDR3)

Address(es): MTU1.TMDR3 000C 1391h

|                    | b7 | b6 | b5 | b4 | b3 | b2 | b1      | b0  |
|--------------------|----|----|----|----|----|----|---------|-----|
|                    | —  | —  | —  | —  | —  | —  | PHCKSEL | LWA |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 1       | 0   |

| Bit      | Symbol  | Bit Name                                      | Description                                                                                                                        | R/W |
|----------|---------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0       | LWA     | MTU1/MTU2 Combination Longword Access Control | 0: 16-bit access is enabled.<br>1: 32-bit access is enabled.                                                                       | R/W |
| b1       | PHCKSEL | External Input Phase Clock Select             | 0: MTCLKA and MTCLKB are selected for the external phase clock.<br>1: MTCLKC and MTCLKD are selected for the external phase clock. | R/W |
| b7 to b2 | —       | Reserved                                      | These bits are read as 0. The write value should be 0.                                                                             | R/W |

The TMDR3 register controls longword access to a 32-bit register or counter in a combination of MTU1 and MTU2. There is only one TMDR3 register in MTU1. The counter (TCNTLW), general register A (TGRALW), and general register B (TGRBLW) of MTU1 and MTU2 are accessed in the combinations listed in Table 24.12.

#### LWA Bit (MTU1/MTU2 Combination Longword Access Control)

This bit selects a 32-bit access in a combination of MTU1 and MTU2.

When LWA is set to 0, the MTU1 and MTU2 independently operate as a 16-bit timer. therefore registers TCNTLW, TGRALW, and TGRBLW cannot be accessed.

When LWA is set to 1, MTU1 and MTU2 operate as a 32-bit cascaded timer and the timer is controlled by registers MTU1.TCR, MTU1.TCR2, MTU1.TIOR, and MTU1.TMDR1. The settings of registers MTU2.TCR, MTU2.TCR2, MTU2.TIOR, and MTU2.TMDR1 are disabled and the 16-bit registers (TCNT, TGRA, and TGRB) in MTU1 and MTU2 cannot be accessed. Furthermore, MTU2 input capture and compare match are also disabled, which in turn disables any linked operation with the ELC.

The cascaded connection of MTU1 and MTU2 with the LWA bit set to 1 can only be used in phase counting mode, but not in normal mode, PWM1 mode, or PWM2 mode. Select phase counting mode when setting the LWA bit to 1.

Initialize the registers TCNT, TGRA, and TGRB in MTU1 and MTU2 in advance before setting the LWA bit to 1.

#### PHCKSEL Bit (External Input Phase Clock Select)

When the MTU1 and MTU2 registers are combined for 32-bit phase counting mode or MTU2 phase counting mode, this bit selects either the A- or B-phase signal from the external clock. Refer to Table 24.65, Clock Input Pins in Phase Counting Mode for details.

**Table 24.12 Setting and Combination of the TMDR3 Register**

| Register                   | TMDR3.LWA = 0 |             | TMDR3.LWA = 1 |             |
|----------------------------|---------------|-------------|---------------|-------------|
|                            | Symbol        | Access mode | Symbol        | Access mode |
| Counter in MTU1*1          | MTU1.TCNT     | Word        | MTU1.TCNTLW   | Longword    |
| Counter in MTU2            | MTU2.TCNT     | Word        |               |             |
| General register A in MTU1 | MTU1.TGRA     | Word        | MTU1.TGRALW   | Longword    |
| General register A in MTU2 | MTU2.TGRA     | Word        |               |             |
| General register B in MTU1 | MTU1.TGRB     | Word        | MTU1.TGRBLW   | Longword    |
| General register B in MTU2 | MTU2.TGRB     | Word        |               |             |

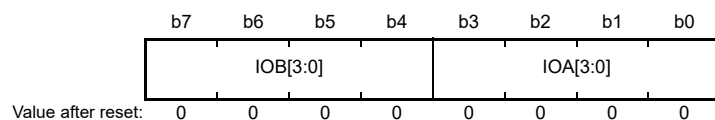
Note 1. When the LWA bit is set to 1, setting the count clock for MTU1 as MTU2.TCNT overflow/underflow is not required.



### 24.2.6 Timer I/O Control Register (TIOR)

- MTU0.TIORH, MTU1.TIOR, MTU2.TIOR, MTU3.TIORH, MTU4.TIORH, MTU6.TIORH, MTU7.TIORH, MTU8.TIORH

Address(es): MTU0.TIORH 000C 1302h, MTU1.TIOR 000C 1382h, MTU2.TIOR 000C 1402h, MTU3.TIORH 000C 1204h, MTU4.TIORH 000C 1206h, MTU6.TIORH 000C 1A04h, MTU7.TIORH 000C 1A06h, MTU8.TIORH 000C 1602h

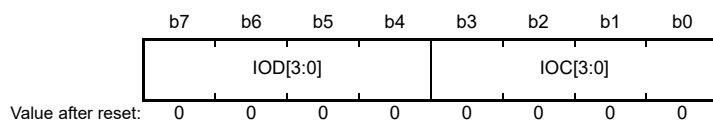


| Bit      | Symbol   | Bit Name                    | Description                                                                                                                                                                                                                                          | R/W |
|----------|----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b3 to b0 | IOA[3:0] | I/O Control A* <sup>1</sup> | Refer to the following tables.<br>MTU0.TIORH: Table 24.27<br>MTU1.TIOR: Table 24.29<br>MTU2.TIOR: Table 24.30<br>MTU3.TIORH: Table 24.31<br>MTU4.TIORH: Table 24.33<br>MTU6.TIORH: Table 24.35<br>MTU7.TIORH: Table 24.37<br>MTU8.TIORH: Table 24.39 | R/W |
| b7 to b4 | IOB[3:0] | I/O Control B* <sup>1</sup> | Refer to the following tables.<br>MTU0.TIORH: Table 24.13<br>MTU1.TIOR: Table 24.15<br>MTU2.TIOR: Table 24.16<br>MTU3.TIORH: Table 24.17<br>MTU4.TIORH: Table 24.19<br>MTU6.TIORH: Table 24.21<br>MTU7.TIORH: Table 24.23<br>MTU8.TIORH: Table 24.25 | R/W |

Note 1. When the value of IOm[3:0] (m = A, B) is changed to the output-prohibited state (0000b or 0100b) during low, high, or toggle output at compare match, this register is in Hi-Z.

- MTU0.TIORL, MTU3.TIORL, MTU4.TIORL, MTU6.TIORL, MTU7.TIORL, MTU8.TIORL

Address(es): MTU0.TIORL 000C 1303h, MTU3.TIORL 000C 1205h, MTU4.TIORL 000C 1207h, MTU6.TIORL 000C 1A05h, MTU7.TIORL 000C 1A07h, MTU8.TIORL 000C 1603h

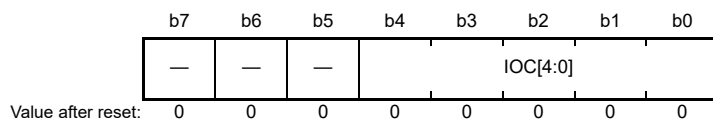


| Bit      | Symbol   | Bit Name        | Description                                                                                                                                                                                      | R/W |
|----------|----------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b3 to b0 | IOC[3:0] | I/O Control C*1 | Refer to the following tables.<br>MTU0.TIORL: Table 24.28<br>MTU3.TIORL: Table 24.32<br>MTU4.TIORL: Table 24.34<br>MTU6.TIORL: Table 24.36<br>MTU7.TIORL: Table 24.38<br>MTU8.TIORL: Table 24.40 | R/W |
| b7 to b4 | IOD[3:0] | I/O Control D*1 | Refer to the following tables.<br>MTU0.TIORL: Table 24.14<br>MTU3.TIORL: Table 24.18<br>MTU4.TIORL: Table 24.20<br>MTU6.TIORL: Table 24.22<br>MTU7.TIORL: Table 24.24<br>MTU8.TIORL: Table 24.26 | R/W |

Note 1. When the value of IOm[3:0] (m = C, D) is changed to the output-prohibited state (0000b or 0100b) during low, high, or toggle output at compare match, this register is in Hi-Z.

- MTU5.TIORU, MTU5.TIORV, MTU5.TIORW

Address(es): MTU5.TIORU 000C 1C86h, MTU5.TIORV 000C 1C96h, MTU5.TIORW 000C 1CA6h



| Bit      | Symbol   | Bit Name      | Description                                                                      | R/W |
|----------|----------|---------------|----------------------------------------------------------------------------------|-----|
| b4 to b0 | IOC[4:0] | I/O Control C | Refer to the following table.<br>MTU5.TIORU, MTU5.TIORV, MTU5.TIORW: Table 24.41 | R/W |
| b7 to b5 | —        | Reserved      | These bits are read as 0. The write value should be 0.                           | R/W |

The TIOR register controls the TGRm register. The MTU has a total of 17 TIOR registers, two each for MTU0, MTU3, MTU4, MTU6, MTU7, and MTU8, one each for MTU1 and MTU2, and three (MTU5.TIORU/TIORV/TIORW) for MTU5. The TIOR register should be set when the TMDR register setting is normal mode, PWM mode, or phase counting mode.

Note that TIOR is affected by the TMDR1 setting.

The initial output specified by TIOR is valid when the counter is stopped (the CSTn bit in TSTRA and the CSTn bit in TSTRB are set to 0). Note also that, in PWM mode 2, the output at the point at which the counter becomes 0000h is specified.

When TGRC or TGRD is designated for buffer operation, this setting is invalid and the register operates as a buffer register.

Table 24.13 TIORH (MTU0)

| Bit 7  | Bit 6  | Bit 5  | Bit 4  | Description             |                                                                                                                                                                       |
|--------|--------|--------|--------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IOB[3] | IOB[2] | IOB[1] | IOB[0] | TGRB Register Function  | MTIOC0B Pin Function                                                                                                                                                  |
| 0      | 0      | 0      | 0      | Output compare register | Output prohibited                                                                                                                                                     |
| 0      | 0      | 0      | 1      |                         | Initial output is low.<br>Low output at compare match.                                                                                                                |
| 0      | 0      | 1      | 0      |                         | Initial output is low.<br>High output at compare match.                                                                                                               |
| 0      | 0      | 1      | 1      |                         | Initial output is low.<br>Toggle output at compare match.                                                                                                             |
| 0      | 1      | 0      | 0      |                         | Output prohibited                                                                                                                                                     |
| 0      | 1      | 0      | 1      |                         | Initial output is high.<br>Low output at compare match.                                                                                                               |
| 0      | 1      | 1      | 0      |                         | Initial output is high.<br>High output at compare match.                                                                                                              |
| 0      | 1      | 1      | 1      |                         | Initial output is high.<br>Toggle output at compare match.                                                                                                            |
| 1      | 0      | 0      | 0      | Input capture register  | Input capture at rising edge.                                                                                                                                         |
| 1      | 0      | 0      | 1      |                         | Input capture at falling edge.                                                                                                                                        |
| 1      | 0      | 1      | x      |                         | Input capture at both edges.                                                                                                                                          |
| 1      | 1      | x      | x      |                         | Capture input source is the clock source for counting in MTU1.<br>Input capture on counting up or down by MTU1.TCNT (LWA = 0) or MTU1.TCNTLW (LWA = 1). <sup>*1</sup> |

x: Don't care

Note 1. When PCLKA/1 is selected as the count clock for MTU1, MTU0 input capture is not generated. Do not select PCLKA/1 as the count clock for MTU1.

Table 24.14 TIORL (MTU0)

| Bit 7  | Bit 6  | Bit 5  | Bit 4  | Description                           |                                                                                                                                                                       |
|--------|--------|--------|--------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IOD[3] | IOD[2] | IOD[1] | IOD[0] | TGRD Register Function                | MTIOC0D Pin Function                                                                                                                                                  |
| 0      | 0      | 0      | 0      | Output compare register <sup>*1</sup> | Output prohibited                                                                                                                                                     |
| 0      | 0      | 0      | 1      |                                       | Initial output is low.<br>Low output at compare match.                                                                                                                |
| 0      | 0      | 1      | 0      |                                       | Initial output is low.<br>High output at compare match.                                                                                                               |
| 0      | 0      | 1      | 1      |                                       | Initial output is low.<br>Toggle output at compare match.                                                                                                             |
| 0      | 1      | 0      | 0      |                                       | Output prohibited                                                                                                                                                     |
| 0      | 1      | 0      | 1      |                                       | Initial output is high.<br>Low output at compare match.                                                                                                               |
| 0      | 1      | 1      | 0      |                                       | Initial output is high.<br>High output at compare match.                                                                                                              |
| 0      | 1      | 1      | 1      |                                       | Initial output is high.<br>Toggle output at compare match.                                                                                                            |
| 1      | 0      | 0      | 0      | Input capture register <sup>*1</sup>  | Input capture at rising edge.                                                                                                                                         |
| 1      | 0      | 0      | 1      |                                       | Input capture at falling edge.                                                                                                                                        |
| 1      | 0      | 1      | x      |                                       | Input capture at both edges.                                                                                                                                          |
| 1      | 1      | x      | x      |                                       | Capture input source is the clock source for counting in MTU1.<br>Input capture on counting up or down by MTU1.TCNT (LWA = 0) or MTU1.TCNTLW (LWA = 1). <sup>*2</sup> |

x: Don't care

Note 1. When the MTU0.TMDR1.BFB bit is set to 1 and MTU0.TGRD register is used as a buffer register, this setting is invalid and input capture/output compare is not generated.

Note 2. When PCLKA/1 is selected as the count clock for MTU1, MTU0 input capture is not generated. Do not select PCLKA/1 as the count clock for MTU1.

Table 24.15 TIOR (MTU1)

| Bit 7  | Bit 6  | Bit 5  | Bit 4  | Description                   |                                                                                         |
|--------|--------|--------|--------|-------------------------------|-----------------------------------------------------------------------------------------|
| IOB[3] | IOB[2] | IOB[1] | IOB[0] | TGRB/TGRBLW Register Function | MTIOC1B Pin Function                                                                    |
| 0      | 0      | 0      | 0      | Output compare register       | Output prohibited                                                                       |
| 0      | 0      | 0      | 1      |                               | Initial output is low.<br>Low output at compare match.                                  |
| 0      | 0      | 1      | 0      |                               | Initial output is low.<br>High output at compare match.                                 |
| 0      | 0      | 1      | 1      |                               | Initial output is low.<br>Toggle output at compare match.                               |
| 0      | 1      | 0      | 0      |                               | Output prohibited                                                                       |
| 0      | 1      | 0      | 1      |                               | Initial output is high.<br>Low output at compare match.                                 |
| 0      | 1      | 1      | 0      |                               | Initial output is high.<br>High output at compare match.                                |
| 0      | 1      | 1      | 1      |                               | Initial output is high.<br>Toggle output at compare match.                              |
| 1      | 0      | 0      | 0      | Input capture register        | Input capture at rising edge.                                                           |
| 1      | 0      | 0      | 1      |                               | Input capture at falling edge.                                                          |
| 1      | 0      | 1      | x      |                               | Input capture at both edges.                                                            |
| 1      | 1      | 0      | 0      |                               | Input capture at occurrence of compare match or input capture in the MTU0.TGRC register |
| 1      | 1      | 1      | x      |                               | Input capture at occurrence of compare match in the MTU8.TGRC register                  |

x: Don't care

Table 24.16 TIOR (MTU2)

| Bit 7  | Bit 6  | Bit 5  | Bit 4  | Description             |                                                            |
|--------|--------|--------|--------|-------------------------|------------------------------------------------------------|
| IOB[3] | IOB[2] | IOB[1] | IOB[0] | TGRB Register Function  | MTIOC2B Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                         | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                         | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                         | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                         | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                         | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                         | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                         | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                         | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                         | Input capture at both edges.                               |

x: Don't care

Table 24.17 TIORH (MTU3)

| Bit 7  | Bit 6  | Bit 5  | Bit 4  | Description             |                                                            |
|--------|--------|--------|--------|-------------------------|------------------------------------------------------------|
| IOB[3] | IOB[2] | IOB[1] | IOB[0] | TGRB Register Function  | MTIOC3B Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                         | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                         | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                         | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                         | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                         | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                         | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                         | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                         | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                         | Input capture at both edges.                               |

x: Don't care

Table 24.18 TIORL (MTU3)

| Bit 7  | Bit 6  | Bit 5  | Bit 4  | Description               |                                                            |
|--------|--------|--------|--------|---------------------------|------------------------------------------------------------|
| IOD[3] | IOD[2] | IOD[1] | IOD[0] | TGRD Register Function    | MTIOC3D Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register*1 | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                           | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                           | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                           | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                           | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                           | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                           | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                           | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register*1  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                           | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                           | Input capture at both edges.                               |

x: Don't care

Note 1. When the MTU3.TMDR1.BFB bit is set to 1 and MTU3.TGRD register is used as a buffer register, this setting is invalid and input capture/output compare is not generated.

Table 24.19 TIORH (MTU4)

| Bit 7  | Bit 6  | Bit 5  | Bit 4  | Description             |                                                            |
|--------|--------|--------|--------|-------------------------|------------------------------------------------------------|
| IOB[3] | IOB[2] | IOB[1] | IOB[0] | TGRB Register Function  | MTIOC4B Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                         | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                         | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                         | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                         | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                         | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                         | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                         | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                         | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                         | Input capture at both edges.                               |

x: Don't care

Table 24.20 TIORL (MTU4)

| Bit 7  | Bit 6  | Bit 5  | Bit 4  | Description               |                                                            |
|--------|--------|--------|--------|---------------------------|------------------------------------------------------------|
| IOD[3] | IOD[2] | IOD[1] | IOD[0] | TGRD Register Function    | MTIOC4D Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register*1 | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                           | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                           | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                           | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                           | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                           | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                           | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                           | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register*1  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                           | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                           | Input capture at both edges.                               |

x: Don't care

Note 1. When the MTU4.TMDR1.BFB bit is set to 1 and MTU4.TGRD register is used as a buffer register, this setting is invalid and input capture/output compare is not generated.

Table 24.21 TIORH (MTU6)

| Bit 7  | Bit 6  | Bit 5  | Bit 4  | Description             |                                                            |
|--------|--------|--------|--------|-------------------------|------------------------------------------------------------|
| IOB[3] | IOB[2] | IOB[1] | IOB[0] | TGRB Register Function  | MTIOC6B Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                         | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                         | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                         | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                         | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                         | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                         | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                         | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                         | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                         | Input capture at both edges.                               |

x: Don't care

Table 24.22 TIORL (MTU6)

| Bit 7  | Bit 6  | Bit 5  | Bit 4  | Description               |                                                            |
|--------|--------|--------|--------|---------------------------|------------------------------------------------------------|
| IOD[3] | IOD[2] | IOD[1] | IOD[0] | TGRD Register Function    | MTIOC6D Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register*1 | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                           | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                           | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                           | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                           | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                           | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                           | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                           | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register*1  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                           | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                           | Input capture at both edges.                               |

x: Don't care

Note 1. When the MTU6.TMDR1.BFB bit is set to 1 and MTU6.TGRD register is used as a buffer register, this setting is invalid and input capture/output compare is not generated.

Table 24.23 TIORH (MTU7)

| Bit 7  | Bit 6  | Bit 5  | Bit 4  | Description             |                                                            |
|--------|--------|--------|--------|-------------------------|------------------------------------------------------------|
| IOB[3] | IOB[2] | IOB[1] | IOB[0] | TGRB Register Function  | MTIOC7B Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                         | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                         | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                         | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                         | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                         | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                         | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                         | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                         | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                         | Input capture at both edges.                               |

x: Don't care

Table 24.24 TIORL (MTU7)

| Bit 7  | Bit 6  | Bit 5  | Bit 4  | Description               |                                                            |
|--------|--------|--------|--------|---------------------------|------------------------------------------------------------|
| IOD[3] | IOD[2] | IOD[1] | IOD[0] | TGRD Register Function    | MTIOC7D Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register*1 | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                           | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                           | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                           | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                           | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                           | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                           | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                           | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register*1  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                           | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                           | Input capture at both edges.                               |

x: Don't care

Note 1. When the MTU7.TMDR1.BFB bit is set to 1 and MTU7.TGRD register is used as a buffer register, this setting is invalid and input capture/output compare is not generated.



Table 24.25 TIORH (MTU8)

| Bit 7  | Bit 6  | Bit 5  | Bit 4  | Description             |                                                                                                                                                            |
|--------|--------|--------|--------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IOB[3] | IOB[2] | IOB[1] | IOB[0] | TGRB Register Function  | MTIOC8B Pin Function                                                                                                                                       |
| 0      | 0      | 0      | 0      | Output compare register | Output prohibited                                                                                                                                          |
| 0      | 0      | 0      | 1      |                         | Initial output is low.<br>Low output at compare match.                                                                                                     |
| 0      | 0      | 1      | 0      |                         | Initial output is low.<br>High output at compare match.                                                                                                    |
| 0      | 0      | 1      | 1      |                         | Initial output is low.<br>Toggle output at compare match.                                                                                                  |
| 0      | 1      | 0      | 0      |                         | Output prohibited                                                                                                                                          |
| 0      | 1      | 0      | 1      |                         | Initial output is high.<br>Low output at compare match.                                                                                                    |
| 0      | 1      | 1      | 0      |                         | Initial output is high.<br>High output at compare match.                                                                                                   |
| 0      | 1      | 1      | 1      |                         | Initial output is high.<br>Toggle output at compare match.                                                                                                 |
| 1      | 0      | 0      | 0      | Input capture register  | Input capture at rising edge.                                                                                                                              |
| 1      | 0      | 0      | 1      |                         | Input capture at falling edge.                                                                                                                             |
| 1      | 0      | 1      | x      |                         | Input capture at both edges.                                                                                                                               |
| 1      | 1      | x      | x      |                         | Capture input source is the clock source for counting in MTU1.<br>Input capture on counting up or down by MTU1.TCNT (LWA = 0) or<br>MTU1.TCNTLW (LWA = 1). |

x: Don't care

Table 24.26 TIORL (MTU8)

| Bit 7  | Bit 6  | Bit 5  | Bit 4  | Description               |                                                            |
|--------|--------|--------|--------|---------------------------|------------------------------------------------------------|
| IOD[3] | IOD[2] | IOD[1] | IOD[0] | TGRD Register Function    | MTIOC8D Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register*1 | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                           | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                           | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                           | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                           | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                           | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                           | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                           | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register    | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                           | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                           | Input capture at both edges.                               |

x: Don't care

Note 1. When the MTU8.TMDR1.BFB bit is set to 1 and the MTU8.TGRD register is used as a buffer register, this setting is invalid and input capture/output compare is not generated.

**Table 24.27 TIORH (MTU0)**

| Bit 3  | Bit 2  | Bit 1  | Bit 0  | Description             |                                                                                                                                                                       |
|--------|--------|--------|--------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IOA[3] | IOA[2] | IOA[1] | IOA[0] | TGRA Register Function  | MTIOC0A Pin Function                                                                                                                                                  |
| 0      | 0      | 0      | 0      | Output compare register | Output prohibited                                                                                                                                                     |
| 0      | 0      | 0      | 1      |                         | Initial output is low.<br>Low output at compare match.                                                                                                                |
| 0      | 0      | 1      | 0      |                         | Initial output is low.<br>High output at compare match.                                                                                                               |
| 0      | 0      | 1      | 1      |                         | Initial output is low.<br>Toggle output at compare match.                                                                                                             |
| 0      | 1      | 0      | 0      |                         | Output prohibited                                                                                                                                                     |
| 0      | 1      | 0      | 1      |                         | Initial output is high.<br>Low output at compare match.                                                                                                               |
| 0      | 1      | 1      | 0      |                         | Initial output is high.<br>High output at compare match.                                                                                                              |
| 0      | 1      | 1      | 1      |                         | Initial output is high.<br>Toggle output at compare match.                                                                                                            |
| 1      | 0      | 0      | 0      | Input capture register  | Input capture at rising edge.                                                                                                                                         |
| 1      | 0      | 0      | 1      |                         | Input capture at falling edge.                                                                                                                                        |
| 1      | 0      | 1      | x      |                         | Input capture at both edges.                                                                                                                                          |
| 1      | 1      | 0      | 0      |                         | Capture input source is the clock source for counting in MTU1.<br>Input capture on counting up or down by MTU1.TCNT (LWA = 0) or MTU1.TCNTLW (LWA = 1). <sup>*1</sup> |
| 1      | 1      | 1      | x      |                         | Input capture on generation of compare match with MTU8.TGRC                                                                                                           |

x: Don't care

Note 1. When PCLKA/1 is selected as the count clock for MTU1, MTU0 input capture is not generated. Do not select PCLKA/1 as the count clock for MTU1.

**Table 24.28 TIORL (MTU0)**

| Bit 3  | Bit 2  | Bit 1  | Bit 0  | Description                           |                                                                                                                                                                       |
|--------|--------|--------|--------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IOC[3] | IOC[2] | IOC[1] | IOC[0] | TGRC Register Function                | MTIOC0C Pin Function                                                                                                                                                  |
| 0      | 0      | 0      | 0      | Output compare register <sup>*1</sup> | Output prohibited                                                                                                                                                     |
| 0      | 0      | 0      | 1      |                                       | Initial output is low.<br>Low output at compare match.                                                                                                                |
| 0      | 0      | 1      | 0      |                                       | Initial output is low.<br>High output at compare match.                                                                                                               |
| 0      | 0      | 1      | 1      |                                       | Initial output is low.<br>Toggle output at compare match.                                                                                                             |
| 0      | 1      | 0      | 0      |                                       | Output prohibited                                                                                                                                                     |
| 0      | 1      | 0      | 1      |                                       | Initial output is high.<br>Low output at compare match.                                                                                                               |
| 0      | 1      | 1      | 0      |                                       | Initial output is high.<br>High output at compare match.                                                                                                              |
| 0      | 1      | 1      | 1      |                                       | Initial output is high.<br>Toggle output at compare match.                                                                                                            |
| 1      | 0      | 0      | 0      | Input capture register <sup>*1</sup>  | Input capture at rising edge.                                                                                                                                         |
| 1      | 0      | 0      | 1      |                                       | Input capture at falling edge.                                                                                                                                        |
| 1      | 0      | 1      | x      |                                       | Input capture at both edges.                                                                                                                                          |
| 1      | 1      | x      | x      |                                       | Capture input source is the clock source for counting in MTU1.<br>Input capture on counting up or down by MTU1.TCNT (LWA = 0) or MTU1.TCNTLW (LWA = 1). <sup>*2</sup> |

x: Don't care

Note 1. When the MTU0.TMDR1.BFB bit is set to 1 and MTU0.TGRC register is used as a buffer register, this setting is invalid and input capture/output compare is not generated.

Note 2. When PCLKA/1 is selected as the count clock for MTU1, MTU0 input capture is not generated. Do not select PCLKA/1 as the count clock for MTU1.

Table 24.29 TIOR (MTU1)

| Bit 3  | Bit 2  | Bit 1  | Bit 0  | Description                   |                                                                       |
|--------|--------|--------|--------|-------------------------------|-----------------------------------------------------------------------|
| IOA[3] | IOA[2] | IOA[1] | IOA[0] | TGRA/TGRALW Register Function | MTIOC1A Pin Function                                                  |
| 0      | 0      | 0      | 0      | Output compare register       | Output prohibited                                                     |
| 0      | 0      | 0      | 1      |                               | Initial output is low.<br>Low output at compare match.                |
| 0      | 0      | 1      | 0      |                               | Initial output is low.<br>High output at compare match.               |
| 0      | 0      | 1      | 1      |                               | Initial output is low.<br>Toggle output at compare match.             |
| 0      | 1      | 0      | 0      |                               | Output prohibited                                                     |
| 0      | 1      | 0      | 1      |                               | Initial output is high.<br>Low output at compare match.               |
| 0      | 1      | 1      | 0      |                               | Initial output is high.<br>High output at compare match.              |
| 0      | 1      | 1      | 1      |                               | Initial output is high.<br>Toggle output at compare match.            |
| 1      | 0      | 0      | 0      | Input capture register        | Input capture at rising edge.                                         |
| 1      | 0      | 0      | 1      |                               | Input capture at falling edge.                                        |
| 1      | 0      | 1      | x      |                               | Input capture at both edges.                                          |
| 1      | 1      | x      | x      |                               | Input capture at generation of MTU0.TGRA compare match/input capture. |

x: Don't care

Table 24.30 TIOR (MTU2)

| Bit 3  | Bit 2  | Bit 1  | Bit 0  | Description             |                                                            |
|--------|--------|--------|--------|-------------------------|------------------------------------------------------------|
| IOA[3] | IOA[2] | IOA[1] | IOA[0] | TGRA Register Function  | MTIOC2A Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                         | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                         | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                         | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                         | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                         | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                         | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                         | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                         | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                         | Input capture at both edges.                               |

x: Don't care

Table 24.31 TIORH (MTU3)

| Bit 3  | Bit 2  | Bit 1  | Bit 0  | Description             |                                                            |
|--------|--------|--------|--------|-------------------------|------------------------------------------------------------|
| IOA[3] | IOA[2] | IOA[1] | IOA[0] | TGRA Register Function  | MTIOC3A Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                         | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                         | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                         | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                         | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                         | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                         | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                         | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                         | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                         | Input capture at both edges.                               |

x: Don't care

Table 24.32 TIORL (MTU3)

| Bit 3  | Bit 2  | Bit 1  | Bit 0  | Description               |                                                            |
|--------|--------|--------|--------|---------------------------|------------------------------------------------------------|
| IOC[3] | IOC[2] | IOC[1] | IOC[0] | TGRC Register Function    | MTIOC3C Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register*1 | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                           | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                           | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                           | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                           | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                           | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                           | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                           | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register*1  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                           | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                           | Input capture at both edges.                               |

x: Don't care

Note 1. When the MTU3.TMDR1.BFB bit is set to 1 and MTU3.TGRC register is used as a buffer register, this setting is invalid and input capture/output compare is not generated.

**Table 24.33 TIORH (MTU4)**

| Bit 3  | Bit 2  | Bit 1  | Bit 0  | Description             |                                                            |
|--------|--------|--------|--------|-------------------------|------------------------------------------------------------|
| IOA[3] | IOA[2] | IOA[1] | IOA[0] | TGRA Register Function  | MTIOC4A Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                         | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                         | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                         | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                         | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                         | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                         | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                         | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                         | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                         | Input capture at both edges.                               |

x: Don't care

**Table 24.34 TIORL (MTU4)**

| Bit 3  | Bit 2  | Bit 1  | Bit 0  | Description               |                                                            |
|--------|--------|--------|--------|---------------------------|------------------------------------------------------------|
| IOC[3] | IOC[2] | IOC[1] | IOC[0] | TGRC Register Function    | MTIOC4C Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register*1 | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                           | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                           | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                           | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                           | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                           | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                           | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                           | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register*1  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                           | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                           | Input capture at both edges.                               |

x: Don't care

Note 1. When the MTU4.TMDR1.BFB bit is set to 1 and MTU4.TGRC register is used as a buffer register, this setting is invalid and input capture/output compare is not generated.

Table 24.35 TIORH (MTU6)

| Bit 3  | Bit 2  | Bit 1  | Bit 0  | Description             |                                                            |
|--------|--------|--------|--------|-------------------------|------------------------------------------------------------|
| IOA[3] | IOA[2] | IOA[1] | IOA[0] | TGRA Register Function  | MTIOC6A Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                         | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                         | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                         | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                         | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                         | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                         | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                         | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                         | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                         | Input capture at both edges.                               |

x: Don't care

Table 24.36 TIORL (MTU6)

| Bit 3  | Bit 2  | Bit 1  | Bit 0  | Description               |                                                            |
|--------|--------|--------|--------|---------------------------|------------------------------------------------------------|
| IOC[3] | IOC[2] | IOC[1] | IOC[0] | TGRC Register Function    | MTIOC6C Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register*1 | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                           | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                           | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                           | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                           | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                           | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                           | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                           | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register*1  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                           | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                           | Input capture at both edges.                               |

x: Don't care

Note 1. When the MTU6.TMDR1.BFB bit is set to 1 and MTU6.TGRC register is used as a buffer register, this setting is invalid and input capture/output compare is not generated.

Table 24.37 TIORH (MTU7)

| Bit 3  | Bit 2  | Bit 1  | Bit 0  | Description             |                                                            |
|--------|--------|--------|--------|-------------------------|------------------------------------------------------------|
| IOA[3] | IOA[2] | IOA[1] | IOA[0] | TGRA Register Function  | MTIOC7A Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                         | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                         | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                         | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                         | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                         | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                         | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                         | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                         | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                         | Input capture at both edges.                               |

x: Don't care

Table 24.38 TIORL (MTU7)

| Bit 3  | Bit 2  | Bit 1  | Bit 0  | Description               |                                                            |
|--------|--------|--------|--------|---------------------------|------------------------------------------------------------|
| IOC[3] | IOC[2] | IOC[1] | IOC[0] | TGRC Register Function    | MTIOC7C Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register*1 | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                           | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                           | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                           | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                           | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                           | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                           | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                           | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register*1  | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                           | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                           | Input capture at both edges.                               |

x: Don't care

Note 1. When the MTU7.TMDR1.BFB bit is set to 1 and MTU7.TGRC register is used as a buffer register, this setting is invalid and input capture/output compare is not generated.

Table 24.39 TIORH (MTU8)

| Bit 3  | Bit 2  | Bit 1  | Bit 0  | Description             |                                                            |
|--------|--------|--------|--------|-------------------------|------------------------------------------------------------|
| IOA[3] | IOA[2] | IOA[1] | IOA[0] | TGRA Register Function  | MTIOC8A Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                         | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                         | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                         | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                         | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                         | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                         | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                         | Initial output is high.<br>Toggle output at compare match. |
| 1      | 0      | 0      | 0      | Input capture register  | Input capture at rising edge.                              |
| 1      | 0      | 0      | 1      |                         | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                         | Input capture at both edges.                               |

x: Don't care

Table 24.40 TIORL (MTU8)

| Bit 3  | Bit 2  | Bit 1  | Bit 0  | Description               |                                                            |
|--------|--------|--------|--------|---------------------------|------------------------------------------------------------|
| IOC[3] | IOC[2] | IOC[1] | IOC[0] | TGRC Register Function    | MTIOC8C Pin Function                                       |
| 0      | 0      | 0      | 0      | Output compare register*1 | Output prohibited                                          |
| 0      | 0      | 0      | 1      |                           | Initial output is low.<br>Low output at compare match.     |
| 0      | 0      | 1      | 0      |                           | Initial output is low.<br>High output at compare match.    |
| 0      | 0      | 1      | 1      |                           | Initial output is low.<br>Toggle output at compare match.  |
| 0      | 1      | 0      | 0      |                           | Output prohibited                                          |
| 0      | 1      | 0      | 1      |                           | Initial output is high.<br>Low output at compare match.    |
| 0      | 1      | 1      | 0      |                           | Initial output is high.<br>High output at compare match.   |
| 0      | 1      | 1      | 1      |                           | Initial output is high.<br>Toggle output at compare match. |
| 1      | x      | 0      | 0      | Input capture register    | Input capture at rising edge.                              |
| 1      | x      | 0      | 1      |                           | Input capture at falling edge.                             |
| 1      | x      | 1      | x      |                           | Input capture at both edges.                               |

x: Don't care

Note 1. When the MTU8.TMDR1.BFA bit is set to 1 and the MTU8.TGRC register is used as a buffer register, this setting is invalid and input capture/output compare is not generated.



Table 24.41 TIORU, TIORV, and TIORW (MTU5)

| Bit 4  | Bit 3  | Bit 2  | Bit 1  | Bit 0  | Description                            |                                                                                                                     |
|--------|--------|--------|--------|--------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| IOC[4] | IOC[3] | IOC[2] | IOC[1] | IOC[0] | TGRU, TGRV, TGRW<br>Registers Function | MTIC5U, MTIC5V, MTIC5W Pin Function                                                                                 |
| 0      | 0      | 0      | 0      | 0      | Output compare register                | No function                                                                                                         |
| 0      | 0      | 0      | 0      | 1      |                                        | Setting prohibited                                                                                                  |
| 0      | 0      | 0      | 1      | x      |                                        | Setting prohibited                                                                                                  |
| 0      | 0      | 1      | x      | x      |                                        | Setting prohibited                                                                                                  |
| 0      | 1      | x      | x      | x      |                                        | Setting prohibited                                                                                                  |
| 1      | 0      | 0      | 0      | 0      | Input capture register*1               | Setting prohibited                                                                                                  |
| 1      | 0      | 0      | 0      | 1      |                                        | Input capture at rising edge.                                                                                       |
| 1      | 0      | 0      | 1      | 0      |                                        | Input capture at falling edge.                                                                                      |
| 1      | 0      | 0      | 1      | 1      |                                        | Input capture at both edges.                                                                                        |
| 1      | 0      | 1      | x      | x      |                                        | Input capture on generation of compare match with MTU8.TGRC                                                         |
| 1      | 1      | 0      | 0      | 0      |                                        | Setting prohibited                                                                                                  |
| 1      | 1      | 0      | 0      | 1      |                                        | Measurement of low pulse width of external input signal.<br>Capture at trough in complementary PWM mode.            |
| 1      | 1      | 0      | 1      | 0      |                                        | Measurement of low pulse width of external input signal.<br>Capture at crest of complementary PWM mode.             |
| 1      | 1      | 0      | 1      | 1      |                                        | Measurement of low pulse width of external input signal.<br>Capture at crest and trough of complementary PWM mode.  |
| 1      | 1      | 1      | 0      | 0      |                                        | Setting prohibited                                                                                                  |
| 1      | 1      | 1      | 0      | 1      |                                        | Measurement of high pulse width of external input signal.<br>Capture at trough in complementary PWM mode.           |
| 1      | 1      | 1      | 1      | 0      |                                        | Measurement of high pulse width of external input signal.<br>Capture at crest of complementary PWM mode.            |
| 1      | 1      | 1      | 1      | 1      |                                        | Measurement of high pulse width of external input signal.<br>Capture at crest and trough of complementary PWM mode. |

x: Don't care

Note 1. Set the IOC[4:0] bits to 19h, 1Ah, 1Bh, 1Dh, 1Eh, or 1Fh only when using external pulse width measurement or only when using dead time compensation linked with MTU6 and MTU7. For details, refer to section 24.3.11, External Pulse Width Measurement and section 24.3.12, Dead Time Compensation.

## 24.2.7 Timer Compare Match Clear Register (TCNTCMPCLR)

Address(es): MTU5.TCNTCMPCLR 000C 1CB6h

|                    | b7 | b6 | b5 | b4 | b3 | b2       | b1       | b0       |
|--------------------|----|----|----|----|----|----------|----------|----------|
|                    | —  | —  | —  | —  | —  | CMPCLR5U | CMPCLR5V | CMPCLR5W |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0        | 0        | 0        |

| Bit      | Symbol   | Bit Name              | Description                                                                                                                                                                                                         | R/W |
|----------|----------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0       | CMPCLR5W | TCNT Compare Clear 5W | 0: Disables MTU5.TCNTW to be cleared to 0000h at MTU5.TCNTW and MTU5.TGRW compare match or input capture<br>1: Enables MTU5.TCNTW to be cleared to 0000h at MTU5.TCNTW and MTU5.TGRW compare match or input capture | R/W |
| b1       | CMPCLR5V | TCNT Compare Clear 5V | 0: Disables MTU5.TCNTV to be cleared to 0000h at MTU5.TCNTV and MTU5.TGRV compare match or input capture<br>1: Enables MTU5.TCNTV to be cleared to 0000h at MTU5.TCNTV and MTU5.TGRV compare match or input capture | R/W |
| b2       | CMPCLR5U | TCNT Compare Clear 5U | 0: Disables MTU5.TCNTU to be cleared to 0000h at MTU5.TCNTU and MTU5.TGRU compare match or input capture<br>1: Enables MTU5.TCNTU to be cleared to 0000h at MTU5.TCNTU and MTU5.TGRU compare match or input capture | R/W |
| b7 to b3 | —        | Reserved              | These bits are read as 0. The write value should be 0.                                                                                                                                                              | R/W |

TCNTCMPCLR specifies requests to clear MTU5.TCNTU, MTU5.TCNTV, and MTU5.TCNTW. The MTU has one TCNTCMPCLR (on MTU5).

### 24.2.8 Timer Interrupt Enable Register (TIER)

- MTU1.TIER, MTU2.TIER

Address(es): MTU1.TIER 000C 1384h, MTU2.TIER 000C 1404h

|                    | b7   | b6 | b5    | b4    | b3 | b2 | b1    | b0    |
|--------------------|------|----|-------|-------|----|----|-------|-------|
|                    | TTGE | —  | TCIEU | TCIEV | —  | —  | TGIEB | TGIEA |
| Value after reset: | 0    | 0  | 0     | 0     | 0  | 0  | 0     | 0     |

- MTU0.TIER, MTU3.TIER, MTU6.TIER

Address(es): MTU0.TIER 000C 1304h, MTU3.TIER 000C 1208h, MTU6.TIER 000C 1A08h

|                    | b7   | b6 | b5 | b4    | b3    | b2    | b1    | b0    |
|--------------------|------|----|----|-------|-------|-------|-------|-------|
|                    | TTGE | —  | —  | TCIEV | TGIED | TGIEC | TGIEB | TGIEA |
| Value after reset: | 0    | 0  | 0  | 0     | 0     | 0     | 0     | 0     |

- MTU4.TIER, MTU7.TIER

Address(es): MTU4.TIER 000C 1209h, MTU7.TIER 000C 1A09h

|                    | b7   | b6    | b5 | b4    | b3    | b2    | b1    | b0    |
|--------------------|------|-------|----|-------|-------|-------|-------|-------|
|                    | TTGE | TTGE2 | —  | TCIEV | TGIED | TGIEC | TGIEB | TGIEA |
| Value after reset: | 0    | 0     | 0  | 0     | 0     | 0     | 0     | 0     |

- MTU8.TIER

Address(es): MTU8.TIER 000C 1604h

|                    | b7 | b6 | b5 | b4    | b3    | b2    | b1    | b0    |
|--------------------|----|----|----|-------|-------|-------|-------|-------|
|                    | —  | —  | —  | TCIEV | TGIED | TGIEC | TGIEB | TGIEA |
| Value after reset: | 0  | 0  | 0  | 0     | 0     | 0     | 0     | 0     |

| Bit | Symbol | Bit Name                              | Description                                                                                                                                                               | R/W |
|-----|--------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0  | TGIEA  | TGR Interrupt Enable A                | 0: Interrupt requests (TGIA) disabled<br>1: Interrupt requests (TGIA) enabled                                                                                             | R/W |
| b1  | TGIEB  | TGR Interrupt Enable B                | 0: Interrupt requests (TGIB) disabled<br>1: Interrupt requests (TGIB) enabled                                                                                             | R/W |
| b2  | TGIEC  | TGR Interrupt Enable C                | 0: Interrupt requests (TGIC) disabled<br>1: Interrupt requests (TGIC) enabled                                                                                             | R/W |
| b3  | TGIED  | TGR Interrupt Enable D                | 0: Interrupt requests (TGID) disabled<br>1: Interrupt requests (TGID) enabled                                                                                             | R/W |
| b4  | TCIEV  | Overflow Interrupt Enable             | 0: Interrupt requests (TCIV) disabled<br>1: Interrupt requests (TCIV) enabled                                                                                             | R/W |
| b5  | TCIEU  | Underflow Interrupt Enable            | 0: Interrupt requests (TCIU) disabled<br>1: Interrupt requests (TCIU) enabled                                                                                             | R/W |
| b6  | TTGE2  | A/D Conversion Start Request Enable 2 | 0: A/D conversion start request generation by MTUn.TCNT underflow (trough) disabled<br>1: A/D conversion start request generation by MTUn.TCNT underflow (trough) enabled | R/W |
| b7  | TTGE   | A/D Conversion Start Request Enable   | 0: A/D conversion start request generation disabled<br>1: A/D conversion start request generation enabled                                                                 | R/W |

n = 4, 7

The TIER register enables or disables interrupt requests from each channel. The MTU has a total of ten TIER registers, two for MTU0 and one each for MTU1 to MTU8.

#### TGIEA and TGIEB Bits (TGR Interrupt Enable A and B)

Each bit enables or disables interrupt requests (TGIm) (m = A, B).

#### TGIEC and TGIED Bits (TGR Interrupt Enable C and D)

Each bit enables or disables an interrupt request (TGIm) (m = C, D).

In MTU1 and MTU2, these bits are reserved. They are read as 0. The write value should be 0.

#### TCIEV Bit (Overflow Interrupt Enable)

This bit enables or disables interrupt requests (TCIV).

#### TCIEU Bit (Underflow Interrupt Enable)

This bit enables or disables interrupt requests (TCIU).

In MTU0, MTU3, MTU4, MTU6, MTU7, and MTU8, this bit is reserved. It is read as 0. The write value should be 0.

#### TTGE2 Bit (A/D Conversion Start Request Enable 2)

This bit enables or disables generation of A/D conversion start requests by MTUn.TCNT underflow (trough) in complementary PWM mode (n = 4, 7).

In MTU0 to MTU3, MTU6, and MTU8, this bit is reserved. It is read as 0. The write value should be 0.

#### TTGE Bit (A/D Conversion Start Request Enable)

This bit enables or disables generation of A/D conversion start requests by TGRA input capture/compare match.

MTU8 is a reserved bit. It is read as 0. The write value should be 0.

- MTU0.TIER2

Address(es): MTU0.TIER2 000C 1324h

|                    | b7    | b6 | b5 | b4 | b3 | b2 | b1    | b0    |
|--------------------|-------|----|----|----|----|----|-------|-------|
|                    | TTGE2 | —  | —  | —  | —  | —  | TGIEF | TGIEE |
| Value after reset: | 0     | 0  | 0  | 0  | 0  | 0  | 0     | 0     |

| Bit      | Symbol | Bit Name                              | Description                                                                                                                                                                                                 | R/W |
|----------|--------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0       | TGIEE  | TGR Interrupt Enable E                | 0: Interrupt requests (TGIE) disabled<br>1: Interrupt requests (TGIE) enabled                                                                                                                               | R/W |
| b1       | TGIEF  | TGR Interrupt Enable F                | 0: Interrupt requests (TGIF) disabled<br>1: Interrupt requests (TGIF) enabled                                                                                                                               | R/W |
| b6 to b2 | —      | Reserved                              | These bits are read as 0. The write value should be 0.                                                                                                                                                      | R/W |
| b7       | TTGE2  | A/D Conversion Start Request Enable 2 | 0: A/D conversion start request generation by compare match between MTU0.TCNT and MTU0.TGRE disabled<br>1: A/D conversion start request generation by compare match between MTU0.TCNT and MTU0.TGRE enabled | R/W |

**TGIEE and TGIEF Bits (TGR Interrupt Enable E and F)**

Each bit enables or disables interrupt requests by compare match between MTU0.TCNT and MTU0.TGRm (m = E, F).

**TTGE2 Bit (A/D Conversion Start Request Enable 2)**

Each bit enables or disables A/D conversion start requests by compare match between MTU0.TCNT and MTU0.TGRE.

- MTU5.TIER

Address(es): MTU5.TIER 000C 1CB2h

|                    | b7 | b6 | b5 | b4 | b3 | b2     | b1     | b0     |
|--------------------|----|----|----|----|----|--------|--------|--------|
|                    | —  | —  | —  | —  | —  | TGIE5U | TGIE5V | TGIE5W |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0      | 0      | 0      |

| Bit      | Symbol | Bit Name                | Description                                                                 | R/W |
|----------|--------|-------------------------|-----------------------------------------------------------------------------|-----|
| b0       | TGIE5W | TGR Interrupt Enable 5W | 0: Interrupt requests TGIW5 disabled<br>1: Interrupt requests TGIW5 enabled | R/W |
| b1       | TGIE5V | TGR Interrupt Enable 5V | 0: Interrupt requests TGIV5 disabled<br>1: Interrupt requests TGIV5 enabled | R/W |
| b2       | TGIE5U | TGR Interrupt Enable 5U | 0: Interrupt requests TGIU5 disabled<br>1: Interrupt requests TGIU5 enabled | R/W |
| b7 to b3 | —      | Reserved                | These bits are read as 0. The write value should be 0.                      | R/W |

**TGIE5m Bits (TGR Interrupt Enable 5m)**

Each bit enables or disables interrupt requests (TGIm5) (m = U, V, W).

### 24.2.9 Timer Status Register (TSR)

- MTU1.TSR, MTU2.TSR

Address(es): MTU1.TSR 000C 1385h, MTU2.TSR 000C 1405h

|                    | b7   | b6 | b5 | b4 | b3 | b2 | b1 | b0 |
|--------------------|------|----|----|----|----|----|----|----|
|                    | TCFD | —  | —  | —  | —  | —  | —  | —  |
| Value after reset: | 1    | 1  | 0  | 0  | 0  | 0  | 0  | 0  |

- MTU3.TSR, MTU4.TSR, MTU6.TSR, MTU7.TSR

Address(es): MTU3.TSR 000C 122Ch, MTU4.TSR 000C 122Dh, MTU6.TSR 000C 1A2Ch, MTU7.TSR 000C 1A2Dh

|                    | b7   | b6 | b5 | b4 | b3 | b2 | b1 | b0 |
|--------------------|------|----|----|----|----|----|----|----|
|                    | TCFD | —  | —  | —  | —  | —  | —  | —  |
| Value after reset: | 1    | 1  | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit      | Symbol | Bit Name             | Description                                               | R/W |
|----------|--------|----------------------|-----------------------------------------------------------|-----|
| b5 to b0 | —      | Reserved             | The read value is undefined. The write value should be 1. | R/W |
| b6       | —      | Reserved             | This bit is read as 1. The write value should be 1.       | R/W |
| b7       | TCFD   | Count Direction Flag | 0: TCNT counts down<br>1: TCNT counts up                  | R   |

TSR indicates the states of each of the channels. The MTU has a total of six TSR registers, one each for MTU1 to MTU4, MTU6, and MTU7.

#### TCFD Flag (Count Direction Flag)

Status flag that indicates the direction in which TCNT is counting in MTU1 to MTU4, MTU6, and MTU7.

### 24.2.10 Timer Buffer Operation Transfer Mode Register (TBTM)

- MTU0.TBTM

Address(es): MTU0.TBTM 000C 1326h

| b7                 | b6 | b5 | b4 | b3 | b2   | b1   | b0   |
|--------------------|----|----|----|----|------|------|------|
| —                  | —  | —  | —  | —  | TTSE | TTSB | TTSA |
| Value after reset: | 0  | 0  | 0  | 0  | 0    | 0    | 0    |

- MTU3.TBTM, MTU4.TBTM, MTU6.TBTM, MTU7.TBTM

Address(es): MTU3.TBTM 000C 1238h, MTU4.TBTM 000C 1239h, MTU6.TBTM 000C 1A38h, MTU7.TBTM 000C 1A39h

| b7                 | b6 | b5 | b4 | b3 | b2 | b1   | b0   |
|--------------------|----|----|----|----|----|------|------|
| —                  | —  | —  | —  | —  | —  | TTSB | TTSA |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0    | 0    |

| Bit      | Symbol | Bit Name        | Description                                                                                                                                                              | R/W |
|----------|--------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0       | TTSA   | Timing Select A | 0: When compare match A occurs in each channel, data is transferred from TGRC to TGRA<br>1: When TCNT is cleared in each channel, data is transferred from TGRC to TGRA  | R/W |
| b1       | TTSB   | Timing Select B | 0: When compare match B occurs in each channel, data is transferred from TGRD to TGRB<br>1: When TCNT is cleared in each channel, data is transferred from TGRD to TGRB  | R/W |
| b2       | TTSE   | Timing Select E | 0: When compare match E occurs in MTU0, data is transferred from MTU0.TGRF to MTU0.TGRE<br>1: When MTU0.TCNT is cleared, data is transferred from MTU0.TGRF to MTU0.TGRE | R/W |
| b7 to b3 | —      | Reserved        | These bits are read as 0. The write value should be 0.                                                                                                                   | R/W |

TBTM specifies the timing for transferring data from the buffer register to the timer general register in PWM mode. The MTU has a total of five TBTM registers, one each for MTU0, MTU3, MTU4, MTU6, and MTU7.

#### TTSA Bit (Timing Select A)

This bit specifies the timing for transferring data from TGRC to TGRA in each channel when they are used together for buffer operation. When a channel is not set to PWM mode, do not set the TTSA bit in the channel to 1.

#### TTSB Bit (Timing Select B)

This bit specifies the timing for transferring data from TGRD to TGRB in each channel when they are used together for buffer operation. When a channel is not set to PWM mode, do not set the TTSB bit in the channel to 1.

#### TTSE Bit (Timing Select E)

This bit specifies the timing for transferring data from MTU0.TGRF to MTU0.TGRE when they are used together for buffer operation.

In MTU3, MTU4, MTU6, and MTU7, this bit is reserved. It is read as 0 and the write value should be 0. When a channel is not set to PWM mode, do not set the TTSE bit in the channel to 1.

### 24.2.11 Timer Input Capture Control Register (TICCR)

Address(es): MTU1.TICCR 000C 1390h

|                    | b7 | b6 | b5 | b4 | b3   | b2   | b1   | b0   |
|--------------------|----|----|----|----|------|------|------|------|
|                    | —  | —  | —  | —  | I2BE | I2AE | I1BE | I1AE |
| Value after reset: | 0  | 0  | 0  | 0  | 0    | 0    | 0    | 0    |

| Bit      | Symbol | Bit Name             | Description                                                                                                                                            | R/W |
|----------|--------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0       | I1AE   | Input Capture Enable | 0: Does not include the MTIOC1A pin in the MTU2.TGRA input capture conditions<br>1: Includes the MTIOC1A pin in the MTU2.TGRA input capture conditions | R/W |
| b1       | I1BE   | Input Capture Enable | 0: Does not include the MTIOC1B pin in the MTU2.TGRB input capture conditions<br>1: Includes the MTIOC1B pin in the MTU2.TGRB input capture conditions | R/W |
| b2       | I2AE   | Input Capture Enable | 0: Does not include the MTIOC2A pin in the MTU1.TGRA input capture conditions<br>1: Includes the MTIOC2A pin in the MTU1.TGRA input capture conditions | R/W |
| b3       | I2BE   | Input Capture Enable | 0: Does not include the MTIOC2B pin in the MTU1.TGRB input capture conditions<br>1: Includes the MTIOC2B pin in the MTU1.TGRB input capture conditions | R/W |
| b7 to b4 | —      | Reserved             | These bits are read as 0. The write value should be 0.                                                                                                 | R/W |

TICCR specifies input capture conditions when MTU1.TCNT and MTU2.TCNT are cascaded. The MTU has one TICCR for MTU1.



## 24.2.12 Timer Synchronous Clear Register (TSYCR)

Address(es): MTU6.TSYCR 000C 1A50h

|                    | b7   | b6   | b5   | b4   | b3   | b2   | b1   | b0   |
|--------------------|------|------|------|------|------|------|------|------|
|                    | CE0A | CE0B | CE0C | CE0D | CE1A | CE1B | CE2A | CE2B |
| Value after reset: | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |

| Bit | Symbol | Bit Name        | Description                                                                                                                                                                             | R/W |
|-----|--------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0  | CE2B   | Clear Enable 2B | 0: Disables counter clearing by the MTU2.TGIB2 interrupt generation timing.<br>1: Enables counter clearing by the MTU2.TGIB2 interrupt generation timing.                               | R/W |
| b1  | CE2A   | Clear Enable 2A | 0: Disables counter clearing by the MTU2.TGIA2 interrupt generation timing* <sup>1</sup> .<br>1: Enables counter clearing by the MTU2.TGIA2 interrupt generation timing* <sup>1</sup> . | R/W |
| b2  | CE1B   | Clear Enable 1B | 0: Disables counter clearing by the MTU1.TGIB1 interrupt generation timing* <sup>1</sup> .<br>1: Enables counter clearing by the MTU1.TGIB1 interrupt generation timing* <sup>1</sup> . | R/W |
| b3  | CE1A   | Clear Enable 1A | 0: Disables counter clearing by the MTU1.TGIA1 interrupt generation timing* <sup>1</sup> .<br>1: Enables counter clearing by the MTU1.TGIA1 interrupt generation timing* <sup>1</sup> . | R/W |
| b4  | CE0D   | Clear Enable 0D | 0: Disables counter clearing by the MTU0.TGID0 interrupt generation timing* <sup>1</sup> .<br>1: Enables counter clearing by the MTU0.TGID0 interrupt generation timing* <sup>1</sup> . | R/W |
| b5  | CE0C   | Clear Enable 0C | 0: Disables counter clearing by the MTU0.TGIC0 interrupt generation timing* <sup>1</sup> .<br>1: Enables counter clearing by the MTU0.TGIC0 interrupt generation timing* <sup>1</sup> . | R/W |
| b6  | CE0B   | Clear Enable 0B | 0: Disables counter clearing by the MTU0.TGIB0 interrupt generation timing* <sup>1</sup> .<br>1: Enables counter clearing by the MTU0.TGIB0 interrupt generation timing* <sup>1</sup> . | R/W |
| b7  | CE0A   | Clear Enable 0A | 0: Disables counter clearing by the MTU0.TGIA0 interrupt generation timing* <sup>1</sup> .<br>1: Enables counter clearing by the MTU0.TGIA0 interrupt generation timing* <sup>1</sup> . | R/W |

Note 1. This does not depend on the TIERn.TGIE<sub>m</sub> bit setting. (n = 0, 1, 2; m = A, B, C, D)

TSYCR specifies synchronous clear conditions for MTU6.TCNT and MTU7.TCNT. The MTU has one TSYCR for MTU1.

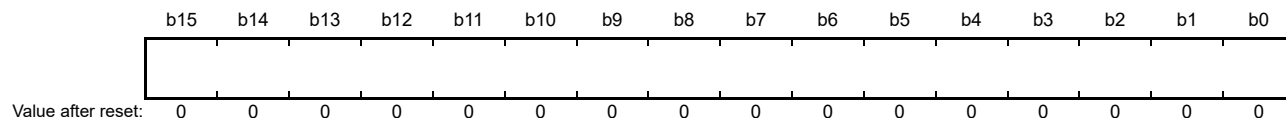
**CE<sub>n</sub>m Bits (Clear Enable nm; n = 0, 1, 2; m = A, B, C, D)**

These bits enable or disable counter clearing by the MTU<sub>n</sub>.TGIE<sub>m</sub> interrupt generation timing.

### 24.2.13 Timer Counter (TCNT)

- MTU0.TCNT to MTU7.TCNT

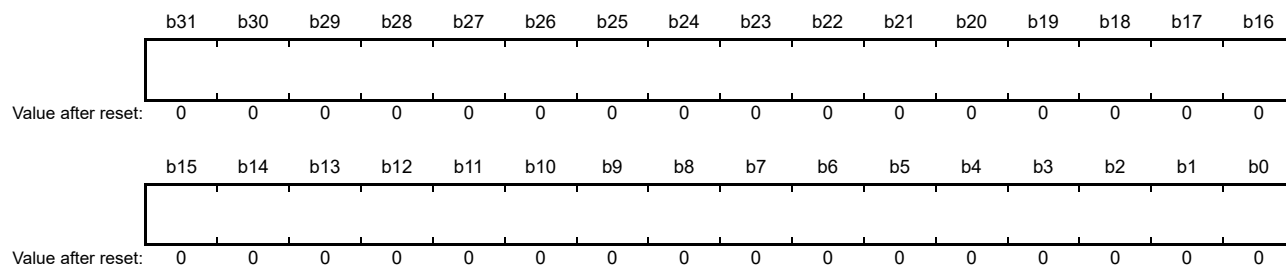
Address(es): MTU0.TCNT 000C 1306h, MTU1.TCNT 000C 1386h, MTU2.TCNT 000C 1406h, MTU3.TCNT 000C 1210h,  
MTU4.TCNT 000C 1212h, MTU5.TCNTU 000C 1C80h, MTU5.TCNTV 000C 1C90h, MTU5.TCNTW 000C 1CA0h,  
MTU6.TCNT 000C 1A10h, MTU7.TCNT 000C 1A12h



Note: TCNT must not be accessed in 8 bits; it should be accessed in 16 bits.

- MTU8.TCNT

Address(es): MTU8.TCNT 000C 1608h



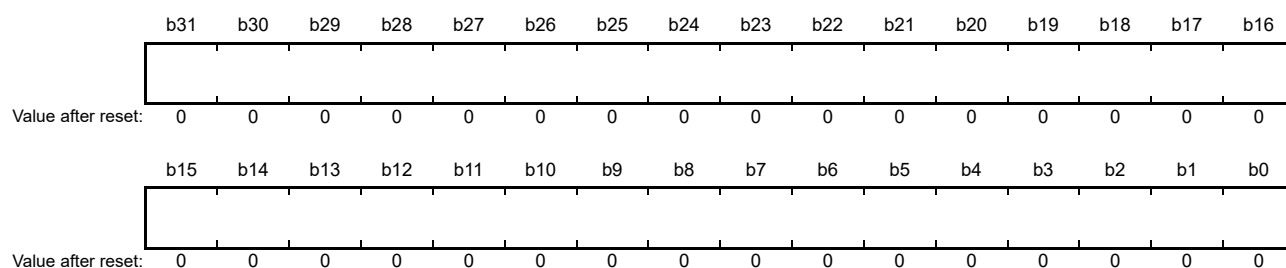
Note: MTU8.TCNT must not be accessed in 8 or 16 bits; it should be accessed in 32 bits.

MTU0.TCNT to MTU7.TCNT are 16-bit readable/writable counters and MTU8.TCNT is a 32-bit readable/writable counter. The MTU has a total of 11 TCNT counters, one each for MTU0 to MTU4 and MTU6 to MTU8 and three (MTU5.TCNTU, TCNTV, and TCNTW) for MTU5. The TCNT counters in MTU0 to MTU4, MTU6, and MTU7 are initialized to 0000h by a reset, and the MTU8.TCNT counter is initialized to 00000000h by a reset. MTU5.TCNTU, MTU5.TCNTV, and MTU5.TCNTW are initialized to 0000h by a reset.

In MTU0 to MTU4, MTU6, and MTU7, the TCNT counters must not be accessed in 8-bit units; they should be accessed in 16-bit units. The MTU8.TCNT counter must not be accessed in 8- or 16-bit units; it should be accessed in 32-bit units. The MTU1.TCNT and MTU2.TCNT counters are read as 0000h when TMDR3.LWA is 1. Refer to section 24.2.5, Timer Mode Register 3 (TMDR3) for details.

### 24.2.14 Timer Longword Counter (TCNTLW)

Address(es): MTU1.TCNTLW 000C 13A0h



Note: TCNTLW must not be accessed in 8 or 16 bits; it should be accessed in 32 bits.

The TCNTLW counter is a 32-bit readable/writable counter. Only one counter of this type is provided, and is formed by combining MTU1.TCNT and MTU2.TCNT. Such operation is only effective when TMDR3.LWA is 1. The TCNTLW counter is initialized to 0000 0000h by a reset. This counter is read as 0000 0000h when TMDR3.LWA is 0. Refer to section 24.2.5, Timer Mode Register 3 (TMDR3) for details. This register can only be used in 32-bit phase counting mode.

## 24.2.15 Timer General Register m (TGRm) (m = A, B, C, D, E, F, U, V, W)

- MTU0.TGRm to MTU7.TGRm

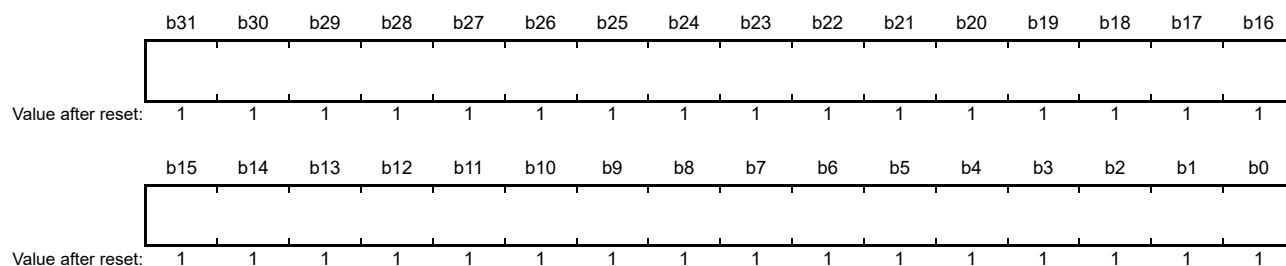
Address(es): MTU0.TGRA 000C 1308h, MTU0.TGRB 000C 130Ah, MTU0.TGRC 000C 130Ch, MTU0.TGRD 000C 130Eh, MTU0.TGRE 000C 1320h, MTU0.TGRF 000C 1322h, MTU1.TGRA 000C 1388h, MTU1.TGRB 000C 138Ah, MTU2.TGRA 000C 1408h, MTU2.TGRB 000C 140Ah, MTU3.TGRA 000C 1218h, MTU3.TGRB 000C 121Ah, MTU3.TGRC 000C 1224h, MTU3.TGRD 000C 1226h, MTU3.TGRE 000C 1272h, MTU4.TGRA 000C 121Ch, MTU4.TGRB 000C 121Eh, MTU4.TGRC 000C 1228h, MTU4.TGRD 000C 122Ah, MTU4.TGRE 000C 1274h, MTU4.TGRF 000C 1276h, MTU5.TGRU 000C 1C82h, MTU5.TGRV 000C 1C92h, MTU5.TGRW 000C 1CA2h, MTU6.TGRA 000C 1A18h, MTU6.TGRB 000C 1A1Ah, MTU6.TGRC 000C 1A24h, MTU6.TGRD 000C 1A26h, MTU6.TGRE 000C 1A72h, MTU7.TGRA 000C 1A1Ch, MTU7.TGRB 000C 1A1Eh, MTU7.TGRC 000C 1A28h, MTU7.TGRD 000C 1A2Ah, MTU7.TGRE 000C 1A74h, MTU7.TGRF 000C 1A76h



Note: The TGRm registers must not be accessed in 8 bits; They should be accessed in 16 bits. The initial value of the TGRm register is FFFFh.

- MTU8.TGRm

Address(es): MTU8.TGRA 000C 160Ch, MTU8.TGRB 000C 1610h, MTU8.TGRC 000C 1614h, MTU8.TGRD 000C 1618h



Note: MTU8.TGRm must not be accessed in 8 or 16 bits; it should be accessed in 32 bits.

The MTU0.TGRm to MTU7.TGRm registers are 16-bit readable/writable registers; the MTU8.TGRm register is a 32-bit readable/writable register. The MTU has a total of 39 TGRm registers, six for MTU0, two each for MTU1 and MTU2, five each for MTU3 and MTU6, six each for MTU4 and MTU7, and three for MTU5, and four for MTU8.

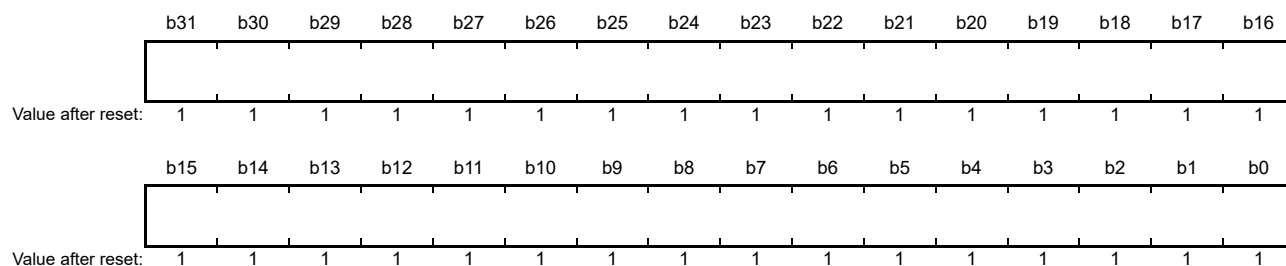
The TGRA, TGRB, TGRC, and TGRD registers function as either output compare or input capture registers. The TGRC and TGRD registers for MTU0, MTU3, MTU4, MTU6, MTU7, and MTU8 can also be designated for operation as buffer registers. TGRm buffer register combinations are TGRA and TGRC, and TGRB and TGRD.

MTU0.TGRE and MTU0.TGRF function as compare registers. When the MTU0.TCNT count matches the MTU0.TGRE value, an A/D conversion start request can be issued. The TGRF register can also be designated for operation as a buffer register. TGRm buffer register combination is TGRE and TGRF. MTU5.TGRU, MTU5.TGRV, and MTU5.TGRW function as compare match, input capture, or external pulse width measurement registers.

The MTU1.TGRA, MTU2.TGRA, MTU1.TGRB, and MTU2.TGRB registers are read as 0000h when TMDR3.LWA is 1. Refer to section 24.2.5, Timer Mode Register 3 (TMDR3) for details.

### 24.2.16 Timer Longword General Register m (TGRmLW) (m = A, B)

Address(es): MTU1.TGRALW 000C 13A4h, MTU1.TGRBLW 000C 13A8h



Note: TGRALW and TGRBLW must not be accessed in 8 or 16 bits; it should be accessed in 32 bits.

The TGRmLW register is a 32-bit readable/writable register. Two general registers of this type are provided, and are formed by combining MTU1.TGRm and MTU2.TGRm. Such operation is only effective when TMDR3.LWA is 1. The TGRmLW register is initialized to FFFF FFFFh by a reset, but it is read as 0000 0000h when TMDR3.LWA is 0. Refer to section 24.2.5, Timer Mode Register 3 (TMDR3) for details.

The TGRALW and TGRBLW registers function as compare match or input capture registers which can only be used in 32-bit phase counting mode.

### 24.2.17 Timer Start Registers (TSTRA, TSTRB, TSTR)

- MTU.TSTRA (for MTU0, MTU1, MTU2, MTU3, MTU4, and MTU8)

Address(es): MTU.TSTRA 000C 1280h

| b7   | b6   | b5 | b4 | b3   | b2   | b1   | b0   |
|------|------|----|----|------|------|------|------|
| CST4 | CST3 | —  | —  | CST8 | CST2 | CST1 | CST0 |

Value after reset: 0 0 0 0 0 0 0 0

| Bit    | Symbol | Bit Name        | Description                                                                | R/W |
|--------|--------|-----------------|----------------------------------------------------------------------------|-----|
| b0     | CST0   | Counter Start 0 | 0: MTU0.TCNT counting is stopped<br>1: MTU0.TCNT performs count operation  | R/W |
| b1     | CST1   | Counter Start 1 | 0: MTU1.TCNT counting is stopped<br>1: MTU1.TCNT performs count operation  | R/W |
| b2     | CST2   | Counter Start 2 | 0: MTU2.TCNT counting is stopped<br>1: MTU2.TCNT performs count operation  | R/W |
| b3     | CST8   | Counter start 8 | 0: MTU8.TCNT counting is stopped<br>1: MTU8.TCNT performs count operation. | R/W |
| b5, b4 | —      | Reserved        | These bits are read as 0. The write value should be 0.                     | R/W |
| b6     | CST3   | Counter Start 3 | 0: MTU3.TCNT counting is stopped<br>1: MTU3.TCNT performs count operation  | R/W |
| b7     | CST4   | Counter Start 4 | 0: MTU4.TCNT counting is stopped<br>1: MTU4.TCNT performs count operation  | R/W |

Note: When 1 is written to a bit in TCSYSTR, the corresponding bit in TSTRA is also set to 1 automatically.

The TSTRA register starts or stops TCNT operation in MTU0 to MTU4 and MTU8.

TSTRB starts or stops TCNT operation in MTU6 and MTU7.

TSTR starts or stops TCNT operation in MTU5.

Before setting the operating mode in TMDR1 or setting the TCNT count clock in TCR, be sure to stop the TCNT counter.

#### CSTn Bits (Counter Start n) (n = 0, 1, 2, 3, 4, 8)

Each bit starts or stops TCNT in the corresponding channel.

If 0 is written to the CSTn bit during operation with the MTIOC pin designated for output, the counter stops. At this time, initial output level specified in the TOCR1A or TOCR2A register is output from the MTIOC pin in complementary PWM mode or reset-synchronized PWM mode. In any mode other than complementary PWM mode and reset synchronous PWM mode, the output compare signal level from the MTIOC pin is retained. If TIOR is written to while the CSTn bit is 0, the pin output level will be changed to the specified initial output value.

- MTU.TSTRB (for MTU6 and MTU7)

Address(es): MTU.TSTRB 000C 1A80h

|                    | b7   | b6   | b5 | b4 | b3 | b2 | b1 | b0 |
|--------------------|------|------|----|----|----|----|----|----|
|                    | CST7 | CST6 | —  | —  | —  | —  | —  | —  |
| Value after reset: | 0    | 0    | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit      | Symbol | Bit Name        | Description                                                               | R/W |
|----------|--------|-----------------|---------------------------------------------------------------------------|-----|
| b5 to b0 | —      | Reserved        | These bits are read as 0. The write value should be 0.                    | R/W |
| b6       | CST6   | Counter Start 6 | 0: MTU6.TCNT counting is stopped<br>1: MTU6.TCNT performs count operation | R/W |
| b7       | CST7   | Counter Start 7 | 0: MTU7.TCNT counting is stopped<br>1: MTU7.TCNT performs count operation | R/W |

Note: When 1 is written to a bit in TCSYSTR, the corresponding bit in TSTRB is also set to 1 automatically.

### CSTn Bits (Counter Start n) (n = 6, 7)

Each bit starts or stops TCNT in the corresponding channel.

If 0 is written to the CSTn bit during operation with the MTIOC pin designated for output, the counter stops. At this time the MTIOC pin output the initial output level set in the TOCR1B or TOCR2B register in complementary PWM mode or reset-synchronized PWM mode, but the output compare signal level from the MTIOC pin is retained in the other modes. If TIOR is written to while the CSTn bit is 0, the pin output level will be changed to the specified initial output value.

- MTU5.TSTR

Address(es): MTU5.TSTR 000C 1CB4h

|                    | b7 | b6 | b5 | b4 | b3 | b2    | b1    | b0    |
|--------------------|----|----|----|----|----|-------|-------|-------|
|                    | —  | —  | —  | —  | —  | CSTU5 | CSTV5 | CSTW5 |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0     | 0     | 0     |

| Bit      | Symbol | Bit Name         | Description                                                                 | R/W |
|----------|--------|------------------|-----------------------------------------------------------------------------|-----|
| b0       | CSTW5  | Counter Start W5 | 0: MTU5.TCNTW counting is stopped<br>1: MTU5.TCNTW performs count operation | R/W |
| b1       | CSTV5  | Counter Start V5 | 0: MTU5.TCNTV counting is stopped<br>1: MTU5.TCNTV performs count operation | R/W |
| b2       | CSTU5  | Counter Start U5 | 0: MTU5.TCNTU counting is stopped<br>1: MTU5.TCNTU performs count operation | R/W |
| b7 to b3 | —      | Reserved         | These bits are read as 0. The write value should be 0.                      | R/W |

## 24.2.18 Timer Synchronous Register m (TSYRm) (m = A, B)

- MTU.TSYRA (for MTU0 to MTU4)

Address(es): MTU.TSYRA 000C 1281h

| b7                   | b6    | b5 | b4 | b3 | b2    | b1    | b0    |
|----------------------|-------|----|----|----|-------|-------|-------|
| SYNC4                | SYNC3 | —  | —  | —  | SYNC2 | SYNC1 | SYNC0 |
| Value after reset: 0 | 0     | 0  | 0  | 0  | 0     | 0     | 0     |

| Bit      | Symbol | Bit Name                      | Description                                                                                                                                                                                                  | R/W |
|----------|--------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0       | SYNC0  | Timer Synchronous Operation 0 | 0: MTU0.TCNT operates independently (TCNT setting/clearing is not related to other channels).<br>1: MTU0.TCNT performs synchronous operation.<br>(TCNT synchronous setting/synchronous clearing is enabled.) | R/W |
| b1       | SYNC1  | Timer Synchronous Operation 1 | 0: MTU1.TCNT operates independently (TCNT setting/clearing is not related to other channels).<br>1: MTU1.TCNT performs synchronous operation.<br>(TCNT synchronous setting/synchronous clearing is enabled.) | R/W |
| b2       | SYNC2  | Timer Synchronous Operation 2 | 0: MTU2.TCNT operates independently (TCNT setting/clearing is not related to other channels).<br>1: MTU2.TCNT performs synchronous operation.<br>(TCNT synchronous setting/synchronous clearing is enabled.) | R/W |
| b5 to b3 | —      | Reserved                      | These bits are read as 0. The write value should be 0.                                                                                                                                                       | R/W |
| b6       | SYNC3  | Timer Synchronous Operation 3 | 0: MTU3.TCNT operates independently (TCNT setting/clearing is not related to other channels).<br>1: MTU3.TCNT performs synchronous operation.<br>(TCNT synchronous setting/synchronous clearing is enabled.) | R/W |
| b7       | SYNC4  | Timer Synchronous Operation 4 | 0: MTU4.TCNT operates independently (TCNT setting/clearing is not related to other channels).<br>1: MTU4.TCNT performs synchronous operation.<br>(TCNT synchronous setting/synchronous clearing is enabled.) | R/W |

TSYRA selects independent operation or synchronous operation of TCNT in MTU0 to MTU4.

TSYRB selects independent operation or synchronous operation of TCNT in MTU6 and MTU7.

A channel performs synchronous operation when the corresponding bit in TSYR is set to 1.

**SYNCn Bits (Timer Synchronous Operation n) (n = 0, 1, 2, 3, 4)**

Each bit selects whether operation is independent of or synchronized with other channels.

When synchronous operation is selected, the TCNT synchronous setting of multiple channels and synchronous clearing by counter clearing on another channel are possible.

To set synchronous operation, the SYNC bits for at least two channels must be set to 1. To set synchronous clearing, in addition to the SYNC bit, the TCNT clearing source must also be set by means of the TCR.CCLR[2:0] bits.



- MTU.TSYRB (for MTU6 and MTU7)

Address(es): MTU.TSYRB 000C 1A81h

| b7    | b6    | b5 | b4 | b3 | b2 | b1 | b0 |
|-------|-------|----|----|----|----|----|----|
| SYNC7 | SYNC6 | —  | —  | —  | —  | —  | —  |

Value after reset: 0 0 0 0 0 0 0 0

| Bit      | Symbol | Bit Name                      | Description                                                                                                                                                                                                  | R/W |
|----------|--------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b5 to b0 | —      | Reserved                      | These bits are read as 0. The write value should be 0.                                                                                                                                                       | R/W |
| b6       | SYNC6  | Timer Synchronous Operation 6 | 0: MTU6.TCNT operates independently (TCNT setting/clearing is not related to other channels).<br>1: MTU6.TCNT performs synchronous operation.<br>(TCNT synchronous setting/synchronous clearing is enabled.) | R/W |
| b7       | SYNC7  | Timer Synchronous Operation 7 | 0: MTU7.TCNT operates independently (TCNT setting/clearing is not related to other channels).<br>1: MTU7.TCNT performs synchronous operation.<br>(TCNT synchronous setting/synchronous clearing is enabled.) | R/W |

**SYNCn Bits (Timer Synchronous Operation n) (n = 6, 7)**

Each bit selects whether operation is independent of or synchronized with other channels.

When synchronous operation is selected, the TCNT synchronous setting of multiple channels and synchronous clearing by counter clearing on another channel are possible.

To set synchronous operation, the SYNC bits for at least two channels must be set to 1. To set synchronous clearing, in addition to the SYNC bit, the TCNT clearing source must also be set by means of bits TCR.CCLR[2:0].

## 24.2.19 Timer Counter Synchronous Start Register (TCSYSTR)

Address(es): MTU.TCSYSTR 000C 1282h

| b7   | b6   | b5   | b4   | b3   | b2 | b1   | b0   |
|------|------|------|------|------|----|------|------|
| SCH0 | SCH1 | SCH2 | SCH3 | SCH4 | —  | SCH6 | SCH7 |
| 0    | 0    | 0    | 0    | 0    | 0  | 0    | 0    |

Value after reset:

| Bit | Symbol | Bit Name            | Description                                                                                         | R/W     |
|-----|--------|---------------------|-----------------------------------------------------------------------------------------------------|---------|
| b0  | SCH7   | Synchronous Start 7 | 0: Does not specify synchronous start for MTU7.TCNT<br>1: Specifies synchronous start for MTU7.TCNT | R/(W)*1 |
| b1  | SCH6   | Synchronous Start 6 | 0: Does not specify synchronous start for MTU6.TCNT<br>1: Specifies synchronous start for MTU6.TCNT | R/(W)*1 |
| b2  | —      | Reserved            | This bit is read as 0. The write value should be 0.                                                 | R       |
| b3  | SCH4   | Synchronous Start 4 | 0: Does not specify synchronous start for MTU4.TCNT<br>1: Specifies synchronous start for MTU4.TCNT | R/(W)*1 |
| b4  | SCH3   | Synchronous Start 3 | 0: Does not specify synchronous start for MTU3.TCNT<br>1: Specifies synchronous start for MTU3.TCNT | R/(W)*1 |
| b5  | SCH2   | Synchronous Start 2 | 0: Does not specify synchronous start for MTU2.TCNT<br>1: Specifies synchronous start for MTU2.TCNT | R/(W)*1 |
| b6  | SCH1   | Synchronous Start 1 | 0: Does not specify synchronous start for MTU1.TCNT<br>1: Specifies synchronous start for MTU1.TCNT | R/(W)*1 |
| b7  | SCH0   | Synchronous Start 0 | 0: Does not specify synchronous start for MTU0.TCNT<br>1: Specifies synchronous start for MTU0.TCNT | R/(W)*1 |

Note 1. Only 1 can be written to this bit. This bit is automatically cleared when the corresponding counter starts.

TCSYSTR specifies synchronous start of the counters.

### SCH7 Bit (Synchronous Start 7)

This bit controls synchronous start of MTU7.TCNT.

[Clearing condition]

- When 1 is set to the TSTRB.CST7 bit while SCH7 = 1

### SCH6 Bit (Synchronous Start 6)

This bit controls synchronous start of MTU6.TCNT.

[Clearing condition]

- When 1 is set to the TSTRB.CST6 bit while SCH6 = 1

### SCH4 Bit (Synchronous Start 4)

This bit controls synchronous start of MTU4.TCNT.

[Clearing condition]

- When 1 is set to the TSTRA.CST4 bit while SCH4 = 1

### SCH3 Bit (Synchronous Start 3)

This bit controls synchronous start of MTU3.TCNT.

[Clearing condition]

- When 1 is set to the TSTRA.CST3 bit while SCH3 = 1

**SCH2 Bit (Synchronous Start 2)**

This bit controls synchronous start of MTU2.TCNT.

[Clearing condition]

- When 1 is set to the TSTRA.CST2 bit while SCH2 = 1

**SCH1 Bit (Synchronous Start 1)**

This bit controls synchronous start of MTU1.TCNT.

[Clearing condition]

- When 1 is set to the TSTRA.CST1 bit while SCH1 = 1

**SCH0 Bit (Synchronous Start 0)**

This bit controls synchronous start of MTU0.TCNT.

[Clearing condition]

- When 1 is set to the TSTRA.CST0 bit while SCH0 = 1

## 24.2.20 Timer Read/Write Enable Register m (TRWERm) (m = A, B)

Address(es): MTU.TRWERA 000C 1284h, MTU.TRWERB 000C 1A84h

|                    | b7 | b6 | b5 | b4 | b3 | b2 | b1 | b0  |
|--------------------|----|----|----|----|----|----|----|-----|
|                    | —  | —  | —  | —  | —  | —  | —  | RWE |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1   |

| Bit      | Symbol | Bit Name          | Description                                                                                           | R/W |
|----------|--------|-------------------|-------------------------------------------------------------------------------------------------------|-----|
| b0       | RWE    | Read/Write Enable | 0: Read/write access to the registers is disabled<br>1: Read/write access to the registers is enabled | R/W |
| b7 to b1 | —      | Reserved          | These bits are read as 0. The write value should be 0.                                                | R/W |

TRWERA enables or disables access to the registers and counters that have write-protection capability against accidental modification in MTU3 and MTU4.

TRWERB enables or disables access to the registers and counters that have write-protection capability against accidental modification in MTU6 and MTU7.

**RWE Bit (Read/Write Enable)**

This bit enables or disables access to the registers that have write-protection capability against accidental modification.  
[Clearing condition]

- When 0 is written to the RWE bit after reading RWE = 1
- Registers and Counters having Write-Protection Capability against Accidental Modification (TRWERA)  
24 registers: MTUn.TCR, MTUn.TCR2, MTUn.TMDR1, MTUn.TIORH, MTUn.TIORL, MTUn.TIER, MTUn.TGRA, MTUn.TGRB, MTU.TOERA, MTU.TOCR1A, MTU.TOCR2A, MTU.TGCRA, MTU.TCDRA, MTU.TDDRA, and MTUn.TCNT (n = 3, 4)
- Registers and Counters having Write-Protection Capability against Accidental Modification (TRWERB)  
23 registers: MTUn.TCR, MTUn.TCR2, MTUn.TMDR1, MTUn.TIORH, MTUn.TIORL, MTUn.TIER, MTUn.TGRA, MTUn.TGRB, MTU.TOERB, MTU.TOCR1B, MTU.TOCR2B, MTU.TCDRB, MTU.TDDRB, and MTUn.TCNT (n = 6, 7)

## 24.2.21 Timer Output Master Enable Register m (TOERm) (m = A, B)

- MTU.TOERA

Address(es): MTU.TOERA 000C 120Ah

|                    | b7 | b6 | b5   | b4   | b3   | b2   | b1   | b0   |
|--------------------|----|----|------|------|------|------|------|------|
|                    | —  | —  | OE4D | OE4C | OE3D | OE4B | OE4A | OE3B |
| Value after reset: | 1  | 1  | 0    | 0    | 0    | 0    | 0    | 0    |

| Bit    | Symbol | Bit Name              | Description                                             | R/W |
|--------|--------|-----------------------|---------------------------------------------------------|-----|
| b0     | OE3B   | Master Enable MTIOC3B | 0: MTU output is disabled*1<br>1: MTU output is enabled | R/W |
| b1     | OE4A   | Master Enable MTIOC4A | 0: MTU output is disabled*1<br>1: MTU output is enabled | R/W |
| b2     | OE4B   | Master Enable MTIOC4B | 0: MTU output is disabled*1<br>1: MTU output is enabled | R/W |
| b3     | OE3D   | Master Enable MTIOC3D | 0: MTU output is disabled*1<br>1: MTU output is enabled | R/W |
| b4     | OE4C   | Master Enable MTIOC4C | 0: MTU output is disabled*1<br>1: MTU output is enabled | R/W |
| b5     | OE4D   | Master Enable MTIOC4D | 0: MTU output is disabled*1<br>1: MTU output is enabled | R/W |
| b7, b6 | —      | Reserved              | These bits are read as 1. The write value should be 1.  | R/W |

Note 1. To output the inactive level from each pin when the MTU output is set to disabled, first set the data direction register (PDR) and port output data register (PODR) of I/O ports to output the inactive level from general I/O ports, and then set the port mode register (PMR) to use general I/O ports. For details, refer to section 22, I/O Ports.

TOERA enables or disables output settings for output pins MTIOC4D, MTIOC4C, MTIOC3D, MTIOC4B, MTIOC4A, and MTIOC3B.

These pins do not output correctly if the bits in the TOERA register have not been set. In MTU3 and MTU4, set TOERA prior to setting TIOR.

Set MTU.TOERA after setting the CST3 and CST4 bits in MTU.TSTRA to 0 (refer to Figure 24.44 and Figure 24.48).

- MTU.TOERB

Address(es): MTU.TOERB 000C 1A0Ah

|                    | b7 | b6 | b5   | b4   | b3   | b2   | b1   | b0   |
|--------------------|----|----|------|------|------|------|------|------|
|                    | —  | —  | OE7D | OE7C | OE6D | OE7B | OE7A | OE6B |
| Value after reset: | 1  | 1  | 0    | 0    | 0    | 0    | 0    | 0    |

| Bit    | Symbol | Bit Name              | Description                                             | R/W |
|--------|--------|-----------------------|---------------------------------------------------------|-----|
| b0     | OE6B   | Master Enable MTIOC6B | 0: MTU output is disabled*1<br>1: MTU output is enabled | R/W |
| b1     | OE7A   | Master Enable MTIOC7A | 0: MTU output is disabled*1<br>1: MTU output is enabled | R/W |
| b2     | OE7B   | Master Enable MTIOC7B | 0: MTU output is disabled*1<br>1: MTU output is enabled | R/W |
| b3     | OE6D   | Master Enable MTIOC6D | 0: MTU output is disabled*1<br>1: MTU output is enabled | R/W |
| b4     | OE7C   | Master Enable MTIOC7C | 0: MTU output is disabled*1<br>1: MTU output is enabled | R/W |
| b5     | OE7D   | Master Enable MTIOC7D | 0: MTU output is disabled*1<br>1: MTU output is enabled | R/W |
| b7, b6 | —      | Reserved              | These bits are read as 1. The write value should be 1.  | R/W |

Note 1. To output the inactive level from each pin when the MTU output is set to disabled, first set the data direction register (PDR) and port output data register (PODR) of I/O ports to output the inactive level from general I/O ports, and then set the port mode register (PMR) to use general I/O ports. For details, refer to section 22, I/O Ports.

TOERB enables or disables output settings for output pins MTIOC7D, MTIOC7C, MTIOC6D, MTIOC7B, MTIOC7A, and MTIOC6B.

These pins do not output correctly if the bits in the TOERB register have not been set. In MTU6, and MTU7, set TOERB prior to setting TIOR.

Set MTU.TOERB after setting the CST6 and CST7 bits in MTU.TSTRB to 0 (refer to Figure 24.44 and Figure 24.48).

## 24.2.22 Timer Output Control Register 1m (TOCR1m) (m = A, B)

Address(es): MTU.TOCR1A 000C 120Eh, MTU.TOCR1B 000C 1A0Eh

| b7                 | b6   | b5 | b4 | b3   | b2   | b1   | b0   |
|--------------------|------|----|----|------|------|------|------|
| —                  | PSYE | —  | —  | TOCL | TOCS | OLSN | OLSP |
| Value after reset: | 0    | 0  | 0  | 0    | 0    | 0    | 0    |

| Bit    | Symbol | Bit Name                            | Description                                                                                                                 | R/W |
|--------|--------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----|
| b0     | OLSP   | Output Level Select P*1, *3         | Refer to Table 24.42.                                                                                                       | R/W |
| b1     | OLSN   | Output Level Select N*1, *3         | Refer to Table 24.43.                                                                                                       | R/W |
| b2     | TOCS   | TOC Select                          | 0: TOCR1m setting is selected (m = A, B)<br>1: TOCR2m setting is selected                                                   | R/W |
| b3     | TOCL   | TOC Register Write Protection*2, *4 | 0: Write access to the TOCS, OLSN, and OLSP bits is enabled<br>1: Write access to the TOCS, OLSN, and OLSP bits is disabled | R/W |
| b5, b4 | —      | Reserved                            | These bits are read as 0. The write value should be 0.                                                                      | R/W |
| b6     | PSYE   | PWM Synchronous Output Enable       | 0: Toggle output is disabled<br>1: Toggle output is enabled                                                                 | R/W |
| b7     | —      | Reserved                            | This bit is read as 0. The write value should be 0.                                                                         | R/W |

Note 1. Setting the TOCR1m.TOCS bit to 0 makes this bit setting valid.

Note 2. Setting the TOCR1m.TOCL bit to 1 prevents accidental modification when the CPU goes out of control.

Note 3. If dead-time is not generated, the negative-phase output is the exact inverse of the positive-phase output. In this case, only the OLSP bit is valid.

Note 4. This bit can be set to 1 only once after a reset. After 1 is written, 0 cannot be written to the bit.

TOCR1A and TOCR1B enable or disable PWM-synchronized toggle output in complementary PWM mode and reset-synchronized PWM mode, and control inversion of PWM output level.

### OLSP Bit (Output Level Select P)

This bit selects the positive-phase output level in reset-synchronized PWM mode and complementary PWM mode. The initial output is selected while the counter is stopped.

### OLSN Bit (Output Level Select N)

This bit selects the negative-phase output level in reset-synchronized PWM mode and complementary PWM mode. The initial output is selected while the counter is stopped.

### TOCS Bit (TOC Select)

This bit selects either the TOCR1m or TOCR2m (m = A, B) setting to be used for the output level in complementary PWM mode and reset-synchronized PWM mode.

### TOCL Bit (TOC Register Write Protection)

This bit enables or disables write access to the TOCS, OLSN, and OLSP bits in TOCR1m (m = A, B).

### PSYE Bit (PWM Synchronous Output Enable)

This bit enables or disables toggle output synchronized with the PWM period from the MTIOC3A or MTIOC6A pin.

**Table 24.42 Output Level Select Function**

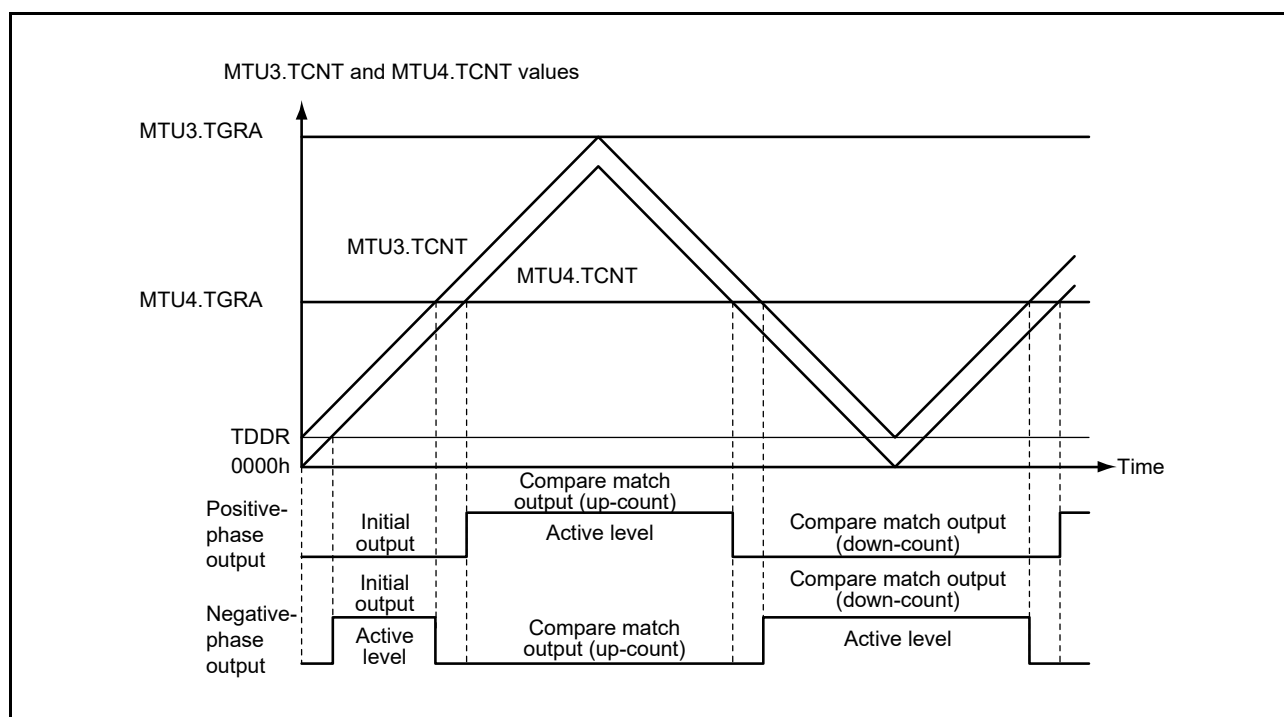
| Bit 0 | Function       |              |                      |               |
|-------|----------------|--------------|----------------------|---------------|
| OLSP  | Initial Output | Active Level | Compare Match Output |               |
|       |                |              | Up-Counting          | Down-Counting |
| 0     | High level     | Low level    | Low level            | High level    |
| 1     | Low level      | High level   | High level           | Low level     |

**Table 24.43 Output Level Select Function**

| Bit 1 | Function       |              |                      |               |
|-------|----------------|--------------|----------------------|---------------|
| OLSN  | Initial Output | Active Level | Compare Match Output |               |
|       |                |              | Up-Counting          | Down-Counting |
| 0     | High level     | Low level    | High level           | Low level     |
| 1     | Low level      | High level   | Low level            | High level    |

Note: The initial output value of negative-phase waveform changes to an active level after the dead time has passed since counting starts.

Figure 24.3 shows an example of output in complementary PWM mode (one phase) when OLSN = 1 and OLSP = 1.

**Figure 24.3 Example of Output in Complementary PWM Mode**



### 24.2.23 Timer Output Control Register 2m (TOCR2m) (m = A, B)

Address(es): MTU.TOCR2A 000C 120Fh, MTU.TOCR2B 000C 1A0Fh

| b7                 | b6    | b5    | b4    | b3    | b2    | b1    | b0 |
|--------------------|-------|-------|-------|-------|-------|-------|----|
| BF[1:0]            | OLS3N | OLS3P | OLS2N | OLS2P | OLS1N | OLS1P |    |
| Value after reset: | 0     | 0     | 0     | 0     | 0     | 0     | 0  |

| Bit    | Symbol  | Bit Name                            | Description                                                                                                                              | R/W |
|--------|---------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0     | OLS1P   | Output Level Select 1P*1, *2        | This bit selects the output level on MTIOC3B or MTIOC6B in reset-synchronized PWM mode and complementary PWM mode. Refer to Table 24.44. | R/W |
| b1     | OLS1N   | Output Level Select 1N*1, *2        | This bit selects the output level on MTIOC3D or MTIOC6D in reset-synchronized PWM mode and complementary PWM mode. Refer to Table 24.45. | R/W |
| b2     | OLS2P   | Output Level Select 2P*1, *2        | This bit selects the output level on MTIOC4A or MTIOC7A in reset-synchronized PWM mode and complementary PWM mode. Refer to Table 24.46. | R/W |
| b3     | OLS2N   | Output Level Select 2N*1, *2        | This bit selects the output level on MTIOC4C or MTIOC7C in reset-synchronized PWM mode and complementary PWM mode. Refer to Table 24.47. | R/W |
| b4     | OLS3P   | Output Level Select 3P*1, *2        | This bit selects the output level on MTIOC4B or MTIOC7B in reset-synchronized PWM mode and complementary PWM mode. Refer to Table 24.48. | R/W |
| b5     | OLS3N   | Output Level Select 3N*1, *2        | This bit selects the output level on MTIOC4D or MTIOC7D in reset-synchronized PWM mode and complementary PWM mode. Refer to Table 24.49. | R/W |
| b7, b6 | BF[1:0] | TOLBR Buffer Transfer Timing Select | These bits select the timing for transferring data from TOLBRm to TOCR2m. Refer to Table 24.50 for details.                              | R/W |

m = A, B

Note 1. Setting the TOCR1m.TOCS bit to 1 makes this bit setting valid.

Note 2. If dead-time is not generated, the negative-phase output is the exact inverse of the positive-phase output. In this case, only the OLSiP bits are valid (i = 1 to 3).

TOCR2A and TOCR2B control inversion of PWM output level in complementary PWM mode and reset-synchronized PWM mode.

The initial output is selected while the counter is stopped.

**Table 24.44 MTIOCNB Output Level Select Function**

| Bit 0 | Function       |              |                      |               |
|-------|----------------|--------------|----------------------|---------------|
| OLS1P | Initial Output | Active Level | Compare Match Output |               |
|       |                |              | Up-Counting          | Down-Counting |
| 0     | High level     | Low level    | Low level            | High level    |
| 1     | Low level      | High level   | High level           | Low level     |

n = 3, 6

**Table 24.45 MTIOCnD Output Level Select Function**

| Bit 1 | Function       |              |                      |               |
|-------|----------------|--------------|----------------------|---------------|
| OLS1N | Initial Output | Active Level | Compare Match Output |               |
|       |                |              | Up-Counting          | Down-Counting |
| 0     | High level     | Low level    | High level           | Low level     |
| 1     | Low level      | High level   | Low level            | High level    |

n = 3, 6

Note: The initial output value of negative-phase waveform changes to an active level after the dead time has passed since counting starts.

**Table 24.46 MTIOCnA Output Level Select Function**

| Bit 2 | Function       |              |                      |               |
|-------|----------------|--------------|----------------------|---------------|
| OLS2P | Initial Output | Active Level | Compare Match Output |               |
|       |                |              | Up-Counting          | Down-Counting |
| 0     | High level     | Low level    | Low level            | High level    |
| 1     | Low level      | High level   | High level           | Low level     |

n = 4, 7

**Table 24.47 MTIOCnC Output Level Select Function**

| Bit 3 | Function       |              |                      |               |
|-------|----------------|--------------|----------------------|---------------|
| OLS2N | Initial Output | Active Level | Compare Match Output |               |
|       |                |              | Up-Counting          | Down-Counting |
| 0     | High level     | Low level    | High level           | Low level     |
| 1     | Low level      | High level   | Low level            | High level    |

n = 4, 7

Note: The initial output value of negative-phase waveform changes to an active level after the dead time has passed since counting starts.

**Table 24.48 MTIOCnB Output Level Select Function**

| Bit 4 | Function       |              |                      |               |
|-------|----------------|--------------|----------------------|---------------|
| OLS3P | Initial Output | Active Level | Compare Match Output |               |
|       |                |              | Up-Counting          | Down-Counting |
| 0     | High level     | Low level    | Low level            | High level    |
| 1     | Low level      | High level   | High level           | Low level     |

n = 4, 7

**Table 24.49 MTIOCnD Output Level Select Function**

| Bit 5 | Function       |              |                      |               |
|-------|----------------|--------------|----------------------|---------------|
| OLS3N | Initial Output | Active Level | Compare Match Output |               |
|       |                |              | Up-Counting          | Down-Counting |
| 0     | High level     | Low level    | High level           | Low level     |
| 1     | Low level      | High level   | Low level            | High level    |

n = 4, 7

Note: The initial output value of negative-phase waveform changes to an active level after the dead time has passed since counting starts.

**Table 24.50 Setting of TOCR2m.BF[1:0] Bits**

| Bit 7 | Bit 6 | Description                                                                                                |                                                                                                    |
|-------|-------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| BF[1] | BF[0] | Complementary PWM Mode                                                                                     | Reset-Synchronized PWM Mode                                                                        |
| 0     | 0     | Does not transfer data from the buffer register (TOLBRm) to TOCR2m.                                        | Does not transfer data from the buffer register (TOLBRm) to TOCR2m.                                |
| 0     | 1     | Transfers data from the buffer register (TOLBRm) to TOCR2m at the crest of the MTUn.TCNT count.            | Transfers data from the buffer register (TOLBRm) to TOCR2m when MTUk.TCNT or MTUn.TCNT is cleared. |
| 1     | 0     | Transfers data from the buffer register (TOLBRm) to TOCR2m at the trough of the MTUn.TCNT count.           | Setting prohibited                                                                                 |
| 1     | 1     | Transfers data from the buffer register (TOLBRm) to TOCR2m at the crest and trough of the MTUn.TCNT count. | Setting prohibited                                                                                 |

n = 4, 7; k = 3, 6; m = A, B

### 24.2.24 Timer Output Level Buffer Register m (TOLBRm) (m = A, B)

Address(es): MTU.TOLBRA 000C 1236h, MTU.TOLBRB 000C 1A36h

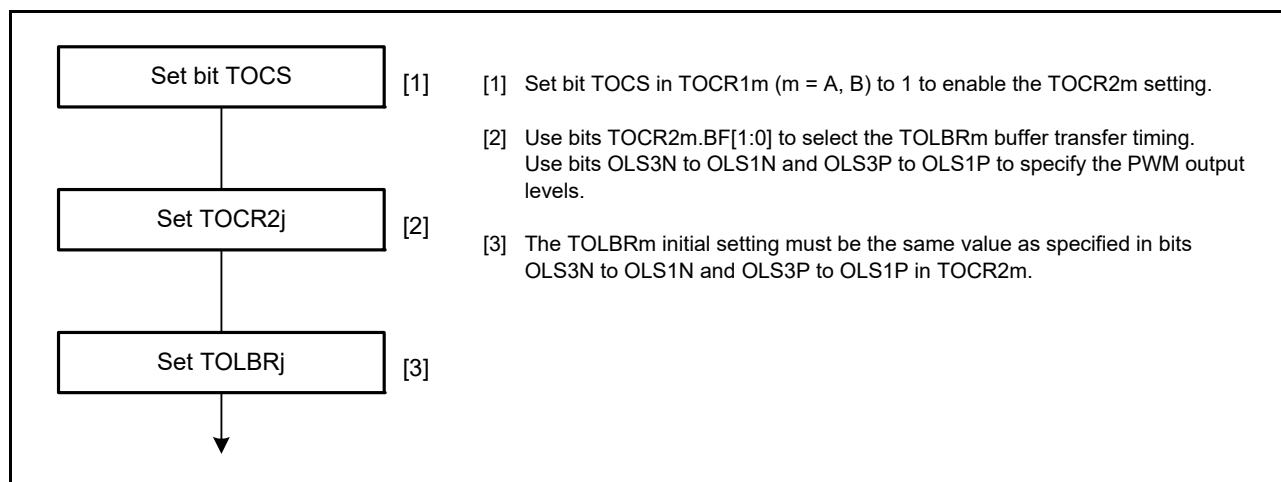
|                    | b7 | b6 | b5    | b4    | b3    | b2    | b1    | b0    |
|--------------------|----|----|-------|-------|-------|-------|-------|-------|
|                    | —  | —  | OLS3N | OLS3P | OLS2N | OLS2P | OLS1N | OLS1P |
| Value after reset: | 0  | 0  | 0     | 0     | 0     | 0     | 0     | 0     |

| Bit    | Symbol | Bit Name               | Description                                                            | R/W |
|--------|--------|------------------------|------------------------------------------------------------------------|-----|
| b0     | OLS1P  | Output Level Select 1P | Specify the buffer value to be transferred to the OLS1P bit in TOCR2m. | R/W |
| b1     | OLS1N  | Output Level Select 1N | Specify the buffer value to be transferred to the OLS1N bit in TOCR2m. | R/W |
| b2     | OLS2P  | Output Level Select 2P | Specify the buffer value to be transferred to the OLS2P bit in TOCR2m. | R/W |
| b3     | OLS2N  | Output Level Select 2N | Specify the buffer value to be transferred to the OLS2N bit in TOCR2m. | R/W |
| b4     | OLS3P  | Output Level Select 3P | Specify the buffer value to be transferred to the OLS3P bit in TOCR2m. | R/W |
| b5     | OLS3N  | Output Level Select 3N | Specify the buffer value to be transferred to the OLS3N bit in TOCR2m. | R/W |
| b7, b6 | —      | Reserved               | These bits are read as 0. The write value should be 0.                 | R/W |

m = A, B

TOLBRA and TOLBRB are buffer registers for TOCR2A and TOCR2B and specify the PWM output level in complementary PWM mode and reset-synchronized PWM mode.

Figure 24.4 shows an example of the PWM output level setting procedure in buffer operation.



**Figure 24.4 Example of PWM Output Level Setting Procedure in Buffer Operation**

### 24.2.25 Timer Gate Control Register A (TGCRA)

Address(es): MTU.TGCRA 000C 120Dh

|                    | b7 | b6  | b5 | b4 | b3 | b2 | b1 | b0 |
|--------------------|----|-----|----|----|----|----|----|----|
|                    | —  | BDC | N  | P  | FB | WF | VF | UF |
| Value after reset: | 1  | 0   | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit | Symbol | Bit Name                          | Description                                                                                                                                                                                                                                 | R/W |
|-----|--------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0  | UF     | Output Phase Switch               | These bits turn on or off the positive-phase/negative-phase output. The setting of these bits is valid only when the FB bit is set to 1. In this case, the setting of b0 to b2 is used instead of the external input. Refer to Table 24.51. | R/W |
| b1  | VF     |                                   |                                                                                                                                                                                                                                             | R/W |
| b2  | WF     |                                   |                                                                                                                                                                                                                                             | R/W |
| b3  | FB     | External Feedback Signal Enable   | 0: Output is switched by external input (input sources are TGRA, TGRB, and TGRC input capture signals in MTU0)<br>1: Output is switched by software (TGCRA's UF, VF, and WF settings)                                                       | R/W |
| b4  | P      | Positive-Phase Output (P) Control | 0: Level output<br>1: Reset-synchronized PWM or complementary PWM output                                                                                                                                                                    | R/W |
| b5  | N      | Negative-Phase Output (N) Control | 0: Level output<br>1: Reset-synchronized PWM or complementary PWM output                                                                                                                                                                    | R/W |
| b6  | BDC    | Brushless DC Motor                | 0: Ordinary output<br>1: Functions of this register are made effective                                                                                                                                                                      | R/W |
| b7  | —      | Reserved                          | This bit is read as 1. The write value should be 1.                                                                                                                                                                                         | R/W |

TGCRA controls the output waveform necessary for brushless DC motor control in reset-synchronized PWM mode and complementary PWM mode. TGCRA register settings are ineffective for anything other than complementary PWM mode and reset-synchronized PWM mode.

#### UF, VF, and WF Bits (Output Phase Switch)

The setting of these bits is valid only when the FB bit is set to 1. In this case, the setting of b0 to b2 is used instead of the external input. Refer to Table 24.51 for details.

#### FB Bit (External Feedback Signal Enable)

This bit selects whether the positive-/negative-phase output is switched automatically with the TGRA, TGRB, and TGRC input capture signals in MTU0 or by writing 0 or 1 to bits 2 to 0 in TGCRA.

When the TGCRA.FB bit is 0, output of MTU3 and MTU4 can be switched with the TGRA, TGRB, and TGRC input capture signals in MTU0.

#### P Bit (Positive-Phase Output (P) Control)

This bit selects the level output or the reset-synchronized PWM/complementary PWM output for the positive-phase output pins (MTIOC3B, MTIOC4A, and MTIOC4B pins).

#### N Bit (Negative-Phase Output (N) Control)

This bit selects the level output or the reset-synchronized PWM/complementary PWM output for the negative-phase output pins (MTIOC3D, MTIOC4C, and MTIOC4D pins).

#### BDC Bit (Brushless DC Motor)

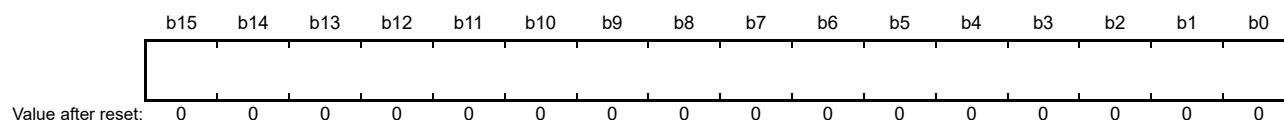
This bit selects whether to make the functions of TGCRA effective or ineffective.

**Table 24.51 Output Level Select Function**

| Bit 2 | Bit 1 | Bit 0 | Function |         |         |         |         |         |
|-------|-------|-------|----------|---------|---------|---------|---------|---------|
| WF    | VF    | UF    | MTIOC3B  | MTIOC4A | MTIOC4B | MTIOC3D | MTIOC4C | MTIOC4D |
|       |       |       | U Phase  | V Phase | W Phase | U Phase | V Phase | W Phase |
| 0     | 0     | 0     | OFF      | OFF     | OFF     | OFF     | OFF     | OFF     |
| 0     | 0     | 1     | ON       | OFF     | OFF     | OFF     | OFF     | ON      |
| 0     | 1     | 0     | OFF      | ON      | OFF     | ON      | OFF     | OFF     |
| 0     | 1     | 1     | OFF      | ON      | OFF     | OFF     | OFF     | ON      |
| 1     | 0     | 0     | OFF      | OFF     | ON      | OFF     | ON      | OFF     |
| 1     | 0     | 1     | ON       | OFF     | OFF     | OFF     | ON      | OFF     |
| 1     | 1     | 0     | OFF      | OFF     | ON      | ON      | OFF     | OFF     |
| 1     | 1     | 1     | OFF      | OFF     | OFF     | OFF     | OFF     | OFF     |

### 24.2.26 Timer Subcounter m (TCNTSm) (m = A, B)

Address(es): MTU.TCNTSA 000C 1220h, MTU.TCNTSB 000C 1A20h



Note: TCNTSA and TCNTSB must not be accessed in 8 bits; it should be accessed in 16 bits.

TCNTSA and TCNTSB are 16-bit read-only counters used only in complementary PWM mode.

The initial value of TCNTSA and TCNTSB after a reset is 0000h.

### 24.2.27 Timer Period Data Register m (TCDRm) (m = A, B)

Address(es): MTU.TCDRA 000C 1214h, MTU.TCDRB 000C 1A14h

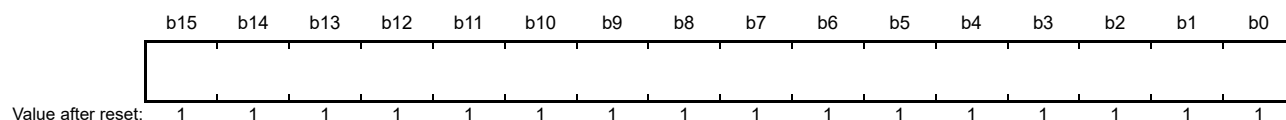


Note: TCDRA and TCDRB must not be accessed in 8 bits; it should be accessed in 16 bits.

TCDRA and TCDRB are 16-bit readable/writable registers used only in complementary PWM mode. Set half the PWM carrier period as the TCDRA and TCDRB values. The TCDRA and TCDRB registers are constantly compared with the TCNTSA and TCNTSB counters in complementary PWM mode, respectively. When a match occurs, the TCNTSA and TCNTSB counters switch the count direction (down-count to up-count). The initial value of TCDRA and TCDRB after a reset is FFFFh.

### 24.2.28 Timer Period Buffer Register m (TCBRm) (m = A, B)

Address(es): MTU.TCBRA 000C 1222h, MTU.TCBRB 000C 1A22h



Note: TCBRA and TCBRB must not be accessed in 8 bits; it should be accessed in 16 bits.

TCBRA and TCBRB are 16-bit readable/writable registers, used only in complementary PWM mode, that function as buffer registers for TCDRA and TCDRB. The TCBRA and TCBRB values are transferred to TCDRA and TCDRB with the transfer timing set in TMDR1. The initial value of TCBRA and TCBRB after a reset is FFFFh.

### 24.2.29 Timer Dead Time Data Register m (TDDRm) (m = A, B)

Address(es): MTU.TDDRA 000C 1216h, MTU.TDDRB 000C 1A16h



Note: TDDRA and TDDRB must not be accessed in 8 bits; it should be accessed in 16 bits.

TDDRA and TDDRB are 16-bit readable/writable registers, used only in complementary PWM mode, that specify the MTU3.TCNT (MTU6.TCNT) and MTU4.TCNT (MTU7.TCNT) counter offset value. In complementary PWM mode, when the MTU3.TCNT (MTU6.TCNT) and MTU4.TCNT (MTU7.TCNT) counters are cleared and then restarted, the TDDRA (TDDRB) value is loaded into the MTU3.TCNT (MTU6.TCNT) counter and the count operation starts. The initial value of TDDRA and TDDRB after a reset is FFFFh.

### 24.2.30 Timer Dead Time Enable Register m (TDERm) (m = A, B)

Address(es): MTU.TDERA 000C 1234h, MTU.TDERB 000C 1A34h

| b7 | b6 | b5 | b4 | b3 | b2 | b1 | b0   |
|----|----|----|----|----|----|----|------|
| —  | —  | —  | —  | —  | —  | —  | TDER |
| 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1    |

Value after reset:

| Bit      | Symbol | Bit Name         | Description                                                 | R/W   |
|----------|--------|------------------|-------------------------------------------------------------|-------|
| b0       | TDER   | Dead Time Enable | 0: No dead time is generated<br>1: Dead time is generated*1 | R/(W) |
| b7 to b1 | —      | Reserved         | These bits are read as 0. The write value should be 0.      | R/W   |

Note 1. TDDRA and TDDRb must be set to 1 or a larger value.

TDERA and TDERB control dead time generation in complementary PWM mode. The MTU has one TDER each for MTU3 and MTU6. TDERA and TDERB should be modified only while TCNT stops.

#### TDER Bit (Dead Time Enable)

This bit specifies whether to generate dead time.

[Clearing condition]

- When 0 is written to TDER after reading TDER = 1



## 24.2.31 Timer Buffer Transfer Set Register m (TBTERm) (m = A, B)

Address(es): MTU.TBTERA 000C 1232h, MTU.TBTERB 000C 1A32h

| b7 | b6 | b5 | b4 | b3 | b2 | b1       | b0 |
|----|----|----|----|----|----|----------|----|
| —  | —  | —  | —  | —  | —  | BTE[1:0] |    |
| 0  | 0  | 0  | 0  | 0  | 0  | 0        | 0  |

Value after reset:

| Bit      | Symbol   | Bit Name                                                    | Description                                                                                                                                                                                                                                                 | R/W |
|----------|----------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b1, b0   | BTE[1:0] | Buffer Transfer Disable and Interrupt Skipping Link Setting | These bits enable or disable transfer from the buffer registers* <sup>1</sup> used in complementary PWM mode to the temporary registers, and specify whether to link the transfer with interrupt skipping function 1.<br>For details, refer to Table 24.52. | R/W |
| b7 to b2 | —        | Reserved                                                    | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                      | R/W |

Note 1. Applicable buffer registers (TBTERA):  
MTU3.TGRC, MTU3.TGRD, MTU4.TGRC, MTU4.TGRD, and TCBRA  
Applicable buffer registers (TBTERB):  
MTU6.TGRC, MTU6.TGRD, MTU7.TGRC, MTU7.TGRD, and TCBRB

TBTERA and TBTERB enable or disable transfer from the buffer registers used in complementary PWM mode to the temporary registers, and specify whether to link the transfer with interrupt skipping 1 operation.

**Table 24.52 Setting of TBTERA.BTE[1:0] Bits and TBTERB.BTE[1:0] Bits**

| Bit 1  | Bit 0  | Description                                                                                                                                            |
|--------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| BTE[1] | BTE[0] |                                                                                                                                                        |
| 0      | 0      | Enables transfer from the buffer registers to the temporary registers* <sup>1</sup> and does not link the transfer with interrupt skipping function 1. |
| 0      | 1      | Disables transfer from the buffer registers to the temporary registers.                                                                                |
| 1      | 0      | Links transfer from the buffer registers to the temporary registers with interrupt skipping function 1.* <sup>2</sup>                                  |
| 1      | 1      | Setting prohibited                                                                                                                                     |

Note 1. Data is transferred according to the MD3 to MD0 bit setting in TMDR1. For details, refer to section 24.3.8, Complementary PWM Mode.

Note 2. When interrupt skipping is disabled the T3AEN and T4VEN (T6AEN and T7VEN) bits are set to 0 in the timer interrupt skipping set register (TITCR1A (TITCR1B)) or the skipping count set bits (T3ACOR and T4VCOR (T6ACOR and T7VCOR)) in TITCR1A (TITCR1B) are set to 0), be sure to disable link of buffer transfer with interrupt skipping (set the BTE1 bit in the timer buffer transfer set register (TBTERA (TBTERB)) to 0). If link with interrupt skipping is enabled while interrupt skipping is disabled, buffer transfer will not be performed.

### 24.2.32 Timer Waveform Control Register m (TWCRm) (m = A, B)

Address(es): MTU.TWCRA 000C 1260h, MTU.TWCRB 000C 1A60h

| b7                 | b6 | b5 | b4 | b3 | b2 | b1  | b0  |
|--------------------|----|----|----|----|----|-----|-----|
| CCE                | —  | —  | —  | —  | —  | SCC | WRE |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0   | 0   |

| Bit      | Symbol | Bit Name                               | Description                                                                                                                                                                                                                                                                | R/W         |
|----------|--------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| b0       | WRE    | Waveform Retain Enable                 | 0: Initial values specified in TOCR1A and TOCR2A (TOCR1B and TOCR2B) are output<br>1: Initial output is inhibited                                                                                                                                                          | R/(W)<br>*3 |
| b1       | SCC    | Synchronous Clearing Control<br>*1, *3 | (Only valid in TWCRB)<br>0: Clearing of MTU6.TCNT and MTU7.TCNT in response to synchronous clearing for MTU0, MTU1, MTU2–MTU6, MTU7 is enabled.<br>1: Clearing of MTU6.TCNT and MTU7.TCNT in response to synchronous clearing for MTU0, MTU1, MTU2–MTU6, MTU7 is disabled. | R/(W)       |
| b6 to b2 | —      | Reserved                               | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                     | R/W         |
| b7       | CCE    | Compare Match Clear Enable<br>*2       | 0: Counters are not cleared at MTU3.TGRA (MTU6.TGRA) compare match<br>1: Counters are cleared at MTU3.TGRA (MTU6.TGRA) compare match                                                                                                                                       | R/(W)       |

Note 1. This bit is only valid in register TWCRB and is a reserved bit in register TWCRA.

Note 2. Do not set to 1 when complementary PWM mode 1 is not selected.

Note 3. Do not set to 1 when complementary PWM mode is not selected.

TWCRA and TWCRB control the output waveform when synchronous counter clearing occurs in MTU3.TCNT and MTU4.TCNT (MTU6.TCNT and MTU7.TCNT) in complementary PWM mode and specifies whether to clear the counters at MTU3.TGRA (MTU6.TGRA) compare match.

The CCE bit and WRE bit in TWCRA and TWCRB should be modified only while TCNT stops.

#### WRE Bit (Waveform Retain Enable)

This bit selects the waveform output when synchronous counter clearing occurs in complementary PWM mode.

The initial output is inhibited with this function only when synchronous clearing occurs within the Tb interval at the trough in complementary PWM mode. When synchronous clearing occurs outside this interval, the initial values specified in TOCR1A and TOCR2A (TOCR1B and TOCR2B) are output regardless of the WRE bit setting. The initial values specified in TOCR1A and TOCR2A (TOCR1B and TOCR2B) are also output when synchronous clearing occurs in the Tb interval at the trough immediately after MTU3.TCNT and MTU4.TCNT (MTU6.TCNT and MTU7.TCNT) start operation.

For the Tb interval at the trough in complementary PWM mode, refer to Figure 24.50.

[Setting condition]

- When 1 is written to the WRE bit after reading WRE = 0

**SCC Bit (Synchronous Clearing Control)**

The setting of this bit selects whether MTU6.TCNT and MTU7.TCNT are or are not cleared when counter-synchronous clearing is generated for MTU0, MTU1, MTU2–MTU6, MTU7 in complementary PWM mode.

Make the complementary PWM mode settings for MTU6 and MTU7 when this function is in use. When writing a new value to the SCC bit while the counter is operating, do so in such a way that the values of the CCE and WRE bits are not changed.

Synchronous clearing from the MTU module only becomes disabled due to the setting of the SCC bit when synchronous clearing is generated outside the Tb interval in the trough. If synchronous clearing is generated within the Tb interval in the trough including immediately after the value at which MTU6.TCNT and MTU7.TCNT start, MTU6.TCNT and MTU7.TCNT are cleared.

Regarding the Tb interval in the trough in complementary PWM mode, refer to Figure 24.50.

[Setting condition]

- Writing of 1 to the SCC bit after reading it as 0

The corresponding bit in register TWCRA is reserved and is read as 0. When writing to TWCRA, write 0 to this bit.

**CCE Bit (Compare Match Clear Enable)**

This bit specifies whether to clear counters at MTU3.TGRA (MTU6.TGRA) compare match in complementary PWM mode.

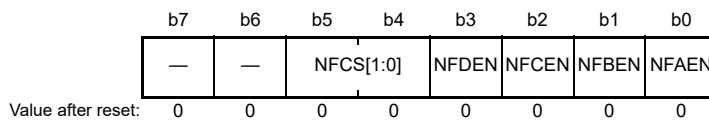
[Setting condition]

- When 1 is written to CCE after reading CCE = 0

### 24.2.33 Noise Filter Control Register n (NFCRn) (n = 0 to 4, 6, 7, 8, C)

- MTU0.NFCR0, MTU1.NFCR1, MTU2.NFCR2, MTU3.NFCR3, MTU4.NFCR4, MTU6.NFCR6, MTU7.NFCR7, MTU8.NFCR8

Address(es): MTU0.NFCR0 000C 1290h, MTU1.NFCR1 000C 1291h, MTU2.NFCR2 000C 1292h, MTU3.NFCR3 000C 1293h, MTU4.NFCR4 000C 1294h, MTU6.NFCR6 000C 1A93h, MTU7.NFCR7 000C 1A94h, MTU8.NFCR8 000C 1298h



| Bit    | Symbol    | Bit Name                  | Description                                                                                                 | R/W |
|--------|-----------|---------------------------|-------------------------------------------------------------------------------------------------------------|-----|
| b0     | NFAEN     | Noise Filter A Enable     | 0: The noise filter for the MTIOCnA pin is disabled.<br>1: The noise filter for the MTIOCnA pin is enabled. | R/W |
| b1     | NFBEN     | Noise Filter B Enable     | 0: The noise filter for the MTIOCnB pin is disabled.<br>1: The noise filter for the MTIOCnB pin is enabled. | R/W |
| b2     | NFCEN     | Noise Filter C Enable*1   | 0: The noise filter for the MTIOCnC pin is disabled.<br>1: The noise filter for the MTIOCnC pin is enabled. | R/W |
| b3     | NFDEN     | Noise Filter D Enable*1   | 0: The noise filter for the MTIOCnD pin is disabled.<br>1: The noise filter for the MTIOCnD pin is enabled. | R/W |
| b5, b4 | NFCS[1:0] | Noise Filter Clock Select | b5 b4<br>0 0: PCLKA/1<br>0 1: PCLKA/8<br>1 0: PCLKA/32<br>1 1: Clock source for counting                    | R/W |
| b7, b6 | —         | Reserved                  | These bits are read as 0. The write value should be 0.                                                      | R/W |

Note 1. These bits are reserved in MTU1 and MTU2. These bits are read as 0 and writing to them has no effect.

The NFCRn register (n = 0 to 4, 6, 7, 8) sets the noise filter function of input capture pins for the corresponding channel.

#### NFAEN Bit (Noise Filter A Enable)

This bit disables or enables the noise filter for input from the MTIOCnA pin. Since changing the value of the bit may lead to the internal generation of an unexpected edge, select the output compare function for the relevant pin in the timer I/O control register or set the TMDR.MD[3:0] bits to a value other than that for normal mode (0000b) before doing so.

#### NFBEN Bit (Noise Filter B Enable)

This bit disables or enables the noise filter for input from the MTIOCnB pin. Since changing the value of the bit may lead to the internal generation of an unexpected edge, select the output compare function for the relevant pin in the timer I/O control register or set the TMDR.MD[3:0] bits to a value other than that for normal mode (0000b) before doing so.

#### NFCEN Bit (Noise Filter C Enable)

This bit disables or enables the noise filter for input from the MTIOCnC pin. Since changing the value of the bit may lead to the internal generation of an unexpected edge, select the output compare function for the relevant pin in the timer I/O control register or set the TMDR.MD[3:0] bits to a value other than that for normal mode (0000b) before doing so.

#### NFDEN Bit (Noise Filter D Enable)

This bit disables or enables the noise filter for input from the MTIOCnD pin. Since changing the value of the bit may lead to the internal generation of an unexpected edge, select the output compare function for the relevant pin in the timer I/O control register or set the TMDR.MD[3:0] bits to a value other than that for normal mode (0000b) before doing so.

**NFCS[1:0] Bits (Noise Filter Clock Select)**

These bits set the sampling interval for the noise filters. When setting the NFCS[1:0] bits, wait for two cycles of the selected sampling interval before setting the input-capture function. When the NFCS[1:0] bits are set to 11b, i.e. selecting the external clock as the source to drive counting, wait for two cycles of the external clock before setting the input capture function.

- MTU0.NFCRC

Address(es): MTU0.NFCRC 000C 1299h

| b7                 | b6 | b5        | b4    | b3    | b2    | b1    | b0 |
|--------------------|----|-----------|-------|-------|-------|-------|----|
| —                  | —  | NFCS[1:0] | NFDEN | NFCEN | NFBEN | NFAEN |    |
| Value after reset: | 0  | 0         | 0     | 0     | 0     | 0     | 0  |

| Bit    | Symbol    | Bit Name                  | Description                                                                                               | R/W |
|--------|-----------|---------------------------|-----------------------------------------------------------------------------------------------------------|-----|
| b0     | NFAEN     | Noise Filter A Enable     | 0: The noise filter for the MTCLKA pin is disabled.<br>1: The noise filter for the MTCLKA pin is enabled. | R/W |
| b1     | NFBEN     | Noise Filter B Enable     | 0: The noise filter for the MTCLKB pin is disabled.<br>1: The noise filter for the MTCLKB pin is enabled. | R/W |
| b2     | NFCEN     | Noise Filter C Enable     | 0: The noise filter for the MTCLKC pin is disabled.<br>1: The noise filter for the MTCLKC pin is enabled. | R/W |
| b3     | NFDEN     | Noise Filter D Enable     | 0: The noise filter for the MTCLKD pin is disabled.<br>1: The noise filter for the MTCLKD pin is enabled. | R/W |
| b5, b4 | NFCS[1:0] | Noise Filter Clock Select | b5 b4<br>0 0: PCLKA/1<br>0 1: PCLKA/2<br>1 0: PCLKA/8<br>1 1: PCLKA/32                                    | R/W |
| b7, b6 | —         | Reserved                  | These bits are read as 0. The write value should be 0.                                                    | R/W |

The NFCRC register sets the noise filter function of external clock pins common to each channel.

**NFAEN Bit (Noise Filter A Enable)**

This bit disables or enables the noise filter for input from the MTCLKA pin. Since changing the value of the bit may lead to the internal generation of an unexpected edge, do so after stopping the internal counter.

**NFBEN Bit (Noise Filter B Enable)**

This bit disables or enables the noise filter for input from the MTCLKB pin. Since changing the value of the bit may lead to the internal generation of an unexpected edge, do so after stopping the internal counter.

**NFCEN Bit (Noise Filter C Enable)**

This bit disables or enables the noise filter for input from the MTCLKC pin. Since changing the value of the bit may lead to the internal generation of an unexpected edge, do so after stopping the internal counter.

**NFDEN Bit (Noise Filter D Enable)**

This bit disables or enables the noise filter for input from the MTCLKD pin. Since changing the value of the bit may lead to the internal generation of an unexpected edge, do so after stopping the internal counter.

**NFCS[1:0] Bits (Noise Filter Clock Select)**

These bits set the sampling interval for the noise filters. After setting the NFCS[1:0] bits, wait for two cycles of the selected sampling interval to set the input capture function.

**24.2.34 Noise Filter Control Register 5 (NFCR5)**

Address(es): MTU5.NFCR5 000C 1A95h

| b7                 | b6 | b5        | b4 | b3 | b2        | b1    | b0    |
|--------------------|----|-----------|----|----|-----------|-------|-------|
| —                  | —  | NFCS[1:0] | —  | —  | NFWE<br>N | NFVEN | NFUEN |
| Value after reset: | 0  | 0         | 0  | 0  | 0         | 0     | 0     |

| Bit    | Symbol    | Bit Name                  | Description                                                                                               | R/W |
|--------|-----------|---------------------------|-----------------------------------------------------------------------------------------------------------|-----|
| b0     | NFUEN     | Noise Filter U Enable     | 0: The noise filter for the MTIC5U pin is disabled.<br>1: The noise filter for the MTIC5U pin is enabled. | R/W |
| b1     | NFVEN     | Noise Filter V Enable     | 0: The noise filter for the MTIC5V pin is disabled.<br>1: The noise filter for the MTIC5V pin is enabled. | R/W |
| b2     | NFWEN     | Noise Filter W Enable     | 0: The noise filter for the MTIC5W pin is disabled.<br>1: The noise filter for the MTIC5W pin is enabled. | R/W |
| b3     | —         | Reserved                  | This bit is read as 0. The write value should be 0.                                                       | R/W |
| b5, b4 | NFCS[1:0] | Noise Filter Clock Select | b5 b4<br>0 0: PCLKA/1<br>0 1: PCLKA/8<br>1 0: PCLKA/32<br>1 1: Clock source for counting                  | R/W |
| b7, b6 | —         | Reserved                  | These bits are read as 0. The write value should be 0.                                                    | R/W |

**NFUEN Bit (Noise Filter U Enable)**

This bit disables or enables the noise filter for input from the MTIC5U pin. Since unexpected edges may be internally generated when the value of this bit is changed, select the output compare function for the relevant pin in the timer I/O control register.

**NFVEN Bit (Noise Filter V Enable)**

This bit disables or enables the noise filter for input from the MTIC5V pin. Since unexpected edges may be internally generated when the value of this bit is changed, select the output compare function for the relevant pin in the timer I/O control register.

**NFWEN Bit (Noise Filter W Enable)**

This bit disables or enables the noise filter for input from the MTIC5W pin. Since unexpected edges may be internally generated when the value of this bit is changed, select the output compare function for the relevant pin in the timer I/O control register.

**NFCS[1:0] Bits (Noise Filter Clock Select)**

These bits set the sampling interval for the noise filters. When setting the NFCS[1:0] bits, wait for two cycles of the selected sampling interval before setting the input-capture function.

## 24.2.35 Timer A/D Conversion Start Request Control Register (TADCR)

## • MTU4.TADCR

Address(es): MTU4.TADCR 000C 1240h

|                    | b15     | b14 | b13 | b12 | b11 | b10 | b9 | b8 | b7    | b6    | b5    | b4    | b3     | b2     | b1     | b0     |
|--------------------|---------|-----|-----|-----|-----|-----|----|----|-------|-------|-------|-------|--------|--------|--------|--------|
|                    | BF[1:0] | —   | —   | —   | —   | —   | —  | —  | UT4AE | DT4AE | UT4BE | DT4BE | ITA3AE | ITA4VE | ITB3AE | ITB4VE |
| Value after reset: | 0       | 0   | 0   | 0   | 0   | 0   | 0  | 0  | 0     | 0     | 0     | 0     | 0      | 0      | 0      | 0      |

| Bit       | Symbol  | Bit Name                                       | Description                                                                                                                                                                         | R/W |
|-----------|---------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0        | ITB4VE  | TCIV4 Interrupt Skipping Link Enable*1, *2, *3 | 0: A/D conversion start request TRG4BN and TCIV4 interrupt skipping 1 are not linked<br>1: A/D conversion start request TRG4BN and TCIV4 interrupt skipping 1 are linked            | R/W |
| b1        | ITB3AE  | TGIA3 Interrupt Skipping Link Enable*1, *2, *3 | 0: A/D conversion start request TRG4BN and TGIA3 interrupt skipping 1 are not linked<br>1: A/D conversion start request TRG4BN and TGIA3 interrupt skipping 1 are linked            | R/W |
| b2        | ITA4VE  | TCIV4 Interrupt Skipping Link Enable*1, *2, *3 | 0: A/D conversion start request TRG4AN and TCIV4 interrupt skipping 1 are not linked<br>1: A/D conversion start request TRG4AN and TCIV4 interrupt skipping 1 are linked            | R/W |
| b3        | ITA3AE  | TGIA3 Interrupt Skipping Link Enable*1, *2, *3 | 0: A/D conversion start request TRG4AN and TGIA3 interrupt skipping 1 are not linked<br>1: A/D conversion start request TRG4AN and TGIA3 interrupt skipping 1 are linked            | R/W |
| b4        | DT4BE   | Down-Count TRG4BN Enable*3                     | 0: A/D conversion start requests (TRG4BN) disabled during MTU4.TCNT down-count operation<br>1: A/D conversion start requests (TRG4BN) enabled during MTU4.TCNT down-count operation | R/W |
| b5        | UT4BE   | Up-Count TRG4BN Enable                         | 0: A/D conversion start requests (TRG4BN) disabled during MTU4.TCNT up-count operation<br>1: A/D conversion start requests (TRG4BN) enabled during MTU4.TCNT up-count operation     | R/W |
| b6        | DT4AE   | Down-Count TRG4AN Enable*3                     | 0: A/D conversion start requests (TRG4AN) disabled during MTU4.TCNT down-count operation<br>1: A/D conversion start requests (TRG4AN) enabled during MTU4.TCNT down-count operation | R/W |
| b7        | UT4AE   | Up-Count TRG4AN Enable                         | 0: A/D conversion start requests (TRG4AN) disabled during MTU4.TCNT up-count operation<br>1: A/D conversion start requests (TRG4AN) enabled during MTU4.TCNT up-count operation     | R/W |
| b13 to b8 | —       | Reserved                                       | These bits are read as 0. The write value should be 0.                                                                                                                              | R/W |
| b15, b14  | BF[1:0] | MTU4.TADCOBRA/TADCOBRB Transfer Timing Select  | Refer to Table 24.53 for details.<br>These bits specify the transfer timing from MTU4.TADCOBRA and MTU4.TADCOBRB to MTU4.TADCORA and MTU4.TADCORB.                                  | R/W |

Note: MTU4.TADCR must not be accessed in 8 bits; it should be accessed in 16 bits.

Note 1. Set to 0 when interrupt skipping is disabled (the T3AEN and T4VEN bits in TITCR1A are set to 0 or the T3ACOR and T4VCOR bits in TITCR1A are set to 0).

Note 2. If link with interrupt skipping is enabled while interrupt skipping is disabled, A/D conversion start requests will not be issued.

Note 3. Set to 0 when complementary PWM mode is not selected.

TADCR enables or disables A/D conversion start requests and specifies whether to link A/D conversion start requests with interrupt skipping function. The MTU has one TADCR each for MTU4 and MTU7.

**Table 24.53 Setting of Transfer Timing by TADCR.BF[1:0] Bits (MTU4)**

| Bit 15 | Bit 14 | Description                                                                                                                                                                            |                                                                                                                                                                                                           |                                                                                                                                                                                                           |                                                                                                                                                                                                           |
|--------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BF[1]  | BF[0]  | In Complementary PWM Mode                                                                                                                                                              | In Reset-Synchronized PWM Mode                                                                                                                                                                            | In PWM Mode 1                                                                                                                                                                                             | In Normal Mode                                                                                                                                                                                            |
| 0      | 0      | Data is not transferred from the cycle set buffer register (MTU4.TADCOBRA, MTU4.TADCOBRB) to the cycle set register (MTU4.TADCORA, MTU4.TADCORB).                                      | Data is not transferred from the cycle set buffer register (MTU4.TADCOBRA, MTU4.TADCOBRB) to the cycle set register (MTU4.TADCORA, MTU4.TADCORB).                                                         | Data is not transferred from the cycle set buffer register (MTU4.TADCOBRA, MTU4.TADCOBRB) to the cycle set register (MTU4.TADCORA, MTU4.TADCORB).                                                         | Data is not transferred from the cycle set buffer register (MTU4.TADCOBRA, MTU4.TADCOBRB) to the cycle set register (MTU4.TADCORA, MTU4.TADCORB).                                                         |
| 0      | 1      | Data is transferred from the cycle set buffer register (MTU4.TADCOBRA, MTU4.TADCOBRB) to the cycle set register (MTU4.TADCORA, MTU4.TADCORB) at the crest of the MTU4.TCNT.            | Data is transferred from the cycle set buffer register (MTU4.TADCOBRA, MTU4.TADCOBRB) to the cycle set register (MTU4.TADCORA, MTU4.TADCORB) when a compare match occurs between MTU3.TCNT and MTU3.TGRA. | Data is transferred from the cycle set buffer register (MTU4.TADCOBRA, MTU4.TADCOBRB) to the cycle set register (MTU4.TADCORA, MTU4.TADCORB) when a compare match occurs between MTU4.TCNT and MTU4.TGRA. | Data is transferred from the cycle set buffer register (MTU4.TADCOBRA, MTU4.TADCOBRB) to the cycle set register (MTU4.TADCORA, MTU4.TADCORB) when a compare match occurs between MTU4.TCNT and MTU4.TGRA. |
| 1      | 0      | Data is transferred from the cycle set buffer register (MTU4.TADCOBRA, MTU4.TADCOBRB) to the cycle set register (MTU4.TADCORA, MTU4.TADCORB) at the trough of the MTU4.TCNT.           | Setting prohibited                                                                                                                                                                                        | Setting prohibited                                                                                                                                                                                        | Setting prohibited                                                                                                                                                                                        |
| 1      | 1      | Data is transferred from the cycle set buffer register (MTU4.TADCOBRA, MTU4.TADCOBRB) to the cycle set register (MTU4.TADCORA, MTU4.TADCORB) at the crest and trough of the MTU4.TCNT. | Setting prohibited                                                                                                                                                                                        | Setting prohibited                                                                                                                                                                                        | Setting prohibited                                                                                                                                                                                        |



- MTU7.TADCR

Address(es): MTU7.TADCR 000C 1A40h

|                    | b15     | b14 | b13 | b12 | b11 | b10 | b9 | b8 | b7    | b6    | b5    | b4    | b3     | b2     | b1     | b0     |
|--------------------|---------|-----|-----|-----|-----|-----|----|----|-------|-------|-------|-------|--------|--------|--------|--------|
|                    | BF[1:0] | —   | —   | —   | —   | —   | —  | —  | UT7AE | DT7AE | UT7BE | DT7BE | ITA6AE | ITA7VE | ITB6AE | ITB7VE |
| Value after reset: | 0       | 0   | 0   | 0   | 0   | 0   | 0  | 0  | 0     | 0     | 0     | 0     | 0      | 0      | 0      | 0      |

| Bit       | Symbol  | Bit Name                                       | Description                                                                                                                                                                         | R/W |
|-----------|---------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0        | ITB7VE  | TCIV7 Interrupt Skipping Link Enable*1, *2, *3 | 0: A/D conversion start request TRG7BN and TCIV7 interrupt skipping 1 are not linked<br>1: A/D conversion start request TRG7BN and TCIV7 interrupt skipping 1 are linked            | R/W |
| b1        | ITB6AE  | TGIA6 Interrupt Skipping Link Enable*1, *2, *3 | 0: A/D conversion start request TRG7BN and TGIA6 interrupt skipping 1 are not linked<br>1: A/D conversion start request TRG7BN and TGIA6 interrupt skipping 1 are linked            | R/W |
| b2        | ITA7VE  | TCIV7 Interrupt Skipping Link Enable*1, *2, *3 | 0: A/D conversion start request TRG7AN and TCIV7 interrupt skipping 1 are not linked<br>1: A/D conversion start request TRG7AN and TCIV7 interrupt skipping 1 are linked            | R/W |
| b3        | ITA6AE  | TGIA6 Interrupt Skipping Link Enable*1, *2, *3 | 0: A/D conversion start request TRG7AN and TGIA6 interrupt skipping 1 are not linked<br>1: A/D conversion start request TRG7AN and TGIA6 interrupt skipping 1 are linked            | R/W |
| b4        | DT7BE   | Down-Count TRG7BN Enable*3                     | 0: A/D conversion start requests (TRG7BN) disabled during MTU7.TCNT down-count operation<br>1: A/D conversion start requests (TRG7BN) enabled during MTU7.TCNT down-count operation | R/W |
| b5        | UT7BE   | Up-Count TRG7BN Enable                         | 0: A/D conversion start requests (TRG7BN) disabled during MTU7.TCNT up-count operation<br>1: A/D conversion start requests (TRG7BN) enabled during MTU7.TCNT up-count operation     | R/W |
| b6        | DT7AE   | Down-Count TRG7AN Enable*3                     | 0: A/D conversion start requests (TRG7AN) disabled during MTU7.TCNT down-count operation<br>1: A/D conversion start requests (TRG7AN) enabled during MTU7.TCNT down-count operation | R/W |
| b7        | UT7AE   | Up-Count TRG7AN Enable                         | 0: A/D conversion start requests (TRG7AN) disabled during MTU7.TCNT up-count operation<br>1: A/D conversion start requests (TRG7AN) enabled during MTU7.TCNT up-count operation     | R/W |
| b13 to b8 | —       | Reserved                                       | These bits are read as 0. The write value should be 0.                                                                                                                              | R/W |
| b15, b14  | BF[1:0] | MTU7.TADCOBRA/TADCOBRB Transfer Timing Select  | Refer to Table 24.54 for details.<br>These bits specify the transfer timing from MTU7.TADCOBRA and MTU7.TADCOBRB to MTU7.TADCORA and MTU7.TADCORB.                                  | R/W |

Note: MTU7.TADCR must not be accessed in 8 bits; it should be accessed in 16 bits.

Note 1. Set to 0 when interrupt skipping is disabled (the T6AEN and T7VEN bits in TITCR1B are set to 0 or the T6ACOR and T7VCOR bits in TITCR1B are set to 0).

Note 2. If link with interrupt skipping is enabled while interrupt skipping is disabled, A/D conversion start requests will not be issued.

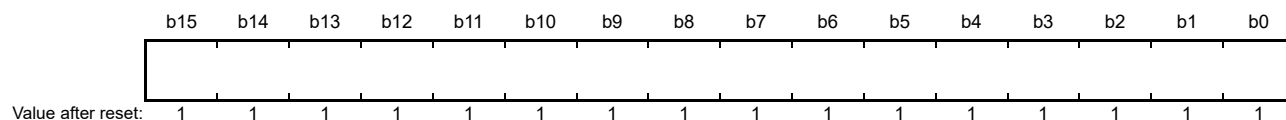
Note 3. Set to 0 when complementary PWM mode is not selected.

**Table 24.54 Setting of Transfer Timing by TADCR.BF[1:0] Bits (MTU7)**

| Bit 15 | Bit 14 | Description                                                                                                                                                                            |                                                                                                                                                                                                           |                                                                                                                                                                                                           |                                                                                                                                                                                                           |
|--------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BF[1]  | BF[0]  | In Complementary PWM Mode                                                                                                                                                              | In Reset-Synchronized PWM Mode                                                                                                                                                                            | In PWM Mode 1                                                                                                                                                                                             | In Normal Mode                                                                                                                                                                                            |
| 0      | 0      | Data is not transferred from the cycle set buffer register (MTU7.TADCOBRA, MTU7.TADCOBRB) to the cycle set register (MTU7.TADCORA, MTU7.TADCORB).                                      | Data is not transferred from the cycle set buffer register (MTU7.TADCOBRA, MTU7.TADCOBRB) to the cycle set register (MTU7.TADCORA, MTU7.TADCORB).                                                         | Data is not transferred from the cycle set buffer register (MTU7.TADCOBRA, MTU7.TADCOBRB) to the cycle set register (MTU7.TADCORA, MTU7.TADCORB).                                                         | Data is not transferred from the cycle set buffer register (MTU7.TADCOBRA, MTU7.TADCOBRB) to the cycle set register (MTU7.TADCORA, MTU7.TADCORB).                                                         |
| 0      | 1      | Data is transferred from the cycle set buffer register (MTU7.TADCOBRA, MTU7.TADCOBRB) to the cycle set register (MTU7.TADCORA, MTU7.TADCORB) at the crest of the MTU7.TCNT.            | Data is transferred from the cycle set buffer register (MTU7.TADCOBRA, MTU7.TADCOBRB) to the cycle set register (MTU7.TADCORA, MTU7.TADCORB) when a compare match occurs between MTU6.TCNT and MTU6.TGRA. | Data is transferred from the cycle set buffer register (MTU7.TADCOBRA, MTU7.TADCOBRB) to the cycle set register (MTU7.TADCORA, MTU7.TADCORB) when a compare match occurs between MTU7.TCNT and MTU7.TGRA. | Data is transferred from the cycle set buffer register (MTU7.TADCOBRA, MTU7.TADCOBRB) to the cycle set register (MTU7.TADCORA, MTU7.TADCORB) when a compare match occurs between MTU7.TCNT and MTU7.TGRA. |
| 1      | 0      | Data is transferred from the cycle set buffer register (MTU7.TADCOBRA, MTU7.TADCOBRB) to the cycle set register (MTU7.TADCORA, MTU7.TADCORB) at the trough of the MTU7.TCNT.           | Setting prohibited                                                                                                                                                                                        | Setting prohibited                                                                                                                                                                                        | Setting prohibited                                                                                                                                                                                        |
| 1      | 1      | Data is transferred from the cycle set buffer register (MTU7.TADCOBRA, MTU7.TADCOBRB) to the cycle set register (MTU7.TADCORA, MTU7.TADCORB) at the crest and trough of the MTU7.TCNT. | Setting prohibited                                                                                                                                                                                        | Setting prohibited                                                                                                                                                                                        | Setting prohibited                                                                                                                                                                                        |

### 24.2.36 Timer A/D Conversion Start Request Cycle Set Register m (TADCORm) (m = A, B)

Address(es): MTU4.TADCORA 000C 1244h, MTU4.TADCORB 000C 1246h, MTU7.TADCORA 000C 1A44h, MTU7.TADCORB 000C 1A46h



Note: TADCORA and TADCORB must not be accessed in 8 bits; it should be accessed in 16 bits.

Note 1. When the A/D conversion start request delaying function linked with skipping function 1 (for details, refer to section 24.3.9 (5), A/D Conversion Start Request Delaying Function Linked with Interrupt Skipping Function 1) is used, the value of this register should be 0002h to TCDRA setting – 2 in MTU4 and 0002h to TCDRB setting – 2 in MTU7.

Note 2. When interrupt skipping function 2 is used and the difference between the TADCORA value and the TADCORB value is small, the skipping count may not be counted correctly and the A/D conversion start request may not be generated with the expected timing in some cases. The TADCORA and TADCORB values should satisfy the following conditions.

(1) When skipping function 2 is specified with the skipping count set to 0

- The difference between the TADCORA and TADCORB values should be equal to or greater than 4.
- The TADCORA compare interval should be equal to or greater than 4 PCLKA cycles (the TADCORA update value should be the previous value + 4 or greater, or previous value – 4 or smaller).
- The TADCORB compare interval should be equal to or greater than 4 PCLKA cycles (the TADCORB update value should be the previous value + 4 or greater, or previous value – 4 or smaller).

(2) When skipping function 2 is specified with the skipping count set to 1 or greater

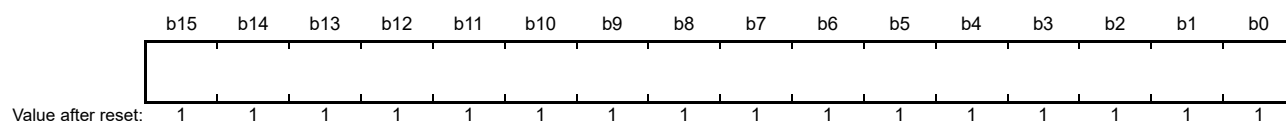
- The difference between the TADCORA and TADCORB values should be equal to or greater than 2.
- The TADCORB compare interval should be equal to or greater than 2 PCLKA cycles (the TADCORB update value should be the previous value + 2 or greater, or previous value – 2 or smaller)

TADCORA and TADCORB are 16-bit readable/writable registers that issue a corresponding A/D conversion start request when the MTUn.TCNT (n = 4, 7) count reaches the value in TADCORA or TADCORB.

MTUn.TADCORA and TADCORB are initialized to FFFFh by a reset.

### 24.2.37 Timer A/D Conversion Start Request Cycle Set Buffer Register m (TADCOBRm) (m = A, B)

Address(es): MTU4.TADCOBRA 000C 1248h, MTU4.TADCOBRB 000C 124Ah, MTU7.TADCOBRA 000C 1A48h, MTU7.TADCOBRB 000C 1A4Ah



Note: TADCOBRA and TADCOBRB must not be accessed in 8 bits; it should be accessed in 16 bits.

TADCOBRA and TADCOBRB are 16-bit readable/writable registers whose values are transferred to TADCORA and TADCORB, respectively, when the crest or trough of the MTUn.TCNT count is reached.

TADCOBRA and TADCOBRB are initialized to FFFFh by a reset.

### 24.2.38 Timer Interrupt Skipping Mode Register m (TITMRm) (m = A, B)

Address(es): MTU.TITMRA 000C 123Ah, MTU.TITMRB 000C 1A3Ah

|                    | b7 | b6 | b5 | b4 | b3 | b2 | b1 | b0   |
|--------------------|----|----|----|----|----|----|----|------|
|                    | —  | —  | —  | —  | —  | —  | —  | TITM |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0    |

| Bit      | Symbol | Bit Name                           | Description                                                                                                                                                                       | R/W |
|----------|--------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0       | TITM   | Interrupt Skipping Function Select | Selects one of the two types of interrupt skipping functions.<br>0: Selects interrupt skipping function 1* <sup>1</sup><br>1: Selects interrupt skipping function 2* <sup>2</sup> | R/W |
| b7 to b1 | —      | Reserved                           | These bits are read as 0. The write value should be 0.                                                                                                                            | R/W |

Note 1. Setting the TITCR1A or TITCR1B register enables interrupt skipping function 1.

Note 2. Setting the TITCR2A or TITCR2B register enables interrupt skipping function 2.

TITMRA and TITMRB are used to select either of two skipping functions for the TITMRA and TITMRB registers.

## 24.2.39 Timer Interrupt Skipping Set Register 1m (TITCR1m) (m = A, B)

- MTU.TITCR1A

Address(es): MTU.TITCR1A 000C 1230h

| b7                 | b6          | b5 | b4 | b3    | b2          | b1 | b0 |
|--------------------|-------------|----|----|-------|-------------|----|----|
| T3AEN              | T3ACOR[2:0] |    |    | T4VEN | T4VCOR[2:0] |    |    |
| Value after reset: | 0           | 0  | 0  | 0     | 0           | 0  | 0  |

| Bit      | Symbol      | Bit Name                               | Description                                                                                                                 | R/W |
|----------|-------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----|
| b2 to b0 | T4VCOR[2:0] | TCIV4 Interrupt Skipping Count Setting | These bits specify the TCIV4 interrupt skipping count within the range from 0 to 7.*1<br>For details, refer to Table 24.55. | R/W |
| b3       | T4VEN       | T4VEN                                  | 0: TCIV4 interrupt skipping disabled<br>1: TCIV4 interrupt skipping enabled                                                 | R/W |
| b6 to b4 | T3ACOR[2:0] | TGIA3 Interrupt Skipping Count Setting | These bits specify the TGIA3 interrupt skipping count within the range from 0 to 7.*1<br>For details, refer to Table 24.56. | R/W |
| b7       | T3AEN       | T3AEN                                  | 0: TGIA3 interrupt skipping disabled<br>1: TGIA3 interrupt skipping enabled                                                 | R/W |

Note 1. When 0 is specified for the interrupt skipping count, no interrupt skipping will be performed.  
Before changing the interrupt skipping count, be sure to set the TITCR1A.T3AEN and TITCR1A.T4VEN bits to 0 to clear the skipping counter (TITCNT1A).

- MTU.TITCR1B

Address(es): MTU.TITCR1B 000C 1A30h

| b7                 | b6          | b5 | b4 | b3    | b2          | b1 | b0 |
|--------------------|-------------|----|----|-------|-------------|----|----|
| T6AEN              | T6ACOR[2:0] |    |    | T7VEN | T7VCOR[2:0] |    |    |
| Value after reset: | 0           | 0  | 0  | 0     | 0           | 0  | 0  |

| Bit      | Symbol      | Bit Name                               | Description                                                                                                                 | R/W |
|----------|-------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----|
| b2 to b0 | T7VCOR[2:0] | TCIV7 Interrupt Skipping Count Setting | These bits specify the TCIV7 interrupt skipping count within the range from 0 to 7.*1<br>For details, refer to Table 24.57. | R/W |
| b3       | T7VEN       | T7VEN                                  | 0: TCIV7 interrupt skipping disabled<br>1: TCIV7 interrupt skipping enabled                                                 | R/W |
| b6 to b4 | T6ACOR[2:0] | TGIA6 Interrupt Skipping Count Setting | These bits specify the TGIA6 interrupt skipping count within the range from 0 to 7.*1<br>For details, refer to Table 24.58. | R/W |
| b7       | T6AEN       | T6AEN                                  | 0: TGIA6 interrupt skipping disabled<br>1: TGIA6 interrupt skipping enabled                                                 | R/W |

Note 1. When 0 is specified for the interrupt skipping count, no interrupt skipping will be performed.  
Before changing the interrupt skipping count, be sure to set the TITCR1B.T6AEN and TITCR1B.T7VEN bits to 0 to clear the skipping counter (TITCNT1B).

Registers TITCR1A and TITCR1B enable or disable interrupt skipping and specify the interrupt skipping count. This setting is valid only while the TITMRA.TITM or TITMRB.TITM bit is set to 0; when the TITMRA.TITM (TITMRB.TITM) bit is set to 1, the setting in the TITCR1A (TITCR1B) register is cleared.

**Table 24.55 Setting of Interrupt Skipping Count by T4VCOR[2:0] Bits**

| Bit 2     | Bit 1     | Bit 0     | Description                                   |
|-----------|-----------|-----------|-----------------------------------------------|
| T4VCOR[2] | T4VCOR[1] | T4VCOR[0] |                                               |
| 0         | 0         | 0         | Does not skip TCIV4 interrupts.               |
| 0         | 0         | 1         | Sets the TCIV4 interrupt skipping count to 1. |
| 0         | 1         | 0         | Sets the TCIV4 interrupt skipping count to 2. |
| 0         | 1         | 1         | Sets the TCIV4 interrupt skipping count to 3. |
| 1         | 0         | 0         | Sets the TCIV4 interrupt skipping count to 4. |
| 1         | 0         | 1         | Sets the TCIV4 interrupt skipping count to 5. |
| 1         | 1         | 0         | Sets the TCIV4 interrupt skipping count to 6. |
| 1         | 1         | 1         | Sets the TCIV4 interrupt skipping count to 7. |

**Table 24.56 Setting of Interrupt Skipping Count by T3ACOR[2:0] Bits**

| Bit 6     | Bit 5     | Bit 4     | Description                                   |
|-----------|-----------|-----------|-----------------------------------------------|
| T3ACOR[2] | T3ACOR[1] | T3ACOR[0] |                                               |
| 0         | 0         | 0         | Does not skip TGIA3 interrupts.               |
| 0         | 0         | 1         | Sets the TGIA3 interrupt skipping count to 1. |
| 0         | 1         | 0         | Sets the TGIA3 interrupt skipping count to 2. |
| 0         | 1         | 1         | Sets the TGIA3 interrupt skipping count to 3. |
| 1         | 0         | 0         | Sets the TGIA3 interrupt skipping count to 4. |
| 1         | 0         | 1         | Sets the TGIA3 interrupt skipping count to 5. |
| 1         | 1         | 0         | Sets the TGIA3 interrupt skipping count to 6. |
| 1         | 1         | 1         | Sets the TGIA3 interrupt skipping count to 7. |

**Table 24.57 Setting of Interrupt Skipping Count by T7VCOR[2:0] Bits**

| Bit 2     | Bit 1     | Bit 0     | Description                                   |
|-----------|-----------|-----------|-----------------------------------------------|
| T7VCOR[2] | T7VCOR[1] | T7VCOR[0] |                                               |
| 0         | 0         | 0         | Does not skip TCIV7 interrupts.               |
| 0         | 0         | 1         | Sets the TCIV7 interrupt skipping count to 1. |
| 0         | 1         | 0         | Sets the TCIV7 interrupt skipping count to 2. |
| 0         | 1         | 1         | Sets the TCIV7 interrupt skipping count to 3. |
| 1         | 0         | 0         | Sets the TCIV7 interrupt skipping count to 4. |
| 1         | 0         | 1         | Sets the TCIV7 interrupt skipping count to 5. |
| 1         | 1         | 0         | Sets the TCIV7 interrupt skipping count to 6. |
| 1         | 1         | 1         | Sets the TCIV7 interrupt skipping count to 7. |

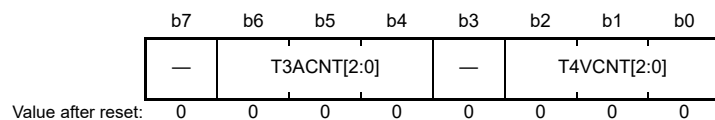
**Table 24.58 Setting of Interrupt Skipping Count by T6ACOR[2:0] Bits**

| Bit 6     | Bit 5     | Bit 4     | Description                                   |
|-----------|-----------|-----------|-----------------------------------------------|
| T6ACOR[2] | T6ACOR[1] | T6ACOR[0] |                                               |
| 0         | 0         | 0         | Does not skip TGIA6 interrupts.               |
| 0         | 0         | 1         | Sets the TGIA6 interrupt skipping count to 1. |
| 0         | 1         | 0         | Sets the TGIA6 interrupt skipping count to 2. |
| 0         | 1         | 1         | Sets the TGIA6 interrupt skipping count to 3. |
| 1         | 0         | 0         | Sets the TGIA6 interrupt skipping count to 4. |
| 1         | 0         | 1         | Sets the TGIA6 interrupt skipping count to 5. |
| 1         | 1         | 0         | Sets the TGIA6 interrupt skipping count to 6. |
| 1         | 1         | 1         | Sets the TGIA6 interrupt skipping count to 7. |

### 24.2.40 Timer Interrupt Skipping Counter 1m (TITCNT1m) (m = A, B)

- MTU.TITCNT1A

Address(es): MTU.TITCNT1A 000C 1231h



| Bit      | Symbol      | Bit Name                | Description                                                                                                             | R/W |
|----------|-------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------|-----|
| b2 to b0 | T4VCNT[2:0] | TCIV4 Interrupt Counter | While the T4VEN bit in TITCR1A is set to 1, the count in these bits is incremented every time a TCIV4 interrupt occurs. | R   |
| b3       | —           | Reserved                | This bit is read as 0.                                                                                                  | R   |
| b6 to b4 | T3ACNT[2:0] | TGIA3 Interrupt Counter | While the T3AEN bit in TITCR1A is set to 1, the count in these bits is incremented every time a TGIA3 interrupt occurs. | R   |
| b7       | —           | Reserved                | This bit is read as 0.                                                                                                  | R   |

Note: To clear the TITCNT1A, set the TITCR1A.T3AEN and TITCR1A.T4VEN bits to 0.

TITCNT1A and TITCNT1B are 8-bit readable/writable counters. TITCNT1A and TITCNT1B retain their values even after stopping the count operation of MTU3.TCNT and MTU4.TCNT (MTU6.TCNT and MTU7.TCNT).

#### T4VCNT[2:0] Bits (TCIV4 Interrupt Counter)

[Clearing conditions]

- When the TITM bit in TITMRA is 1
- When the T4VEN bit in TITCR1A is set to 0
- When the T4VCOR[2:0] bits in TITCR1A are set to 000b
- When the T4VCNT[2:0] bits in TITCNT1A match the T4VCOR[2:0] bits in TITCR1A

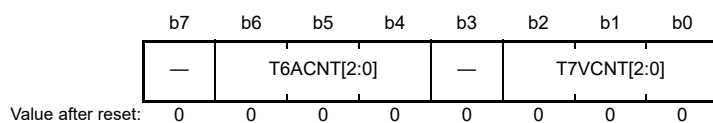
#### T3ACNT[2:0] Bits (TGIA3 Interrupt Counter)

[Clearing conditions]

- When the TITM bit in TITMRA is 1
- When the T3AEN bit in TITCR1A is set to 0
- When the T3ACOR[2:0] bits in TITCR1A are set to 000b
- When the T3ACNT[2:0] bits in TITCNT1A match the T3ACOR[2:0] bits in TITCR1A

- MTU.TITCNT1B

Address(es): MTU.TITCNT1B 000C 1A31h



| Bit      | Symbol      | Bit Name                | Description                                                                                                             | R/W |
|----------|-------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------|-----|
| b2 to b0 | T7VCNT[2:0] | TCIV7 Interrupt Counter | While the T7VEN bit in TITCR1B is set to 1, the count in these bits is incremented every time a TCIV7 interrupt occurs. | R   |
| b3       | —           | Reserved                | This bit is read as 0.                                                                                                  | R   |
| b6 to b4 | T6ACNT[2:0] | TGIA6 Interrupt Counter | While the T6AEN bit in TITCR1B is set to 1, the count in these bits is incremented every time a TGIA6 interrupt occurs. | R   |
| b7       | —           | Reserved                | This bit is read as 0.                                                                                                  | R   |

Note: To clear the TITCNT1B, set the TITCR1B.T6AEN and TITCR1B.T7VEN bits to 0.

### T7VCNT[2:0] Bits (TCIV7 Interrupt Counter)

[Clearing conditions]

- When the TITM bit in TITMRB is 1
- When the T7VEN bit in TITCR1B is set to 0
- When the T7VCOR[2:0] bits in TITCR1B are set to 000b
- When the T7VCNT[2:0] bits in TITCNT1B match the T7VCOR[2:0] bits in TITCR1B

### T6ACNT[2:0] Bits (TGIA6 Interrupt Counter)

[Clearing conditions]

- When the TITM bit in TITMRB is 1
- When the T6AEN bit in TITCR1B is set to 0
- When the T6ACOR[2:0] bits in TITCR1B are set to 000b
- When the T6ACNT[2:0] bits in TITCNT1B match the T6ACOR[2:0] bits in TITCR1B



## 24.2.41 Timer Interrupt Skipping Set Register 2m (TITCR2m) (m = A, B)

- MTU.TITCR2A

Address(es): MTU.TITCR2A 000C 123Bh

| b7                 | b6 | b5 | b4 | b3 | b2           | b1 | b0 |
|--------------------|----|----|----|----|--------------|----|----|
| —                  | —  | —  | —  | —  | TRG4COR[2:0] |    |    |
| Value after reset: | 0  | 0  | 0  | 0  | 0            | 0  | 0  |

| Bit      | Symbol       | Bit Name                                       | Description                                                                                                                    | R/W |
|----------|--------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----|
| b2 to b0 | TRG4COR[2:0] | TRG4AN/TRG4BN Interrupt Skipping Count Setting | These bits specify the TRG4AN/TRG4BN interrupt skipping count within the range from 0 to 7. For details, refer to Table 24.59. | R/W |
| b7 to b3 | —            | Reserved                                       | These bits are read as 0. The write value should be 0.                                                                         | R/W |

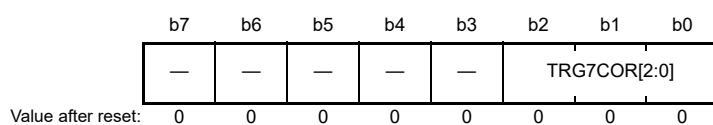
TITCR2A and TITCR2B specify the interrupt skipping count for TRG4AN and TRG4BN (TRG7AN and TRG7BN). This setting is valid only while TITMRA or TITMRB is set to 1.

**Table 24.59 Setting of Interrupt Skipping Count by TRG4COR[2:0] Bits**

| Bit 2      | Bit 1      | Bit 0      |                                                           |
|------------|------------|------------|-----------------------------------------------------------|
| TRG4COR[2] | TRG4COR[1] | TRG4COR[0] | Description                                               |
| 0          | 0          | 0          | Does not skip TRG4AN and TRG4BN interrupts.               |
| 0          | 0          | 1          | Sets the TRG4AN and TRG4BN interrupt skipping count to 1. |
| 0          | 1          | 0          | Sets the TRG4AN and TRG4BN interrupt skipping count to 2. |
| 0          | 1          | 1          | Sets the TRG4AN and TRG4BN interrupt skipping count to 3. |
| 1          | 0          | 0          | Sets the TRG4AN and TRG4BN interrupt skipping count to 4. |
| 1          | 0          | 1          | Sets the TRG4AN and TRG4BN interrupt skipping count to 5. |
| 1          | 1          | 0          | Sets the TRG4AN and TRG4BN interrupt skipping count to 6. |
| 1          | 1          | 1          | Sets the TRG4AN and TRG4BN interrupt skipping count to 7. |

- MTU.TITCR2B

Address(es): MTU.TITCR2B 000C 1A3Bh



| Bit      | Symbol       | Bit Name                                       | Description                                                                                                                       | R/W |
|----------|--------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----|
| b2 to b0 | TRG7COR[2:0] | TRG7AN/TRG7BN Interrupt Skipping Count Setting | These bits specify the TRG7AN/TRG7BN interrupt skipping count within the range from 0 to 7.<br>For details, refer to Table 24.60. | R/W |
| b7 to b3 | —            | Reserved                                       | These bits are read as 0. The write value should be 0.                                                                            | R/W |

**Table 24.60** Setting of Interrupt Skipping Count by TRG7COR[2:0] Bits

| Bit 2      | Bit 1      | Bit 0      | Description                                               |
|------------|------------|------------|-----------------------------------------------------------|
| TRG7COR[2] | TRG7COR[1] | TRG7COR[0] |                                                           |
| 0          | 0          | 0          | Does not skip TRG7AN and TRG7BN interrupts.               |
| 0          | 0          | 1          | Sets the TRG7AN and TRG7BN interrupt skipping count to 1. |
| 0          | 1          | 0          | Sets the TRG7AN and TRG7BN interrupt skipping count to 2. |
| 0          | 1          | 1          | Sets the TRG7AN and TRG7BN interrupt skipping count to 3. |
| 1          | 0          | 0          | Sets the TRG7AN and TRG7BN interrupt skipping count to 4. |
| 1          | 0          | 1          | Sets the TRG7AN and TRG7BN interrupt skipping count to 5. |
| 1          | 1          | 0          | Sets the TRG7AN and TRG7BN interrupt skipping count to 6. |
| 1          | 1          | 1          | Sets the TRG7AN and TRG7BN interrupt skipping count to 7. |

## 24.2.42 Timer Interrupt Skipping Counter 2m (TITCNT2m) (m = A, B)

- MTU.TITCNT2A

Address(es): MTU.TITCNT2A 000C 123Ch



| Bit      | Symbol       | Bit Name                        | Description                                                                                                                                                                                                              | R/W |
|----------|--------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b2 to b0 | TRG4CNT[2:0] | TRG4AN/TRG4BN Interrupt Counter | These bits start counting from the value set in TRG4COR[2:0] and the count decrements every time TRG4AN or TRG4BN is generated. When the count reaches 0 and is reloaded, the TRG4AN and TRG4BN interrupts become valid. | R   |
| b7 to b3 | —            | Reserved                        | These bits are read as 0.                                                                                                                                                                                                | R   |

TITCNT2A and TITCNT2B start counting from the values set in the TRG4COR[2:0] and TRG7COR[2:0] bits and the count decrements every time TRG4AN or TRG4BN (TITCNT2A) is generated or TRG7AN or TRG7BN (TITCNT2B) is generated. When the count reaches 0 and is reloaded, the TRG4AN and TRG4BN interrupts or the TRG7AN and TRG7BN interrupts become valid.

**TRG4CNT[2:0] Bits (TRG4AN/TRG4BN Interrupt Counter)**

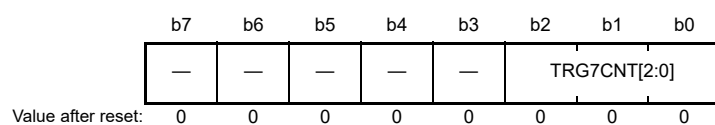
These bits start counting from the value set in the TRG4COR[2:0] bits and the count decrements every time a TRG4AN or TRG4BN interrupt is generated. When the count reaches 0 and is reloaded, the TRG4AN and TRG4BN interrupts become valid.

[Clearing conditions]

- When the TITM bit in TITMRA is 0
- When the TRG4COR[2:0] bits in TITCR2A are set to 000b
- When the count of TRG4AN and TRG4BN occurrence matches the TRG4COR[2:0] value in TITCR2A

- MTU.TITCNT2B

Address(es): MTU.TITCNT2B 000C 1A3Ch



| Bit      | Symbol       | Bit Name                           | Description                                                                                                                                                                                                              | R/W |
|----------|--------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b2 to b0 | TRG7CNT[2:0] | TRG7AN/TRG7BN<br>Interrupt Counter | These bits start counting from the value set in TRG7COR[2:0] and the count decrements every time TRG7AN or TRG7BN is generated. When the count reaches 0 and is reloaded, the TRG7AN and TRG7BN interrupts become valid. | R   |
| b7 to b3 | —            | Reserved                           | These bits are read as 0.                                                                                                                                                                                                | R   |

### TRG7CNT[2:0] Bits (TRG7AN/TRG7BN Interrupt Counter)

These bits start counting from the value set in the TRG7COR[2:0] bits and the count decrements every time a TRG7AN or TRG7BN interrupt is generated. When the count reaches 0 and is reloaded, the TRG7AN and TRG7BN interrupts become valid.

[Clearing conditions]

- When the TITM bit in TITMRB is 0
- When the TRG7COR[2:0] bits in TITCR2B are set to 000b
- When the count of TRG7AN and TRG7BN occurrence matches the TRG7COR[2:0] value in TITCR2B

## 24.3 Operation

### 24.3.1 Basic Functions

Each channel has TCNT and TGR register. TCNT performs up-counting, and is also capable of free-running operation, periodic counting, and external event counting.

Each TGR register can be used as an input capture register or an output compare register.

#### (1) Counter Operation

When one of bits CST0 to CST4 and CST8 in the TSTRA register, bits CST6 and CST7 in the TSTRB register, and bits CSTU5, CSTV5, and CSTW5 in the MTU5.TSTR register is set to 1, TCNT for the corresponding channel begins counting. TCNT can operate as a free-running counter, periodic counter, for example.

#### (a) Example of Count Operation Setting Procedure

Figure 24.5 shows an example of the count operation setting procedure.

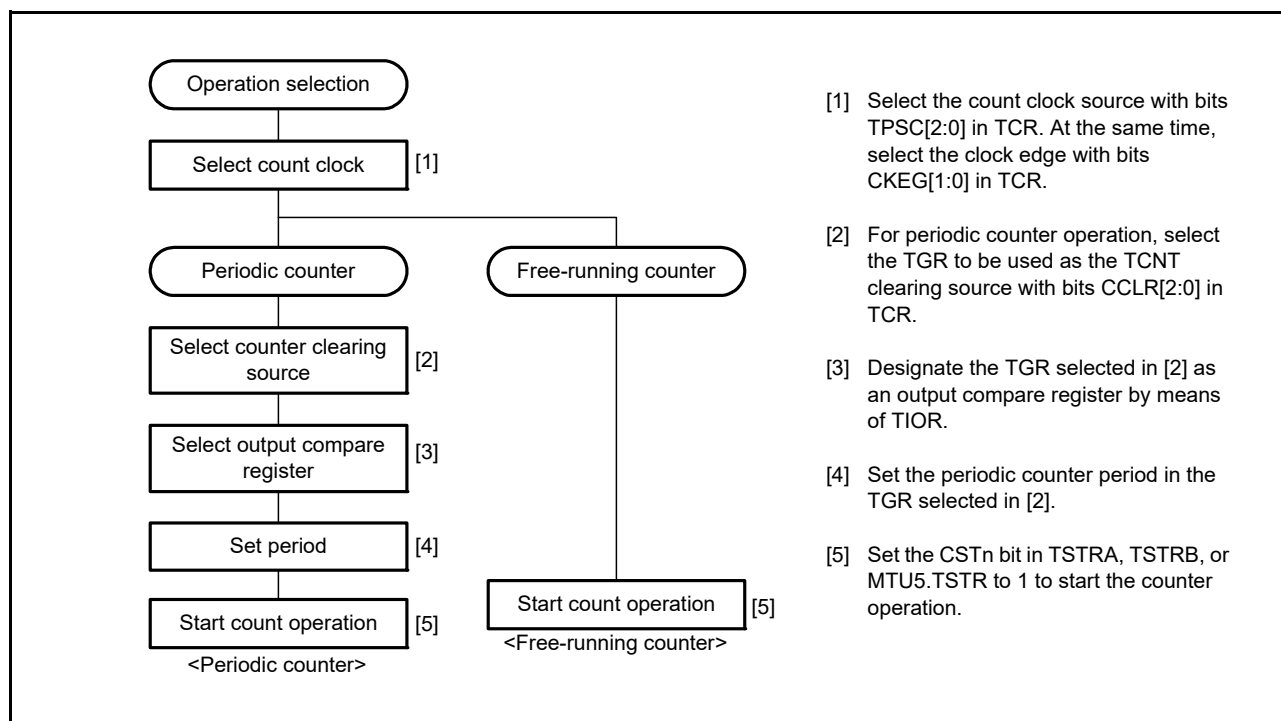
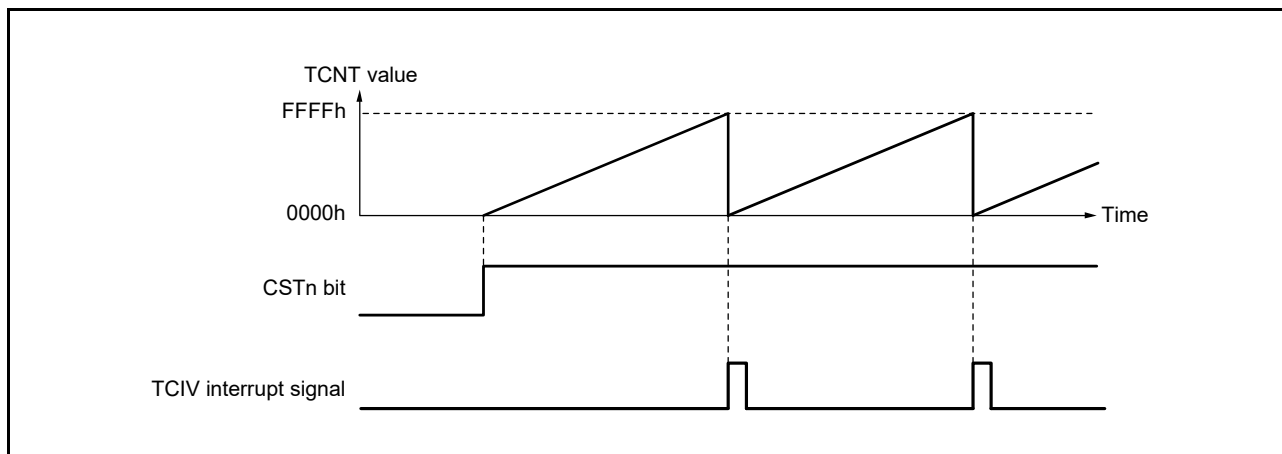


Figure 24.5 Example of Count Operation Setting Procedure

### (b) Free-Running Count Operation and Periodic Count Operation

Immediately after a reset, the TCNT counters are all designated as free-running counters. When the CSTn bit in TSTRA, TSTRB, or MTU5.TSTR is set to 1, the corresponding TCNT counter starts up-count operation as a free-running counter. When TCNT overflows (from FFFFh to 0000h), an interrupt request is issued to the CPU if the corresponding TIER.TCIEV bit is 1. After an overflow, TCNT starts counting up again from 0000h.

Figure 24.6 illustrates free-running counter operation.

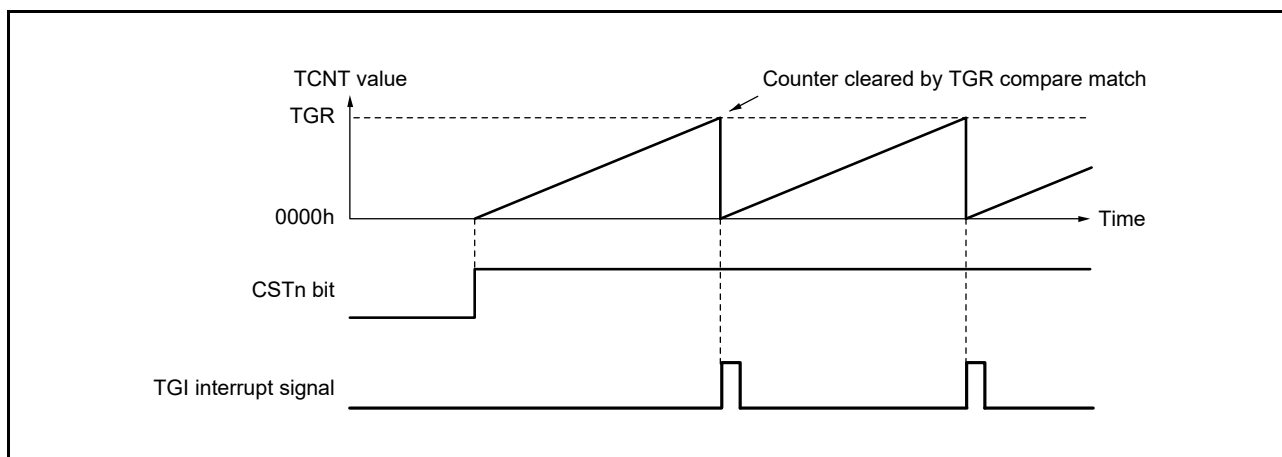


**Figure 24.6 Free-Running Counter Operation**

When compare match is selected as the TCNT clearing source, TCNT for the relevant channel performs periodic count operation. TGR for setting the period is designated as an output compare register, and counter clearing by compare match is selected by means of bits CCLR[2:0] in TCR. After the settings have been made, TCNT starts up-count operation as a periodic counter when the CSTn bit in TSTRA, TSTRB or MTU5.TSTR is set to 1. When the count matches the value in TGR, TCNT becomes 0000h.

If the value of the corresponding TIER.TGIE bit is 1 at this point, an interrupt request is issued to the CPU. After a compare match, TCNT starts counting up again from 0000h.

Figure 24.7 illustrates periodic counter operation.



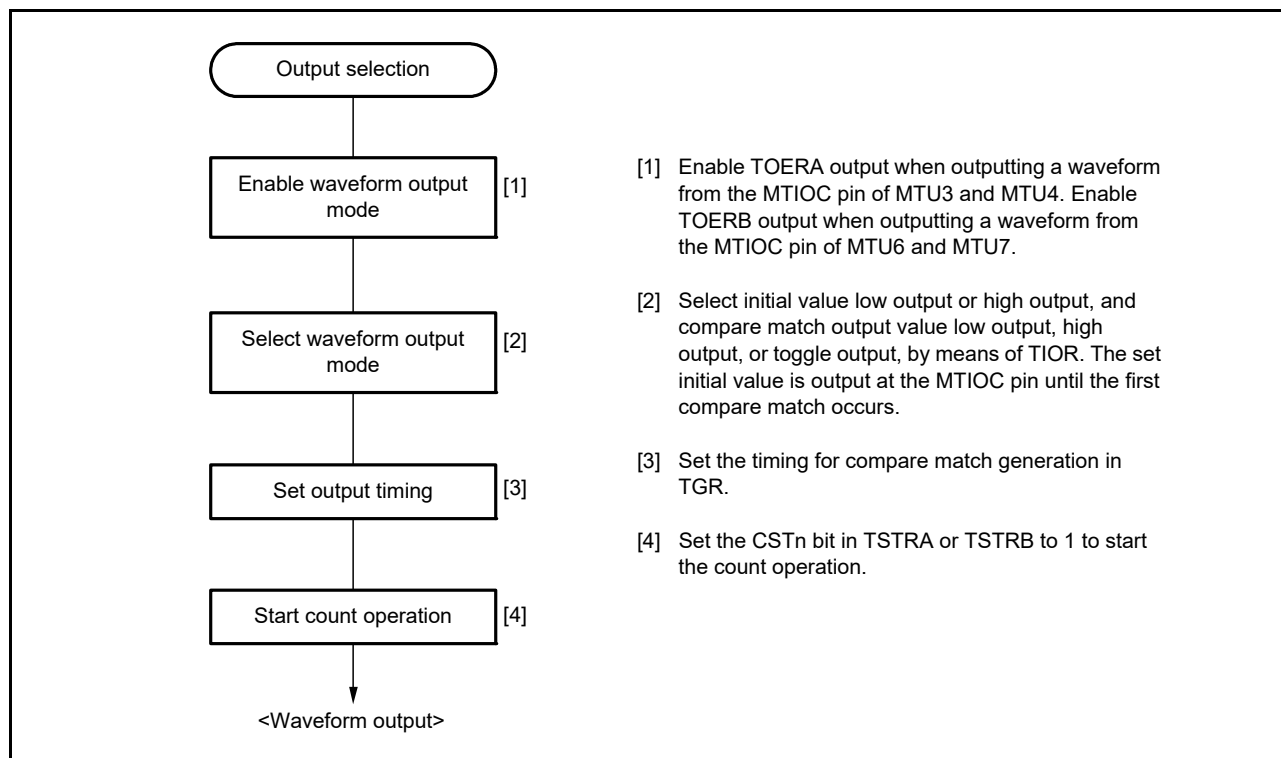
**Figure 24.7 Periodic Counter Operation**

## (2) Waveform Output by Compare Match

Upon compare match, low, high, or toggle output from the corresponding pin can be performed.

### (a) Example of Procedure for Setting Waveform Output by Compare Match

Figure 24.8 shows an example of the procedure for setting waveform output by compare match

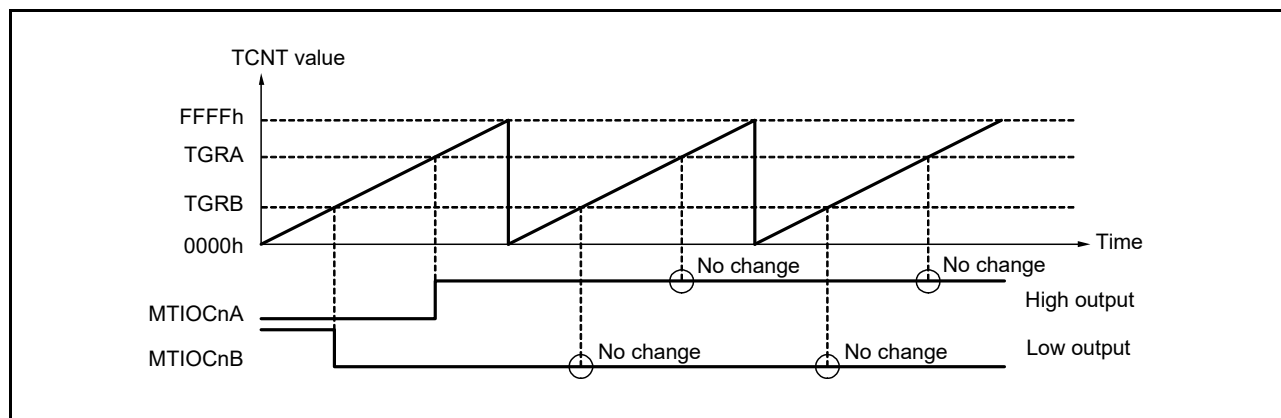


**Figure 24.8** Example of Procedure for Setting Waveform Output by Compare Match

## (b) Examples of Waveform Output Operation

Figure 24.9 shows an example of low output and high output.

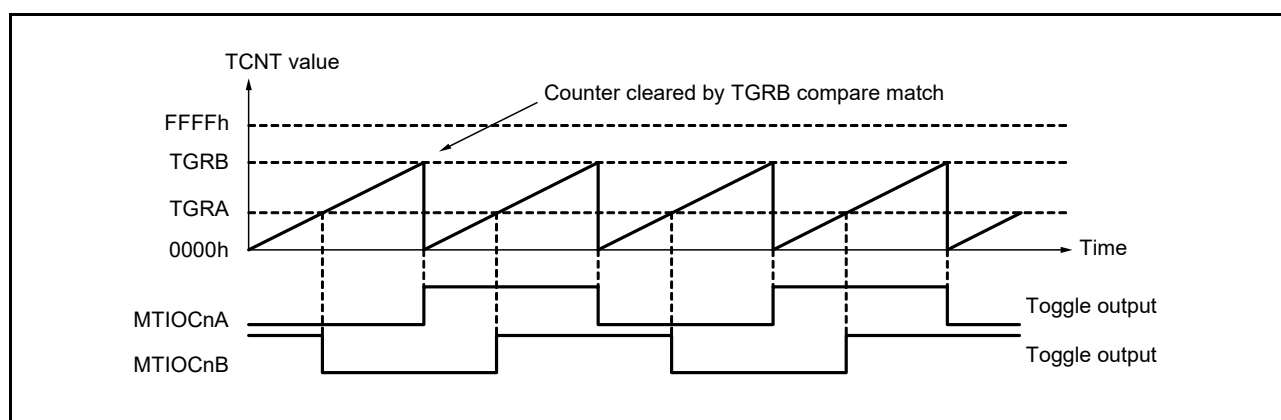
In this example, TCNT has been designated as a free-running counter, and settings have been made so that high is output by compare match A and low is output by compare match B. When the pin level is the same as the specified level, the pin level does not change.



**Figure 24.9** Example of low output and high output Operation (n = 0 to 4, 6, 7, 8)

Figure 24.10 shows an example of toggle output.

In this example, TCNT has been designated as a periodic counter (with counter clearing on compare match B), and settings have been made so that the output is toggled by both compare match A and compare match B.



**Figure 24.10** Example of Toggle Output Operation (n = 0 to 4, 6, 7, 8)



### (3) Input Capture Function

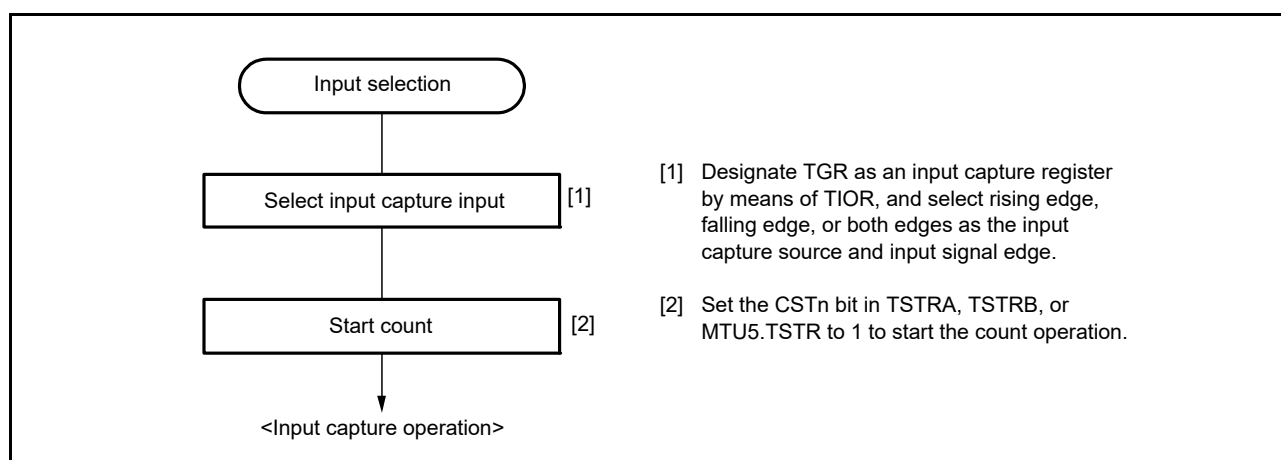
The TCNT value can be transferred to TGR on detection of the MTIOCnm pin (n = 0 to 4, 6, 7, 8; m = A to D) or MTIC5m pin (m = U, V, W) input edge.

The rising edge, falling edge, or both edges can be selected as the detection edge. For MTU0 and MTU1, another channel's count clock or compare match signal can also be specified as the input capture source.

**Note:** When another channel's count clock is used as the input capture input for MTU0 and MTU1, PCLKA/1 should not be selected as the count clock used for input capture input. Input capture will not be generated if PCLKA/1 is selected.

#### (a) Example of Input Capture Operation Setting Procedure

Figure 24.11 shows an example of the input capture operation setting procedure.



**Figure 24.11** Example of Input Capture Operation Setting Procedure

## (b) Example of Input Capture Operation

Figure 24.12 shows an example of input capture operation.

In this example, both rising and falling edges have been selected as the MTIOCnA pin input capture input edge, the falling edge has been selected as the MTIOCnB pin input capture input edge, and counter clearing by TGRB input capture has been designated for TCNT. (n = 0 to 4, 6, 7, 8)

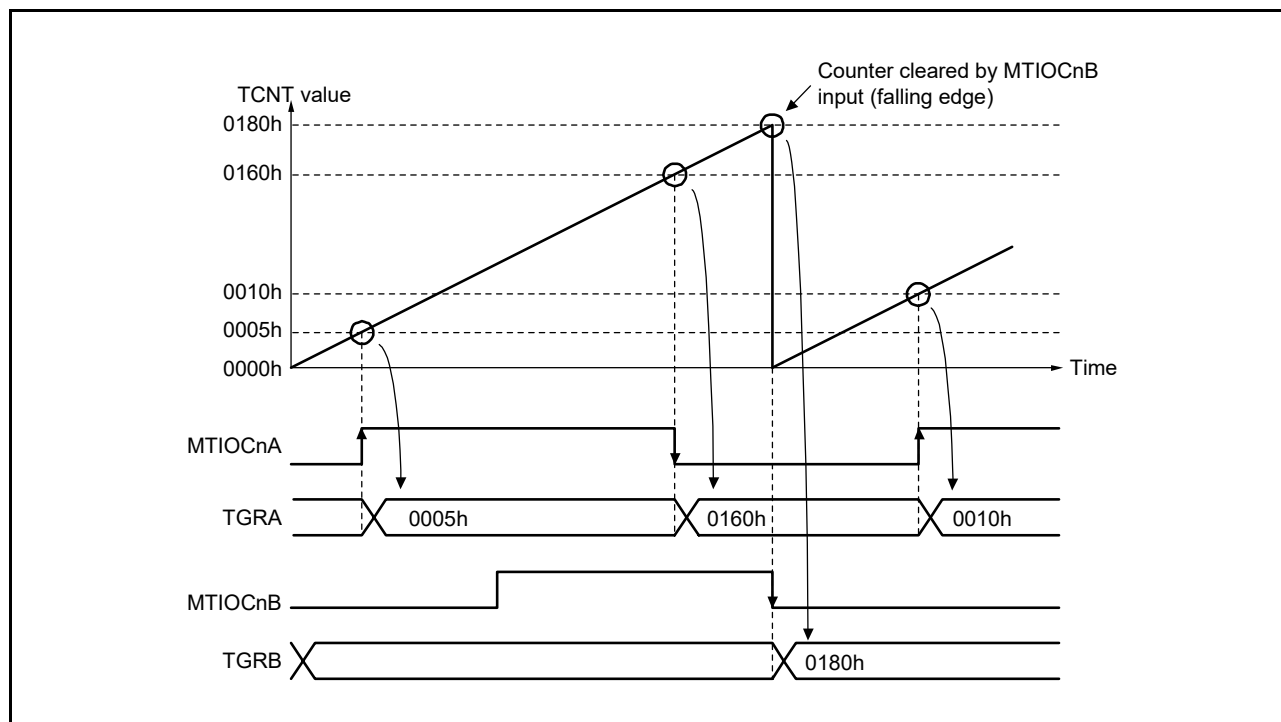


Figure 24.12 Example of Input Capture Operation (n = 0 to 4, 6, 7, 8)

### 24.3.2 Synchronous Operation

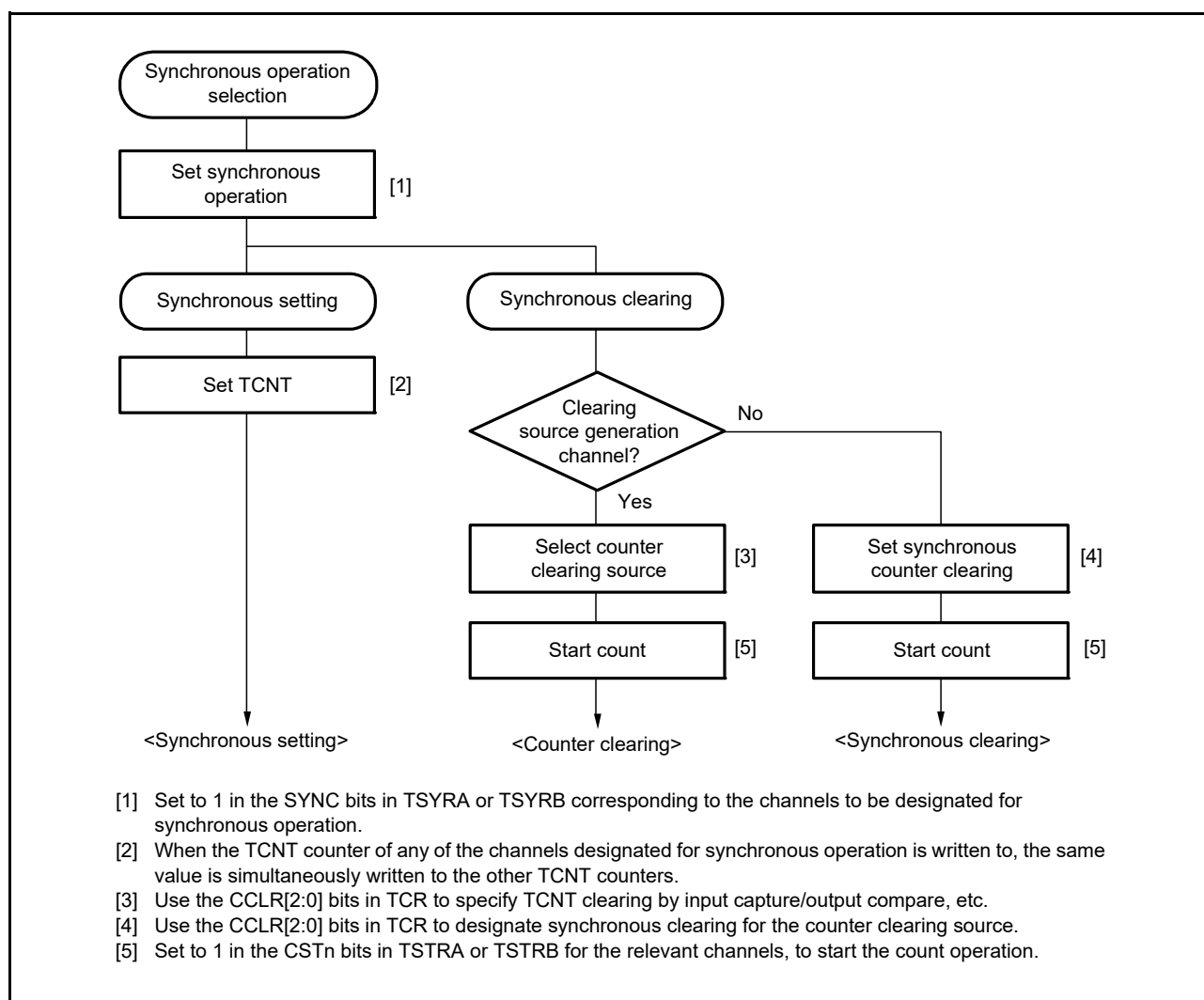
In synchronous operation, the values in multiple TCNT counters can be modified simultaneously (synchronous setting). In addition, multiple TCNT counters can be cleared simultaneously (synchronous clearing) by making the appropriate setting in TCR.

Synchronous operation can increase the number of TGR registers assigned to the same time base.

MTU0 to MTU4, MTU6, and MTU7 can all be designated for synchronous operation. MTU5 and MTU8 cannot be used for synchronous operation.

#### (1) Example of Synchronous Operation Setting Procedure

Figure 24.13 shows an example of the synchronous operation setting procedure.



**Figure 24.13 Example of Synchronous Operation Setting Procedure**

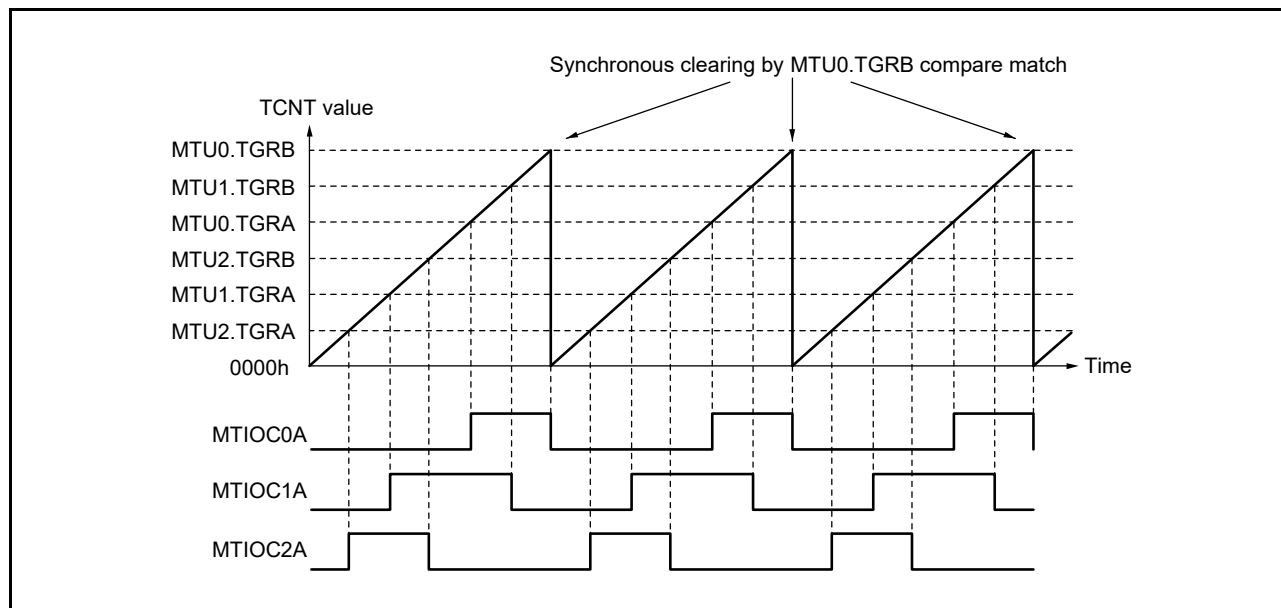
## (2) Example of Synchronous Operation

Figure 24.14 shows an example of synchronous operation.

In this example, synchronous operation and PWM mode 1 have been designated for MTU0 to MTU 2, MTU0.TGRB compare match has been set as the counter clearing source in MTU0, and synchronous clearing has been set for the counter clearing source in MTU1 and MTU2.

Three-phase PWM waveforms are output from pins MTIOC0A, MTIOC1A, and MTIOC2A. At this time, synchronous setting and synchronous clearing by MTU0.TGRB compare match are performed for the TCNT counters in MTU0 to MTU2, and the data set in MTU0.TGRB is used as the PWM period.

For details of PWM modes, refer to section 24.3.5, PWM Modes.



**Figure 24.14 Example of Synchronous Operation**

### 24.3.3 Buffer Operation

Buffer operation, provided for MTU0, MTU3, MTU4, MTU6, MTU7, and MTU8, enables TGRC and TGRD to be used as buffer registers. In MTU0, TGRF can also be used as a buffer register.

Buffer operation differs depending on whether TGR has been designated as an input capture register or as a compare match register.

Note: MTU0.TGRE cannot be designated as an input capture register and can only operate as a compare match register.

Table 24.61 shows the register combinations used in buffer operation.

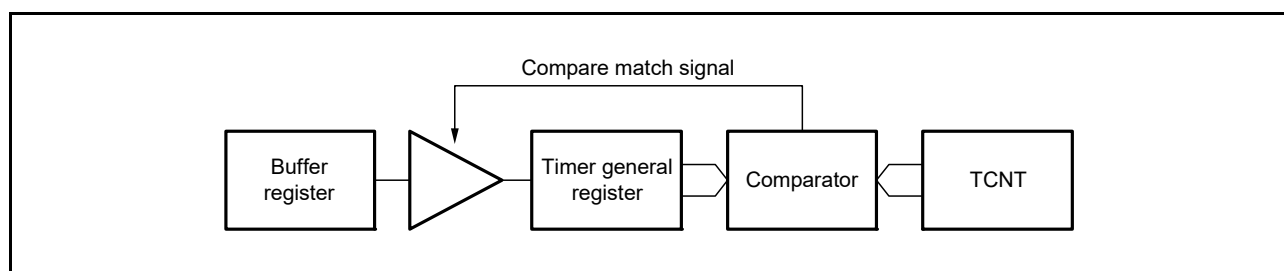
**Table 24.61 Register Combinations in Buffer Operation**

| Channel | Timer General Register | Buffer Register |
|---------|------------------------|-----------------|
| MTU0    | TGRA                   | TGRC            |
|         | TGRB                   | TGRD            |
|         | TGRE                   | TGRF            |
| MTU3    | TGRA                   | TGRC            |
|         | TGRB                   | TGRD            |
| MTU4    | TGRA                   | TGRC            |
|         | TGRB                   | TGRD            |
| MTU6    | TGRA                   | TGRC            |
|         | TGRB                   | TGRD            |
| MTU7    | TGRA                   | TGRC            |
|         | TGRB                   | TGRD            |
| MTU8    | TGRA                   | TGRC            |
|         | TGRB                   | TGRD            |

- When TGR is an output compare register

When a compare match occurs, the value in the buffer register for the corresponding channel is transferred to the timer general register.

This operation is illustrated in Figure 24.15.

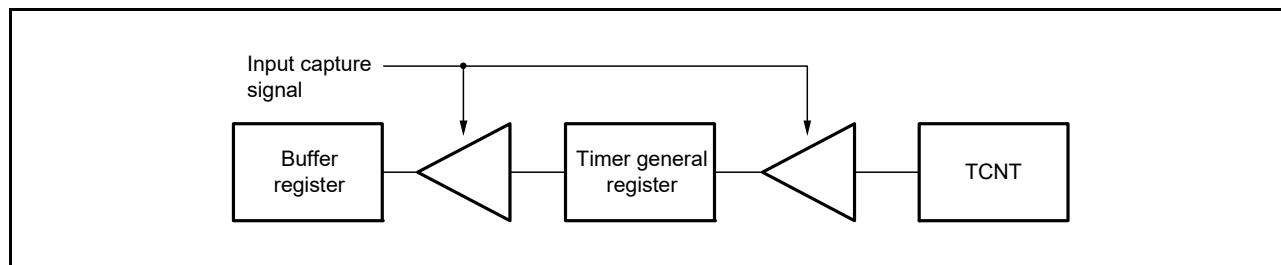


**Figure 24.15 Compare Match Buffer Operation**

- When TGR is an input capture register

When an input capture occurs, the value in TCNT is transferred to TGR and the value previously held in TGR is transferred to the buffer register.

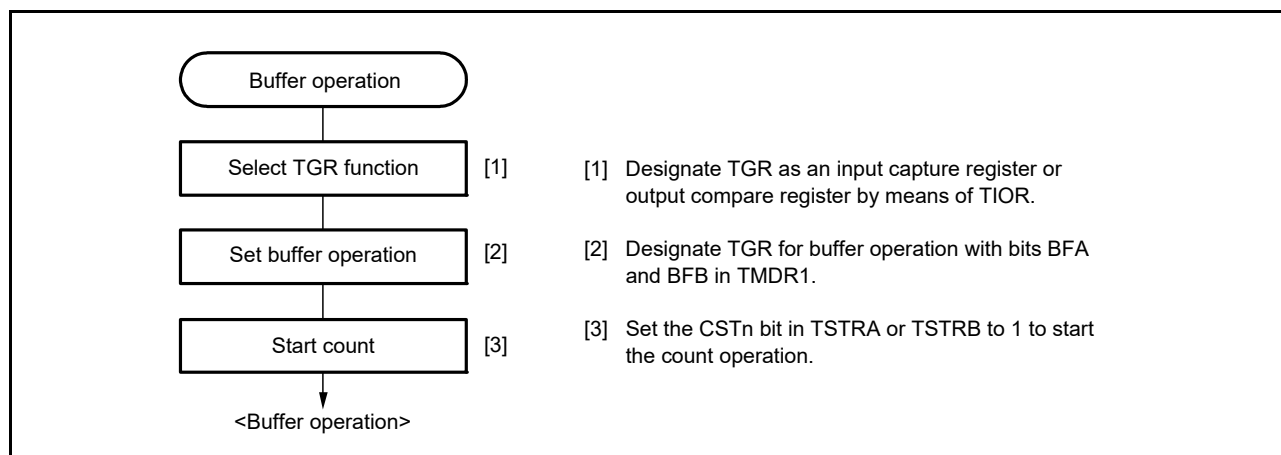
This operation is illustrated in Figure 24.16.



**Figure 24.16** Input Capture Buffer Operation

### (1) Example of Buffer Operation Setting Procedure

Figure 24.17 shows an example of the buffer operation setting procedure.



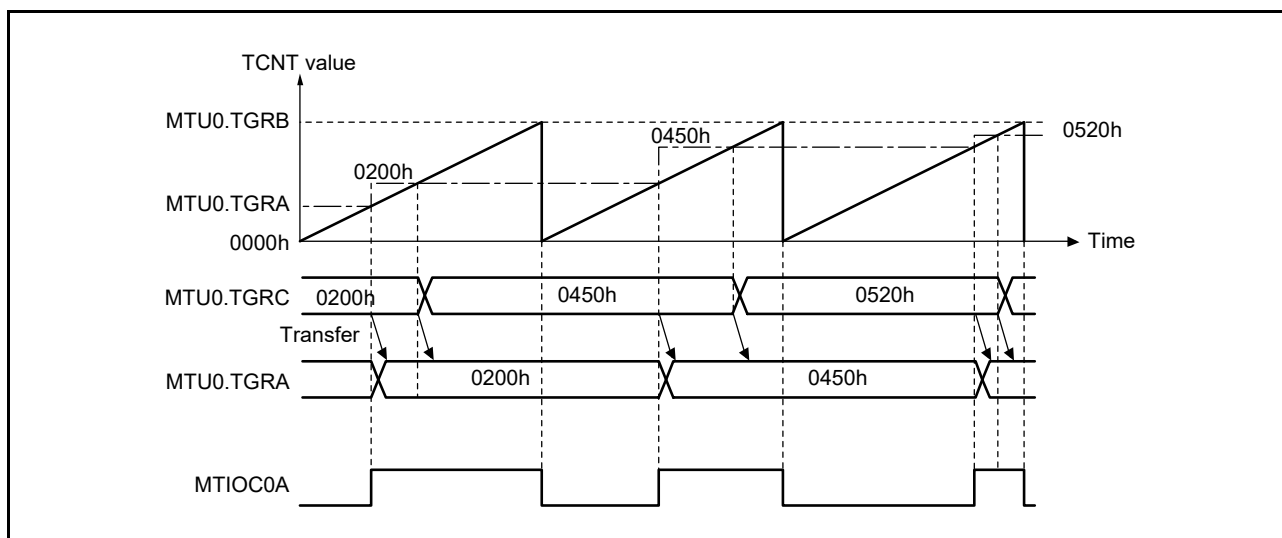
**Figure 24.17** Example of Buffer Operation Setting Procedure

### (2) Examples of Buffer Operation

#### (a) When TGR is an Output Compare Register

Figure 24.18 shows an operation example in which PWM mode 1 has been designated for MTU0, and buffer operation has been designated for TGRA and TGRC. The settings used in this example are TCNT clearing by compare match B, high output at compare match A, and low output at compare match B. In this example, the TTSA bit in TBTM is set to 0. As buffer operation has been set, when compare match A occurs, the output changes and the value in buffer register TGRC is simultaneously transferred to timer general register TGRA. This operation is repeated each time compare match A occurs.

For details of PWM modes, refer to section 24.3.5, PWM Modes.



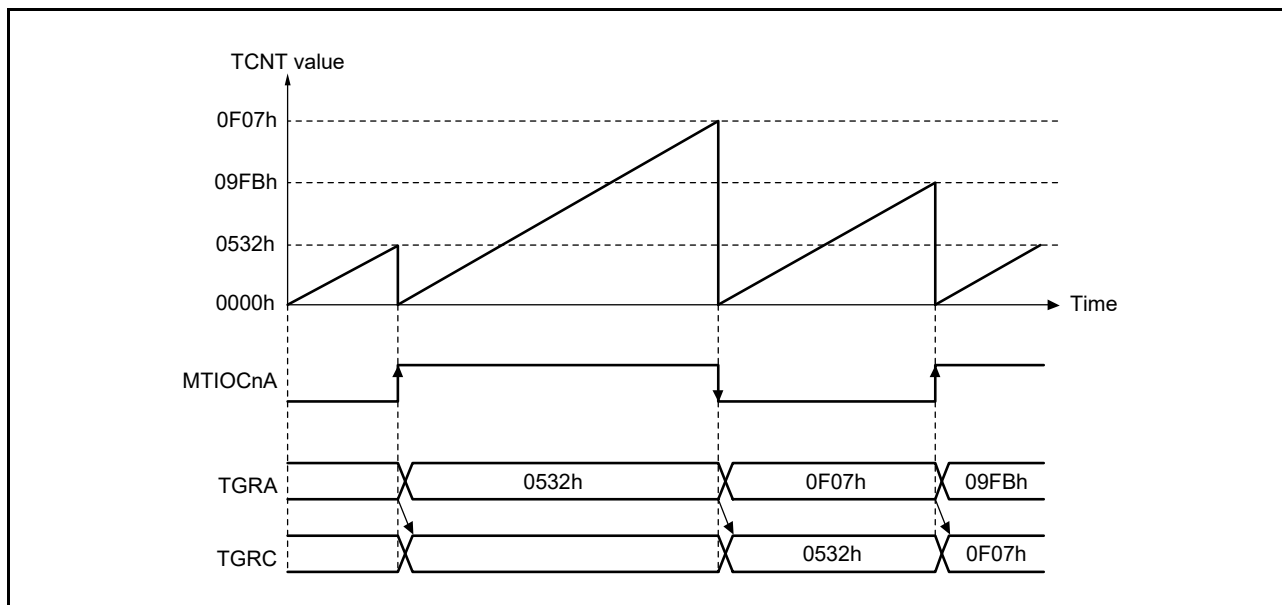
**Figure 24.18 Example of Buffer Operation (1)**

**(b) When TGR is an Input Capture Register**

Figure 24.19 shows an operation example in which TGRA has been designated as an input capture register, and buffer operation has been designated for TGRA and TGRC.

Counter clearing by TGRA input capture has been set for TCNT, and both rising and falling edges have been selected as the MTIOCnA pin input capture input edge. (n = 0 to 4, 6, 7, 8)

As buffer operation has been set, when the TCNT value is transferred to TGRA upon occurrence of input capture A, the value previously stored in TGRA is simultaneously transferred to TGRC.



**Figure 24.19 Example of Buffer Operation (2) (n = 0 to 4, 6, 7, 8)**

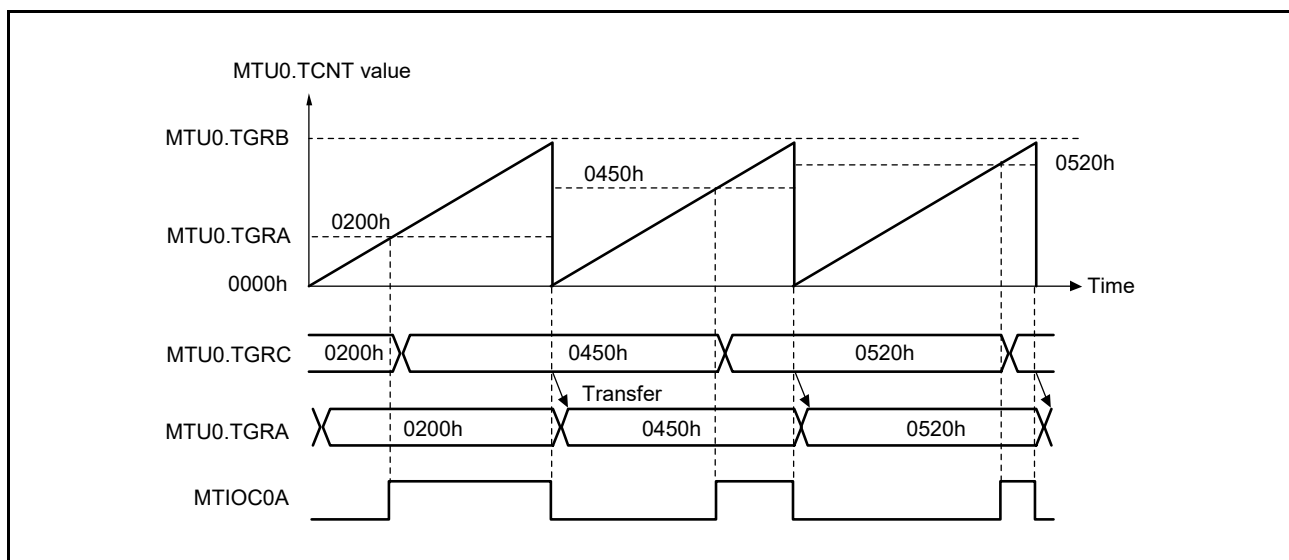
### (3) Selecting Timing for Transfer from Buffer Registers to Timer General Registers in Buffer Operation

The timing for transfer from buffer registers to timer general registers can be selected in PWM mode 1 or 2 for MTU0 or in PWM mode 1 for MTU3, MTU4, MTU6, and MTU7 by setting the buffer operation transfer mode registers (MTUn.TBTM (n = 0, 3, 4, 6, 7)). Either compare match (value after reset) or TCNT clearing can be selected for the transfer timing. TCNT clearing as transfer timing is one of the following cases.

- When TCNT overflows (FFFFh to 0000h)
- When 0000h is written to TCNT during counting
- When TCNT becomes 0000h under the condition specified in the CCLR[2:0] bits in TCR

Note: TBTM must be modified only while TCNT stops.

Figure 24.20 shows an operation example in which PWM mode 1 is designated for MTU0 and buffer operation is designated for MTU0.TGRA and MTU0.TGRC. The settings used in this example are MTU0.TCNT clearing by compare match B, high output at compare match A, and low output at compare match B. The TTSA bit in MTU0.TBTM is set to 1.



**Figure 24.20** Example of Buffer Operation When MTU0.TCNT Clearing is Selected for MTU0.TGRC to MTU0.TGRA Transfer Timing



### 24.3.4 Cascaded Operation

In cascaded operation, two 16-bit counters in different channels are used together as a 32-bit counter.

There are two functions for connecting MTU1 and MTU2 to use as a 32-bit counter: cascade connection to be set when the MTU1.TMDR3.LWA bit is 0, and cascade connection 32-bit phase counting mode to be set when the MTU1.TMDR3.LWA bit is 1. For details on cascade connection 32-bit phase counting mode, refer to [section 24.3.6.2, Cascade Connection 32-Bit Phase Counting Mode](#). This section describes the cascade connection function to be set when the MTU1.TMDR3.LWA bit is 0.

This function operates when the MTU1.TMDR3.LWA bit is set to 0 and the MTU1.TCR.TPSC[2:0] bits are set so that MTU1.TCNT counts at an overflow/underflow of MTU2.TCNT. Underflow occurs only when the MTU2 to which the lower 16 bits allocated is in phase counting mode.

Table 24.62 shows the register combinations used in cascaded operation.

**Note:** When phase counting mode is set for MTU1, the count clock setting is invalid and the counters operate independently in phase counting mode.

**Table 24.62 Cascaded Combinations**

| Combination   | Upper 16 Bits | Lower 16 Bits |
|---------------|---------------|---------------|
| MTU1 and MTU2 | MTU1.TCNT     | MTU2.TCNT     |

For simultaneous input capture of MTU1.TCNT and MTU2.TCNT during cascaded operation, additional input capture input pins can be specified by the input capture control register (TICCR). The input-capture condition is of edges in the signal produced by taking the logical OR of the input level on the main input pin and the input level on the added input pin. Accordingly, if either is at the high level, a change in the level of the other will not produce an edge for detection. For details, refer to (4), Cascaded Operation Example (c). For input capture in cascade connection, refer to [section 24.6.21, Simultaneous Input Capture in MTU1.TCNT and MTU2.TCNT in Cascade Connection](#).

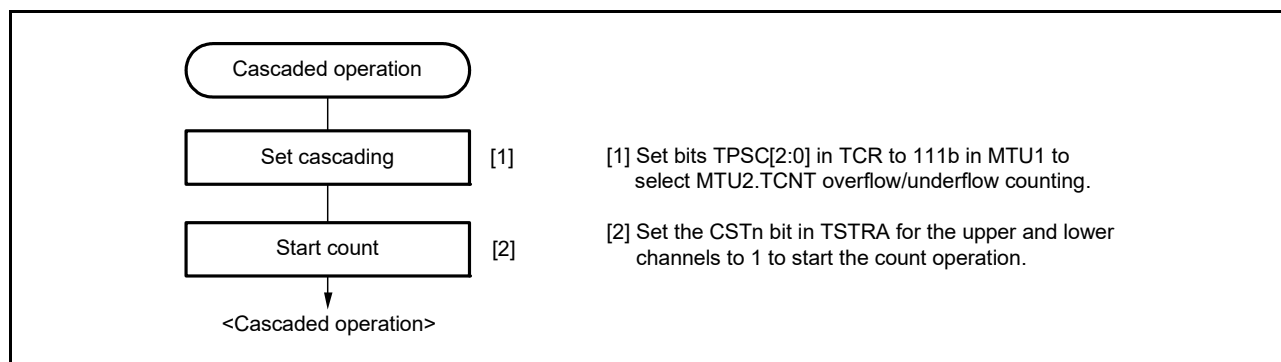
Table 24.63 shows the TICCR setting and input capture input pins.

**Table 24.63 TICCR Setting and Input Capture Input Pins**

| Target Input Capture                      | TICCR Setting                | Input Capture Input Pin |
|-------------------------------------------|------------------------------|-------------------------|
| Input capture from MTU1.TCNT to MTU1.TGRA | I2AE bit = 0 (Initial value) | MTIOC1A                 |
|                                           | I2AE bit = 1                 | MTIOC1A, MTIOC2A        |
| Input capture from MTU1.TCNT to MTU1.TGRB | I2BE bit = 0 (Initial value) | MTIOC1B                 |
|                                           | I2BE bit = 1                 | MTIOC1B, MTIOC2B        |
| Input capture from MTU2.TCNT to MTU2.TGRA | I1AE bit = 0 (Initial value) | MTIOC2A                 |
|                                           | I1AE bit = 1                 | MTIOC2A, MTIOC1A        |
| Input capture from MTU2.TCNT to MTU2.TGRB | I1BE bit = 0 (Initial value) | MTIOC2B                 |
|                                           | I1BE bit = 1                 | MTIOC2B, MTIOC1B        |

## (1) Example of Cascaded Operation Setting Procedure

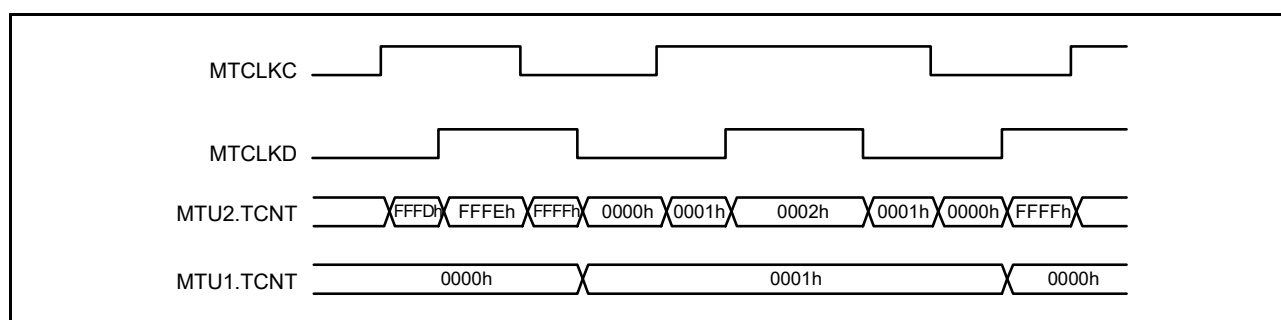
Figure 24.21 shows an example of the cascaded operation setting procedure.



**Figure 24.21 Cascaded Operation Setting Procedure**

## (2) Cascaded Operation Example (a)

Figure 24.22 shows the operation when MTU1.TCNT is set for counting at MTU2.TCNT overflow/underflow and MTU2 is set for phase counting mode 1 while MTU1.TCNT and MTU2.TCNT are cascaded. MTU1.TCNT is incremented by MTU2.TCNT overflow and decremented by MTU2.TCNT underflow.



**Figure 24.22 Cascaded Operation Example (a)**

### (3) Cascaded Operation Example (b)

Figure 24.23 illustrates the operation when MTU1.TCNT and MTU2.TCNT have been cascaded and the I2AE bit in TICCRR has been set to 1 to include the MTIOC2A pin in the MTU1.TGRA input capture conditions. In this example, the MTU1.TIOR.IOA[3:0] bits have selected the MTIOC1A rising edge for the input capture timing while the MTU2.TIOR.IOA[3:0] bits have selected the MTIOC2A rising edge for the input capture timing.

Under these conditions, the rising edge of both MTIOC1A and MTIOC2A is used for the MTU1.TGRA input capture condition. For the MTU2.TGRA input capture condition, the MTIOC2A rising edge is used.

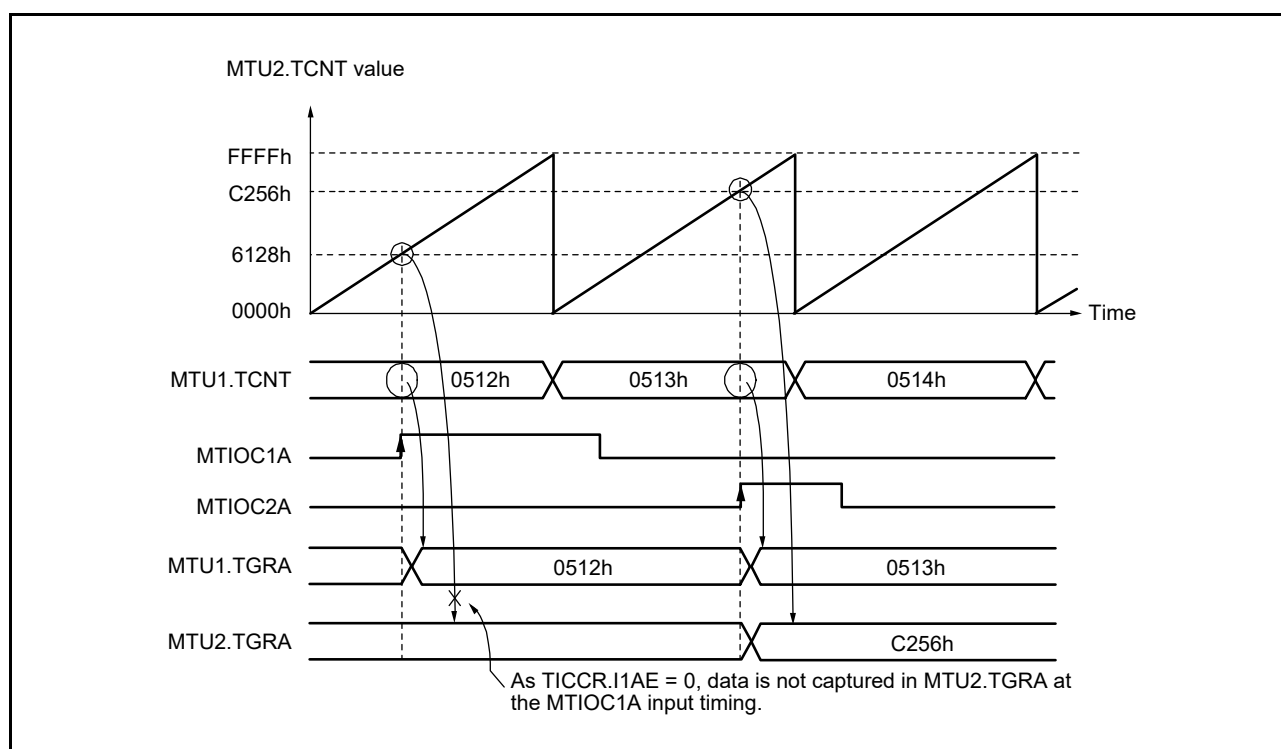


Figure 24.23 Cascaded Operation Example (b)

## (4) Cascaded Operation Example (c)

Figure 24.24 illustrates the operation when MTU1.TCNT and MTU2.TCNT have been cascaded and the TICC.R.I2AE and I1AE bits have been set to 1 to include the MTIOC2A and MTIOC1A pins in the MTU1.TGRA and MTU2.TGRA input capture conditions, respectively. In this example, the IOA[3:0] bits in both MTU1.TIOR and MTU2.TIOR have selected both the rising and falling edges for the input capture timing. Under these conditions, the ORed result of MTIOC1A and MTIOC2A input is used for the MTU1.TGRA and MTU2.TGRA input capture conditions.

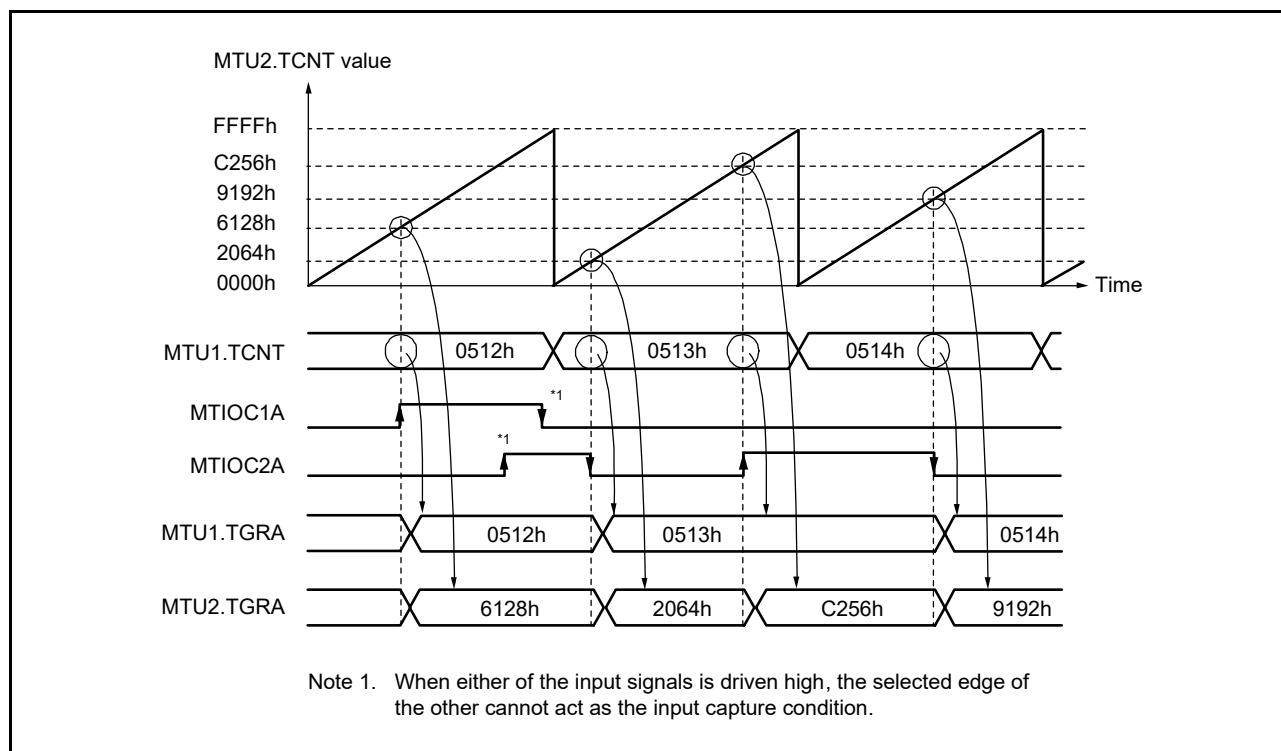


Figure 24.24 Cascaded Operation Example (c)

## (5) Cascaded Operation Example (d)

Figure 24.25 illustrates the operation when MTU1.TCNT and MTU2.TCNT have been cascaded and the TICC.R.I2AE bit has been set to 1 to include the MTIOC2A pin in the MTU1.TGRA input capture conditions. In this example, the IOA[3:0] bits in MTU1.TIOR have selected occurrence of MTU0.TGRA compare match or input capture for the input capture timing while the IOA[3:0] bits in MTU2.TIOR have selected the MTIOC2A rising edge for the input capture timing.

Under these conditions, as MTU1.TIOR has selected occurrence of MTU0.TGRA compare match or input capture for the input capture timing, the MTIOC2A edge is not used for MTU1.TGRA input capture condition although the I2AE bit in TICC.R has been set to 1.

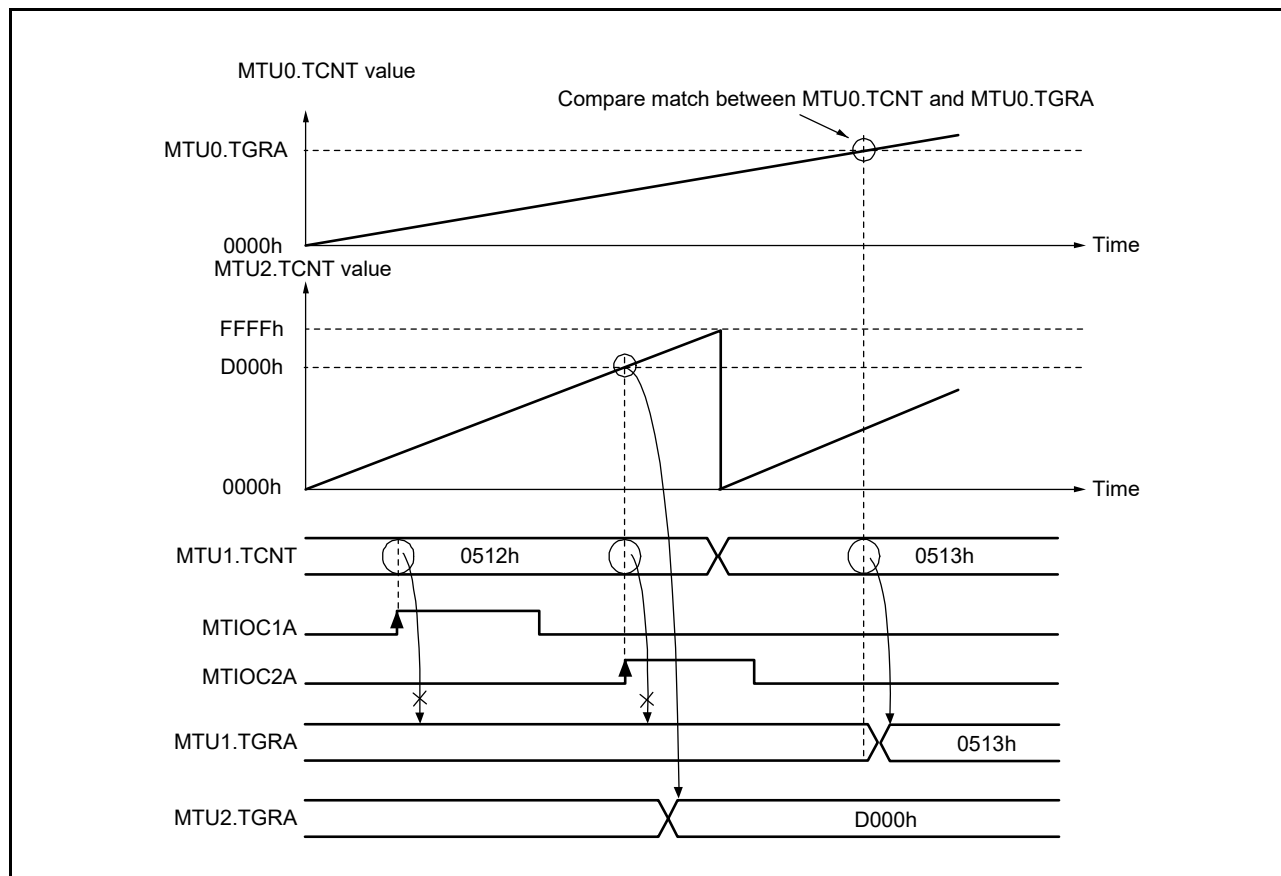


Figure 24.25 Cascaded Operation Example (d)

### 24.3.5 PWM Modes

PWM modes are provided to output PWM waveforms from the external pins. The output level can be selected as low, high, or toggle output in response to a compare match of each TGR.

PWM waveforms in the range of 0% to 100% duty cycle can be output according to the TGR settings.

By designating TGR compare match as the counter clearing source, the PWM period can be specified in that register.

Every channel except MTU5 and MTU8 can be set to PWM mode independently. Channels set to PWM mode can perform synchronous operation with each other or other channels set to any other mode.

There are two PWM modes as described below.

#### (a) PWM Mode 1

PWM waveforms are output from the MTIOCnA and MTIOCnC pins by pairing TGRA with TGRB and TGRC with TGRD. The levels specified by the TIOR.IOA[3:0] and IOC[3:0] bits are output from the MTIOCnA and MTIOCnC pins at compare matches A and C, and the level specified by the TIOR.IOB[3:0] and IOD[3:0] bits are output at compare matches B and D ( $n = 0$  to 4, 6, 7). The initial output value is set in TGRA or TGRC. If the values set in paired TGRs are identical, the output value does not change even when a compare match occurs.

In PWM mode 1, PWM waveforms in up to 12 phases can be output.

**(b) PWM Mode 2**

PWM waveform output is generated using one TGR as the period register and the others as duty registers. The level specified in TIOR is output at compare matches. Upon counter clearing by a period register compare match, the initial value set in TIOR is output from each pin. If the values set in the period and duty registers are identical, the output value does not change even when a compare match occurs.

Up to eight phases of PWM waveforms can be output by combining synchronous clearing of channels that cannot be set to PWM mode 2 as synchronous operation.

The correspondence between PWM output pins and registers is shown in Table 24.64.

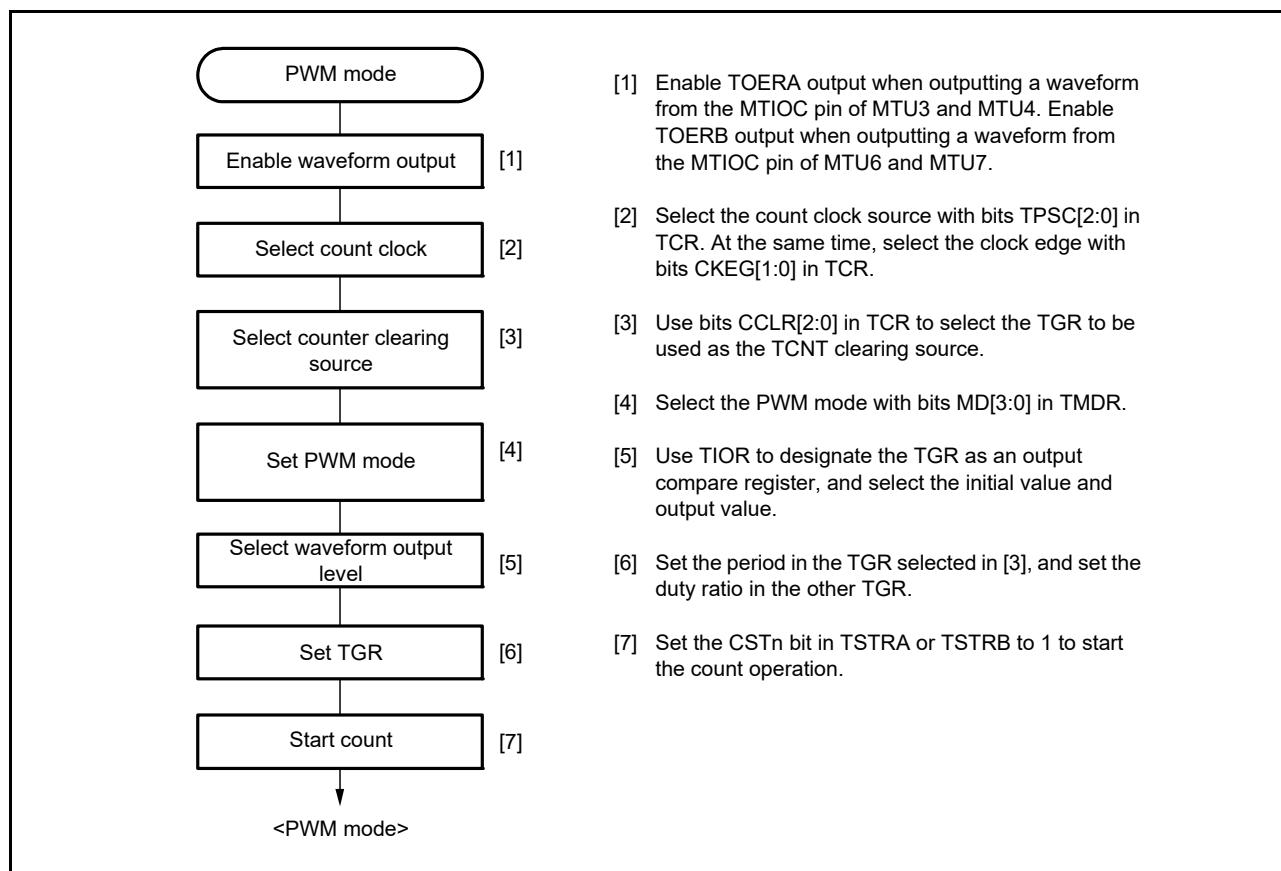
**Table 24.64 PWM Output Registers and Output Pins**

| Channel | Register | Output Pins |                    |
|---------|----------|-------------|--------------------|
|         |          | PWM Mode 1  | PWM Mode 2         |
| MTU0    | TGRA     | MTIOC0A     | MTIOC0A            |
|         | TGRB     |             | MTIOC0B            |
|         | TGRC     | MTIOC0C     | MTIOC0C            |
|         | TGRD     |             | MTIOC0D            |
| MTU1    | TGRA     | MTIOC1A     | MTIOC1A            |
|         | TGRB     |             | MTIOC1B            |
| MTU2    | TGRA     | MTIOC2A     | MTIOC2A            |
|         | TGRB     |             | MTIOC2B            |
| MTU3    | TGRA     | MTIOC3A     | Setting prohibited |
|         | TGRB     |             |                    |
|         | TGRC     | MTIOC3C     |                    |
|         | TGRD     |             |                    |
| MTU4    | TGRA     | MTIOC4A     |                    |
|         | TGRB     |             |                    |
|         | TGRC     | MTIOC4C     |                    |
|         | TGRD     |             |                    |
| MTU6    | TGRA     | MTIOC6A     |                    |
|         | TGRB     |             |                    |
|         | TGRC     | MTIOC6C     |                    |
|         | TGRD     |             |                    |
| MTU7    | TGRA     | MTIOC7A     |                    |
|         | TGRB     |             |                    |
|         | TGRC     | MTIOC7C     |                    |
|         | TGRD     |             |                    |

Note: In PWM mode 2, PWM waveform output is not possible for the TGR register in which the PWM period is set.

### (1) Example of PWM Mode Setting Procedure

Figure 24.26 shows an example of the PWM mode setting procedure.



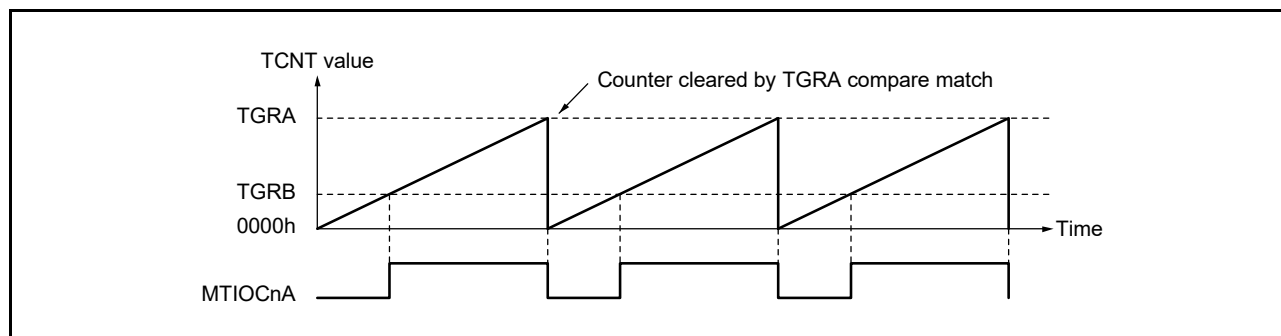
**Figure 24.26 Example of PWM Mode Setting Procedure**

### (2) Examples of PWM Mode Operation

Figure 24.27 shows an example of operation in PWM mode 1.

In this example, TGRA compare match is set as the TCNT clearing source, 0 is set as the initial output value and output value for TGRA, and 1 is set as the output value for TGRB.

In this case, the value set in TGRA is used as the period, and the value set in TGRB is used as the duty ratio.



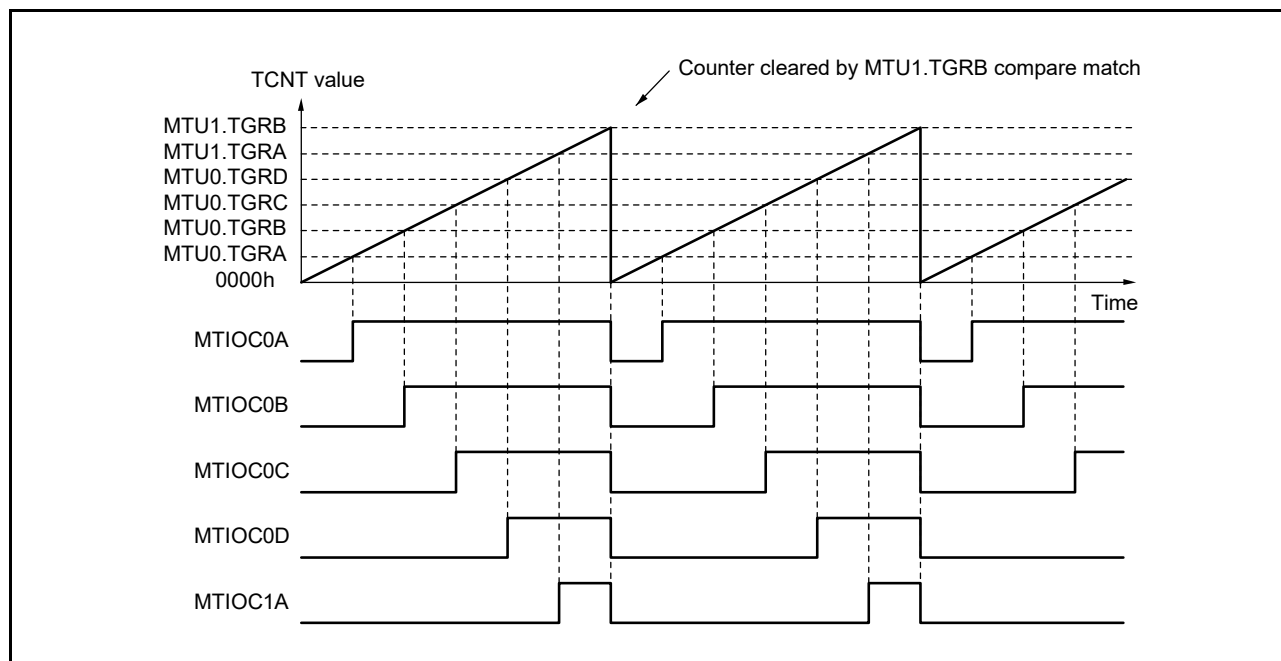
**Figure 24.27 Example of PWM Mode 1 Operation (n = 0 to 4, 6, 7)**



Figure 24.28 shows an example of operation in PWM mode 2.

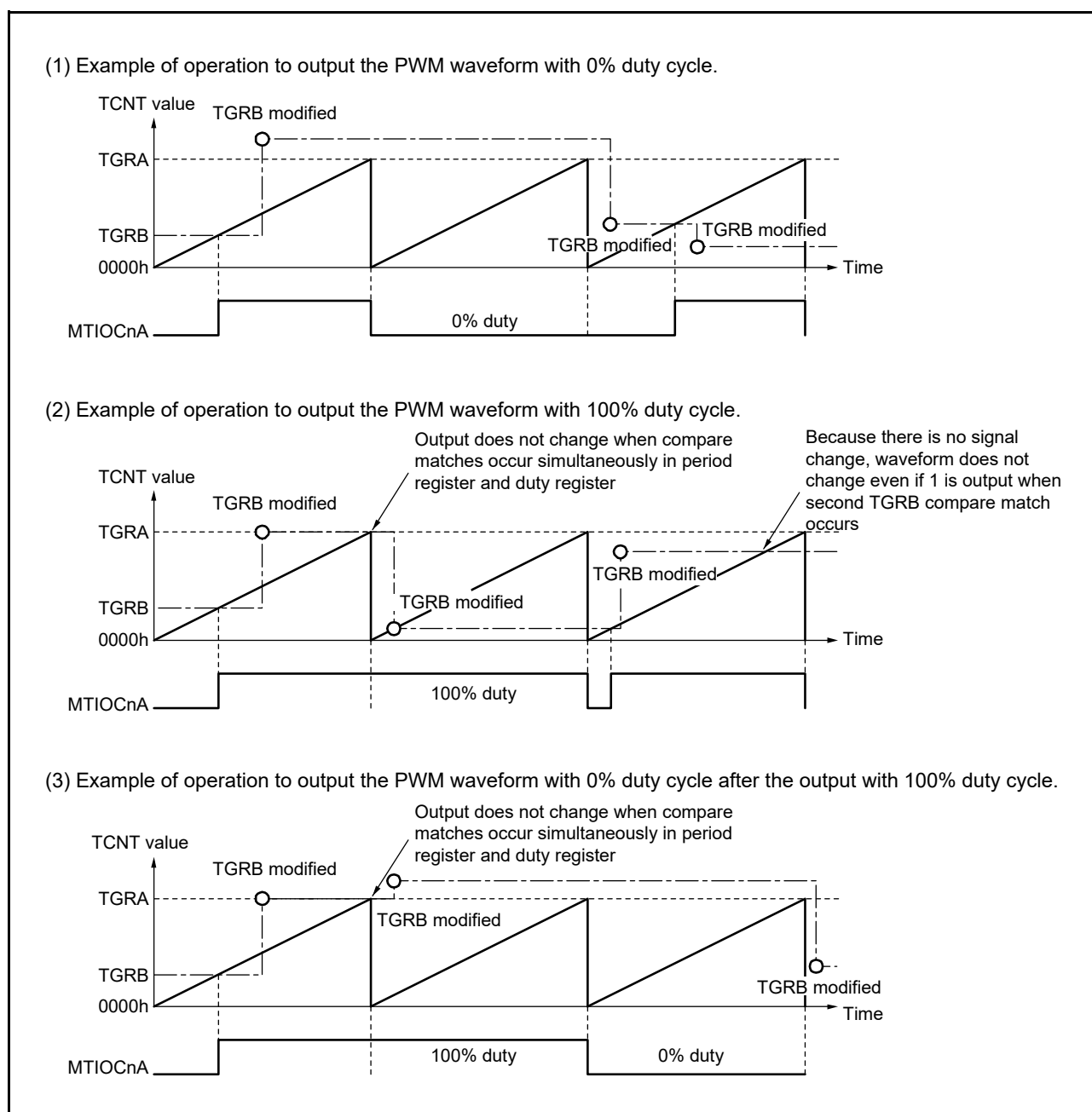
In this example, synchronous operation is designated for MTU0 and MTU1, MTU1.TGRB compare match is set as the TCNT clearing source, and low is set as the initial output value and high as the output value for the other TGR registers (MTU0.TGRA to MTU0.TGRD and MTU1.TGRA), outputting 5-phase PWM waveforms.

In this case, the value set in MTU1.TGRB is used as the period, and the values set in the other TGRs are used as the duty ratio.



**Figure 24.28** Example of PWM Mode 2 Operation

Figure 24.29 shows examples of PWM waveform output with 0% duty and 100% duty in PWM mode 1. In this example, TGRA compare match is set as the TCNT clearing source, a low level is set as the initial output value for TGRA, and a high level is set as the output value for TGRB.



**Figure 24.29** Examples of PWM Mode Operation (PWM Waveform Output with 0% Duty and 100% Duty)  
(n = 0 to 4, 6, 7)

### 24.3.6 Phase Counting Mode

There are two phase counting modes: 16-bit phase counting mode in which MTU1 and MTU2 operate independently, and cascade connection 32-bit phase counting mode in which MTU1 and MTU2 are cascaded.

In phase counting mode, the phase difference between two external input clocks is detected and the corresponding TCNT is incremented or decremented.

Two external clock input pins for each phase counting mode are not affected by the settings of TCR.TPSC[2:0], TCR2.TPSC2[2:0], and CKEG[1:0]. The two external clock input pins used in 16-bit phase counting mode and cascade connection 32-bit phase counting mode of MTU2 can be selected by MTU1.TMDR3.PHCKSEL. In a phase counting mode other than 16-bit phase counting mode and cascade connection 32-bit phase counting mode of MTU2, MTCLKA and MTCLKB are selected for A-phase and B-phase, respectively. In phase counting mode, the external clock pins MTCLKA, MTCLKB, MTCLKC, and MTCLKD are used for two-phase encoder pulse input.

Table 24.65 lists the external clock input pins to be connected in each phase counting mode.

**Table 24.65 Clock Input Pins in Phase Counting Mode**

| Phase Counting Mode                           | TMDR3.PHCKSEL bit | External Clock Input Pins |         |
|-----------------------------------------------|-------------------|---------------------------|---------|
|                                               |                   | A-Phase                   | B-Phase |
| MTU1 16-bit phase counting mode               | x (Don't care)    | MTCLKA                    | MTCLKB  |
| MTU2 16-bit phase counting mode               | 0                 | MTCLKA                    | MTCLKB  |
|                                               | 1 (initial value) | MTCLKC                    | MTCLKD  |
| Cascade connection 32-bit phase counting mode | 0                 | MTCLKA                    | MTCLKB  |
|                                               | 1 (initial value) | MTCLKC                    | MTCLKD  |

#### 24.3.6.1 16-Bit Phase Counting Mode

When the MTU1.TMDR3.LWA is 0, 16-bit phase counting mode can be set individually for MTU1 and MTU2.

In 16-bit phase counting mode, the phase difference between two external input clocks is detected and the 16-bit counter TCNT of the corresponding channel is incremented or decremented.

When 16-bit phase counting mode is specified, an external clock is selected as the count clock and TCNT operates as an up-counter/down-counter regardless of the setting of TCR.TPSC[2:0], TCR2.TPSC2[2:0], and CKEG[1:0]. However, the functions of TCR.CCLR[1:0], TIOR, TIER, and TGR are enabled, and input capture/compare match and interrupt functions can be used.

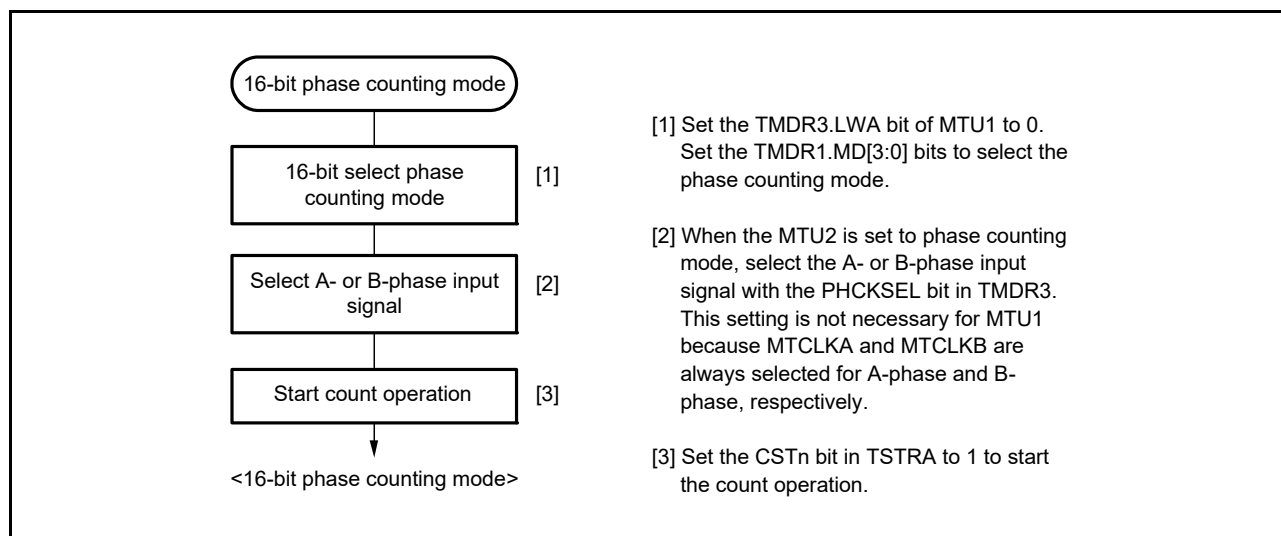
These external input pins can be used for two-phase encoder pulse input.

When an overflow occurs while TCNT is counting up and the corresponding TIER.TCIEV bit is 1, a TCIV interrupt is generated. When an underflow occurs while TCNT is counting down and the corresponding TIER.TCIEU bit is 1, a TCIU interrupt is generated.

The TSR.TCFD flag is the count direction flag. Read the TCFD flag to check whether TCNT is counting up and down.

## (1) Example of 16-Bit Phase Counting Mode Setting Procedure

Figure 24.30 shows an example of the phase counting mode setting procedure.



**Figure 24.30 Example of 16-Bit Phase Counting Mode Setting Procedure**

## (2) Examples of 16-Bit Phase Counting Mode Operation

In phase counting mode, TCNT is incremented or decremented according to the phase difference between two external clocks. There are five modes according to the count conditions. Each mode operates under the condition PHCKSEL = 1, which means the phase clock for MTU1 is input from MTCLKA or MTCLKB and that for MTU2 is input from MTCLKC or MTCLKD.

### (a) Phase Counting Mode 1

Figure 24.31 shows an example of operation in phase counting mode 1, and Table 24.66 summarizes the TCNT up-counting and down-counting conditions.

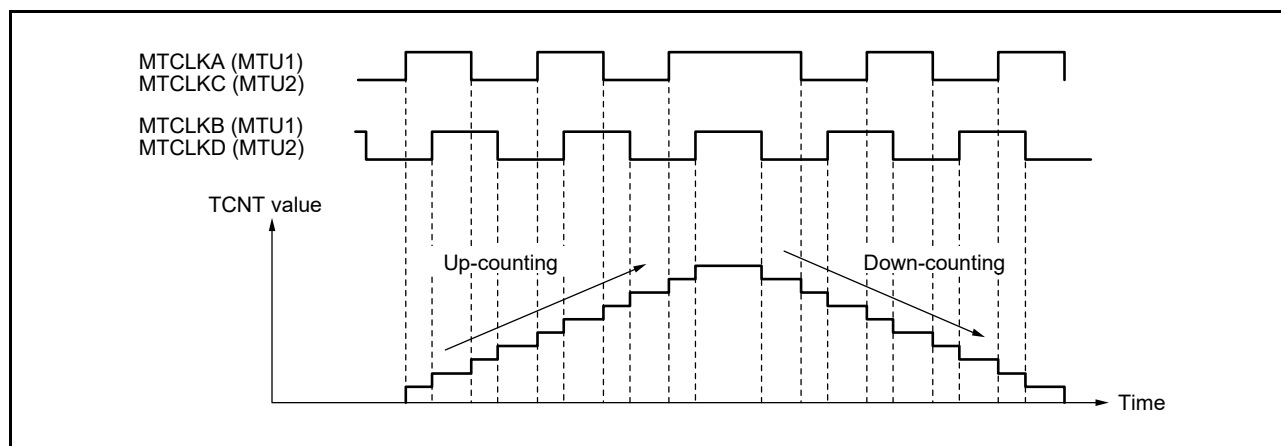


Figure 24.31 Example of Operation in Phase Counting Mode 1

Table 24.66 Up-Counting and Down-Counting Conditions in Phase Counting Mode 1

| MTCLKA (MTU1)<br>MTCLKC (MTU2)                                                      | MTCLKB (MTU1)<br>MTCLKD (MTU2)                                                      | Operation     |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------|
| High                                                                                |  | Up-counting   |
| Low                                                                                 |  |               |
|  | Low                                                                                 |               |
|  | High                                                                                |               |
| High                                                                                |  | Down-counting |
| Low                                                                                 |  |               |
|  | High                                                                                |               |
|  | Low                                                                                 |               |

 : Rising edge  
 : Falling edge

## (b) Phase Counting Mode 2

Figure 24.32 to Figure 24.34 show the examples of operation in phase counting mode 2 and Table 24.67 summarizes the TCNT up-counting and down-counting conditions.

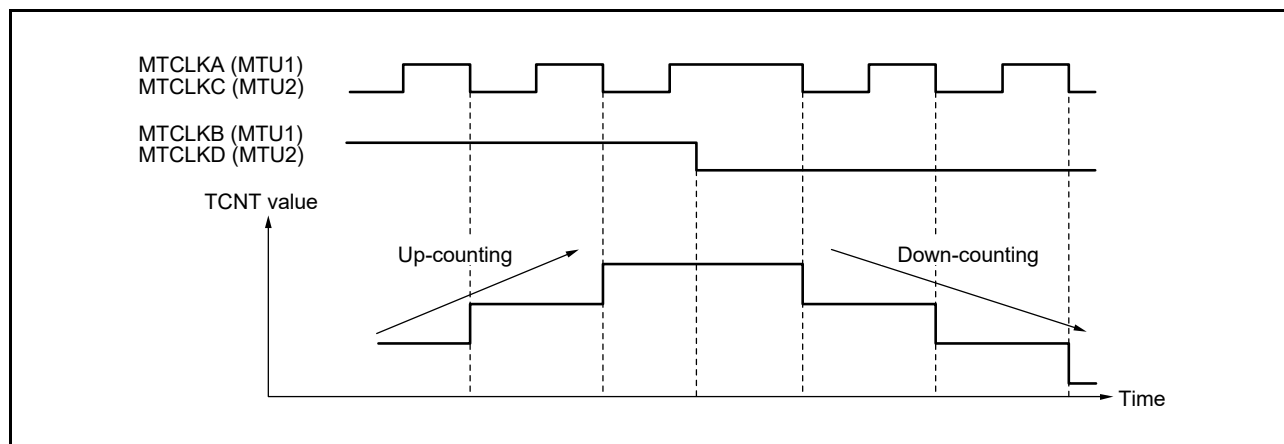


Figure 24.32 Example of Operation in Phase Counting Mode 2 (When MTUn.TCR2.PCB[1:0] = 00b (n = 1, 2))

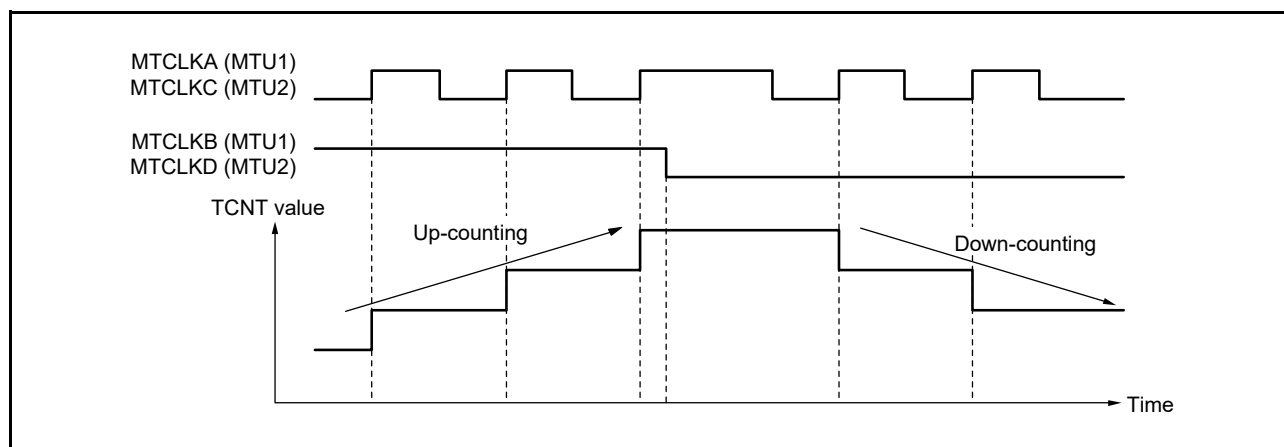


Figure 24.33 Example of Operation in Phase Counting Mode 2 (When MTUn.TCR2.PCB[1:0] = 01b (n = 1, 2))

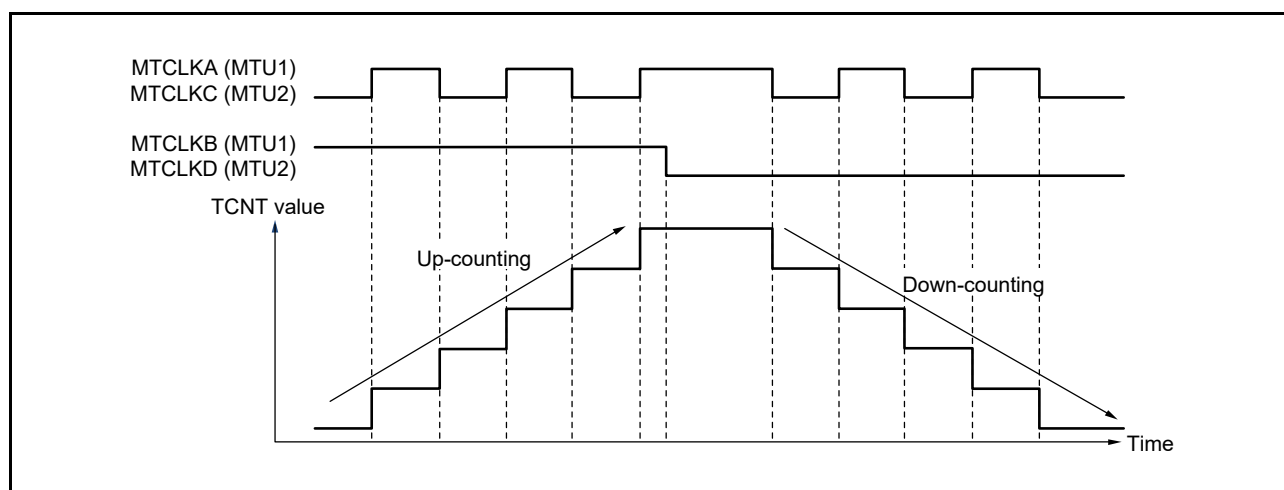


Figure 24.34 Example of Operation in Phase Counting Mode 2 (When MTUn.TCR2.PCB[1:0] = 1xb (n = 1, 2))

**Table 24.67 Up-Counting and Down-Counting Conditions in Phase Counting Mode 2**

| PCB[1:0] | MTCLKA (MTU1)<br>MTCLKC (MTU2)                                                      | MTCLKB (MTU1)<br>MTCLKD (MTU2)                                                      | Operation                |
|----------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------|
| 00b      | High                                                                                |    | Not counted (Don't care) |
|          | Low                                                                                 |    |                          |
|          |    | Low                                                                                 |                          |
|          |    | High                                                                                | Up-counting              |
|          | High                                                                                |    | Not counted (Don't care) |
|          | Low                                                                                 |    |                          |
|          |    | High                                                                                |                          |
|          |    | Low                                                                                 | Down-counting            |
| 01b      | High                                                                                |    | Not counted (Don't care) |
|          | Low                                                                                 |    |                          |
|          |    | Low                                                                                 | Down-counting            |
|          |    | High                                                                                | Not counted (Don't care) |
|          | High                                                                                |    |                          |
|          | Low                                                                                 |    |                          |
|          |    | High                                                                                | Up-counting              |
|          |   | Low                                                                                 | Not counted (Don't care) |
| 1xb      | High                                                                                |  | Not counted (Don't care) |
|          | Low                                                                                 |  |                          |
|          |  | Low                                                                                 | Down-counting            |
|          |  | High                                                                                | Up-counting              |
|          | High                                                                                |  | Not counted (Don't care) |
|          | Low                                                                                 |  |                          |
|          |  | High                                                                                | Up-counting              |
|          |  | Low                                                                                 | Down-counting            |

 : Rising edge  
 : Falling edge

## (c) Phase Counting Mode 3

Figure 24.35 to Figure 24.37 show the examples of operation in phase counting mode 3 and Table 24.68 summarizes the TCNT up-counting and down-counting conditions.

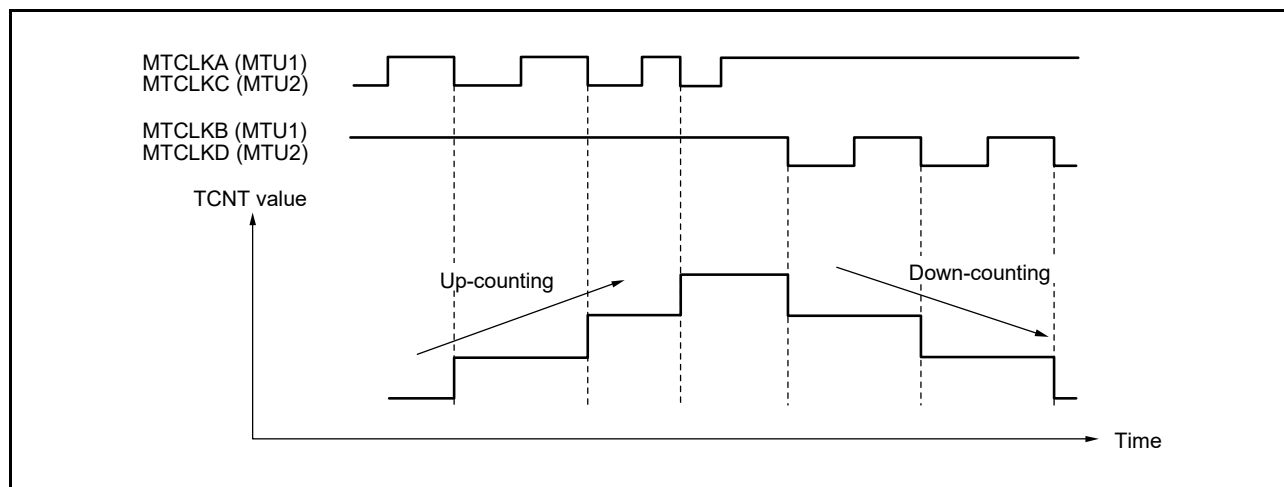


Figure 24.35 Example of Operation in Phase Counting Mode 3 (When MTUn.TCR2.PCB[1:0] = 00b (n = 1, 2))

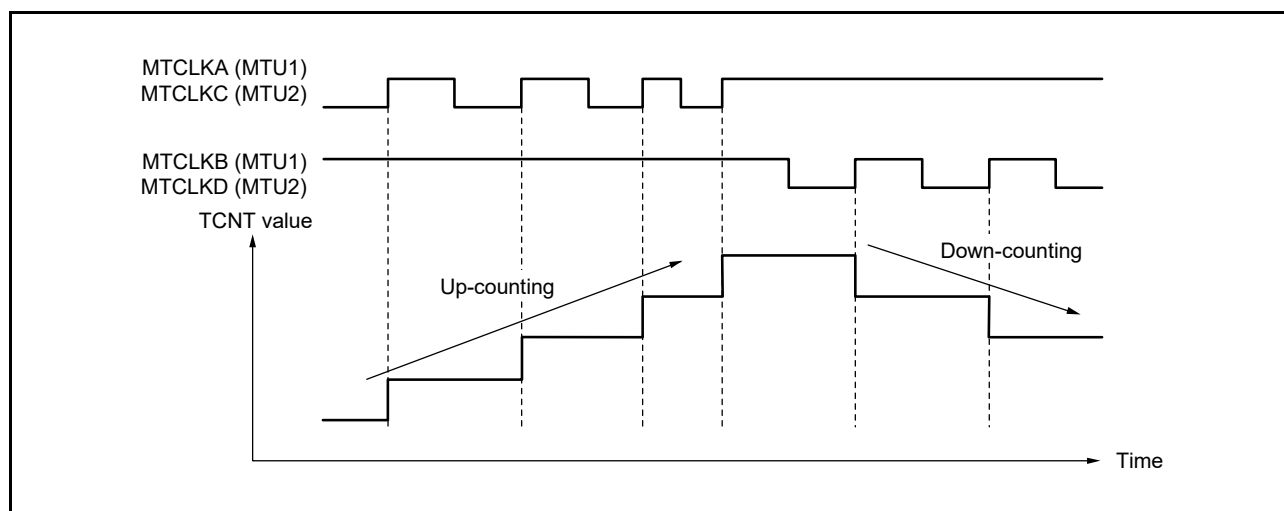


Figure 24.36 Example of Operation in Phase Counting Mode 3 (When MTUn.TCR2.PCB[1:0] = 01b (n = 1, 2))



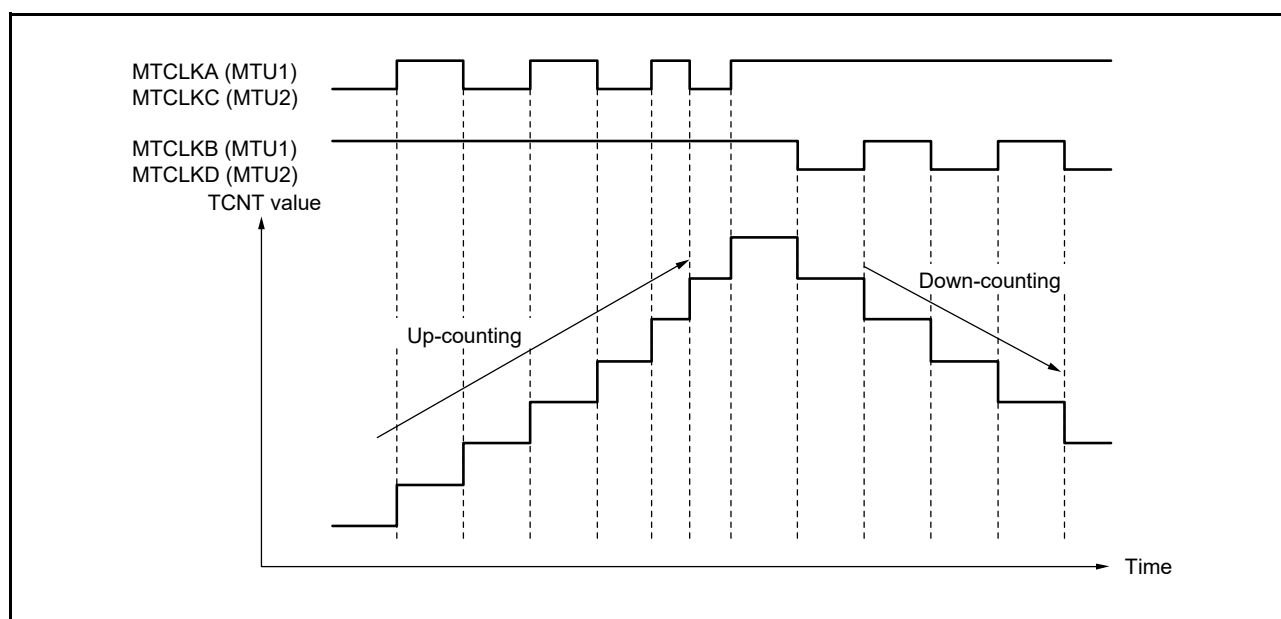


Figure 24.37 Example of Operation in Phase Counting Mode 3 (When MTUn.TCR2.PCB[1:0] = 1xb (n = 1, 2))

Table 24.68 Up-Counting and Down-Counting Conditions in Phase Counting Mode 3

| PCB[1:0] | MTCLKA (MTU1)<br>MTCLKC (MTU2) | MTCLKB (MTU1)<br>MTCLKD (MTU2) | Operation                |
|----------|--------------------------------|--------------------------------|--------------------------|
| 00b      | High                           |                                | Not counted (Don't care) |
|          | Low                            |                                |                          |
|          |                                | Low                            | Up-counting              |
|          |                                | High                           |                          |
|          | High                           |                                | Down-counting            |
|          | Low                            |                                |                          |
|          |                                | High                           | Not counted (Don't care) |
|          |                                | Low                            |                          |
| 01b      | High                           |                                | Down-counting            |
|          | Low                            |                                |                          |
|          |                                | Low                            | Not counted (Don't care) |
|          |                                | High                           |                          |
|          | High                           |                                | Up-counting              |
|          | Low                            |                                |                          |
|          |                                | High                           | Not counted (Don't care) |
|          |                                | Low                            |                          |
| 1xb      | High                           |                                | Down-counting            |
|          | Low                            |                                |                          |
|          |                                | Low                            | Up-counting              |
|          |                                | High                           |                          |
|          | High                           |                                | Down-counting            |
|          | Low                            |                                |                          |
|          |                                | High                           | Up-counting              |
|          |                                | Low                            |                          |

: Rising edge  
 : Falling edge

## (d) Phase Counting Mode 4

Figure 24.38 shows an example of operation in phase counting mode 4, and Table 24.69 summarizes the TCNT up-counting and down-counting conditions.

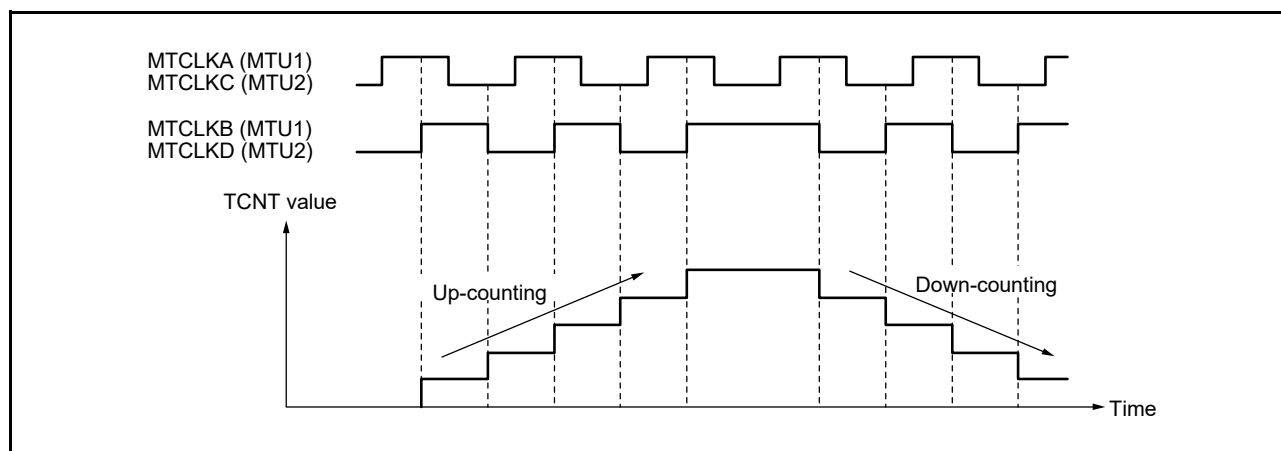


Figure 24.38 Example of Operation in Phase Counting Mode 4

Table 24.69 Up-Counting and Down-Counting Conditions in Phase Counting Mode 4

| MTCLKA (MTU1)<br>MTCLKC (MTU2)                                                      | MTCLKB (MTU1)<br>MTCLKD (MTU2)                                                      | Operation                |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------|
| High                                                                                |  | Up-counting              |
| Low                                                                                 |  | Up-counting              |
|  | Low                                                                                 | Not counted (Don't care) |
|  | High                                                                                | Not counted (Don't care) |
| High                                                                                |  | Down-counting            |
| Low                                                                                 |  | Down-counting            |
|  | High                                                                                | Not counted (Don't care) |
|  | Low                                                                                 | Not counted (Don't care) |

 : Rising edge

 : Falling edge

## (e) Phase Counting Mode 5

Figure 24.39 and Figure 24.40 show the examples of operation in phase counting mode 5 and Table 24.70 summarizes the TCNT up-counting and down-counting conditions.

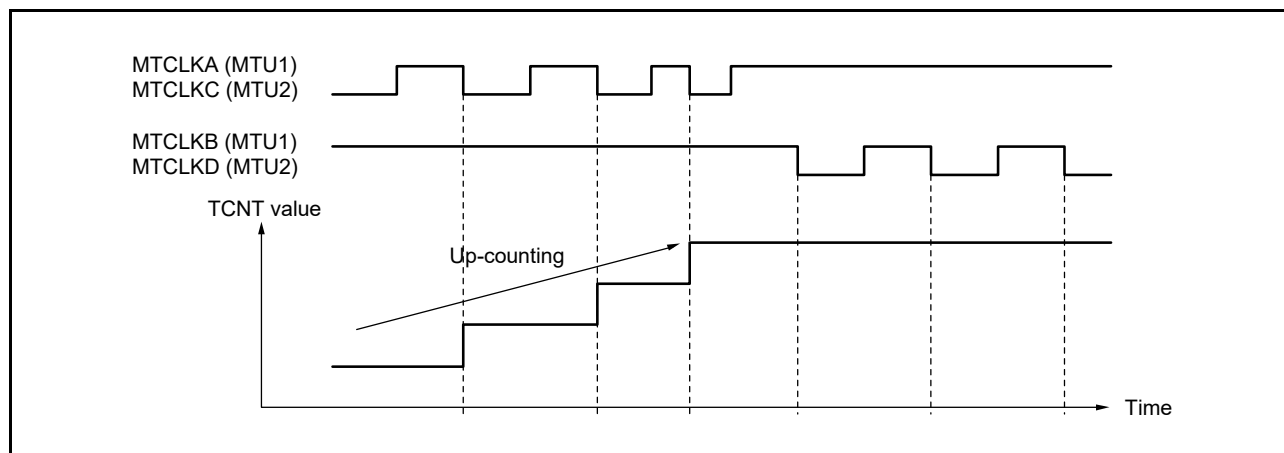


Figure 24.39 Example of Operation in Phase Counting Mode 5 (When MTUn.TCR2.PCB[1:0] = 0xb (n = 1, 2))

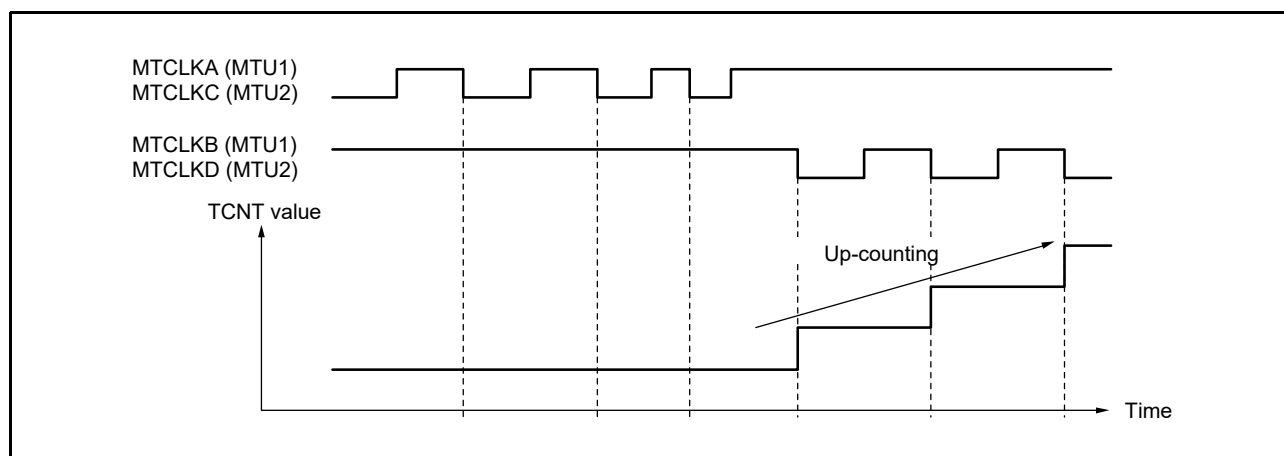


Figure 24.40 Example of Operation in Phase Counting Mode 5 (When MTUn.TCR2.PCB[1:0] = 1xb (n = 1, 2))

**Table 24.70 Up-Counting and Down-Counting Conditions in Phase Counting Mode 5**

| PCB[1:0] | MTCLKA (MTU1)<br>MTCLKC (MTU2)                                                      | MTCLKB (MTU1)<br>MTCLKD (MTU2)                                                    | Operation                |
|----------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------|
| 0xb      | High                                                                                |  | Not counted (Don't care) |
|          | Low                                                                                 |  |                          |
|          |    | Low                                                                               |                          |
|          |    | High                                                                              | Up-counting              |
|          | High                                                                                |  | Not counted (Don't care) |
|          | Low                                                                                 |  |                          |
|          |    | High                                                                              |                          |
|          |    | Low                                                                               | Up-counting              |
| 1xb      | High                                                                                |  | Not counted (Don't care) |
|          | Low                                                                                 |  | Up-counting              |
|          |    | Low                                                                               | Not counted (Don't care) |
|          |    | High                                                                              |                          |
|          | High                                                                                |  | Up-counting              |
|          | Low                                                                                 |  | Not counted (Don't care) |
|          |   | High                                                                              |                          |
|          |  | Low                                                                               |                          |

 : Rising edge

 : Falling edge

### (3) 16-Bit Phase Counting Mode Application Example

Figure 24.41 shows an example in which MTU1 is in phase counting mode, and MTU1 is coupled with MTU0 to input 2-phase encoder pulses of a servo motor in order to detect position or speed.

MTU1 is set to phase counting mode 1, and the encoder pulse A-phase and B-phase are input to MTCLKA and MTCLKB.

In MTU0, MTU0.TGRC compare match is specified as the TCNT clearing source and MTU0.TGRA and MTU0.TGRC are used for the compare match function and are set with the speed control period and position control period.

MTU0.TGRB is used for input capture, with MTU0.TGRB and MTU0.TGRD operating in buffer mode. The MTU1 count clock is designated as the MTU0.TGRB input capture source, and the widths of 2-phase encoder 4-multiplication pulses are detected.

MTU1.TGRA and MTU1.TGRB for MTU1 are designated for the input capture function and MTU0.TGRA and MTU0.TGRC compare matches in MTU0 are selected as the input capture sources to store the up/down-counter values for the control periods.

This procedure enables the accurate detection of position and speed.

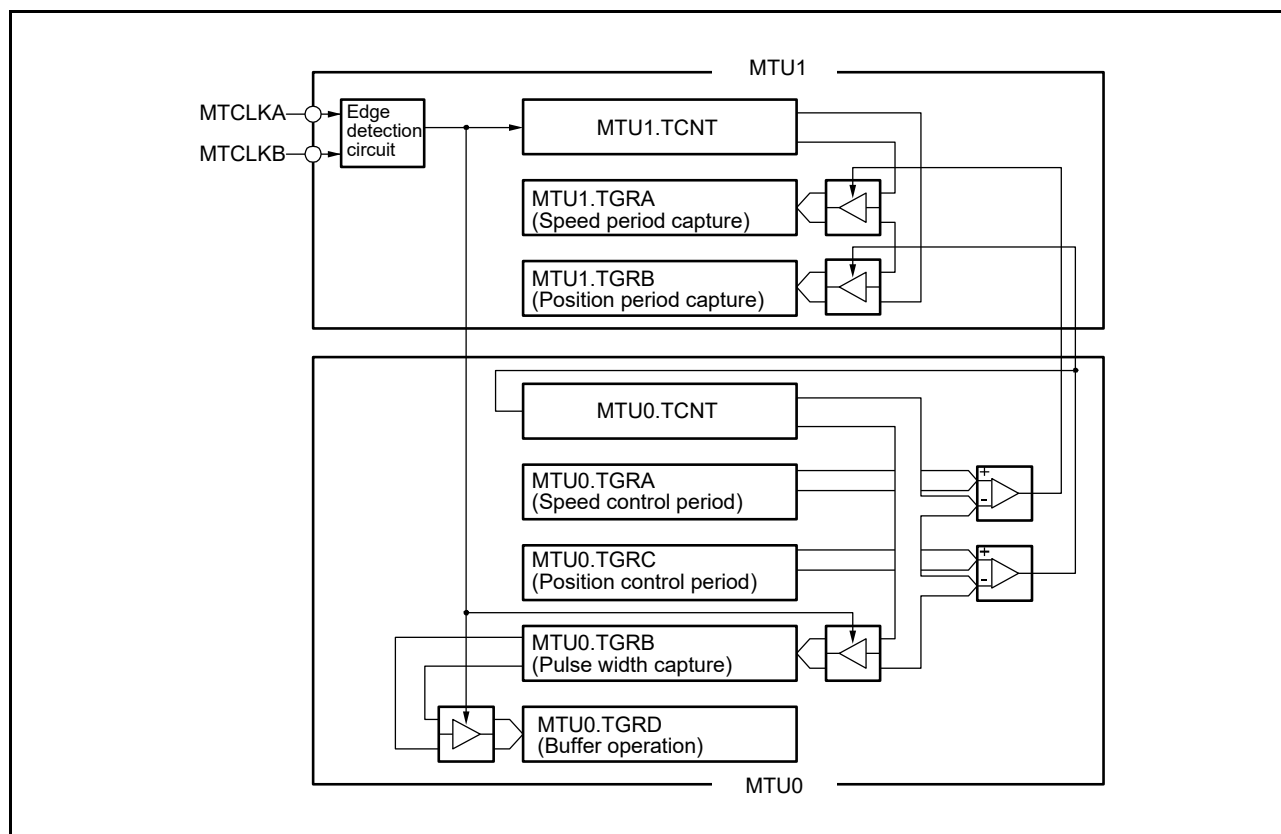


Figure 24.41 16-Bit Phase Counting Mode Application Example

### 24.3.6.2 Cascade Connection 32-Bit Phase Counting Mode

When MTU1 is set to phase counting mode by setting MTU1.TMDR3.LWA = 1, MTU1 and MTU2 are connected to operate in cascade connection 32-bit phase counting mode as shown in Figure 24.42. When this mode is used, the TCR, TCR2, TIOR, TIER, TGR, and TSR registers are controlled by MTU1 and the settings of MTU2 are disabled. Refer to Figure 24.43 for the procedure for setting cascade connection 32-bit phase counting mode.

In this mode, three-phase (A, B, and Z) signals can be input. As an encoder pulse signal, the external input phase clocks MTCLKA and MTCLKB or MTCLKC and MTCLKD can be selected for A-phase and B-phase, and MTIOC1A can be selected for Z-phase, respectively. Refer to Table 24.69 for selecting external clock input of A-phase and B-phase. A counter event is generated using an A-phase or B-phase pulse and counted by the 32-bit counter MTU1.TCNTLW.

An input capture can also be generated using a Z-phase signal; thus angular velocity can be measured using the captured value in the general register.

Furthermore, MTU8 can be used as a channel for measuring a 1-ms interval, and a compare match signal can be output at a 1-ms interval to the MTU1 and MTU2, which operate in cascade connection 32-bit phase counting mode. That is, a compare match signal of MTU8 is used as a capture signal of MTU1 and MTU2, and the number of A-phase and B-phase pulses for a 1-ms period can be measured.

When MTU0 or MTU5 is specified as the channel for measuring a Z-phase signal pulse, the compare match signal of the MTU8.TGRC register can be output as a capture signal or clear signal to MTU0 or MTU5, thus the Z-phase count at a 1-ms interval can be measured.

In addition, a counter event signal of combined MTU1 and MTU2 can be used as a capture signal of the MTU8.TGRD register, and measurement can be performed including the intervals of A, B, or both phase pulses. In this case, the MTU8.TGRD register should be set to buffer operation.

Refer to section 24.3.4, **Cascaded Operation**, for details on the cascade connection function for connecting MTU1 and MTU2 in a mode other than cascade connection 32-bit phase counting mode.

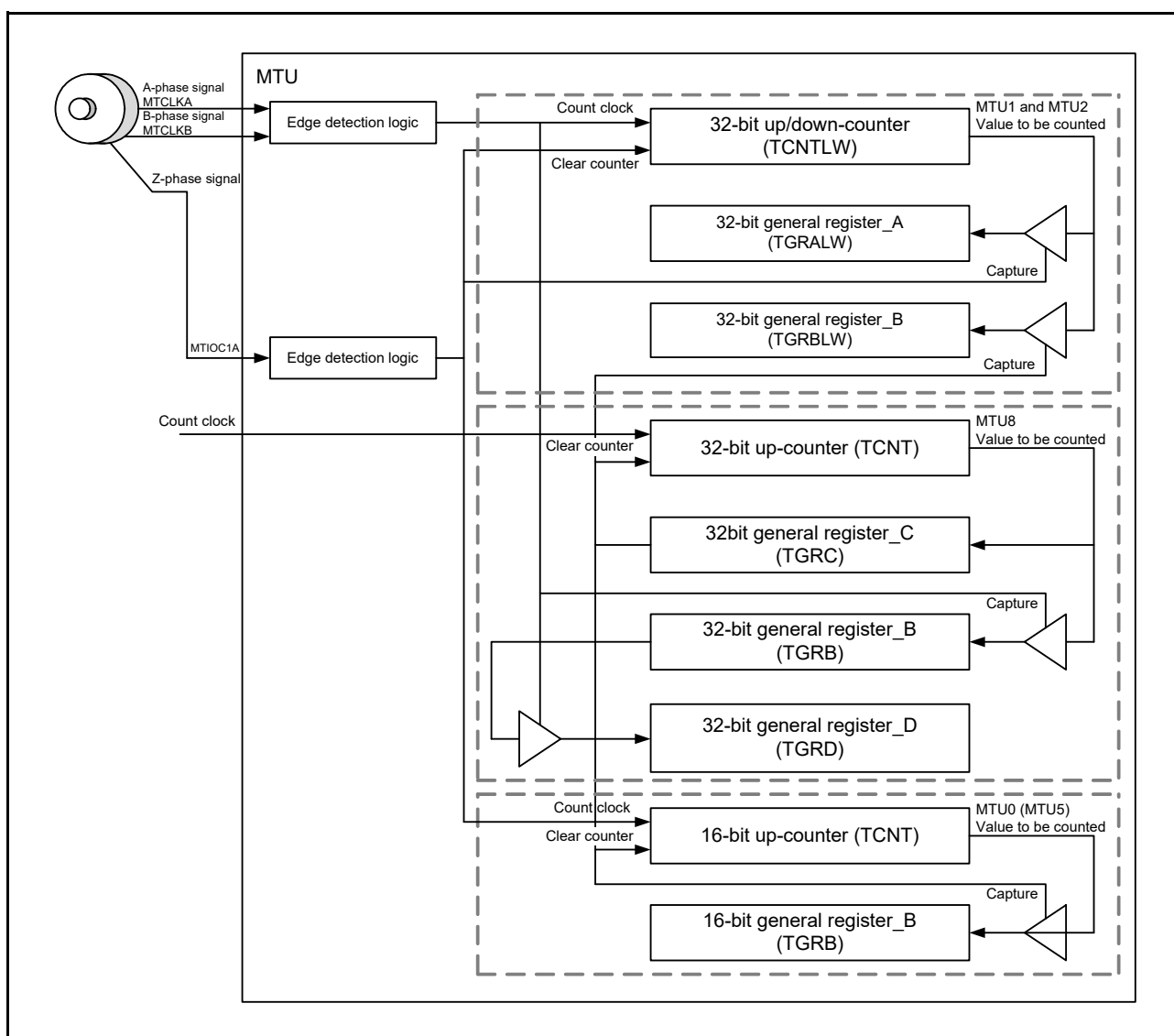


Figure 24.42 Block Diagram for Operation in Cascade Connection 32-Bit Phase Counting Mode

## (1) Example of Setting Cascade Connection 32-Bit Phase Counting Mode

Figure 24.43 shows an example of the procedure for setting cascade connection 32-bit phase counting mode.

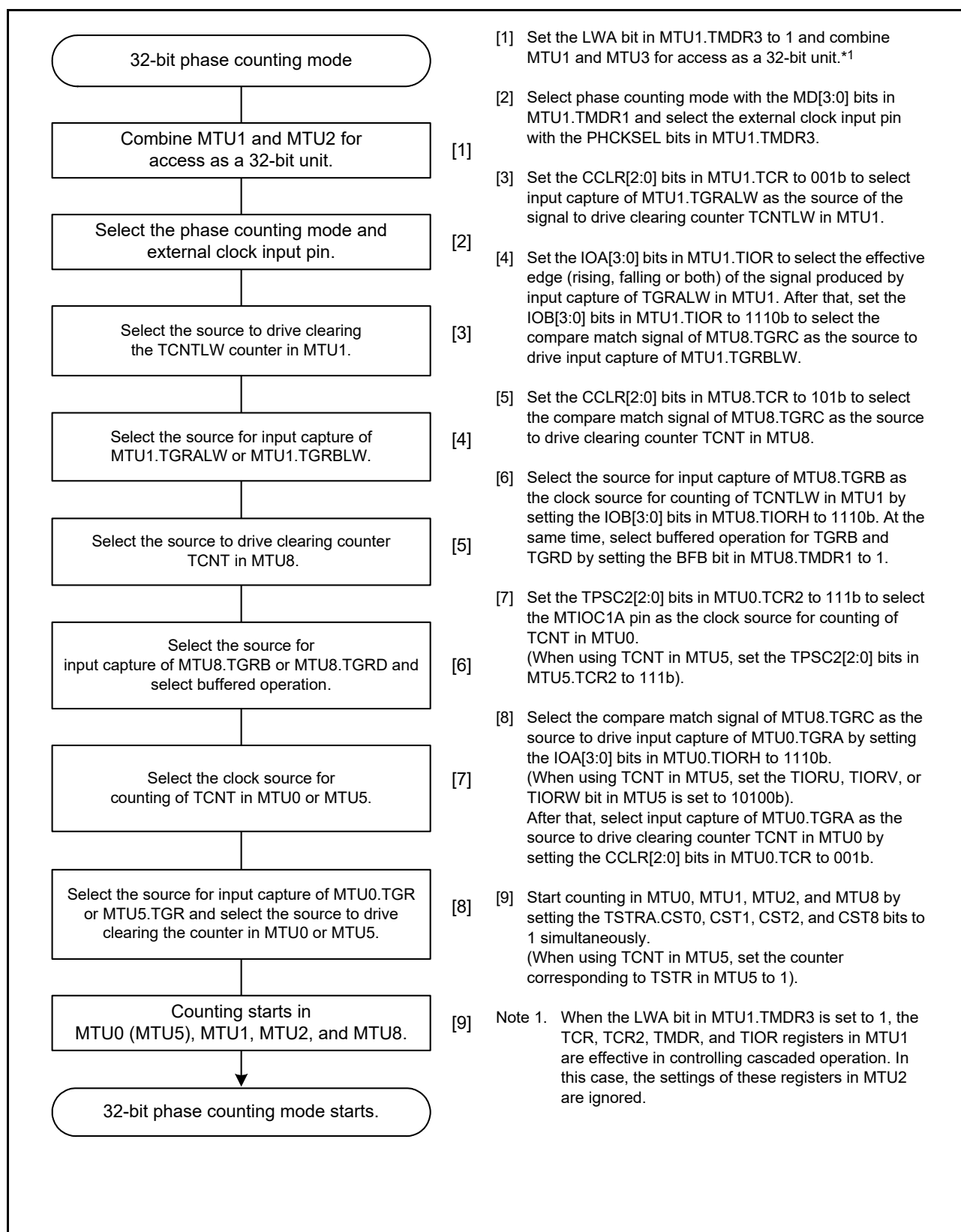


Figure 24.43 Procedure for Setting Cascade Connection 32-Bit Phase Counting Mode



### 24.3.7 Reset-Synchronized PWM Mode

In the reset-synchronized PWM mode, six phases of positive and negative PWM waveforms (12 phases in total) that share a common wave transition point can be output by combining MTU3 and MTU4 and MTU6 and MTU7.

When set for reset-synchronized PWM mode, the MTIOC3B, MTIOC3D, MTIOC4A, MTIOC4C, MTIOC4B, MTIOC4D, MTIOC6B, MTIOC6D, MTIOC7A, MTIOC7C, MTIOC7B, and MTIOC7D pins function as PWM output pins and timer counters 3 and 6 (MTU3.TCNT and MTU6.TCNT) functions as an up-counter.

Table 24.71 shows the PWM output pins used. Table 24.72 shows the settings of the registers.

**Table 24.71 Output Pins for Reset-Synchronized PWM Mode**

| Channel | Output Pin | Description                                                 |
|---------|------------|-------------------------------------------------------------|
| MTU3    | MTIOC3A    | Toggle output synchronized with PWM period (or I/O port)    |
|         | MTIOC3B    | PWM output pin 1                                            |
|         | MTIOC3D    | PWM output pin 1' (negative-phase waveform of PWM output 1) |
| MTU4    | MTIOC4A    | PWM output pin 2                                            |
|         | MTIOC4C    | PWM output pin 2' (negative-phase waveform of PWM output 2) |
|         | MTIOC4B    | PWM output pin 3                                            |
|         | MTIOC4D    | PWM output pin 3' (negative-phase waveform of PWM output 3) |
| MTU6    | MTIOC6A    | Toggle output synchronized with PWM period (or I/O port)    |
|         | MTIOC6B    | PWM output pin 4                                            |
|         | MTIOC6D    | PWM output pin 4' (negative-phase waveform of PWM output 4) |
| MTU7    | MTIOC7A    | PWM output pin 5                                            |
|         | MTIOC7C    | PWM output pin 5' (negative-phase waveform of PWM output 5) |
|         | MTIOC7B    | PWM output pin 6                                            |
|         | MTIOC7D    | PWM output pin 6' (negative-phase waveform of PWM output 6) |

**Table 24.72 Register Settings for Reset-Synchronized PWM Mode**

| Register  | Setting                                                                                     |
|-----------|---------------------------------------------------------------------------------------------|
| MTU3.TCNT | Initial setting (0000h)                                                                     |
| MTU4.TCNT | Initial setting (0000h)                                                                     |
| MTU3.TGRA | Set the count period for MTU3.TCNT                                                          |
| MTU3.TGRB | Set the transition point of the PWM waveform to be output from the MTIOC3B and MTIOC3D pins |
| MTU4.TGRA | Set the transition point of the PWM waveform to be output from the MTIOC4A and MTIOC4C pins |
| MTU4.TGRB | Set the transition point of the PWM waveform to be output from the MTIOC4B and MTIOC4D pins |
| MTU6.TCNT | Initial setting (0000h)                                                                     |
| MTU7.TCNT | Initial setting (0000h)                                                                     |
| MTU6.TGRA | Set the count period for MTU6.TCNT                                                          |
| MTU6.TGRB | Set the transition point of the PWM waveform to be output from the MTIOC6B and MTIOC6D pins |
| MTU7.TGRA | Set the transition point of the PWM waveform to be output from the MTIOC7A and MTIOC7C pins |
| MTU7.TGRB | Set the transition point of the PWM waveform to be output from the MTIOC7B and MTIOC7D pins |

## (1) Example of Procedure for Setting Reset-Synchronized PWM Mode

Figure 24.44 shows an example of procedure for setting the reset-synchronized PWM mode.

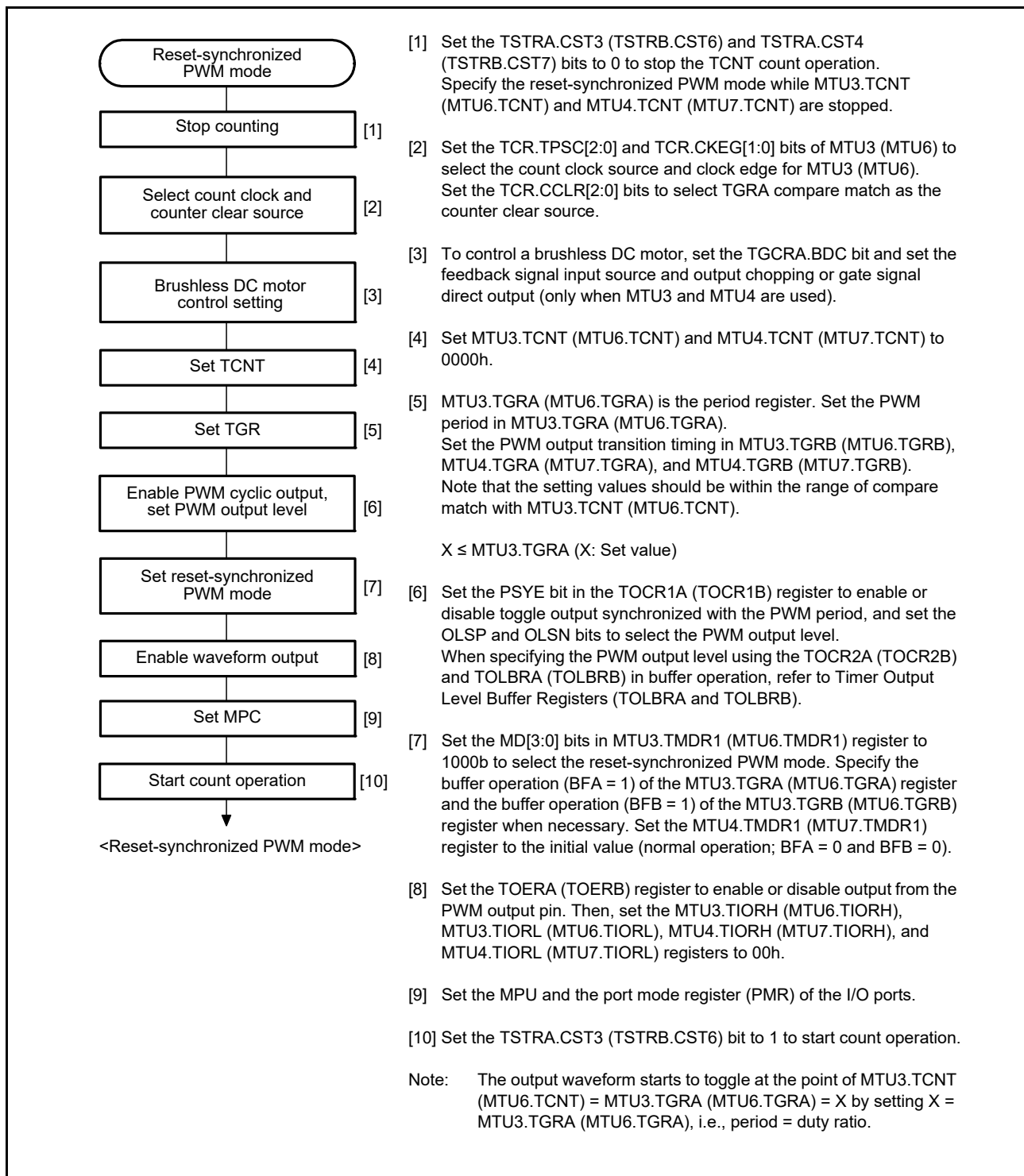
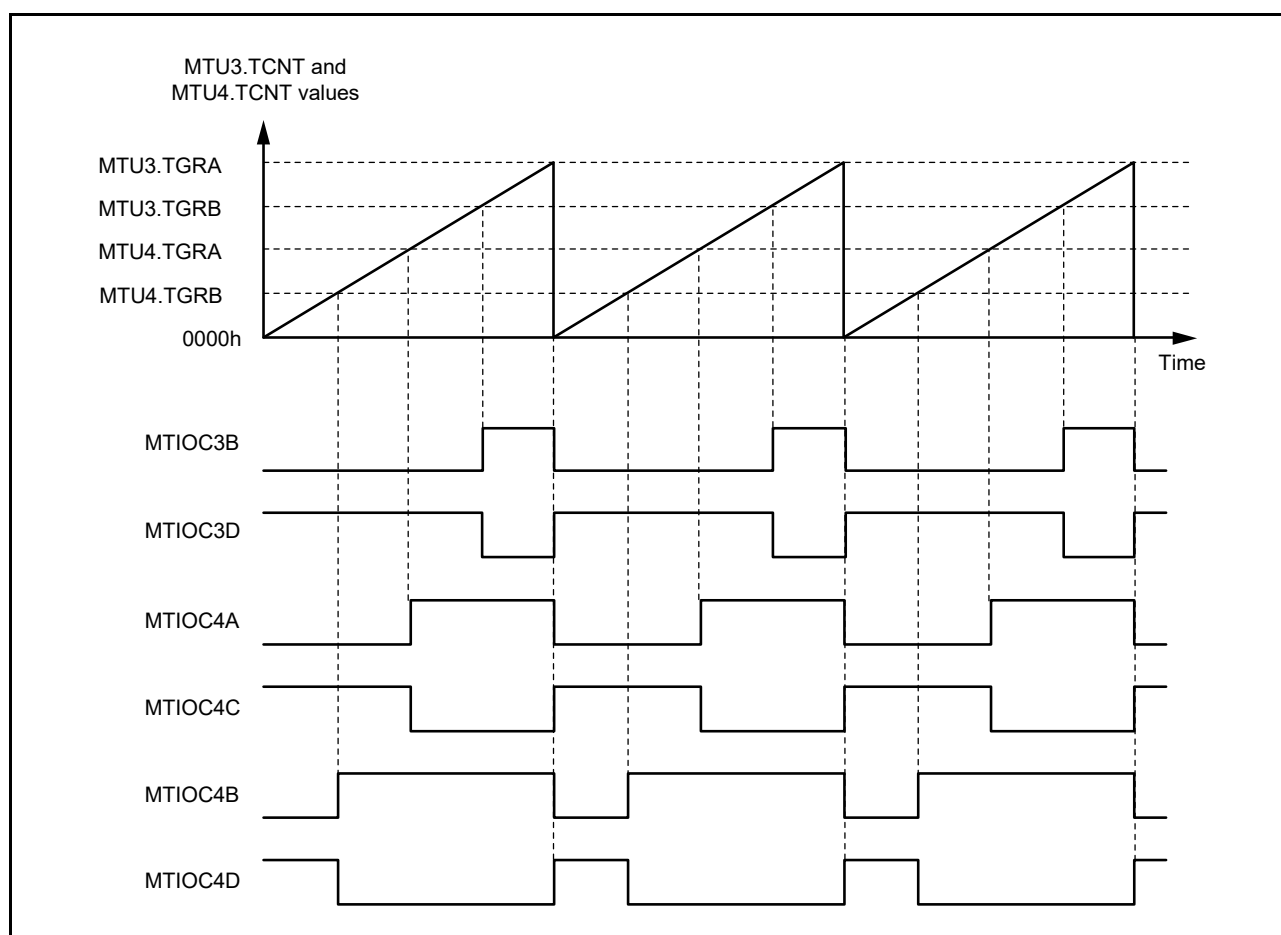


Figure 24.44 Procedure for Selecting Reset-Synchronized PWM Mode

## (2) Example of Reset-Synchronized PWM Mode Operation

Figure 24.45 shows an example of operation in the reset-synchronized PWM mode.

MTU3.TCNT and MTU4.TCNT (MTU6.TCNT and MTU7.TCNT) operate as up-counters. The counters are cleared when a compare match occurs between MTU3.TCNT (MTU6.TCNT) and MTU3.TGRA (MTU6.TGRA), and then begin incrementing from 0000h. The output from the PWM pins toggles every time a compare match occurs in MTU3.TGRB (MTU6.TGRB), MTU4.TGRA (MTU7.TGRA), and MTU4.TGRB (MTU7.TGRB) and the counters are cleared.



**Figure 24.45** Example of Reset-Synchronized PWM Mode Operation  
(When OLSN = 1 and OLSP = 1 in MTU3.TOCR1 and MTU4.TOCR1)

### 24.3.8 Complementary PWM Mode

In complementary PWM mode, dead time can be set for PWM waveforms to be output. The dead time is the period during which the upper and lower arm transistors are set to the inactive level in order to prevent short-circuiting of the arms.

Six positive-phase and six negative-phase PWM waveforms (12 phases in total) with dead time can be output by combining MTU3/ MTU4 and MTU6/MTU7. PWM waveforms without dead time can also be output.

In complementary PWM mode, MTIOC3B, MTIOC3D, MTIOC4A, MTIOC4B, MTIOC4C, MTIOC4D, MTIOC6B, MTIOC6D, MTIOC7A, MTIOC7B, MTIOC7C, and MTIOC7D pins function as PWM output pins, and the MTIOC3A and MTIOC6A pins can be set for toggle output synchronized with the PWM period.

MTU3.TCNT, MTU4.TCNT, MTU6.TCNT, and MTU7.TCNT function as up/down-counters.

Table 24.73 shows the PWM output pins used. Table 24.74 shows the settings of the registers used.

A function to directly cut off the PWM output by using an external signal is supported as a port function.

**Table 24.73 Output Pins for Complementary PWM Mode**

| Channel | Output Pin | Description                                                        |
|---------|------------|--------------------------------------------------------------------|
| MTU3    | MTIOC3A    | Toggle output synchronized with PWM period (or I/O port)           |
|         | MTIOC3B    | PWM output pin 1                                                   |
|         | MTIOC3C    | I/O port*1                                                         |
|         | MTIOC3D    | PWM output pin 1' (negative-phase waveform output of PWM output 1) |
| MTU4    | MTIOC4A    | PWM output pin 2                                                   |
|         | MTIOC4C    | PWM output pin 2' (negative-phase waveform output of PWM output 1) |
|         | MTIOC4B    | PWM output pin 3                                                   |
|         | MTIOC4D    | PWM output pin 3' (negative-phase waveform output of PWM output 1) |
| MTU6    | MTIOC6A    | Toggle output synchronized with PWM period (or I/O port)           |
|         | MTIOC6B    | PWM output pin 4                                                   |
|         | MTIOC6C    | I/O port*1                                                         |
|         | MTIOC6D    | PWM output pin 4' (negative-phase waveform output of PWM output 4) |
| MTU7    | MTIOC7A    | PWM output pin 5                                                   |
|         | MTIOC7C    | PWM output pin 5' (negative-phase waveform output of PWM output 5) |
|         | MTIOC7B    | PWM output pin 6                                                   |
|         | MTIOC7D    | PWM output pin 6' (negative-phase waveform output of PWM output 6) |

Note 1. Avoid setting the MTIOC3C and MTIOC6C pins as timer I/O pins in complementary PWM mode.

**Table 24.74 Register Settings for Complementary PWM Mode (1/2)**

| Channel | Counter/ Register | Description                                                       | Read/Write from CPU          |
|---------|-------------------|-------------------------------------------------------------------|------------------------------|
| MTU3    | TCNT              | Starts up-counting from the value set in the dead time register   | Maskable by TRWERA setting*1 |
|         | TGRA              | Set MTU3.TCNT upper limit value (1/2 carrier period + dead time)  | Maskable by TRWERA setting*1 |
|         | TGRB              | PWM output 1 compare register                                     | Maskable by TRWERA setting*1 |
|         | TGRC              | MTU3.TGRA buffer register                                         | Readable/writable            |
|         | TGRD              | PWM output 1/MTU3.TGRB buffer register                            | Readable/writable            |
|         | TGRE              | MTU3.TGRB buffer register B (when double buffer function is used) | Readable/writable            |
| MTU4    | TCNT              | Starts up-counting after being initialized to 0000h               | Maskable by TRWERA setting*1 |
|         | TGRA              | PWM output 2 compare register                                     | Maskable by TRWERA setting*1 |
|         | TGRB              | PWM output 3 compare register                                     | Maskable by TRWERA setting*1 |
|         | TGRC              | PWM output 2/MTU4.TGRA buffer register                            | Readable/writable            |
|         | TGRD              | PWM output 3/MTU4.TGRB buffer register                            | Readable/writable            |
|         | TGRE              | MTU4.TGRA buffer register B (when double buffer function is used) | Readable/writable            |
| MTU6    | TCNT              | Starts up-counting from the value set in the dead time register   | Maskable by TRWERB setting*2 |
|         | TGRA              | Set MTU6.TCNT upper limit value (1/2 carrier period + dead time)  | Maskable by TRWERB setting*2 |
|         | TGRB              | PWM output 4 compare register                                     | Maskable by TRWERB setting*2 |
|         | TGRC              | MTU6.TGRA buffer register                                         | Readable/writable            |
|         | TGRD              | PWM output 4/MTU6.TGRB buffer register                            | Readable/writable            |
|         | TGRE              | MTU6.TGRB buffer register B (when double buffer function is used) | Readable/writable            |
| MTU7    | TCNT              | Starts up-counting after being initialized to 0000h               | Maskable by TRWERB setting*2 |
|         | TGRA              | PWM output 5 compare register                                     | Maskable by TRWERB setting*2 |
|         | TGRB              | PWM output 6 compare register                                     | Maskable by TRWERB setting*2 |
|         | TGRC              | PWM output 5/MTU7.TGRA buffer register                            | Readable/writable            |
|         | TGRD              | PWM output 6/MTU7.TGRB buffer register                            | Readable/writable            |
|         | TGRE              | MTU7.TGRA buffer register B (when double buffer function is used) | Readable/writable            |
|         | TGRF              | MTU7.TGRB buffer register B (when double buffer function is used) | Readable/writable            |

Note 1. Access can be enabled or disabled according to the setting in TRWERA (timer read/write enable register A).

Note 2. Access can be enabled or disabled according to the setting in TRWERB (timer read/write enable register B).

**Table 24.75 Register Settings for Complementary PWM Mode (2/2)**

| Channel                                 | Counter/ Register | Description                                                                       | Read/Write from CPU          |
|-----------------------------------------|-------------------|-----------------------------------------------------------------------------------|------------------------------|
| Timer dead time data register A (TDDRA) |                   | Set MTU4.TCNT and MTU3.TCNT offset value (dead time value)                        | Maskable by TRWERA setting*1 |
| Timer dead time data register B (TDDRb) |                   | Set MTU7.TCNT and MTU6.TCNT offset value (dead time value)                        | Maskable by TRWERB setting*2 |
| Timer period data register A (TCDRA)    |                   | Set MTU4.TCNT upper limit value (1/2 carrier period)                              | Maskable by TRWERA setting*1 |
| Timer period data register B (TCDRB)    |                   | Set MTU7.TCNT upper limit value (1/2 carrier period)                              | Maskable by TRWERB setting*2 |
| Timer period buffer register A (TCBRA)  |                   | TCDRA buffer register                                                             | Readable/writable            |
| Timer period buffer register B (TCBRB)  |                   | TCDRB buffer register                                                             | Readable/writable            |
| Subcounter A (TCNTSA)                   |                   | Subcounter A for dead time generation                                             | Read-only                    |
| Subcounter B (TCNTSB)                   |                   | Subcounter B for dead time generation                                             | Read-only                    |
| Temporary register 1A (TEMP1A)          |                   | PWM output 1/MTU3.TGRB temporary register A                                       | Not readable/writable        |
| Temporary register 1B (TEMP1B)          |                   | PWM output 1/MTU3.TGRB temporary register B (when double buffer function is used) | Not readable/writable        |
| Temporary register 2A (TEMP2A)          |                   | PWM output 2/MTU4.TGRA temporary register A                                       | Not readable/writable        |
| Temporary register 2B (TEMP2B)          |                   | PWM output 2/MTU4.TGRA temporary register B (when double buffer function is used) | Not readable/writable        |
| Temporary register 3A (TEMP3A)          |                   | PWM output 3/MTU4.TGRB temporary register A                                       | Not readable/writable        |
| Temporary register 3B (TEMP3B)          |                   | PWM output 3/MTU4.TGRB temporary register B (when double buffer function is used) | Not readable/writable        |
| Temporary register 4A (TEMP4A)          |                   | PWM output 4/MTU6.TGRB temporary register A                                       | Not readable/writable        |
| Temporary register 4B (TEMP4B)          |                   | PWM output 4/MTU6.TGRB temporary register B (when double buffer function is used) | Not readable/writable        |
| Temporary register 5A (TEMP5A)          |                   | PWM output 5/MTU7.TGRA temporary register A                                       | Not readable/writable        |
| Temporary register 5B (TEMP5B)          |                   | PWM output 5/MTU7.TGRA temporary register B (when double buffer function is used) | Not readable/writable        |
| Temporary register 6A (TEMP6A)          |                   | PWM output 6/MTU7.TGRB temporary register A                                       | Not readable/writable        |
| Temporary register 6B (TEMP6B)          |                   | PWM output 6/MTU7.TGRB temporary register B (when double buffer function is used) | Not readable/writable        |

Note 1. Access can be enabled or disabled according to the setting in TRWERA (timer read/write enable register A).

Note 2. Access can be enabled or disabled according to the setting in TRWERB (timer read/write enable register B).

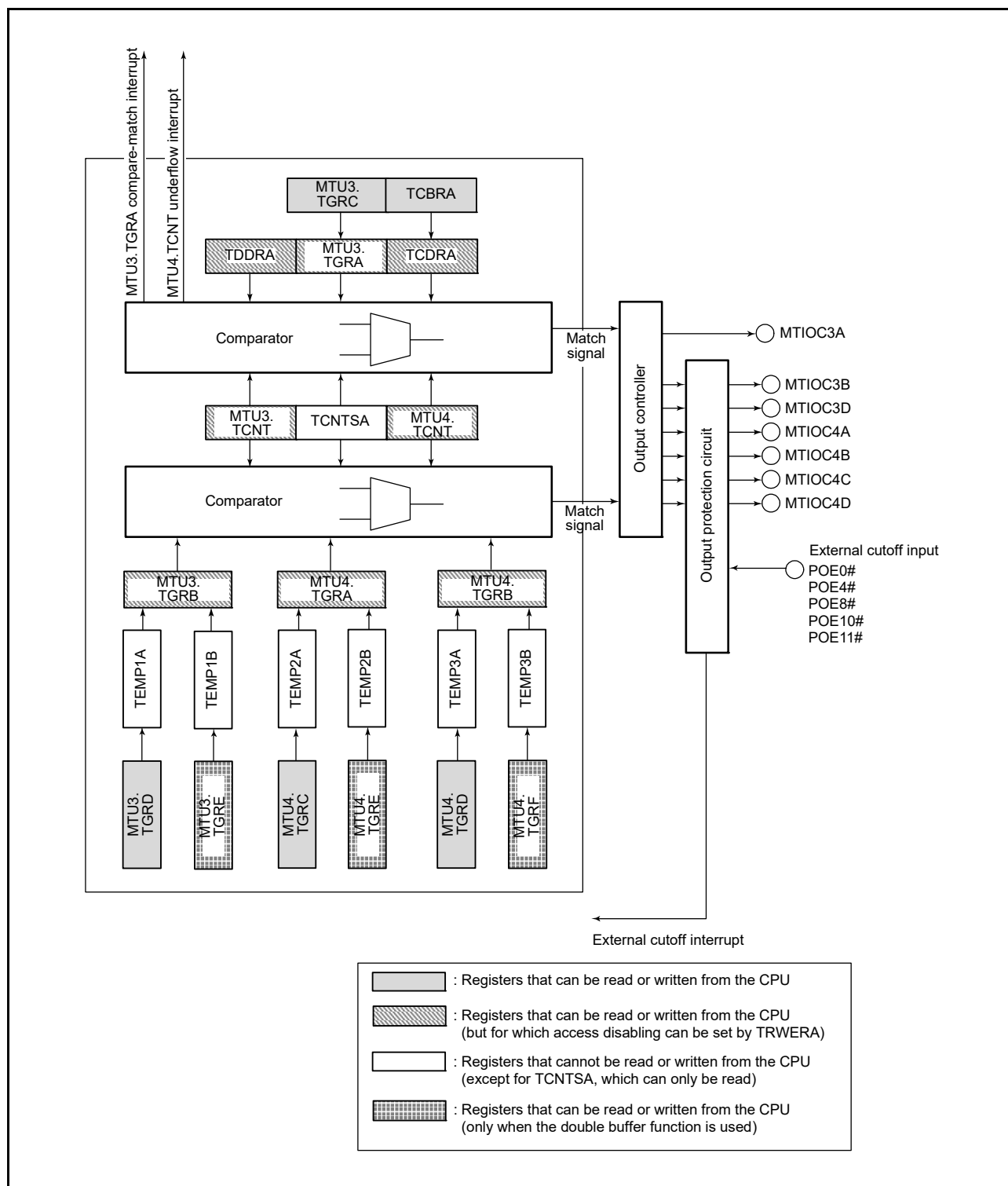


Figure 24.46 Block Diagram of MTU3 and MTU4 in Complementary PWM Mode

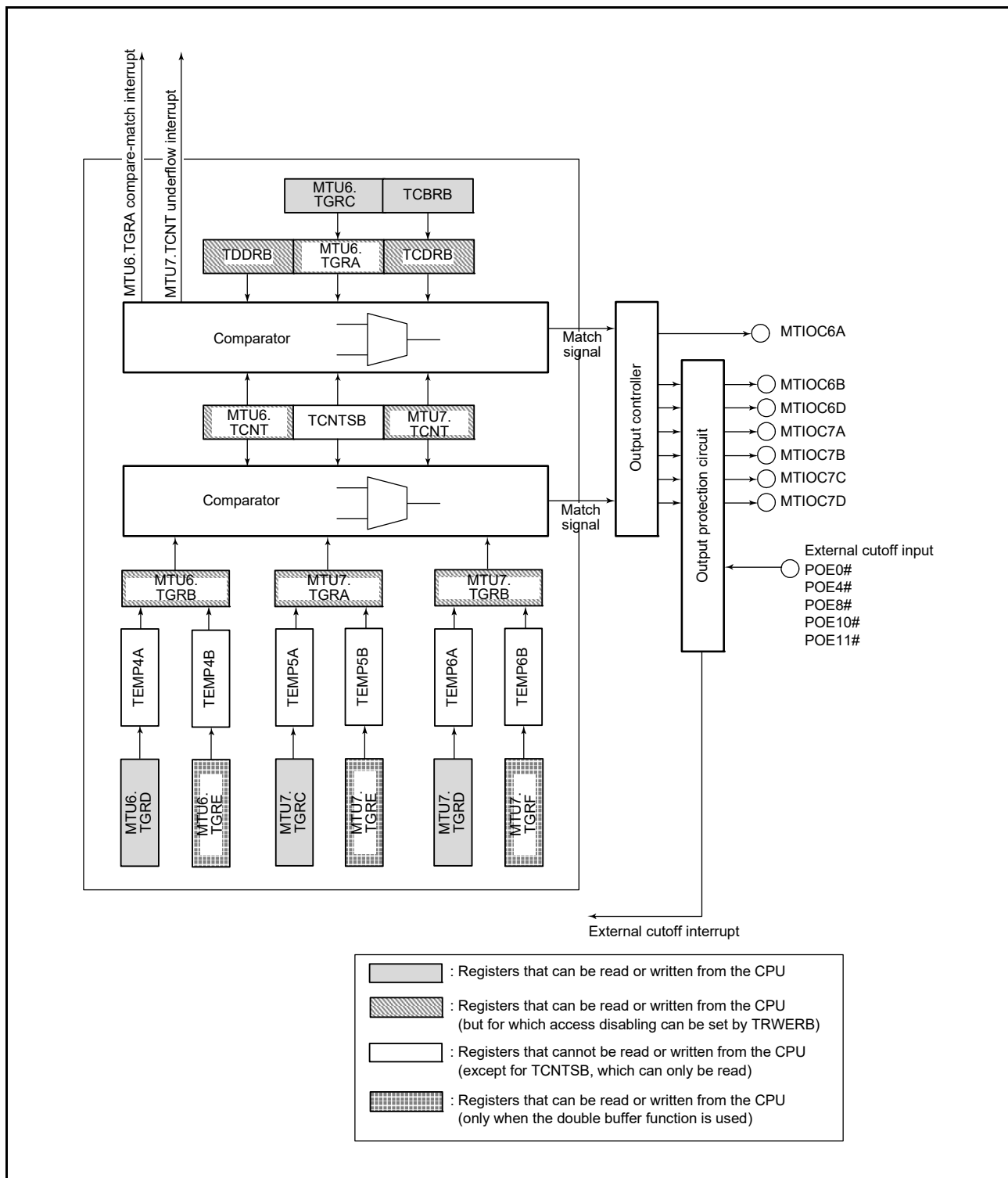


Figure 24.47 Block Diagram of MTU6 and MTU7 in Complementary PWM Mode



## (1) Example of Complementary PWM Mode Setting Procedure

Figure 24.48 shows an example of the complementary PWM mode setting procedure.

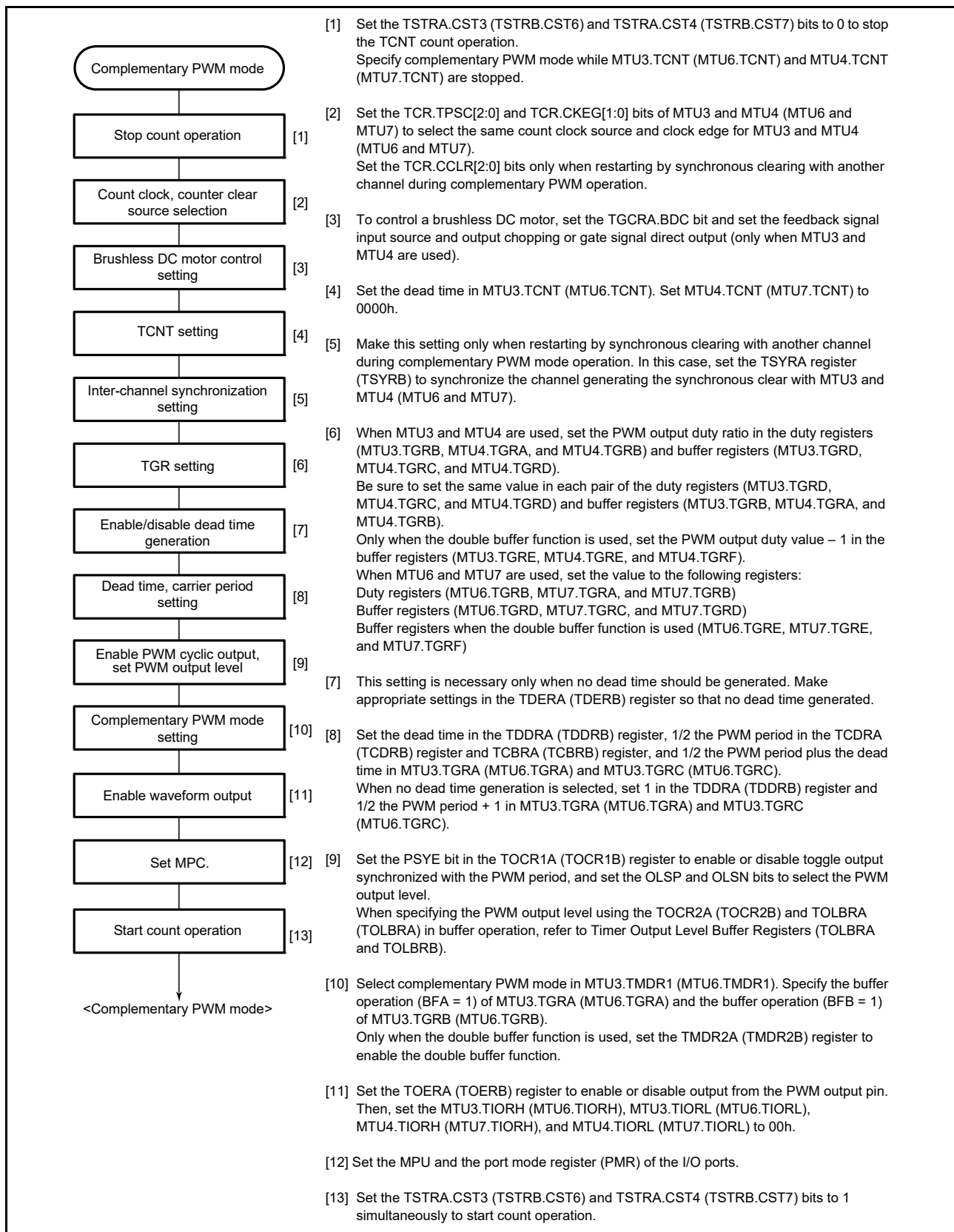


Figure 24.48 Example of Complementary PWM Mode Setting Procedure

## (2) Outline of Complementary PWM Mode Operation

In complementary PWM mode, six phases (three positive and three negative) PWM waveforms can be output. Figure 24.49 illustrates counter operation in complementary PWM mode (MTU3 and MTU4), and Figure 24.50 shows an example of operation in complementary PWM mode.

### (a) Counter Operation

In complementary PWM mode, three counters—MTU3.TCNT, MTU4.TCNT, and TCNTSA (MTU6.TCNT, MTU7.TCNT, and TCNTSB)—in each unit perform up-/down-count operations.

MTU3.TCNT (MTU6.TCNT) is automatically initialized to the value set in TDDRA (TDDRB) when complementary PWM mode is selected and the CST3 bit in TSTRA (TSTRB) is 0. When the CST3 bit is set to 1, MTU3.TCNT (MTU6.TCNT) counts up to the value set in MTU3.TGRA (MTU6.TGRA), then switches to down-counting when it matches MTU3.TGRA (MTU6.TGRA). When the MTU4.TCNT (MTU7.TCNT) value matches 0000h, MTU3.TCNT (MTU6.TCNT) switches to up-counting, and the operation is repeated in this way.

MTU4.TCNT (MTU7.TCNT) should be initialized to 0000h after a reset. When the CST4 bit is set to 1, MTU4.TCNT (MTU7.TCNT) counts up in synchronization with MTU3.TCNT (MTU6.TCNT), and switches to down-counting when MTU3.TCNT (MTU6.TCNT) matches MTU3.TGRA (MTU6.TGRA). On reaching 0000h, MTU4.TCNT (MTU7.TCNT) switches to up-counting, and the operation is repeated in this way. TCNTSA (TCNTSB) is a read-only counter. It does not need to be initialized after a reset.

In counting up by MTU3.TCNT and MTU4.TCNT (or MTU6.TCNT and MTU7.TCNT), MTU3.TCNT (or MTU6.TCNT) starts counting up when it matches TCDRA (or TCDRB) and switches to counting down when it matches MTU3.TGRA (or MTU6.TGRA). Furthermore, when MTU4.TCNT (or MTU7.TCNT) matches TDDRA (or TDDRB), TCNTSA (or TCNTSB) is set to the value in MTU3.TGRA (or MTU6.TGRA) and counting is stopped.

When MTU4.TCNT (MTU7.TCNT) matches TDDRA (TDDRB) during down-counting of MTU3.TCNT and MTU4.TCNT (MTU6.TCNT and MTU7.TCNT), TCNTSA (TCNTSB) starts up-counting, and when MTU4.TCNT (MTU7.TCNT) matches 0000h, the operation switches to down-counting. When MTU3.TCNT (MTU6.TCNT) matches TCDRA (TCDRB), TCNTSA (TCNTSB) becomes 0000h and stops counting.

TCNTSA (TCNTSB) is compared with the compare register and temporary register, in which the PWM duty is specified, only during the count operation.

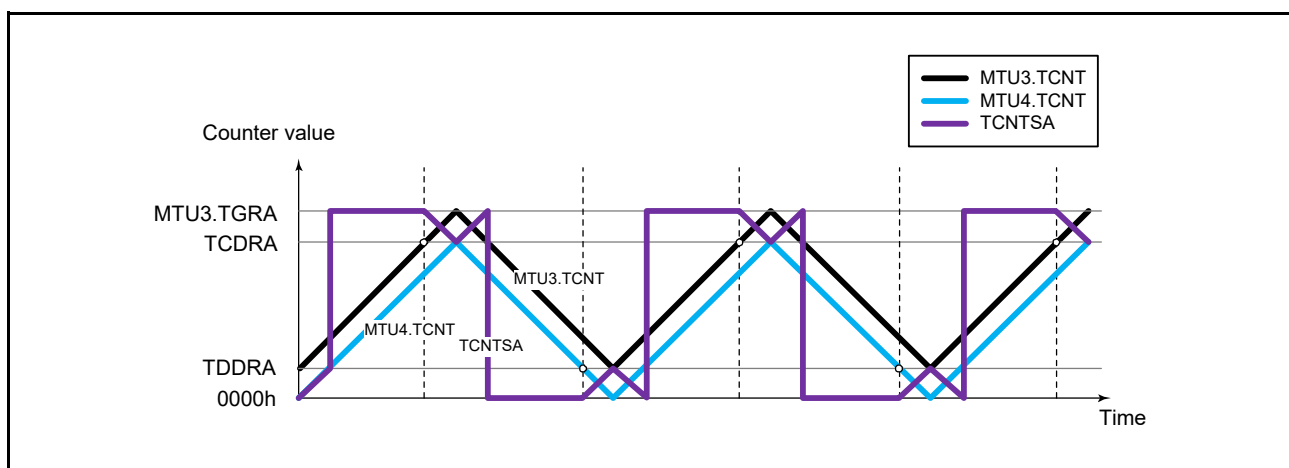


Figure 24.49 Count Operation in Complementary PWM Mode (MTU3 and MTU4)

### (b) Register Operation

In complementary PWM mode, nine registers (compare registers, buffer registers, and temporary registers) are used to control the duty ratio for the PWM output. Figure 24.50 shows an example of operation in complementary PWM mode (MTU3 and MTU4).

MTU3.TGRB, MTU4.TGRA, and MTU4.TGRB (MTU6.TGRB, MTU7.TGRA, and MTU7.TGRB) are constantly compared with the counters to generate PWM waveforms. When these registers match the counter, the value set in the OLSN and OLSP bits in the timer output control registers (TOCR1A and TOCR1B) is output from the PWM output pin. MTU3.TGRD, MTU4.TGRC, and MTU4.TGRD (MTU6.TGRD, MTU7.TGRC, and MTU7.TGRD) are buffer registers for these compare registers.

When the double buffer function is used, MTU3.TGRE, MTU4.TGRE, and MTU4.TGRF (MTU6.TGRE, MTU7.TGRE, and MTU7.TGRF) are also used as buffer registers B. For details of double buffer operation, refer to section 24.3.8 (2) (s), Double Buffer Function in Complementary PWM Mode.

Data in a compare register can be changed by writing new data to the corresponding buffer register. The buffer registers can be read or written at any time.

When modifying data in a buffer register, write to MTU4.TGRD (MTU7.TGRD) last and enable data transfer from the buffer register to a temporary register. At this time, transfer from the TCBRA (TCBRB) register and MTU3.TGRC (MTU6.TGRC) register, which operate as buffer registers for the timer period registers, to temporary registers is also enabled. Data is transferred to all five temporary registers at the same time.

When transfer is enabled in the Ta interval, data written to a buffer register is transferred to the temporary register. The data is not transferred to the temporary register in the Tb1 and Tb2 intervals. Data enabled for transfer in this interval is transferred to the temporary register at the end of this interval.

The value transferred to a temporary register is transferred to the compare register at the end of the Tb1 interval (when matches MTU3.TGRA (MTU6.TGRA) while TCNTSA (TCNTSB) is counting up), or at the end of the Tb2 interval (when matches 0000h while TCNTSA (TCNTSB) is counting down). The timing for transfer from the temporary register to the compare register can be selected with bits MD[3:0] in the timer mode register 1 (TMDR1). Figure 24.50 shows an example in which the trough is selected for the transfer timing.

In the Tb interval in which data is not transferred to the temporary register (Tb1 in Figure 24.50), the temporary register has the same function as the compare register and is compared with the counter. In this interval, therefore, there are two compare match registers for one output phase; the compare register contains the pre-change data and the temporary register contains new data. In this interval, three counters MTU3.TCNT, MTU4.TCNT, and TCNTSA (MTU6.TCNT, MTU7.TCNT, and TCNTSB) and two registers (compare register and temporary register) are compared, and PWM output is controlled accordingly.

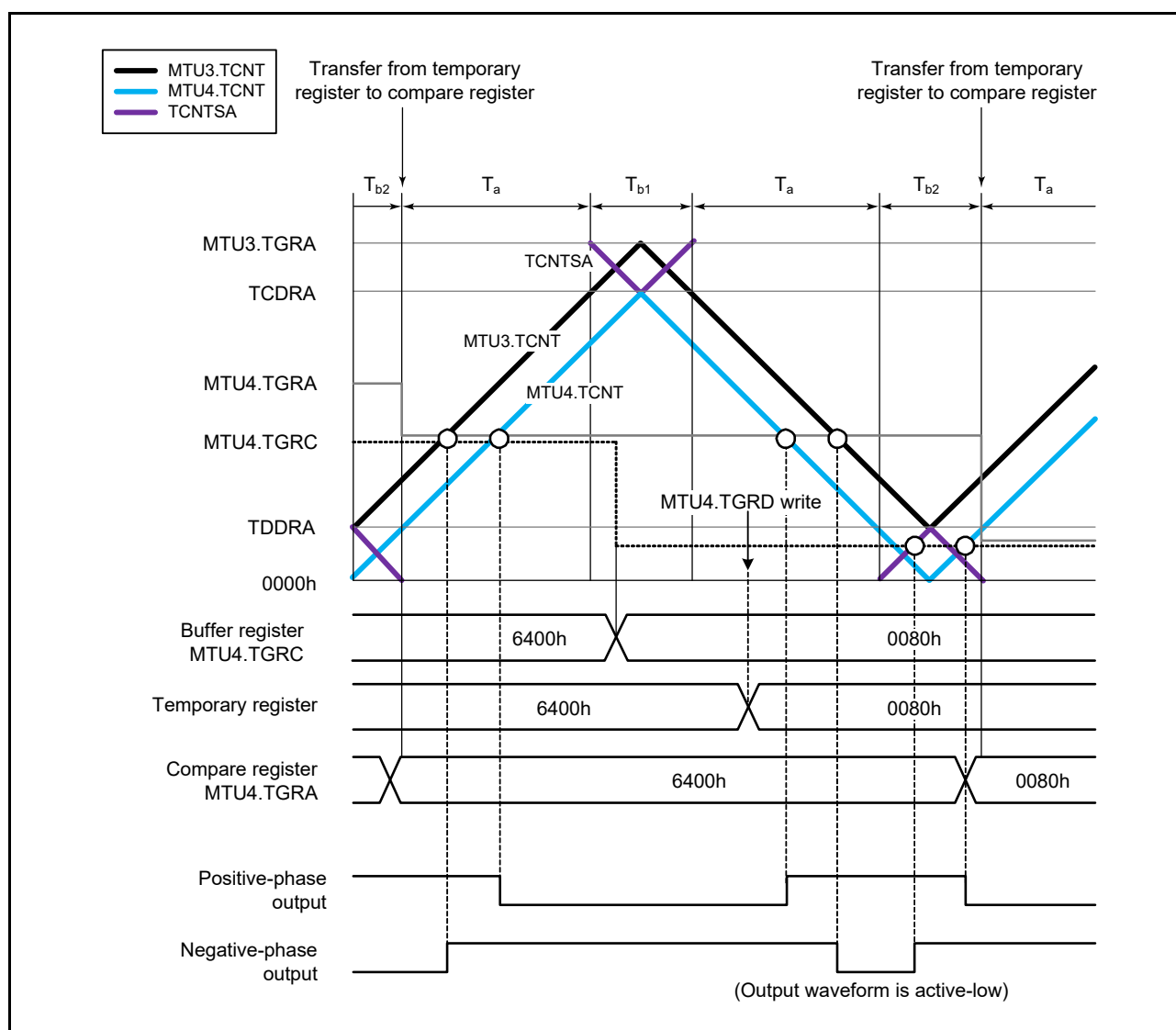


Figure 24.50 Example of Operation in Complementary PWM Mode (MTU3 and MTU4)

**(c) Initial Setting**

In complementary PWM mode, there are nine registers that require initial setting. In addition, there is a register that specifies whether to generate dead time (it should be used only when dead time generation should be disabled).

Before setting complementary PWM mode with MTU3.TMDR1.MD[3:0] (MTU6.TMDR1.MD[3:0]) bits, initial values should be set in the following registers.

The TOCR1A, TOCR2A, TOCR1B, and TOCR2B registers are used to set the PWM output level. MTU3.TGRC (MTU6.TGRC) operates as the buffer register for MTU3.TGRA (MTU6.TGRA), and should be set with  $1/2$  the PWM period + dead time  $T_d$ . The timer period buffer register (TCBRA or TCBRB) operates as the buffer register for the timer period data register (TCDRA or TCDRB), and should be set with  $1/2$  the PWM period. Set dead time  $T_d$  in the timer dead time data register (TDDRA or TDDRB).

When dead time is not needed, the TDER bit in the timer dead time enable register (TDERA or TDERB) should be set to 0, MTU3.TGRC and MTU3.TGRA (MTU6.TGRC and MTU6.TGRA) should be set to  $1/2$  the PWM carrier period + 1, and TDDRA (TDDRB) should be set to 1.

Set the respective initial PWM duty values in three buffer registers A (MTU3.TGRD, MTU4.TGRC, and MTU4.TGRD (MTU6.TGRD, MTU7.TGRC, and MTU7.TGRD)).

Set the respective (initial PWM duty – 1) values in three buffer registers B (MTU3.TGRE, MTU4.TGRE, and MTU4.TGRF (MTU6.TGRE, MTU7.TGRE, and MTU7.TGRF)) only when the double buffer function is used.

The values set in the five buffer registers excluding TDDRA (TDDRB) are transferred to the corresponding compare registers as soon as complementary PWM mode is set.

Set MTU4.TCNT (MTU7.TCNT) to 0000h before setting complementary PWM mode.

**Table 24.76 Registers and Counters Requiring Initial Setting**

| Register and Counter                                                     | Setting                                                                                                               |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| TOCR1A, TOCR2A,<br>TOCR1B, TOCR2B                                        | PWM output level                                                                                                      |
| MTU3.TGRC<br>MTU6.TGRC                                                   | $1/2$ PWM period + dead time $T_d$<br>( $1/2$ PWM period + 1 when dead time generation is disabled by TDERA or TDERB) |
| TDDRA, TDDRB                                                             | Dead time $T_d$ (1 when dead time generation is disabled by TDERA or TDERB)                                           |
| TCBRA, TCBRB                                                             | $1/2$ PWM period                                                                                                      |
| MTU3.TGRD, MTU4.TGRC,<br>MTU4.TGRD<br>MTU6.TGRD, MTU7.TGRC,<br>MTU7.TGRD | Initial PWM duty ratio value for each phase                                                                           |
| MTU3.TGRE, MTU4.TGRE,<br>MTU4.TGRF<br>MTU6.TGRE, MTU7.TGRE,<br>MTU7.TGRF | Initial PWM duty ratio – 1 value for each phase (only when double buffer function is used)                            |
| MTU4.TCNT<br>MTU7.TCNT                                                   | 0000h                                                                                                                 |

Note: The value set in MTU3.TGRC (MTU6.TGRC) should be the sum of  $1/2$  the PWM period set in TCBRA (TCBRB) and dead time  $T_d$  set in TDDRA (TDDRB). When dead time generation is disabled by TDERA (TDERB), TGRC should be set to  $1/2$  the PWM period + 1.

**(d) PWM Output Level Setting**

In complementary PWM mode, the PWM output level is set with bits OLSN and OLSP in timer output control register 1 (TOCR1A or TOCR1B) or bits OLS1P to OLS3P and OLS1N to OLS3N in timer output control register 2 (TOCR2A or TOCR2B).

The output level can be set for each of the three positive phases and three negative phases of 6-phase output.

Complementary PWM mode should be cleared before setting or changing output levels.

### (e) Dead Time Setting

In complementary PWM mode, dead time can be set for PWM output.

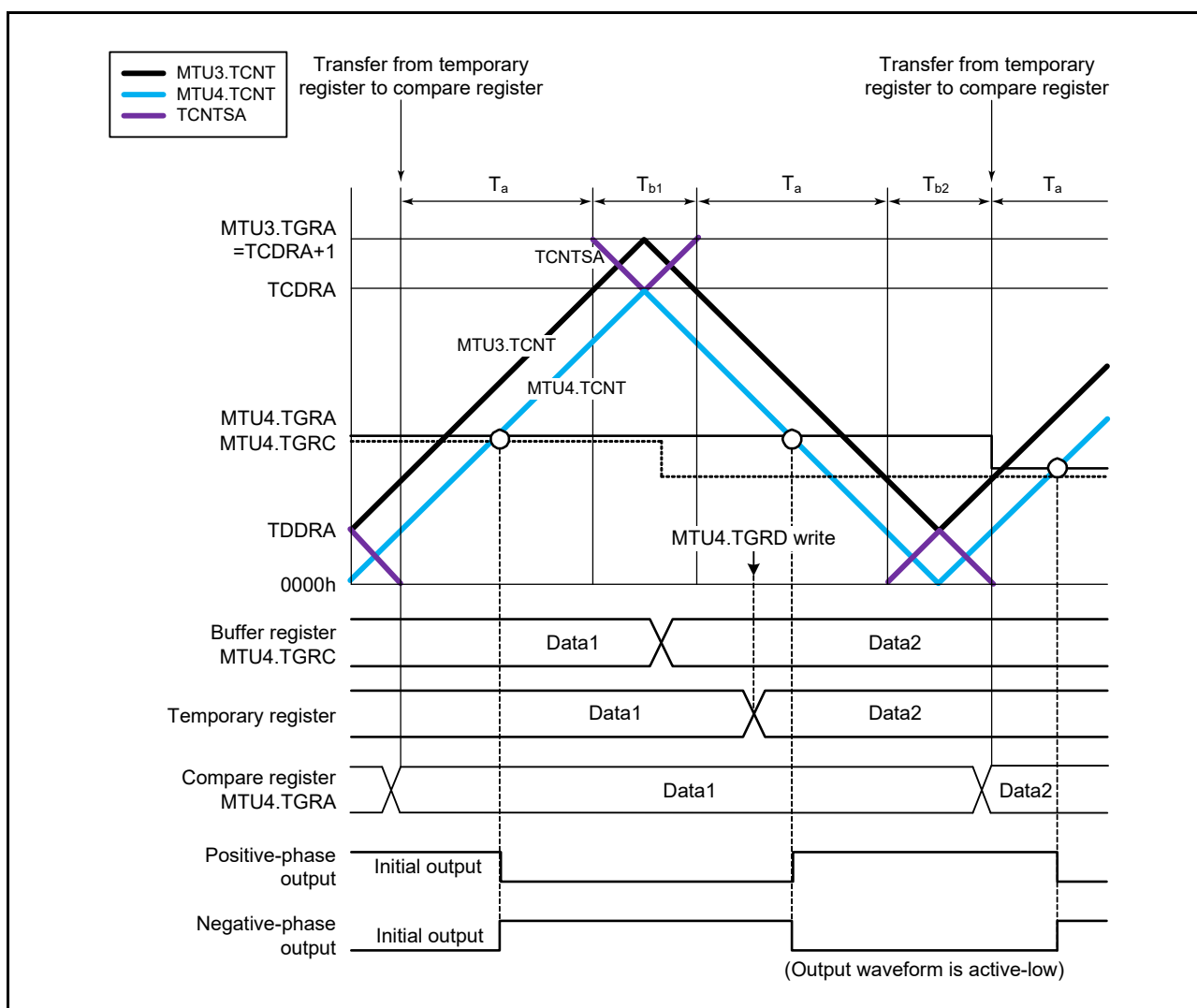
The dead time is set in the timer dead time data register (TDDRA or TDDRB). The value set in TDDRA (TDDRB) is used as the MTU3.TCNT (MTU6.TCNT) counter start value and creates a non-overlapping interval between MTU3.TCNT (MTU6.TCNT) and MTU4.TCNT (MTU7.TCNT). Complementary PWM mode should be cleared before changing the contents of TDDRA (TDDRB).

(f) Dead Time Suppressing

Dead time generation is suppressed by setting the TDER bit in the timer dead time enable register (TDERA or TDERB) to 0. TDERA (TDERB) can be set to 0 only when 0 is written to it after reading TDER = 1.

MTU3.TGRA and MTU3.TGRC (MTU6.TGRA and MTU6.TGRC) should be set to  $1/2$  PWM period + 1 and the timer dead time data register (TDDRA or TDDRB) should be set to 1.

By the above settings, PWM waveforms without dead time can be obtained. Figure 24.51 shows an example of operation without dead time (MTU3 and MTU4).



**Figure 24.51 Example of Operation without Dead Time (MTU3 and MTU4)**

## (g) PWM Period Setting

In complementary PWM mode, the PWM period is set in two registers—MTU3.TGRA (MTU6.TGRA), in which the MTU3.TCNT (MTU6.TCNT) upper limit value is set, and TCDRA (TCDRB), in which the MTU4.TCNT (MTU7.TCNT) upper limit value is set. The settings should be made so as to achieve the following relationship between these two registers:

With dead time: MTU3.TGRA (MTU6.TGRA) setting = TCDRA (TCDRB) setting + TDDRA (TDDRB) setting

Without dead time: MTU3.TGRA (MTU6.TGRA) setting = TCDRA (TCDRB) setting + 1

In addition, the settings should be made so as to achieve the following relationship between the TCDRA (TCDRB) register and the TDDRA (TDDRB) register:

$\text{TCDRA (TCDRB) setting} > \text{TDDRA (TDDRB) setting} \times 2 + 2$

The MTU3.TGRA and TCDRA (MTU6.TGRA and TCDBR) settings are made by setting values in buffer registers MTU3.TGRC and TCBRA (MTU6.TGRC and TCBRB). When data is written to MTU4.TGRD (MTU7.TGRD) to enable transfers, the values set in MTU3.TGRC and TCBRA (MTU6.TGRC and TCBRB) are transferred simultaneously to the MTU3.TGRA and TCDRA (MTU6.TGRA and TCDBR) with the transfer timing selected with the MTU3.TMDR1.MD[3:0] (MTU6.TMDR1.MD[3:0]) bits.

The new PWM period is reflected from the next cycle when data is updated at the crest, or from the current cycle when updated in the trough. Figure 24.52 illustrates the operation when the PWM period is updated at the crest. Refer to the following section, (h), Register Data Updating, for the method of updating the data in each buffer register.

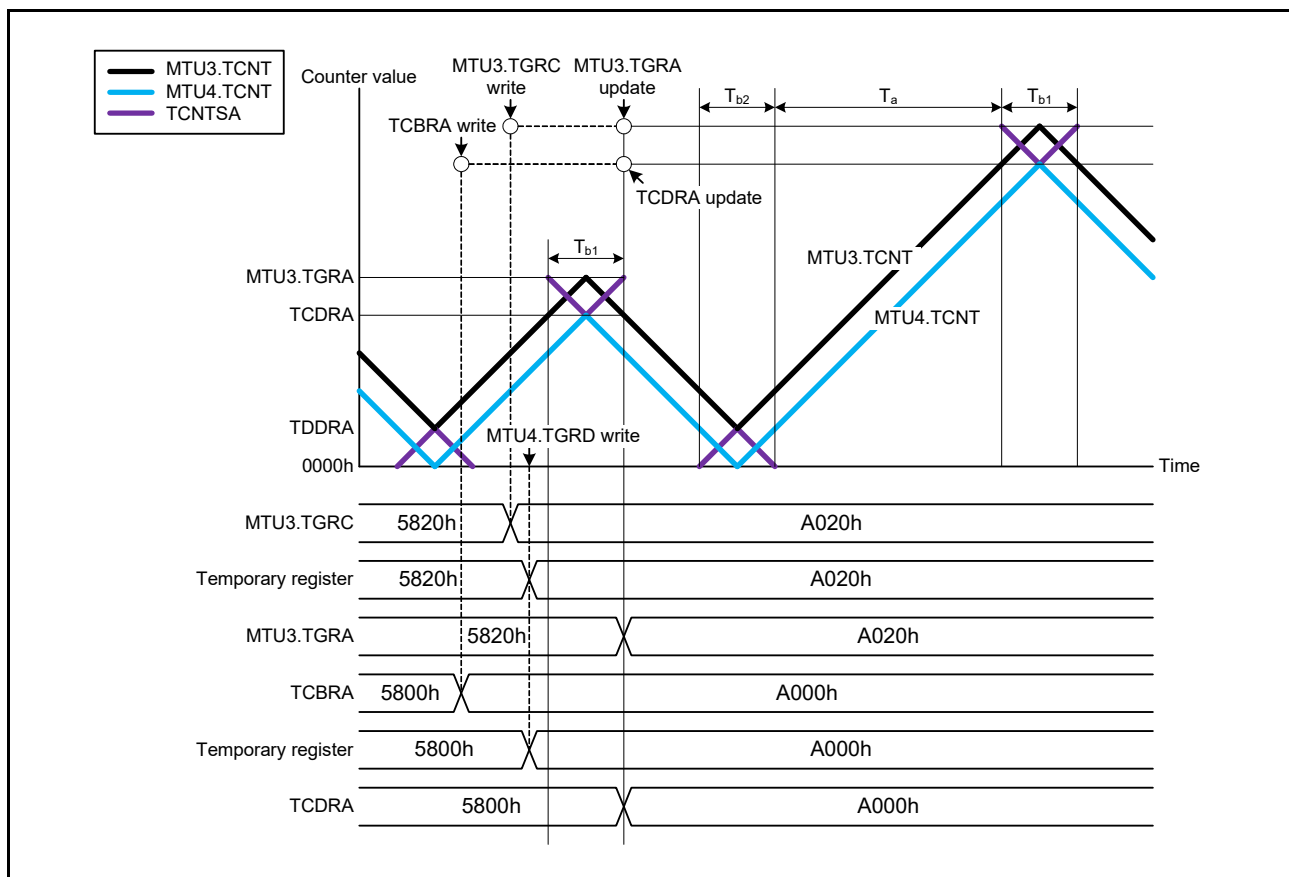


Figure 24.52 Example of PWM Period Updating (MTU3 and MTU4)

#### (h) Register Data Updating

The buffer registers are used to update the data in five compare registers for the PWM duty and PWM period in complementary PWM mode. The update data can be written to the buffer register at any time.

There is a temporary register between each of these registers and its buffer register. While subcounter TCNTSA (TCNTSB) is not counting, if buffer register data is updated, the temporary register value also changes. Data is not transferred from buffer registers to temporary registers while TCNTSA (TCNTSB) is counting; in this case, the value written to a buffer register is transferred after TCNTSA (TCNTSB) halts.

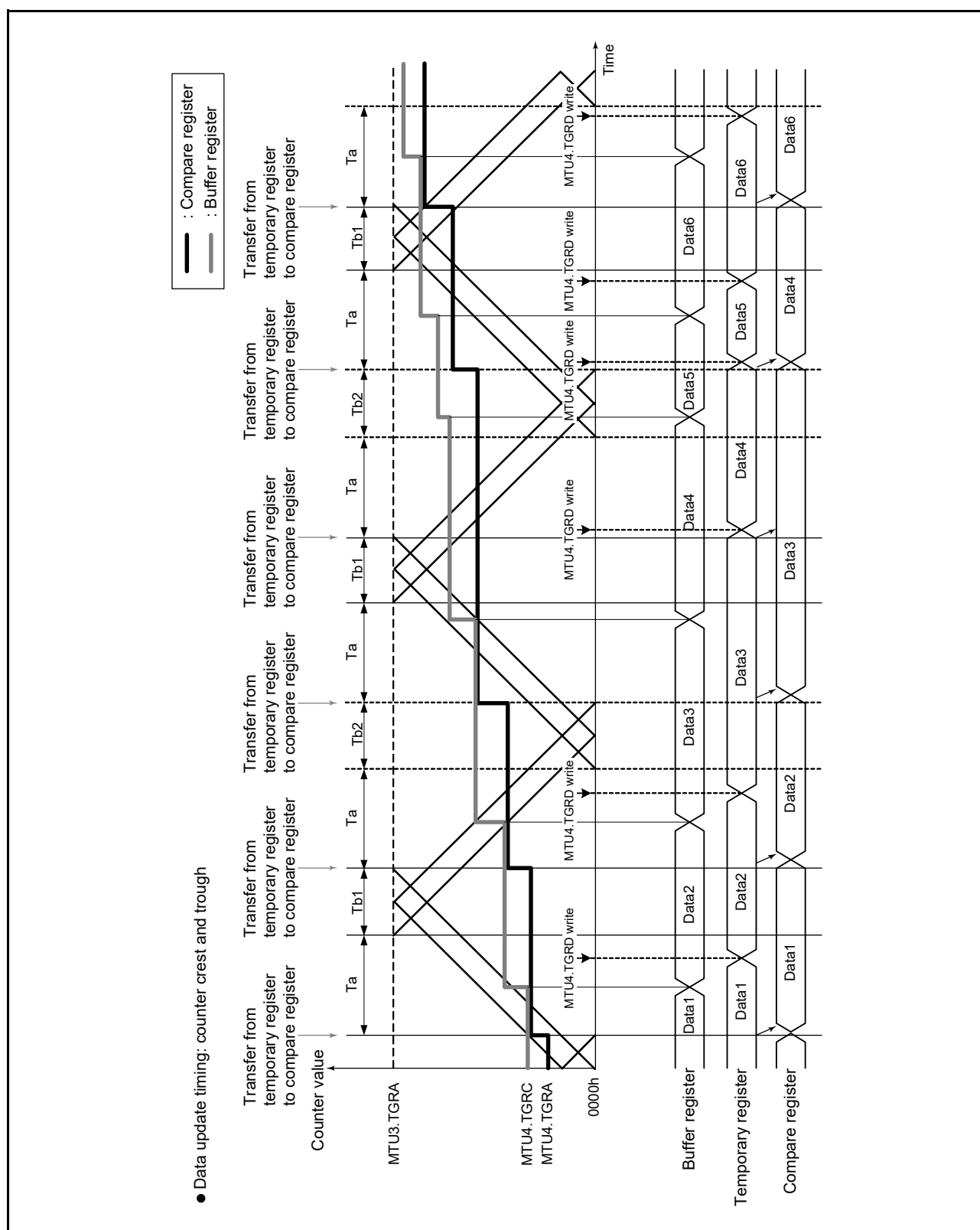
The temporary register value is transferred to the compare register at the data update timing set with MTU3.TMDR1.MD[3:0] (MTU6.TMDR1.MD[3:0]) bits. Figure 24.53 shows an example of data updating in complementary PWM mode (MTU3 and MTU4). This example shows the mode in which data is updated at both the counter crest and trough.

When updating buffer register data, be sure to write to MTU4.TGRD (MTU7.TGRD) at the end of the update. Data is transferred from buffer registers to the temporary registers simultaneously for all five registers after the write to MTU4.TGRD (MTU7.TGRD).

Even when not updating all five registers or when not updating the MTU4.TGRD (MTU7.TGRD) data, be sure to write to MTU4.TGRD (MTU7.TGRD) after writing data to the registers to be updated. In this case, the data written to MTU4.TGRD (MTU7.TGRD) should be the same as the data prior to the write operation.

Refer to section 24.3.8 (2) (s), Double Buffer Function in Complementary PWM Mode, for data updating when the double buffer function is used.





## (i) Initial Output in Complementary PWM Mode

In complementary PWM mode, the initial output is determined by the setting of the OLSN and OLSP bits in the TOCR1A (TOCR1B) register or the OLS1N to OLS3N and OLS1P to OLS3P bits in the TOCR2A register (TOCR2B). This initial output is the non-active level of the PWM output and continues from when complementary PWM mode is set with the MTU3.TMDR1 (MTU6.TMDR1) until MTU4.TCNT (MTU7.TCNT) exceeds the value set in the TDDRA (TDDRB) register. Figure 24.54 shows an example of the initial output in complementary PWM mode. An example of the waveform when the initial PWM duty ratio value is smaller than the TDDRA (TDDRB) value is shown in Figure 24.55.

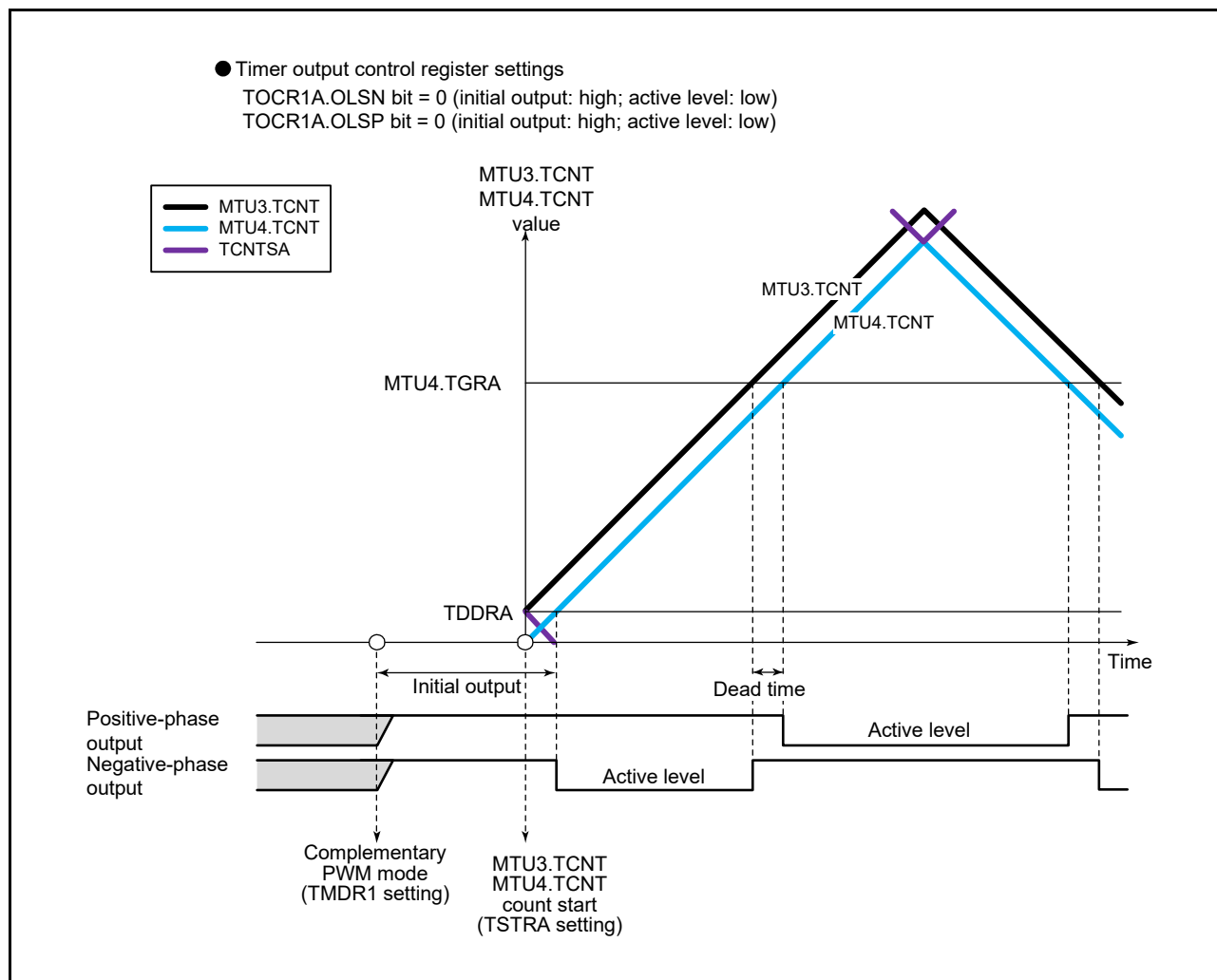


Figure 24.54 Example of Initial Output in Complementary PWM Mode (MTU3 and MTU4) (1))

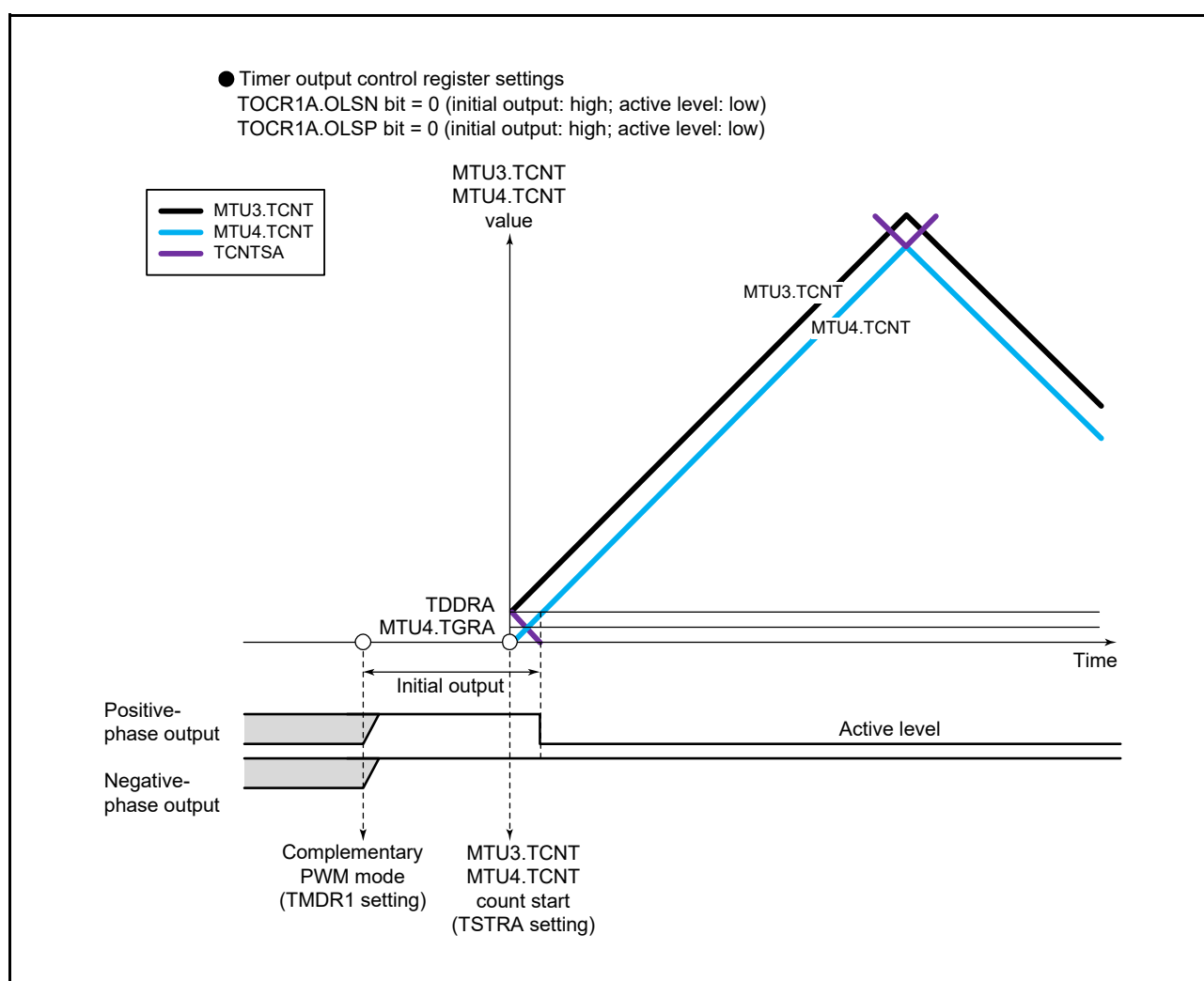


Figure 24.55 Example of Initial Output in Complementary PWM Mode (MTU3 and MTU4) (2)

### (j) Method for Generating PWM Output in Complementary PWM Mode

In complementary PWM mode, six phases (three positive and three negative) PWM waveforms can be output. Dead time can be set for PWM waveforms to be output.

A PWM waveform is generated by output of the level selected in the timer output control register in the event of a compare match between a counter and a compare register. While TCNTSA (TCNTSB) is counting, the compare register and temporary register values are simultaneously compared to create consecutive PWM output from 0 to 100% duty ratio. The relative timing of turn-on and turn-off compare match occurrence may vary, but the compare match that turns off each phase takes precedence to secure the dead time and ensure that the positive-phase and negative-phase turn-on times do not overlap. Figure 24.56 to Figure 24.58 show examples of waveform generation in complementary PWM mode.

The positive-phase and negative-phase turn-off timing is generated by a compare match with the counter indicated by a solid line, and the turn-on timing is generated by a compare match with the counter indicated by a dotted line, which operates with a delay equal to the dead time behind the counter indicated by a solid line. In the T1 period, compare match a that turns off the negative phase has the highest priority, and compare matches before a are ignored. In the T2 period, compare match c that turns off the positive phase has the highest priority, and compare matches before c are ignored.

In most cases, compare matches occur in the order  $a \rightarrow b \rightarrow c \rightarrow d$  (or  $c \rightarrow d \rightarrow a' \rightarrow b'$ ) as shown in Figure 24.56.

If compare matches deviate from the  $a \rightarrow b \rightarrow c \rightarrow d$  order, since the time for which the negative phase is off is shorter than twice the dead time, the positive phase is not turned on. If compare matches deviate from the  $c \rightarrow d \rightarrow a' \rightarrow b'$  order, since the time for which the positive phase is off is shorter than twice the dead time, the negative phase is not turned on.

As shown in Figure 24.57, if compare match c follows compare match a before compare match b, compare match b is ignored and the negative phase is turned on by compare match d. This is because turning off the positive phase has priority due to the occurrence of compare match c (positive-phase off timing) before compare match b (positive-phase on timing) (consequently, the waveform does not change because the positive phase goes from off to off).

Similarly, in the example in Figure 24.58, turning off the negative phase has priority due to the occurrence of compare match a' (negative-phase off timing) before compare match d (negative-phase on timing). As a result, the negative phase is not turned on.

Thus, in complementary PWM mode, compare matches at turn-off timings take precedence, and turn-on timing compare matches that occur before a turn-off timing compare match are ignored.

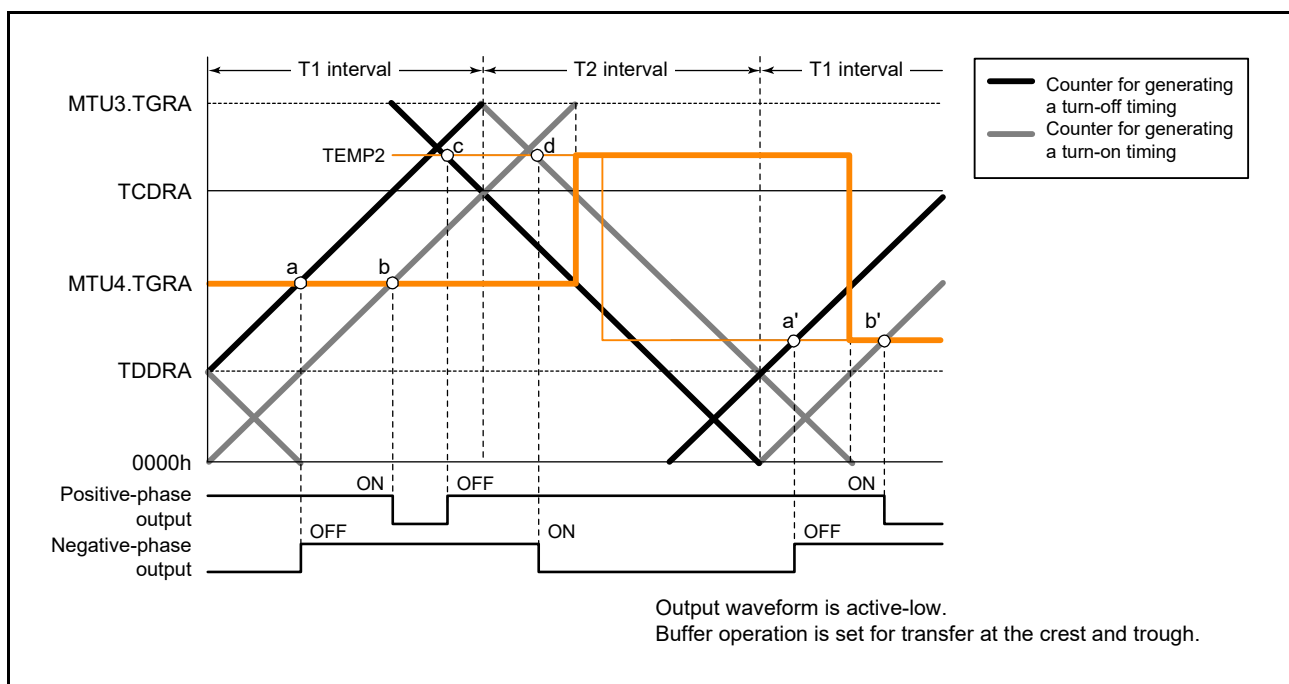


Figure 24.56 Example of Waveform Output in Complementary PWM Mode (MTU3 and MTU4) (1)

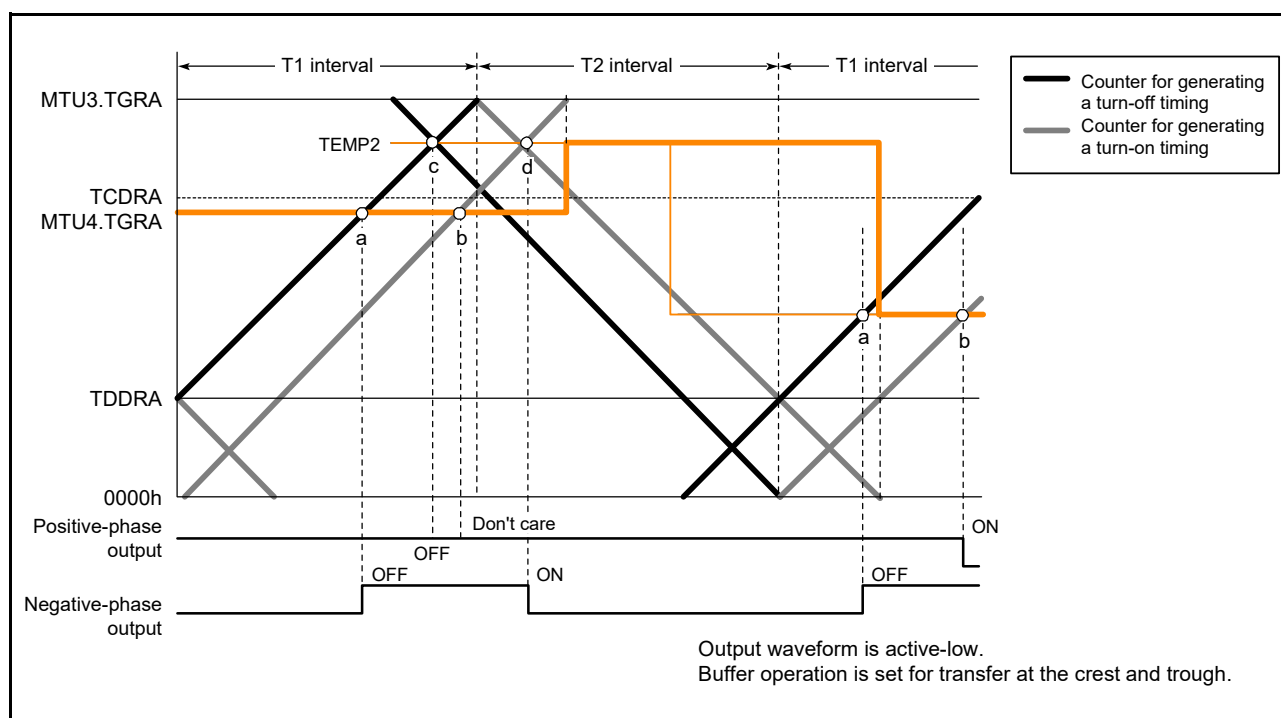


Figure 24.57 Example of Waveform Output in Complementary PWM Mode (MTU3 and MTU4) (2)

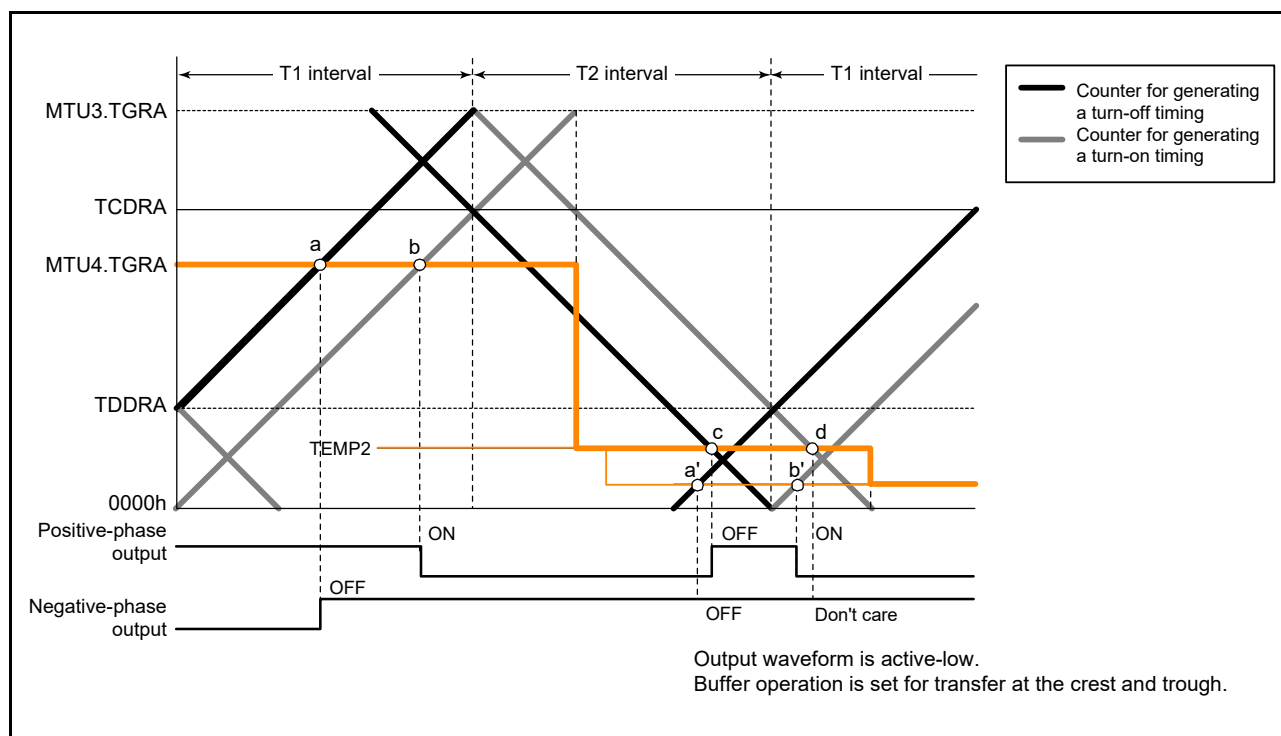


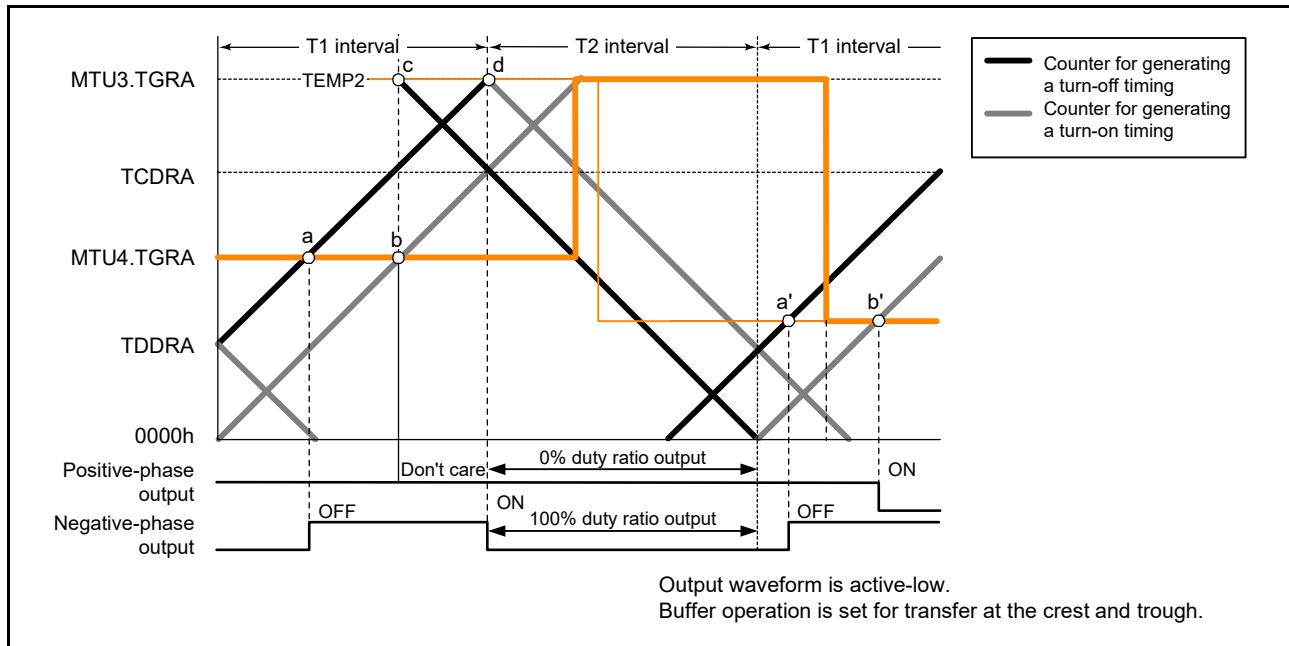
Figure 24.58 Example of Waveform Output in Complementary PWM Mode (MTU3 and MTU4) (3)

## (k) 0% and 100% Duty Ratio Output in Complementary PWM Mode

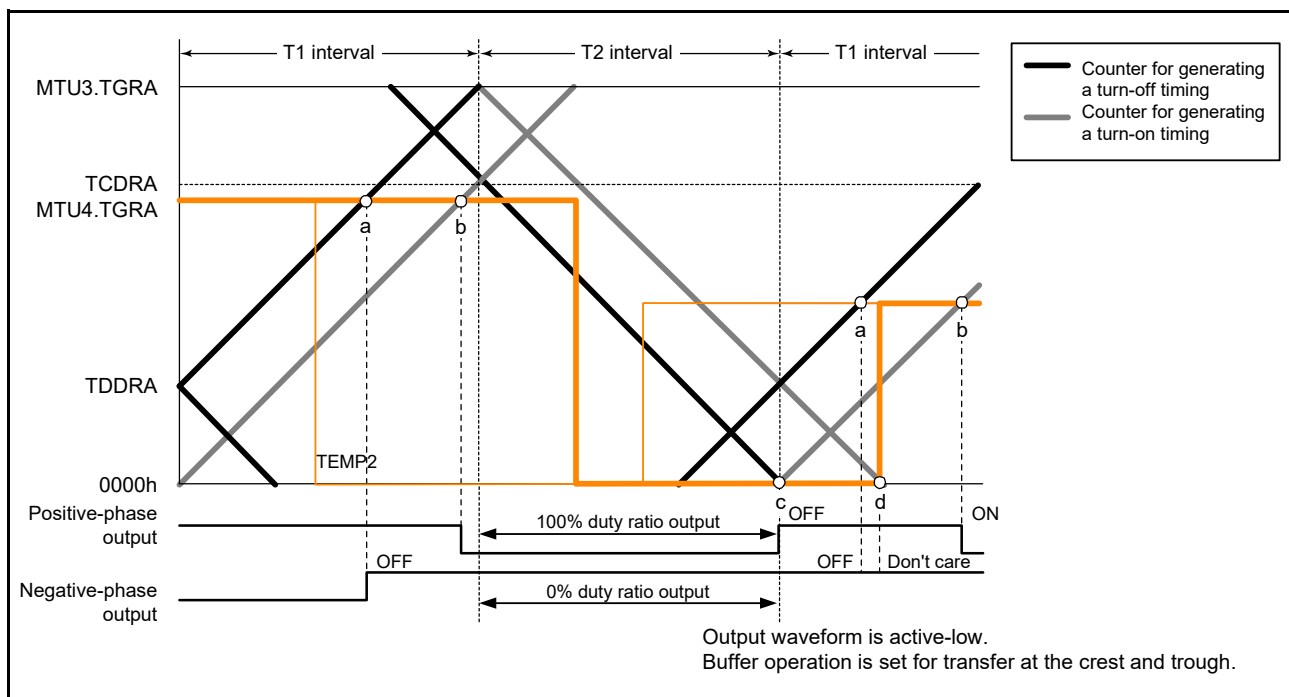
In complementary PWM mode, 0% and 100% duty PWM output can be output as required. Figure 24.59 to Figure 24.63 show output examples.

A 100% duty waveform is output when the compare register value is set to 0000h. The waveform in this case has a positive phase with a 100% on-state. A 0% duty waveform is output when the compare register value is set to the same value as MTU3.TGRA (MTU6.TGRA). The waveform in this case has a positive phase with a 100% off-state.

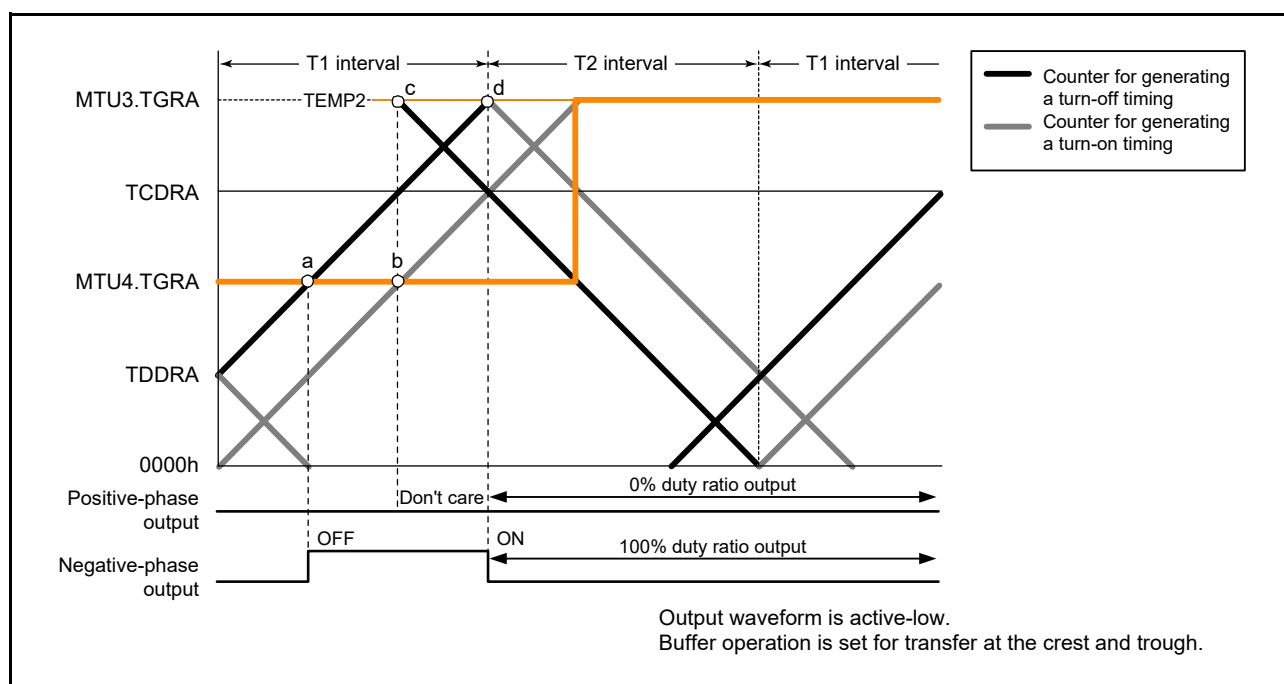
Turn-on and turn-off compare matches occur simultaneously, but if a turn-on compare match and turn-off compare match for the same phase occur simultaneously, both compare matches are ignored and the waveform does not change.



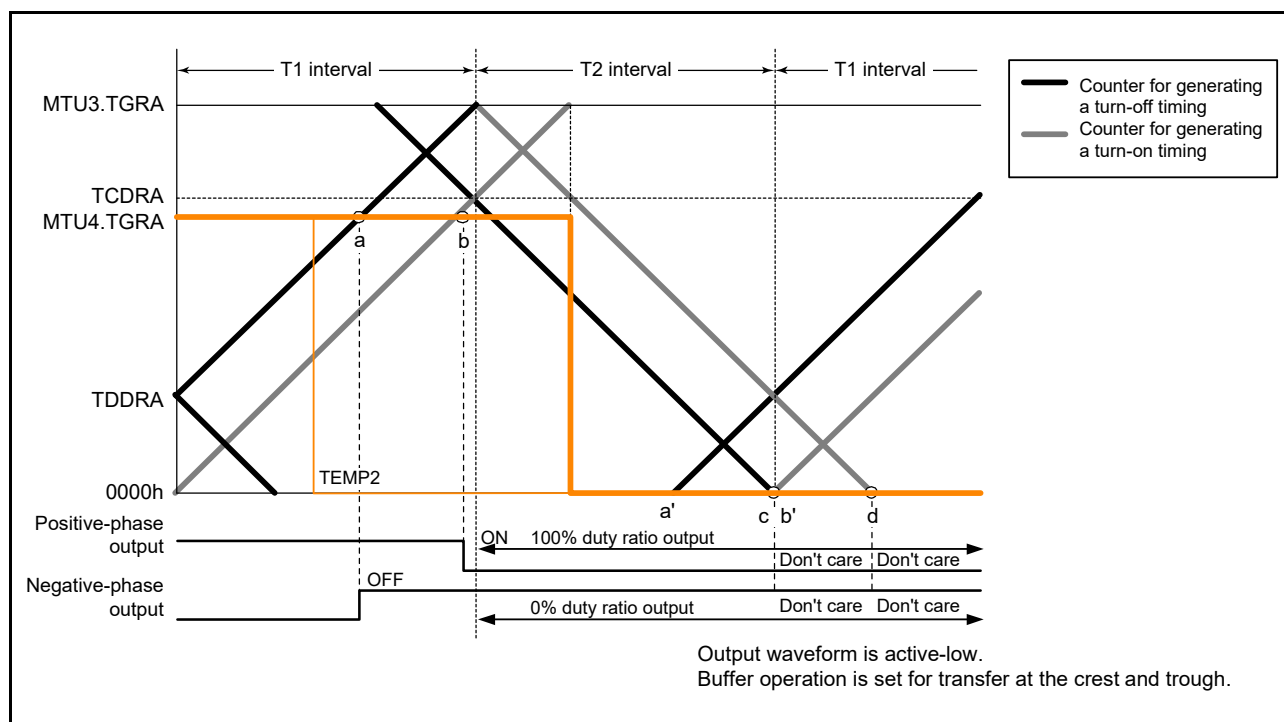
**Figure 24.59** Example of 0% and 100% Waveform Output in Complementary PWM Mode (MTU3 and MTU4) (1)



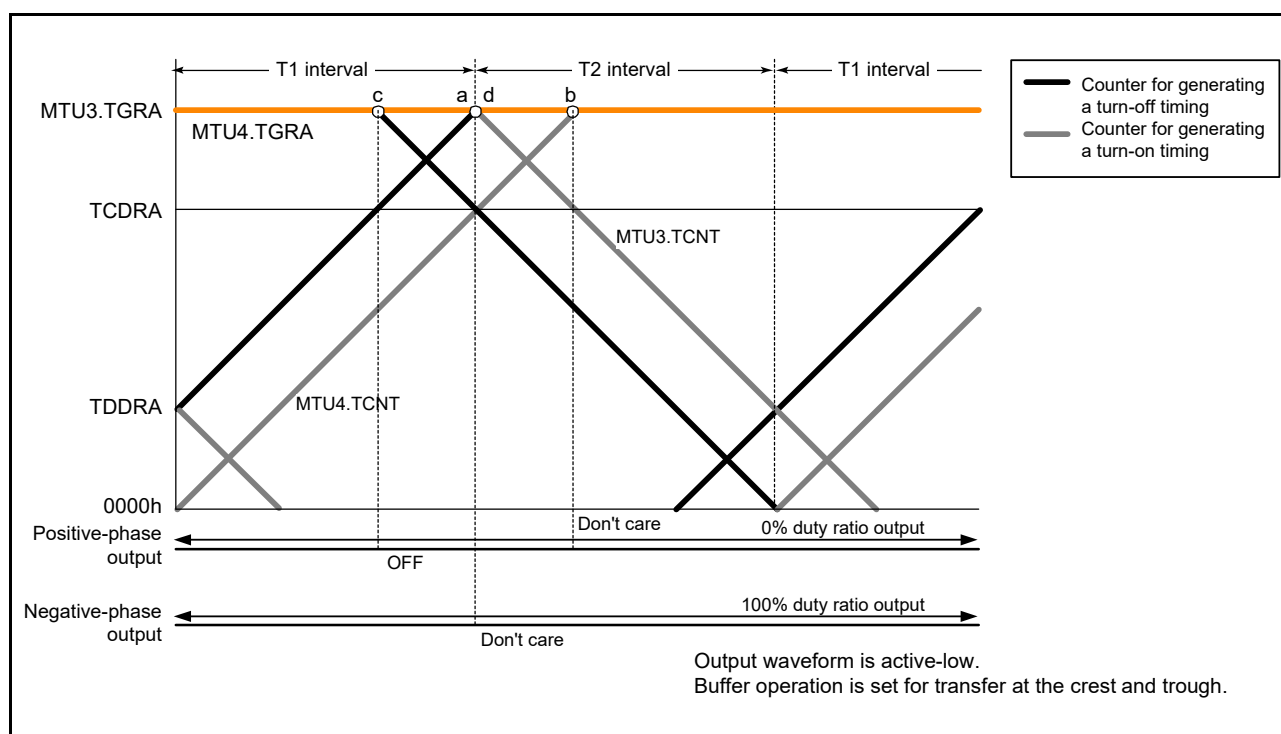
**Figure 24.60** Example of 0% and 100% Waveform Output in Complementary PWM Mode (MTU3 and MTU4) (2)



**Figure 24.61** Example of 0% and 100% Waveform Output in Complementary PWM Mode (MTU3 and MTU4) (3)



**Figure 24.62** Example of 0% and 100% Waveform Output in Complementary PWM Mode (MTU3 and MTU4) (4)



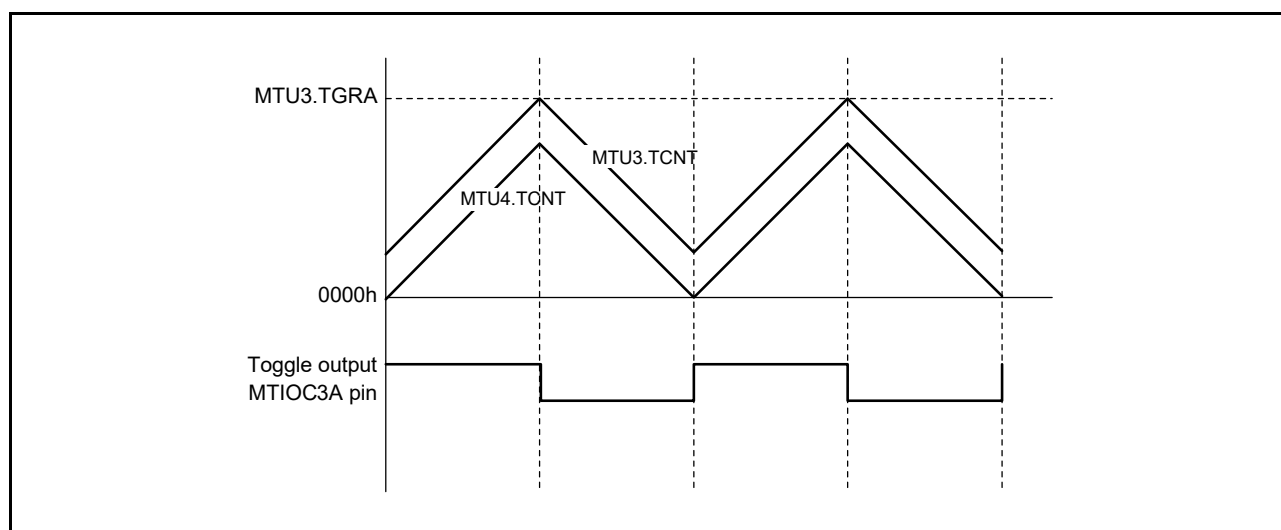
**Figure 24.63** Example of 0% and 100% Waveform Output in Complementary PWM Mode (MTU3 and MTU4) (5)

#### (I) Toggle Output Synchronized with PWM Period

In complementary PWM mode, toggle output from the PWM output pin in synchronization with the PWM period can be enabled by setting the PSYE bit in the TOCR1A (TOCR1B) register to 1. An example of a toggle output waveform is shown in Figure 24.64.

This output is toggled by a compare match between MTU3.TCNT and MTU3.TGRA (MTU6.TCNT and MTU6.TGRA) and a compare match between MTU4.TCNT (MTU7.TCNT) and 0000h.

The MTIOC3A (MTIOC6A) pin is assigned for this toggle output. The initial output is high-level output.



**Figure 24.64** Example of Toggle Output Waveform Synchronized with PWM Output (MTU3 and MTU4)

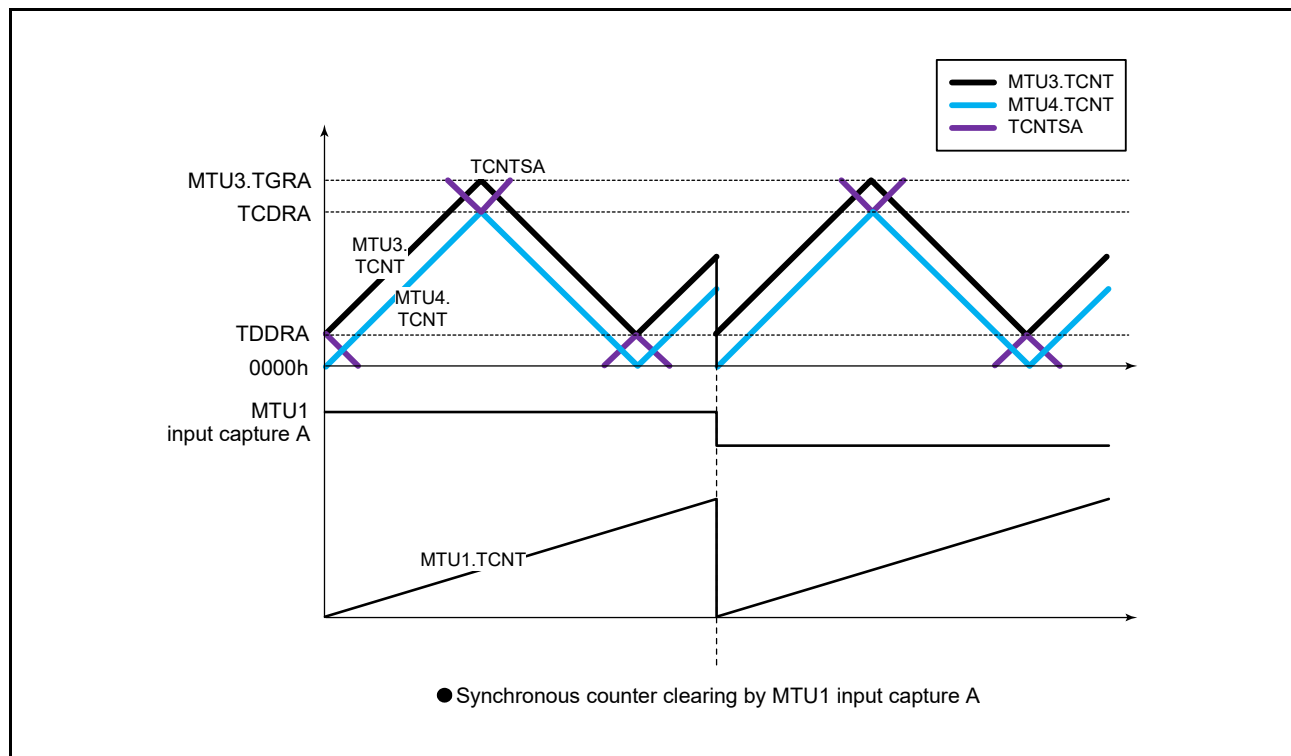


**(m) Counter Clearing by Another Channel**

In complementary PWM mode, MTU3.TCNT, MTU4.TCNT, and TCNTSA (MTU6.TCNT, MTU7.TCNT, and TCNTSB) can be cleared by another channel source when a mode for synchronization with another channel is specified through the TSYRA (TSYRB) register and synchronous clearing is selected with MTU3.TCR.CCLR[2:0] (MTU6.TCR.CCLR[2:0]) bits.

Figure 24.65 illustrates an example of this operation.

Use of this function enables a counter to be cleared and restarted through an external signal.



**Figure 24.65 Counter Clearing Synchronized with Another Channel (MTU3 and MTU4)**

## (n) Output Waveform Control at Synchronous Counter Clearing in Complementary PWM Mode

Setting the WRE bit in TWCRA (TWCRB) to 1 suppresses initial output when synchronous counter clearing occurs in the Tb interval (Tb2 interval) at the trough in complementary PWM mode and controls abrupt change in duty cycle at synchronous counter clearing.

Initial output suppression through the WRE bit = 1 is applicable only when synchronous clearing occurs in the Tb2 interval as indicated by (10) or (11) in Figure 24.66. When synchronous clearing occurs outside that interval, the initial value specified by the OLSN and OLSP bits in TOCR1A (TOCR1B) is output. Even in the Tb2 interval, if synchronous clearing occurs in the initial value output period (indicated by (1) in Figure 24.66) immediately after the counters start operation, initial value output is not suppressed.

This function can be used in both channel combinations of MTU3 and MTU4, and MTU6 and MTU7. In MTU3 and MTU4, synchronous clearing in any of MTU0 to MTU2 can cause counter clearing; in MTU6 and MTU7, compare match or input capture in any of MTU0 to MTU2 can cause counter clearing.

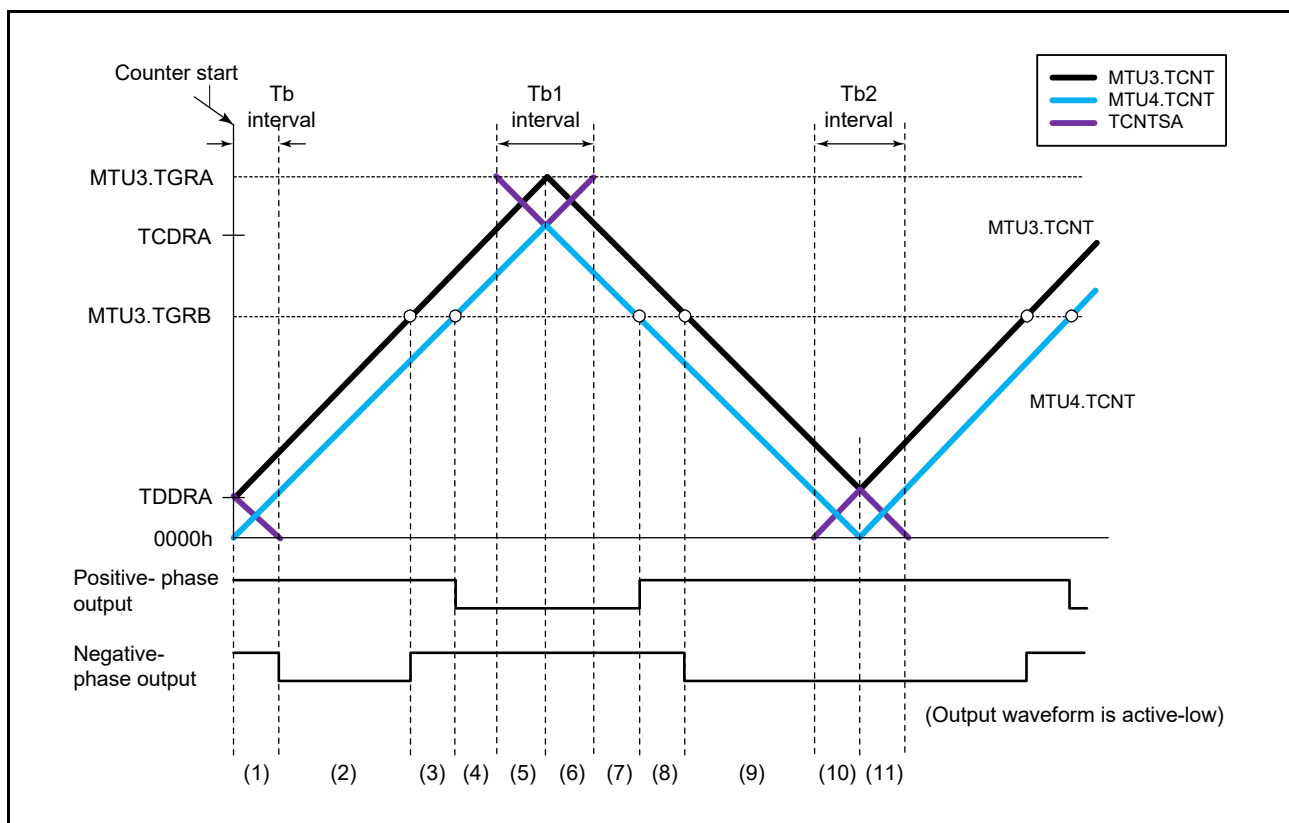
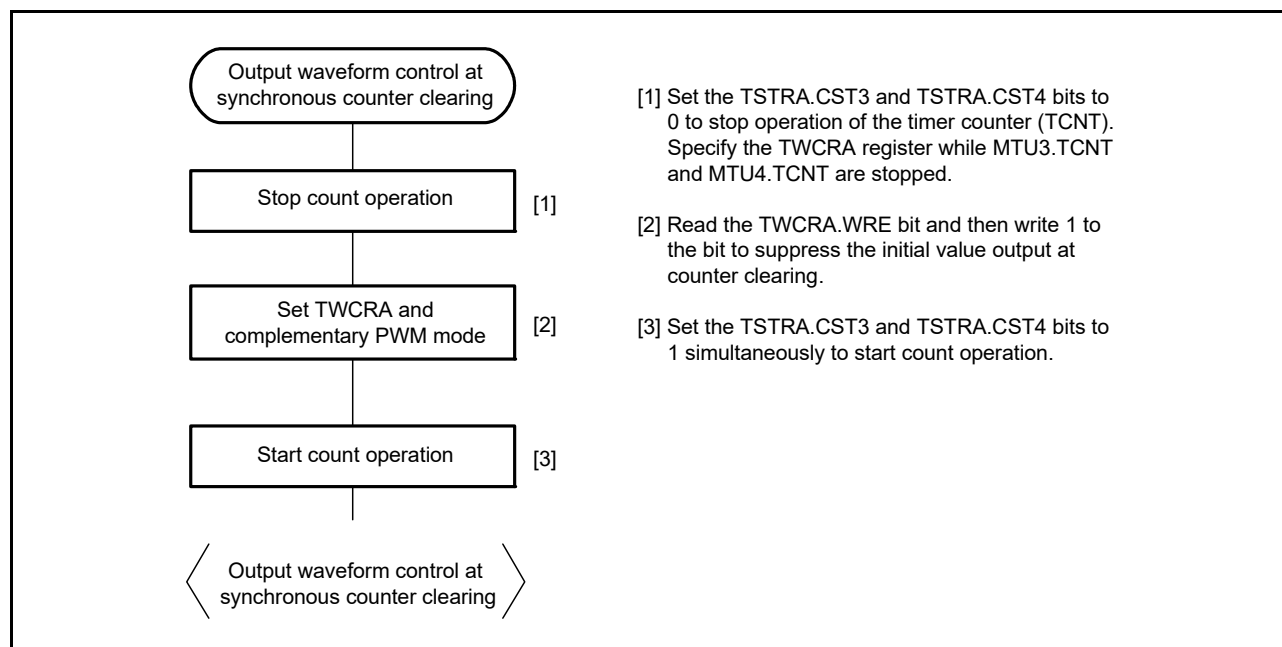


Figure 24.66 Timing for Synchronous Counter Clearing (MTU3 and MTU4)

- Example of Procedure for Setting Output Waveform Control at Synchronous Counter Clearing in Complementary PWM Mode.

An example of the procedure for setting output waveform control at synchronous counter clearing in complementary PWM mode is shown in Figure 24.67.

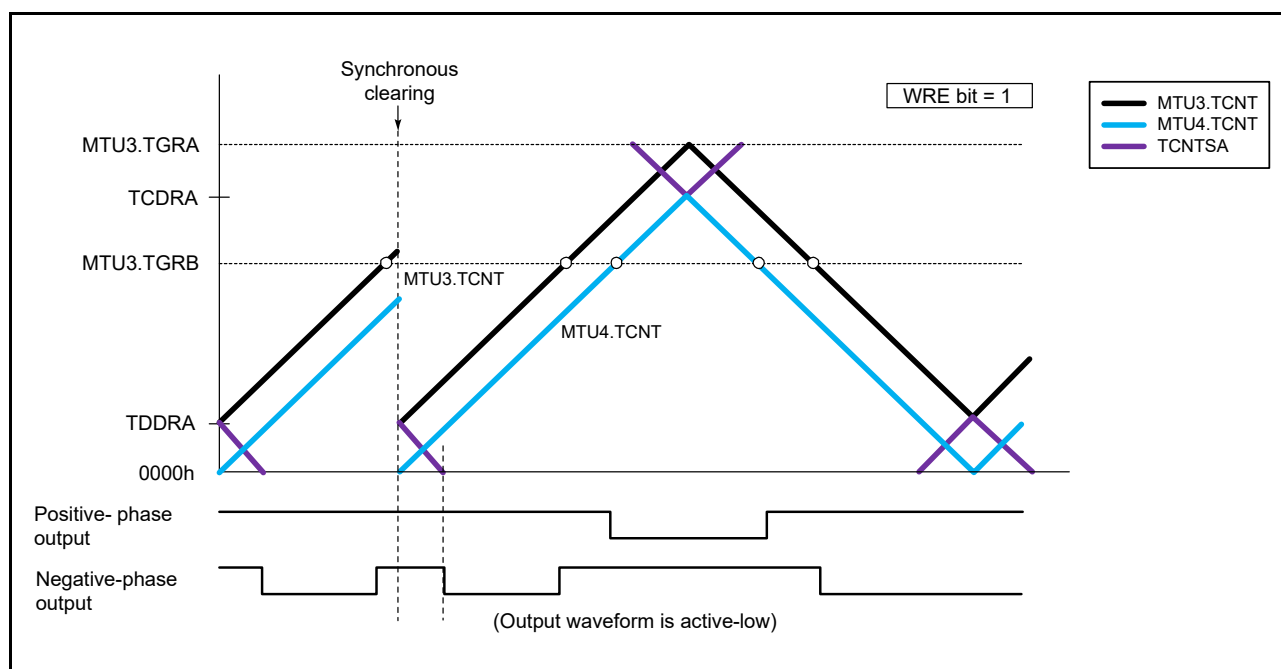


**Figure 24.67** Example of Procedure for Setting Output Waveform Control at Synchronous Counter Clearing in Complementary PWM Mode (MTU3 and MTU4)

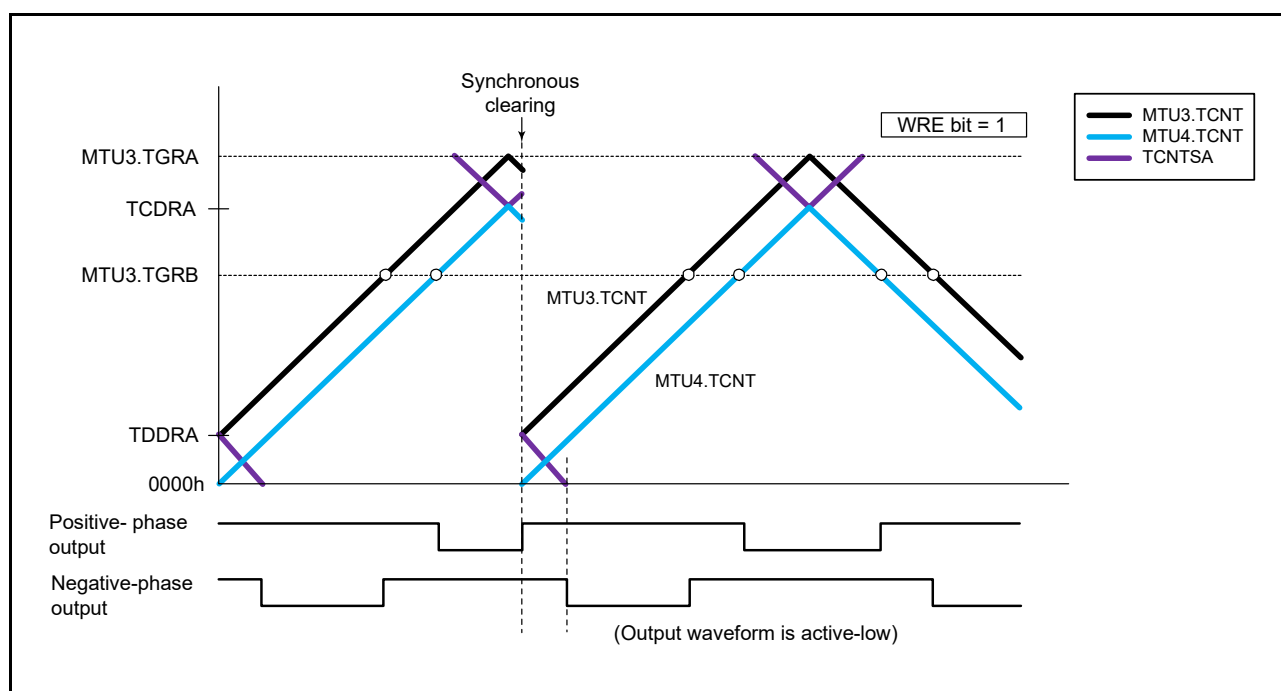
- Examples of Output Waveform Control at Synchronous Counter Clearing in Complementary PWM Mode

Figure 24.68 to Figure 24.71 show examples of output waveform control in which MTU3 and MTU4 operate in complementary PWM mode and synchronous counter clearing is generated while the WRE bit in TWCRA is set to 1. In the examples shown in Figure 24.68 to Figure 24.71, synchronous counter clearing occurs at timing (3), (6), (8), and (11) shown in Figure 24.66, respectively.

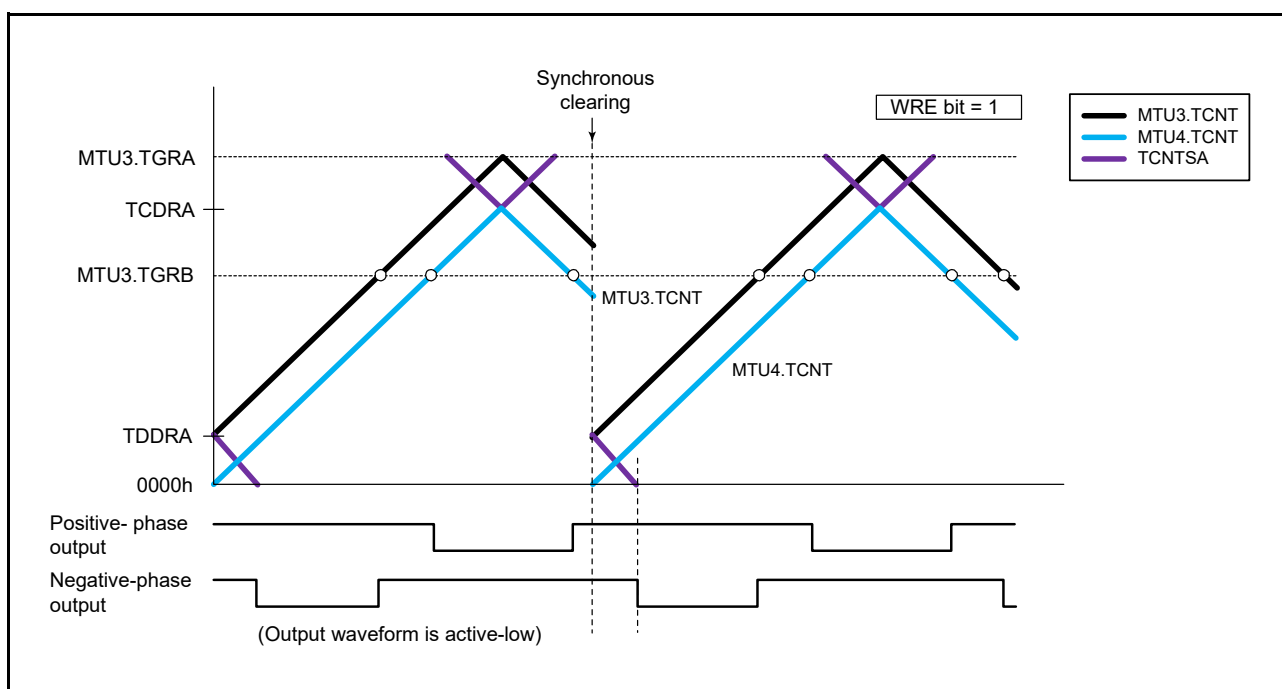
In MTU6 and MTU7, these examples are equivalent to the cases when MTU6 and MTU7 operate in complementary PWM mode and synchronous counter clearing is generated while the SCC bit is set to 0 and the WRE bit is set to 1 in TWCRA.



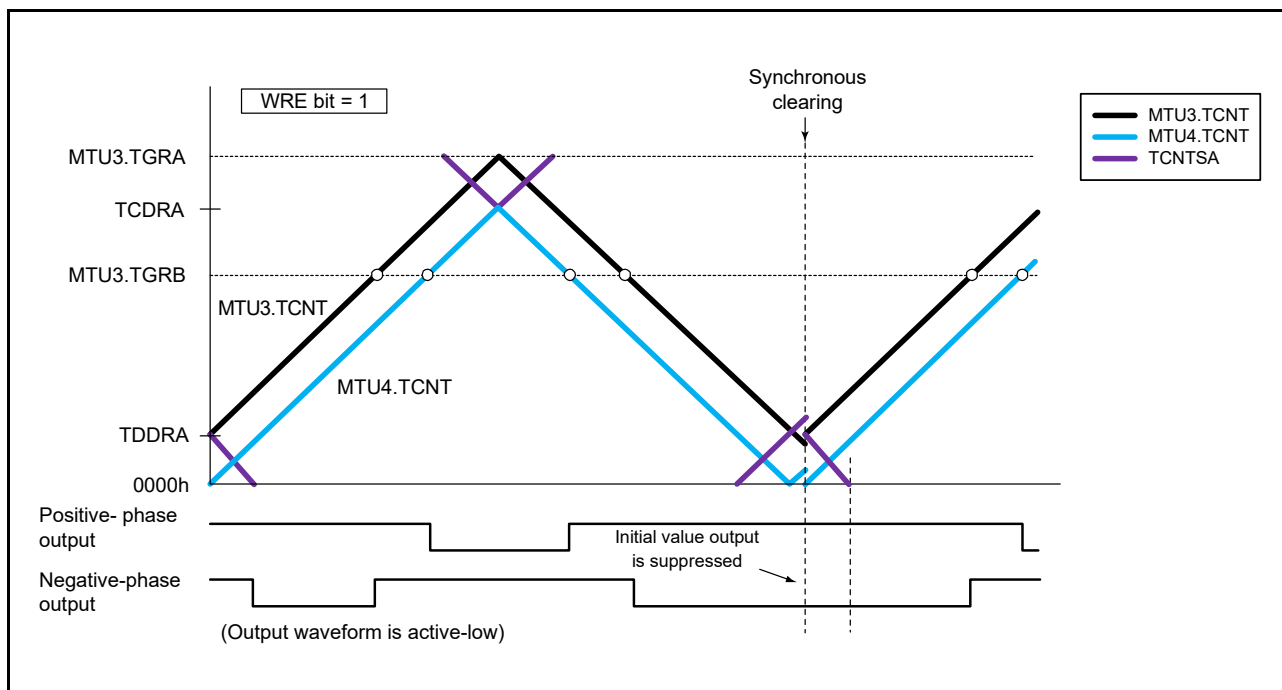
**Figure 24.68** Example of Synchronous Clearing in Dead Time during Up-Counting (Timing (3) in Figure 24.66; TWCRA.WRE Bit is 1)



**Figure 24.69** Example of Synchronous Clearing in Tb1 interval (Timing (6) in Figure 24.66; TWCRA.WRE Bit is 1)



**Figure 24.70** Example of Synchronous Clearing in Dead Time during Down-Counting (Timing (8) in Figure 24.66; TWCRA.WRE Bit is 1)



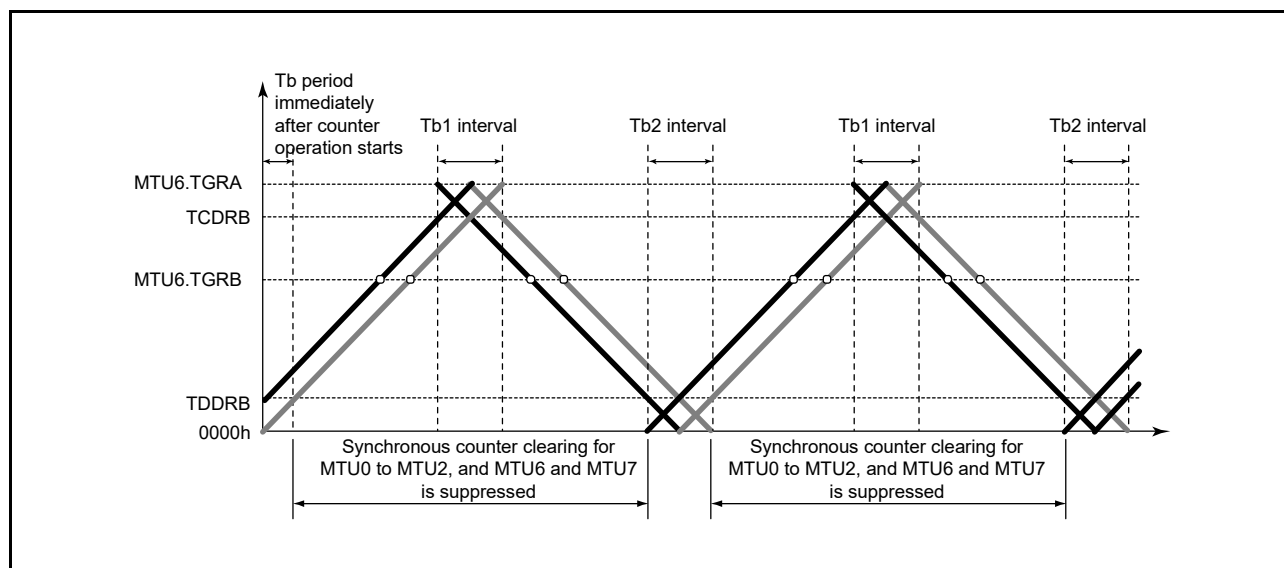
**Figure 24.71** Example of Synchronous Clearing in Tb2 interval (Timing (11) in Figure 24.66; TWCRA.WRE Bit is 1)

## (o) Suppressing Synchronous Counter Clearing for MTU0 to MTU2, and MTU6 and MTU7

In MTU6 and MTU7, setting the SCC bit in TWCRB to 1 suppresses synchronous counter clearing caused by MTU0 to MTU2.

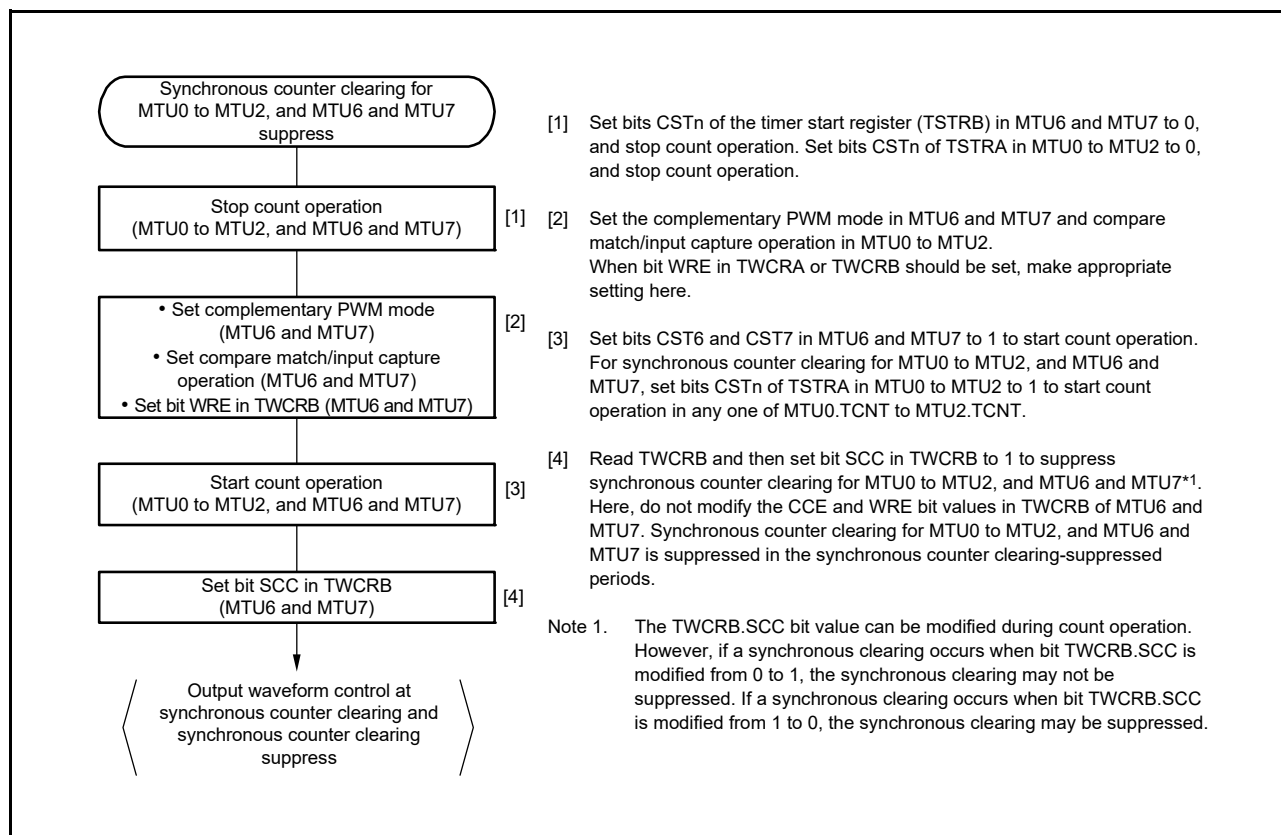
Synchronous counter clearing is suppressed only within the interval shown in Figure 24.72. When using this function, MTU6 and MTU7 should be set to complementary PWM mode.

For details of synchronous clearing caused by MTU0 to MTU2, refer to section 24.3.10 (2), Synchronous Counter Clearing for MTU6 and MTU7.



**Figure 24.72** Synchronous Clearing-Suppressed Interval Specified by TWCRB.SCC Bit for MTU0 to MTU2, and MTU6 and MTU7

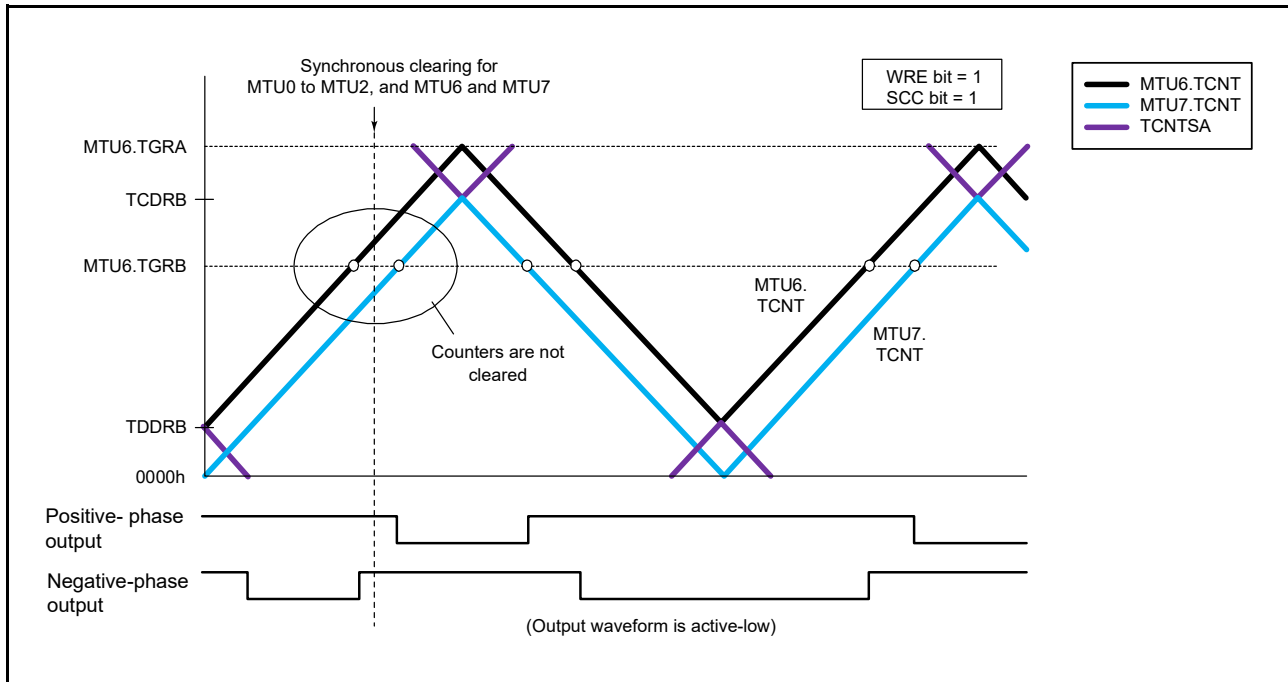
- Example of Procedure for Suppressing Synchronous Counter Clearing for MTU0 to MTU2, and MTU6 and MTU7
- An example of the procedure for suppressing synchronous counter clearing for MTU0 to MTU2, and MTU6 and MTU7 is shown in Figure 24.73.



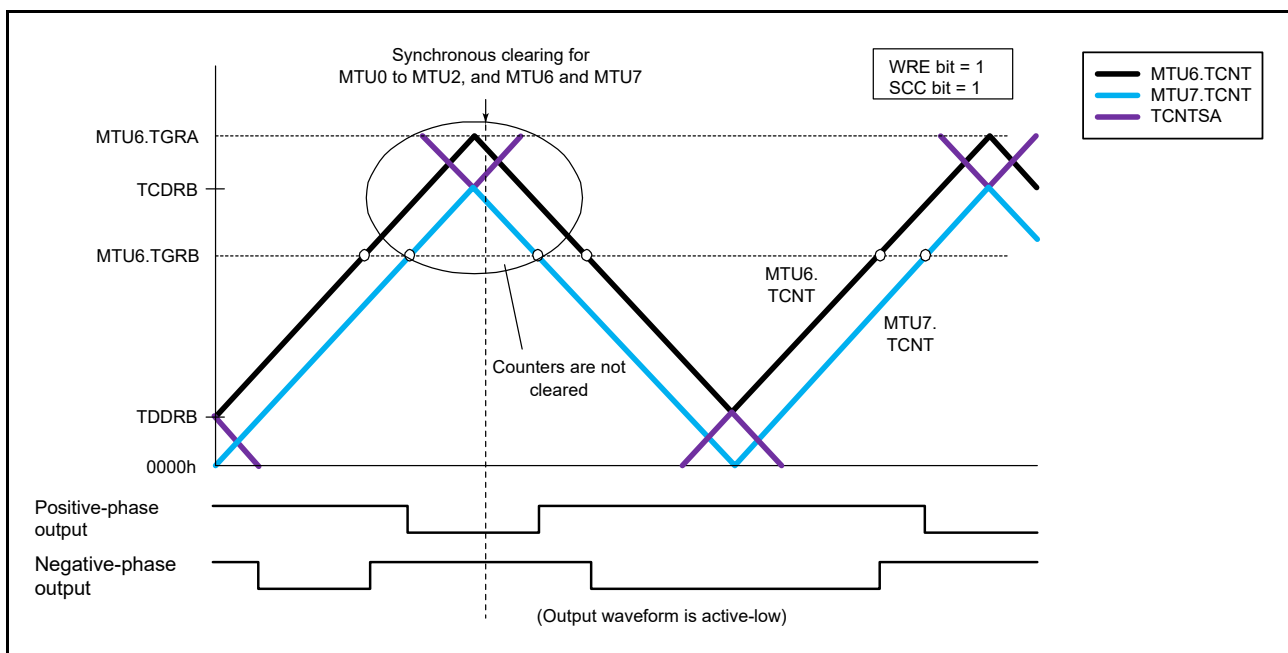
**Figure 24.73 Example of Procedure for Suppressing Synchronous Counter Clearing for MTU0 to MTU2, and MTU6 and MTU7**

- Examples of Suppression of Synchronous Counter Clearing for MTU 0 to MTU2, and MTU6 and MTU7

Figure 24.74 to Figure 24.77 show examples of operation in which MTU6 and MTU7 operate in complementary PWM mode and synchronous counter clearing for MTU 0 to MTU2, and MTU6 and MTU7 is suppressed by setting the SCC bit in TWCRB in MTU6 and MTU7 to 1. In the examples shown in Figure 24.74 to Figure 24.77, synchronous counter clearing occurs at timing (3), (6), (8), and (11) shown in Figure 24.66, respectively. In these examples, the WRE bit in TWCRB in MTU6 and MTU7 is set to 1.

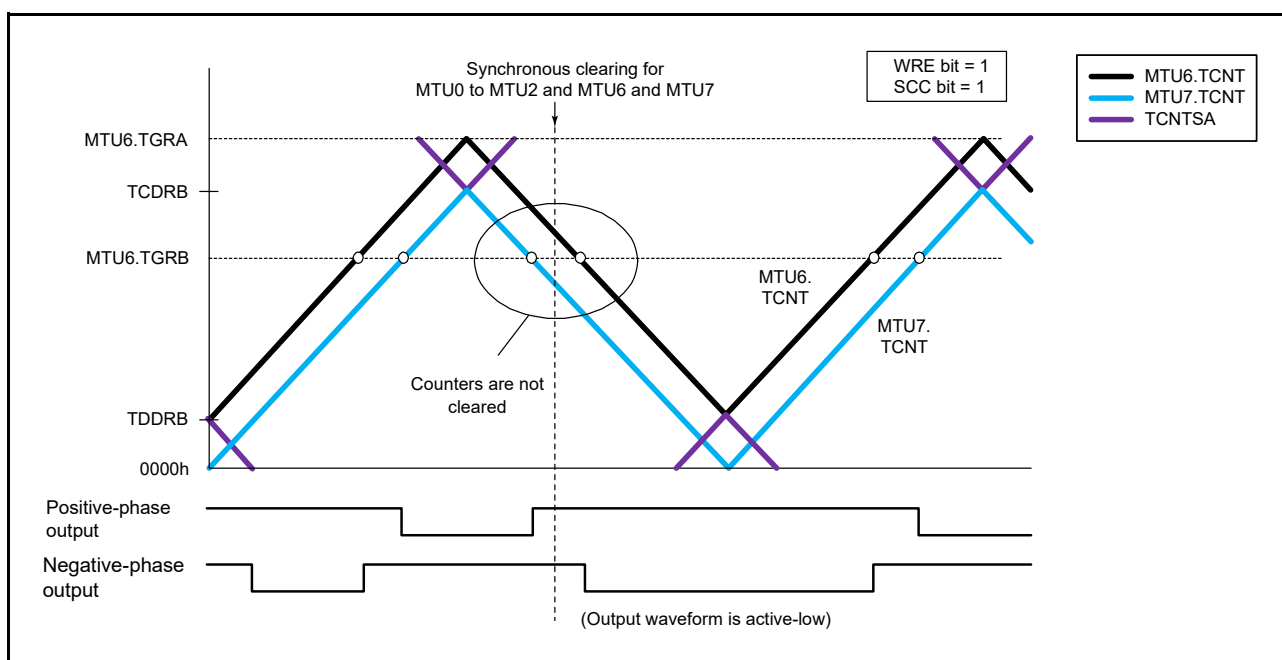


**Figure 24.74** Example of Synchronous Clearing in Dead Time during Up-Counting  
(Timing (3) in Figure 24.66; TWCRB.WRE Bit is 1 and SCC Bit is 1 in MTU6 and MTU7)

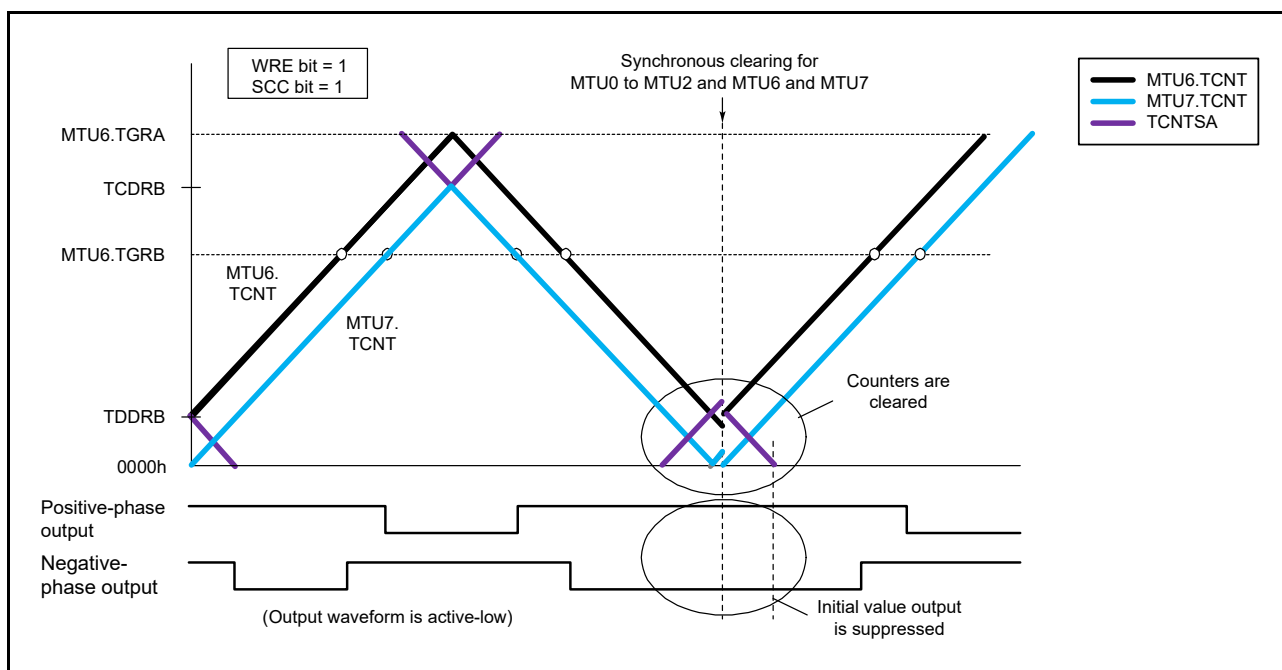


**Figure 24.75** Example of Synchronous Clearing in Tb1 interval  
(Timing (6) in Figure 24.66; TWCRB.WRE Bit is 1 and SCC Bit is 1 in MTU6 and MTU7)





**Figure 24.76** Example of Synchronous Clearing in Dead Time during Down-Counting  
(Timing (8) in Figure 24.66; TWCRB.WRE Bit is 1 and SCC Bit is 1 in MTU6 and MTU7)



**Figure 24.77** Example of Synchronous Clearing in Tb2 interval  
(Timing (11) in Figure 24.66; TWCRB.WRE Bit is 1 and SCC Bit is 1 in MTU6 and MTU7)

**(p) Counter Clearing by MTU3.TGRA (MTU6.TGRA) Compare Match**

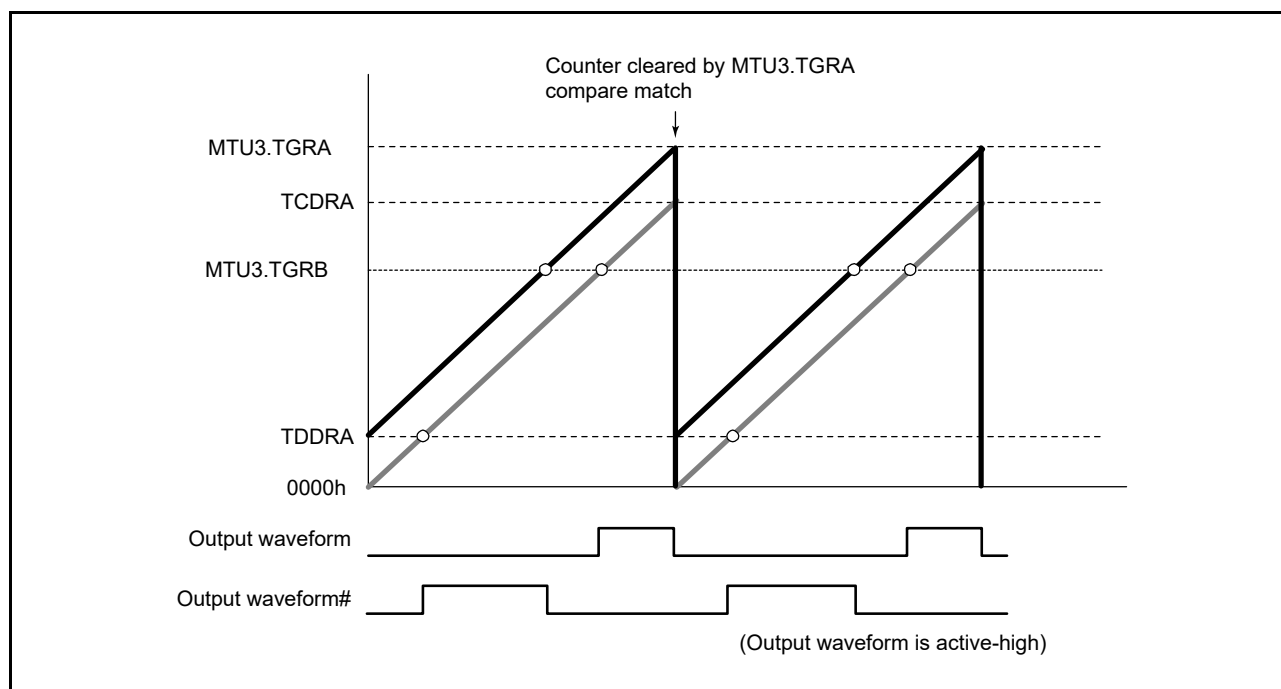
In complementary PWM mode, MTU3.TCNT, MTU4.TCNT, and TCNTSA (MTU6.TCNT, MTU7.TCNT, and TCNTSB) can be cleared by MTU3.TGRA (MTU6.TGRA) compare match when the TWCRA.CCE (TWCRB.CCE) bit. Figure 24.78 illustrates an operation example.

Note 1. Use this function only in complementary PWM mode 1 (transfer at crest).

Note 2. Do not specify synchronous clearing by another channel (do not set 1 in the SYNC0 to SYNC4, or SYNC6 and SYNC7 bits in the timer synchronous register (TSYRA or TSYRB) and the CE0A to CE0D, CE1A, CE1B, CE2A, or CE2B bits in TSYCR).

Note 3. Do not set the PWM duty value to 0000h.

Note 4. Do not set the PSYE bit in timer output control register 1 (TOCR1A or TOCR1B) to 1.



**Figure 24.78 Example of Counter Clearing Operation by MTU3.TGRA Compare Match**

**(q) Example of Waveform Output for Driving AC Synchronous Motor (Brushless DC Motor)**

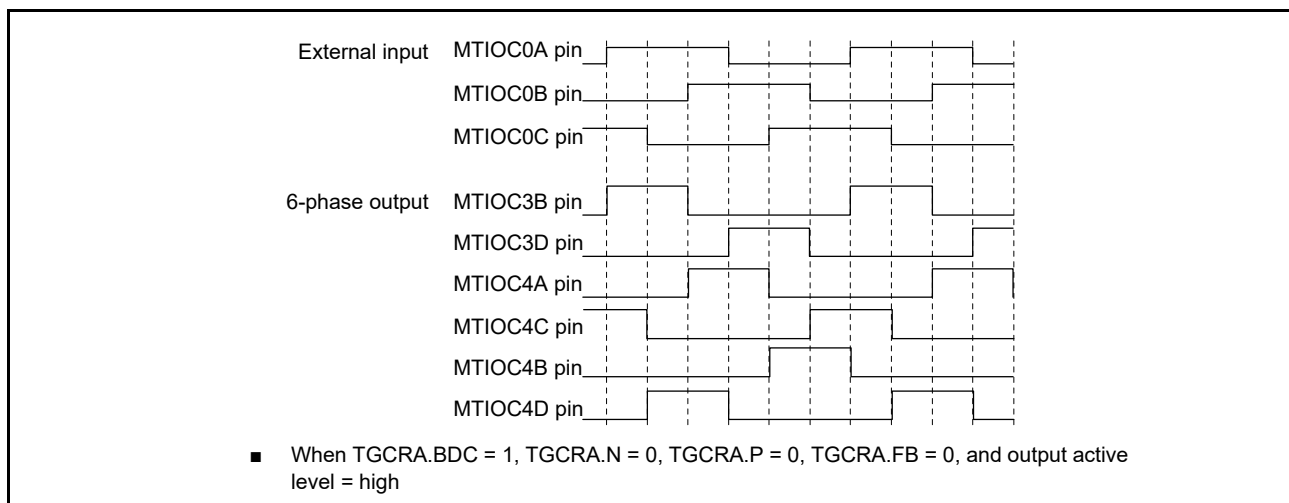
In complementary PWM mode when MTU3 and MTU4 are used, a brushless DC motor can easily be controlled using the TGCRA register. Figure 24.79 to Figure 24.82 show examples of brushless DC motor driving waveforms created using TGCRA.

To switch the output phases for a 3-phase brushless DC motor by means of external signals detected with a Hall element, etc., set the TGCRA.FB bit to 0. In this case, the external signals indicating the magnetic pole position should be input to timer input pins MTIOC0A, MTIOC0B, and MTIOC0C in MTU0 (make appropriate settings with the MPC and port mode registers (PMR) of the I/O ports). When an edge is detected at pin MTIOC0A, MTIOC0B, or MTIOC0C, the output on/off state is switched automatically.

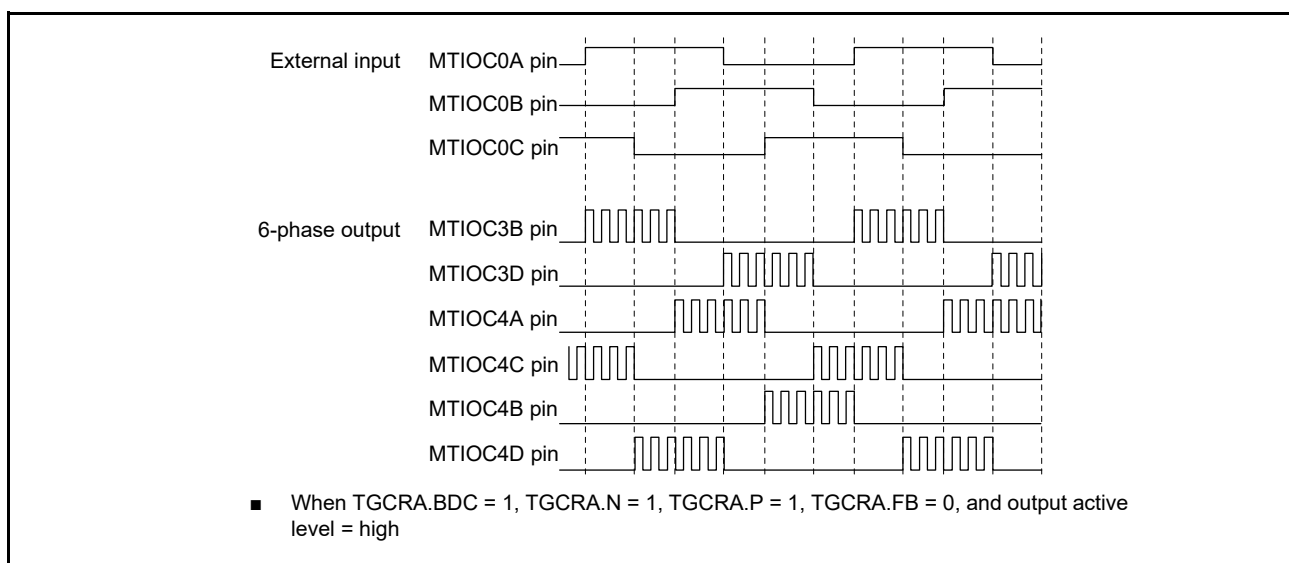
When the TGCRA.FB bit is 1, the output on/off state is switched when the UF, VF, or WF bit in TGCRA is set to 0 or 1. The driving waveforms are output from the 6-phase PWM output pins for complementary PWM mode.

With this 6-phase output, while the output is turned on, chopping output is available through complementary PWM mode output function by setting the N bit or P bit in TGCRA to 1. When the N bit or P bit is 0, the level output is selected.

The active level of the 6-phase output (on output level) can be set with the TOCR1A.OLSN and TOCR1A.OLSP bits regardless of the setting of the N and P bits.



**Figure 24.79 Example of Output Phase Switching by External Input (1)**



**Figure 24.80 Example of Output Phase Switching by External Input (2)**

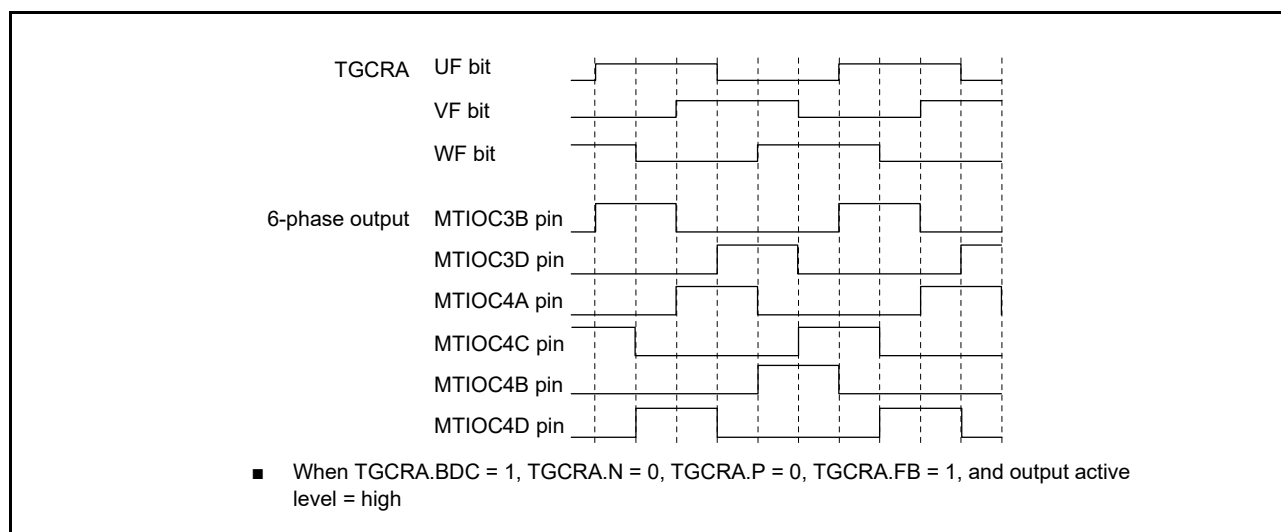


Figure 24.81 Example of Output Phase Switching through UF, VF, and WF Bit Settings (1)

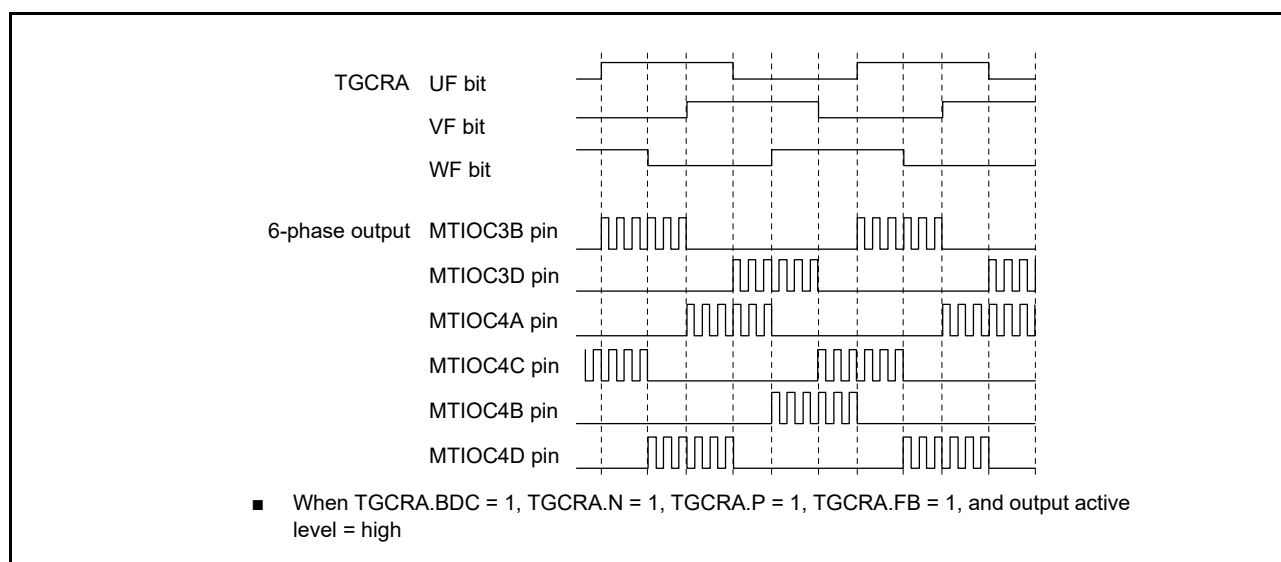


Figure 24.82 Example of Output Phase Switching through UF, VF, and WF Bit Settings (2)

**(r) A/D Conversion Start Request Setting**

In complementary PWM mode, an A/D conversion start request can be issued using MTU3.TGRA (MTU6.TGRA) compare match, MTU4.TCNT (MTU7.TCNT) underflow (trough), or compare match on a channel other than MTU3 and MTU4 (MTU6 and MTU7).

When start requests using MTU3.TGRA (MTU6.TGRA) compare match are specified, A/D conversion can be started at the crest of the MTU3.TCNT (MTU6.TCNT) count.

A/D conversion start requests can be specified by setting the TIER.TTGE bit. To issue an A/D conversion start request at an MTU4.TCNT (MTU7.TCNT) underflow (trough), set the MTU4.TIER.TTGE2 (MTU7.TIER.TTGE2) bit to 1.

**(s) Double Buffer Function in Complementary PWM Mode**

In complementary PWM mode 3 (transfer at the crest and trough), the PWM output setting resolution can be improved from  $\pm 2$  to  $\pm 1$  by setting the TMDR2A.DRS (TMDR2B.DRS) bit to 1.

When setting buffer registers A (MTU3.TGRD, MTU4.TGRC, MTU4.TGRD, MTU6.TGRD, MTU7.TGRC, and MTU7.TGRD), set also buffer registers B (MTU3.TGRE, MTU4.TGRE, MTU4.TGRF, MTU6.TGRE, MTU7.TGRE, and MTU7.TGRF) at the same time. Each buffer register B should be set to the buffer register A value or (buffer register A value – 1). For details of the setting procedure, refer to section 24.3.8 (1), Example of Complementary PWM Mode Setting Procedure

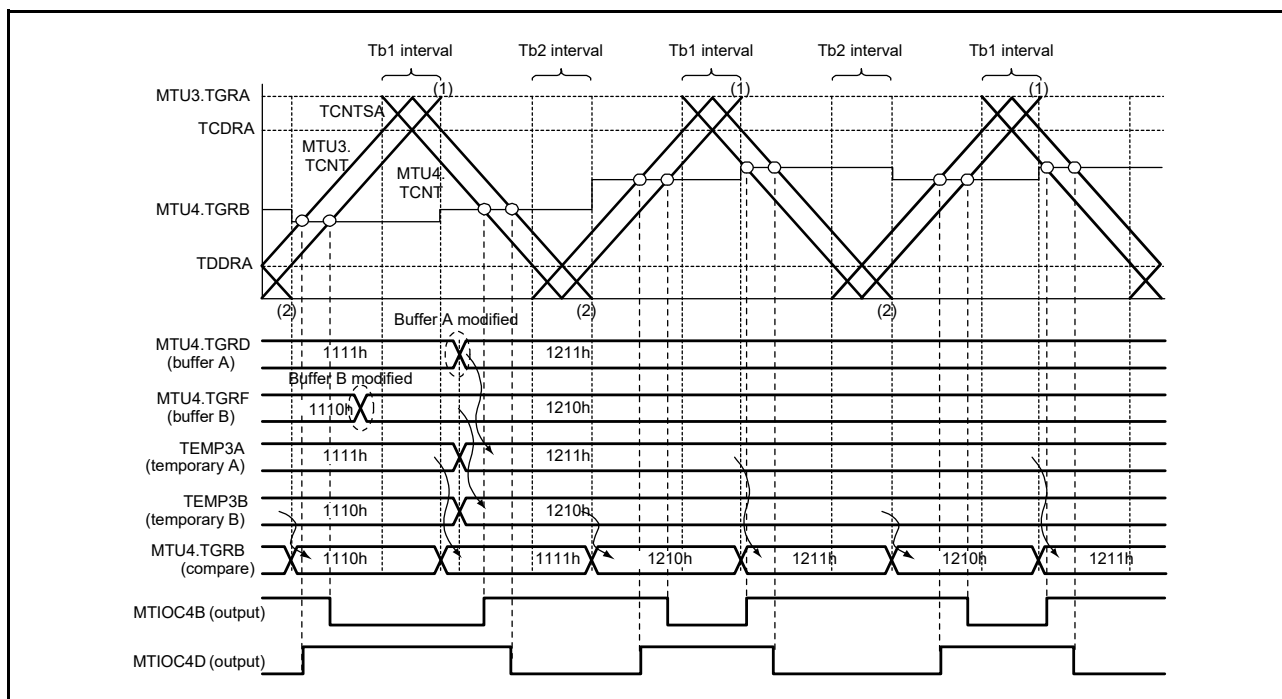
**Note:** When a buffer register B is set to the buffer register A value, symmetric PWM waveforms are output. When a buffer register B is set to (buffer register A value – 1), asymmetric PWM waveforms are output.

Figure 24.83 shows an example of double buffer operation.

Each register data is transferred as follows.

- After MTU4.TGRD or MTU7.TGRD (buffer A) is written to, data is transferred from MTU4.TGRD or MTU7.TGRD (buffer A) to TEMP3A or TEMP6A (temporary A) and from MTU4.TGRF or MTU7.TGRF (buffer B) to TEMP3B or TEMP6B (temporary B).
- With timing (1) in the figure, data is transferred from TEMP3A or TEMP6A (temporary A) to MTU4.TGRB or MTU7.TGRB (compare).
- With timing (2) in the figure, data is transferred from TEMP3B or TEMP6B (temporary B) to MTU4.TGRB or MTU7.TGRB (compare).

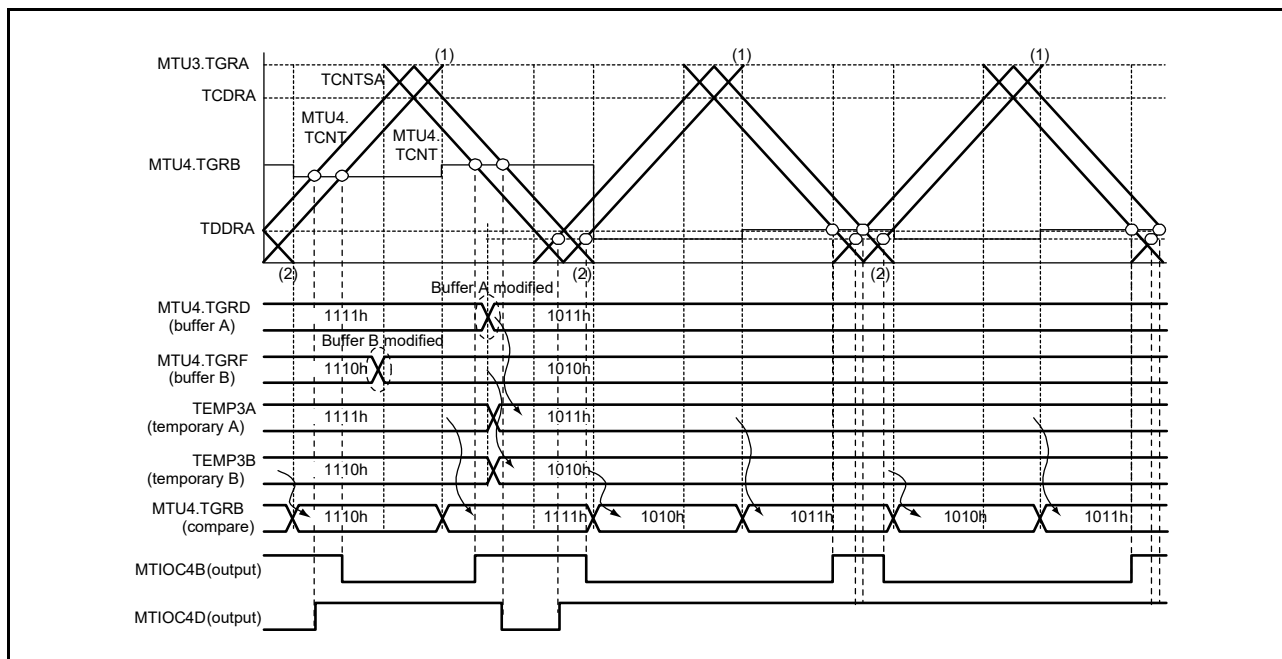
In the crest interval (Tb1 interval), the compare register and temporary register A are valid; in the trough interval (Tb2 interval), the compare register and temporary register B are valid.



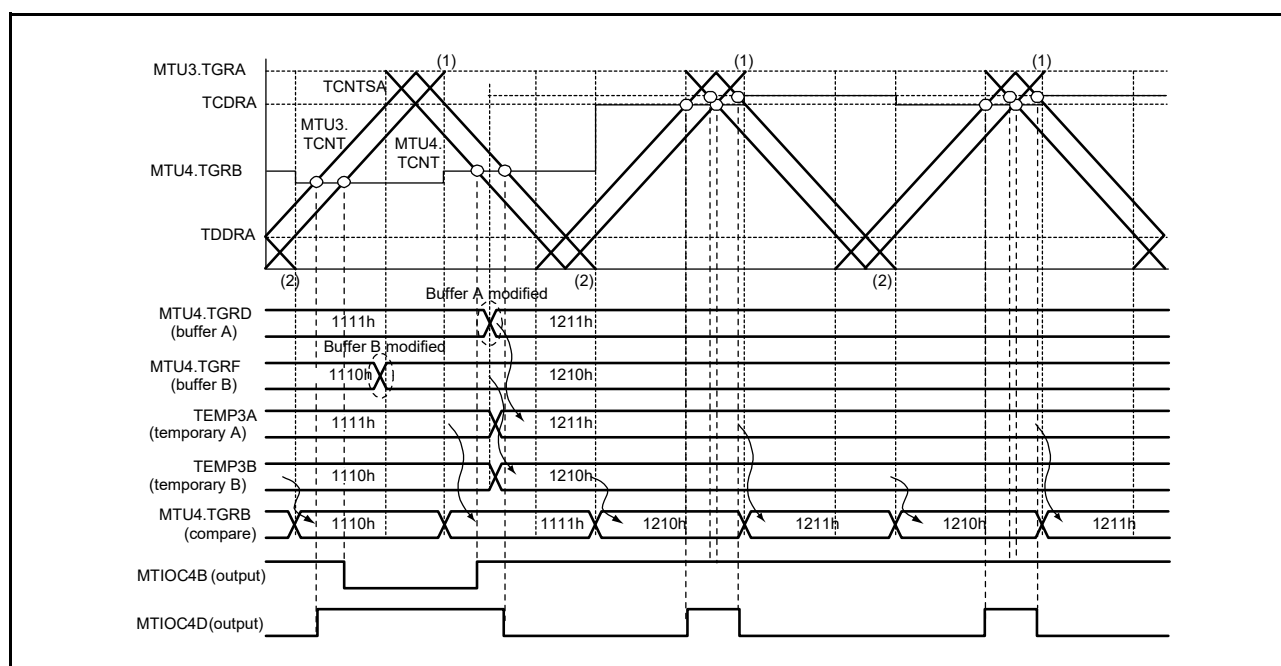
**Figure 24.83 Example of Double Buffer Operation**

Figure 24.84 shows an example when the buffer write value is smaller than the TDDRA (TDDRB) value, and Figure 24.85 shows an example when the write value is greater than TCDRA (TCDRB).

In the crest interval, the output is controlled according to the compare match with the compare register or temporary register A; in the trough interval, the output is controlled according to the compare match with the compare register or temporary register B.



**Figure 24.84 Example of Double Buffer Operation (Buffer Write Value is Smaller than TDDRA)**



**Figure 24.85 Example of Double Buffer Operation (Buffer Write Value is Greater than TCDRA)**

### (3) Interrupt Skipping Function 1 in Complementary PWM Mode

Interrupts TGIA3 (TGIA6) (at the crest) and TCIV4 (TCIV7) (at the trough) in MTU3 and MTU4 (MTU6 and MTU7) can be skipped up to seven times by making settings in the TITCR1A (TITCR1B) register.

Transfers from a buffer register to a temporary register or a compare register can be skipped in coordination with interrupt skipping by making settings in the TBTERA (TBTERB) register. For the linkage with buffer registers, refer to description (c), Buffer Transfer Control Linked with Interrupt Skipping, below.

A/D conversion start requests generated by the A/D conversion start request delaying function can also be skipped in coordination with interrupt skipping by making settings in the MTU4.TADCR (MTU7.TADCR) register. For the linkage with the A/D conversion start request delaying function, refer to section 24.3.9, A/D Conversion Start Request Delaying Function.

The TITCR1A (TITCR1B) register should be set while interrupt skipping function 1 is selected by setting the TITM bit in the timer interrupt skipping mode register (TITMRA or TITMRB) to 0, TGIA3 (TGIA6) interrupt requests are disabled by setting the MTU3.TIER (MTU6.TIER) register, TCIV4 (TCIV7) interrupt requests are disabled by setting the MTU4.TIER (MTU7.TIER) register, and a compare match is not generated. Before changing the skipping count, be sure to set the T3AEN (T6AEN) and T4VEN (T7VEN) bits to 0 to clear the skipping counter.

#### (a) Example of Interrupt Skipping Function 1 Setting Procedure

Figure 24.86 shows an example of the interrupt skipping function 1 setting procedure. Figure 24.87 shows the periods during which interrupt skipping count can be changed.

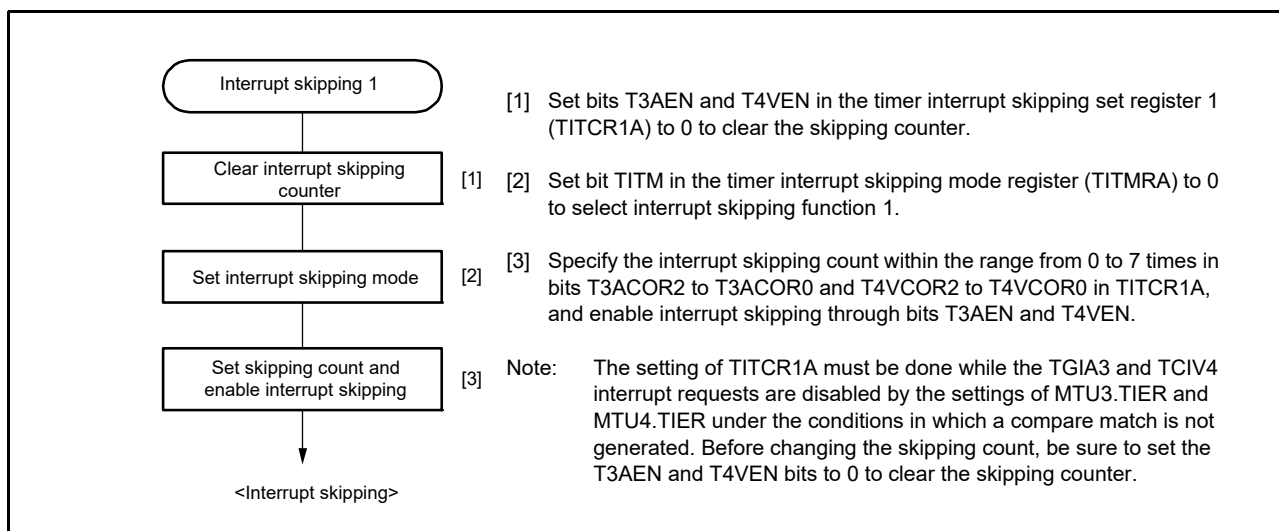


Figure 24.86 Example of Interrupt Skipping Function 1 Setting Procedure

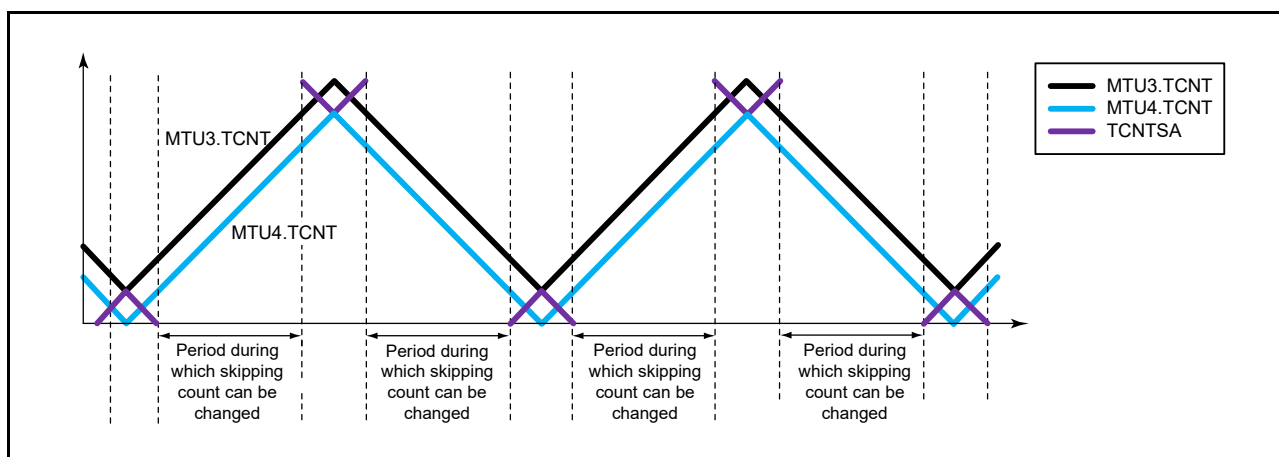
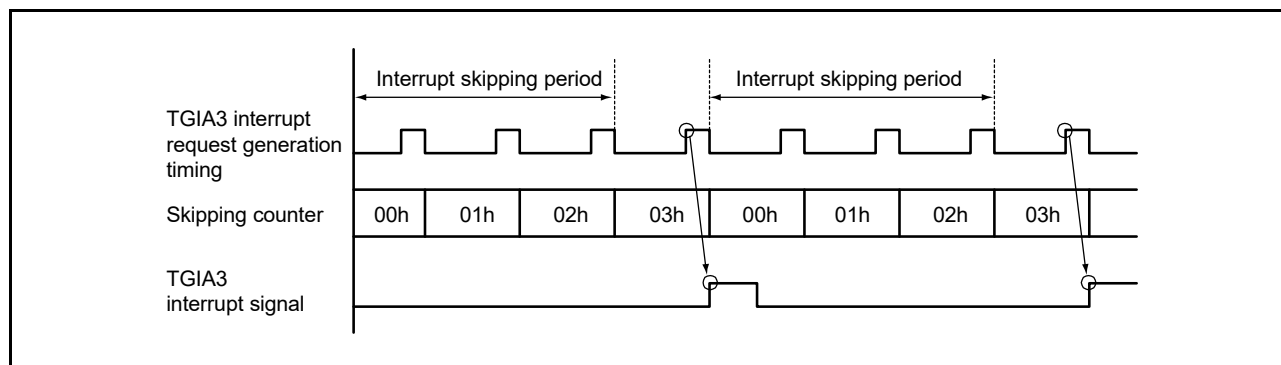


Figure 24.87 Periods during which Interrupt Skipping Count can be Changed



## (b) Example of Interrupt Skipping Function 1

Figure 24.88 shows an example of TGIA3 (TGIA6) interrupt skipping in which the interrupt skipping count is set to three by the T3ACOR (T6ACOR) bits and the T3AEN (T6AEN) bit is set to 1 in the TITCR1A (TITCR1B) register.



**Figure 24.88** Example of Interrupt Skipping Function 1

### (c) Buffer Transfer Control Linked with Interrupt Skipping

In complementary PWM mode, whether to transfer data from a buffer register to a temporary register and whether to link the transfer with interrupt skipping can be specified with the BTE[1:0] bits in the TBTERA (TBTERB) register.

Figure 24.89 shows an example of operation when buffer transfer is disabled (BTE[1:0] = 01b). While this setting is valid, data is not transferred from the buffer register to the temporary register.

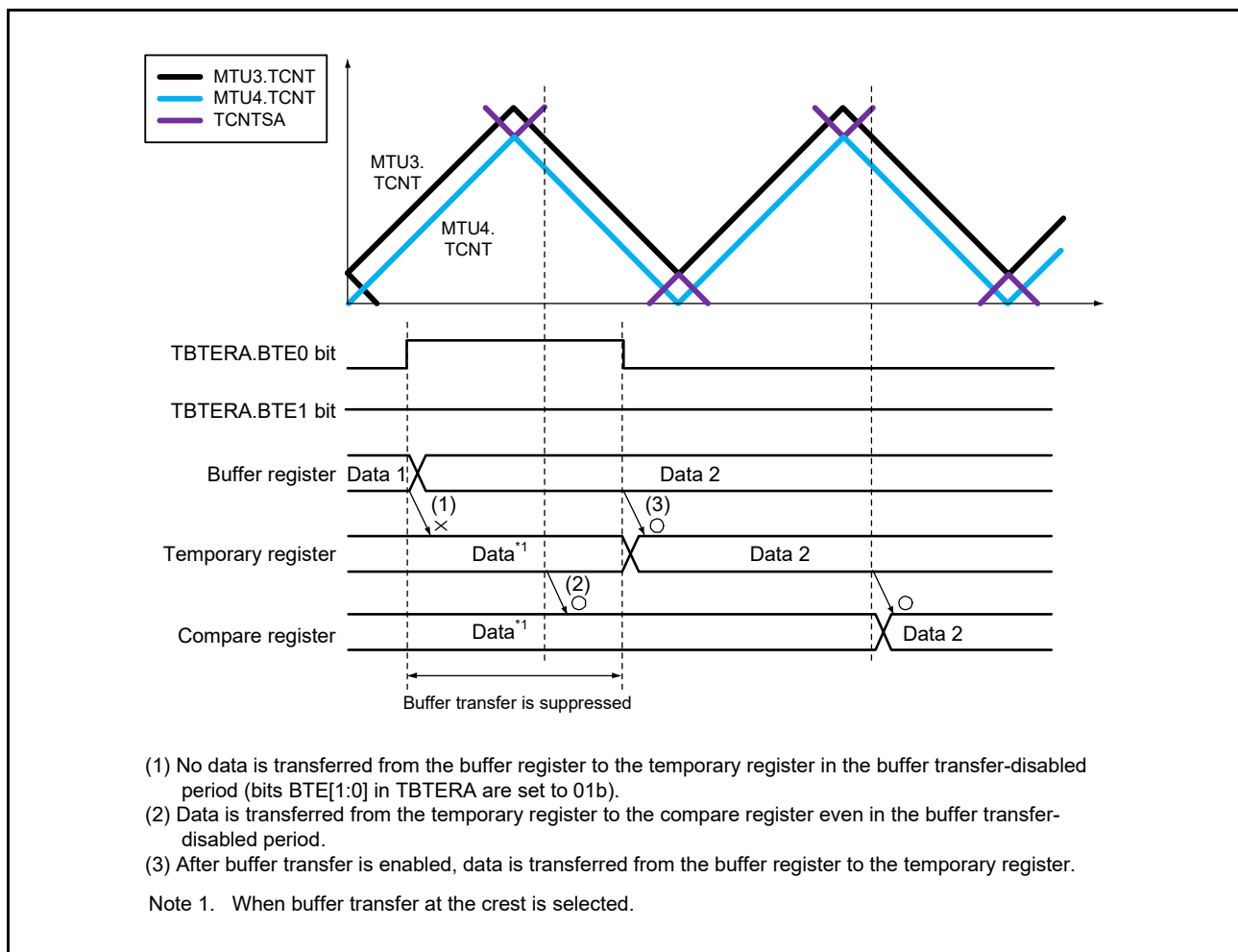
Figure 24.90 shows an example of operation when buffer transfer is linked with interrupt skipping (BTE[1:0] = 10b). While this setting is valid, data is not transferred from the buffer register outside the buffer transfer-enabled period.

Note that the buffer transfer-enabled period differs depending on whether only the T3AEN (T6AEN) bit in the TITCR1A (TITCR1B) register is set to 1, only the T4VEN (T7VEN) bit in the TITCR1A (TITCR1B) register is set to 1, or both the T3AEN and T4VEN (T6AEN and T7VEN) bits are set to 1. Figure 24.91 shows the relationship between the T3AEN (T6AEN) and T4VEN (T7VEN) bit settings in TITCR1A (TITCR1B) and buffer transfer-enabled period.

**Note:** This function must be used in combination with interrupt skipping function 1.

When interrupt skipping is disabled (the T3AEN and T4VEN (T6AEN and T7VEN) bits in the timer interrupt skipping set register 1 (TITCR1A or TITCR1B) are set to 0 or the skipping count set bits (T3ACOR and T4VCOR (T6ACOR and T7VCOR)) in TITCR1A (TITCR1B) are set to 0), make sure that buffer transfer is not linked with interrupt skipping (set the BTE1 bit in TBTERA or TBTERB to 0).

If buffer transfer is linked with interrupt skipping while interrupt skipping is disabled, buffer transfer is never performed.



**Figure 24.89** Example of Operation When Buffer Transfer is Disabled (BTE[1:0] = 01b)

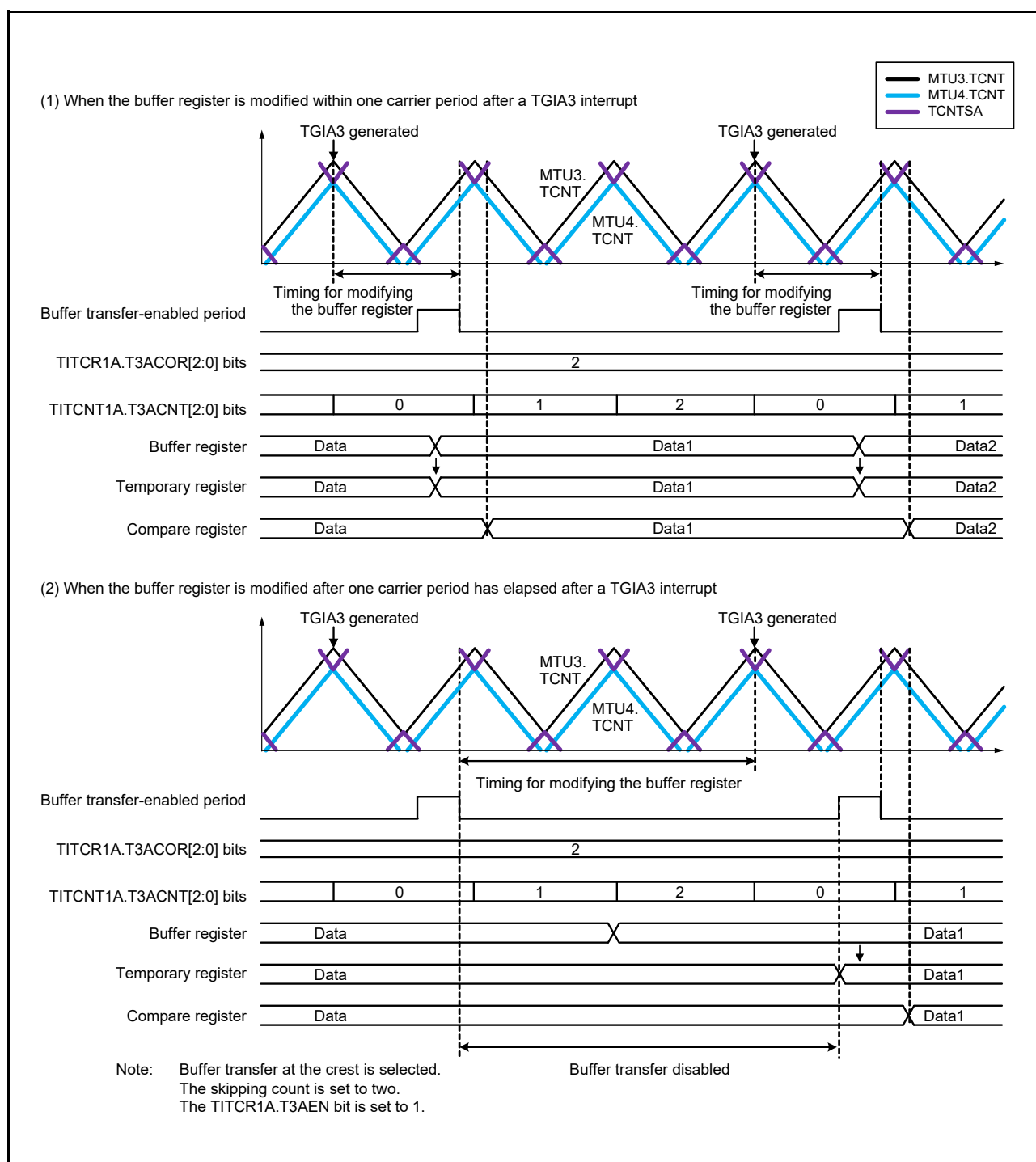
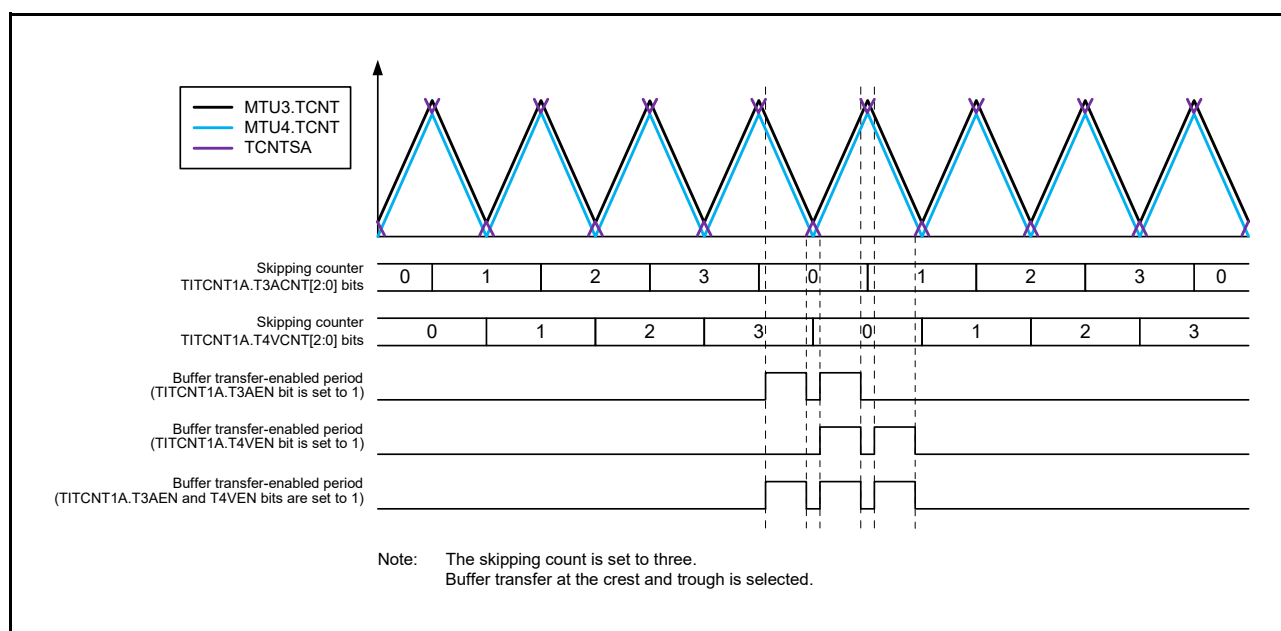


Figure 24.90 Example of Operation When Buffer Transfer is Linked with Interrupt Skipping (BTE[1:0] = 10b)



**Figure 24.91 Relationship between Bits T3AEN and T4VEN in TITCR1A and Buffer Transfer-Enabled Period**

#### (4) Complementary PWM Mode Output Protection Functions

The following output protection functions are provided for complementary PWM mode.

##### (a) Register and Counter Miswrite Prevention Function

Access from the CPU to the mode registers, control registers, compare registers, and counters can be enabled or disabled by setting the RWE bit in the TRWERA (TRWERB) register. The applicable registers are some of the registers in MTU3, MTU4, MTU6, and MTU7 shown below:

47 registers in total

MTU3.TCR, MTU4.TCR, MTU3.TCR2, MTU4.TCR2, MTU3.TMDR1, MTU4.TMDR1, MTU3.TIORH, MTU4.TIORH, MTU3.TIORL, MTU4.TIORL, MTU3.TIER, MTU4.TIER, MTU3.TCNT, MTU4.TCNT, MTU3.TGRA, MTU4.TGRA, MTU3.TGRB, MTU4.TGRB, MTU.TOERA, MTU.TOCR1A, MTU.TOCR2A, MTU.TGCRA, MTU.TCDRA, MTU.TDDRA, MTU6.TCR, MTU7.TCR, MTU6.TCR2, MTU7.TCR2, MTU6.TMDR1, MTU7.TMDR1, MTU6.TIORH, MTU7.TIORH, MTU6.TIORL, MTU7.TIORL, MTU6.TIER, MTU7.TIER, MTU6.TCNT, MTU7.TCNT, MTU6.TGRA, MTU7.TGRA, MTU6.TGRB, MTU7.TGRB, MTU.TOERB, MTU.TOCR1B, MTU.TOCR2B, MTU.TCDRB, and MTU.TDDRB

This function can disable CPU access to the mode registers, control registers, and counters to prevent miswriting due to CPU runaway. In the access-disabled state, the applicable registers are read as undefined and writing to these registers is ignored.

##### (b) Halting of PWM Output by External Signal

The PWM output pins of MTU0, MTU3, MTU4, MTU6, and MTU7 can be set to the high-impedance state automatically.

Refer to section 25, Port Output Enable 3 (POE3a), for details.

### 24.3.9 A/D Conversion Start Request Delaying Function

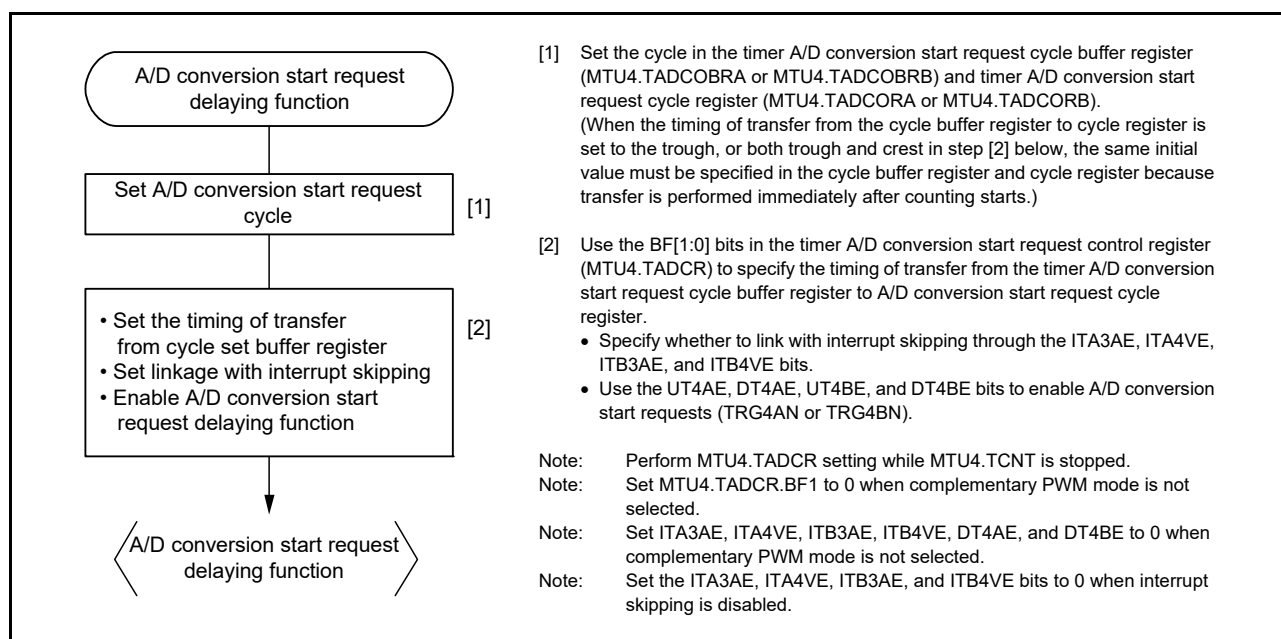
A/D conversion start requests can be issued in MTU4 or MTU7 by making settings in the timer A/D conversion start request control register (MTU4.TADCR or MTU7.TADCR), timer A/D conversion start request cycle set registers (MTU4.TADCORA and MTU4.TADCORB, or MTU7.TADCORA and MTU7.TADCORB), and timer A/D conversion start request cycle set buffer registers (MTU4.TADCOBRA and MTU4.TADCOBRB, or MTU7.TADCOBRA and MTU7.TADCOBRB).

The A/D conversion start request delaying function compares MTU4.TCNT with MTU4.TADCORA or MTU4.TADCORB (MTU7.TCNT with MTU7.TADCORA or MTU7.TADCORB), and when their values match, the function issues a respective A/D conversion start request (TRG4AN or TRG4BN (TRG7AN or TRG7BN)).

A/D conversion start requests (TRG4AN and TRG4BN (TRG7AN and TRG7BN)) can be skipped in coordination with interrupt skipping by making settings in the ITA3AE, ITA4VE, ITB3AE, and ITB4VE bits in MTU4.TADCR (the ITA6AE, ITA7VE, ITB6AE, and ITB7VE bits in MTU7.TADCR).

#### (1) Example of Procedure for Specifying A/D Conversion Start Request Delaying Function

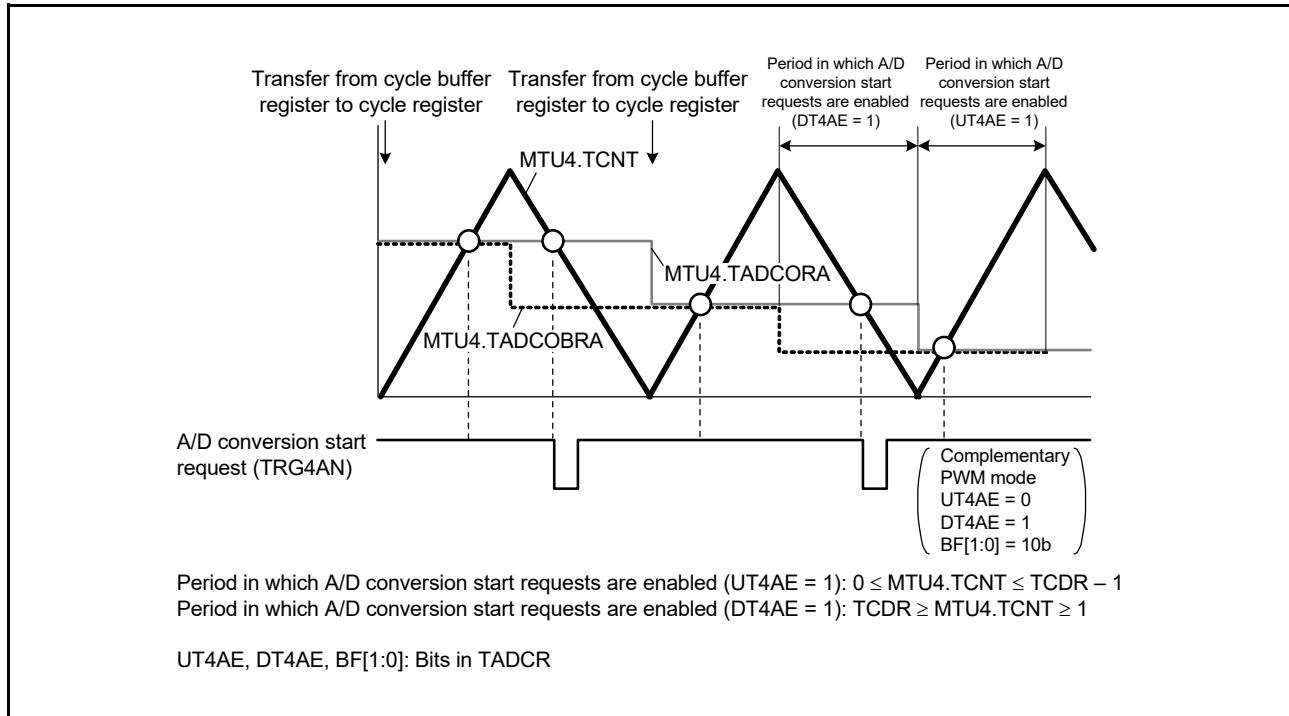
Figure 24.92 shows an example of procedure for specifying the A/D conversion start request delaying function.



**Figure 24.92 Example of Procedure for Specifying A/D Conversion Start Request Delaying Function (MTU3 and MTU4)**

### (2) Basic Example of A/D Conversion Start Request Delaying Function Operation

Figure 24.93 shows a basic example of A/D conversion start request signal (TRG4AN (TRG7AN)) operation when the trough of MTU4.TCNT (MTU7.TCNT) is specified for the buffer transfer timing and an A/D conversion start request signal is output during MTU4.TCNT (MTU7.TCNT) down-counting.



**Figure 24.93 Basic Example of A/D Conversion Start Request Signal (TRG4AN) Operation**

### (3) Period in Which A/D Conversion Start Requests are Enabled

When the MTU4.TCNT (MTU7.TCNT) counter and the MTU4.TADCORA or MTU4.TADCOBR (MTU7.TADCORA or MTU7.TADCOBR) register matches within the period enabled by the UT4AE and UT4BE (UT7AE and UT7BE) bits, the corresponding A/D conversion start request (TRG4AN or TRG4BN) is issued.

When the UT4AE and UT4BE (UT7AE and UT7BE) bits in the MTU4.TADCR (MTU7.TADCR) register are set to 1 in complementary PWM mode, A/D conversion start requests are enabled during the MTU4.TCNT (MTU7.TCNT) up-counting ( $0 \leq \text{MTU4.TCNT (MTU7.TCNT)} \leq \text{TCDR} - 1$ ). When the DT4AE and DT4BE (DT7AE and DT7BE) bits in the MTU4.TADCR (MTU7.TADCR) register are set to 1, A/D conversion start requests are enabled during MTU4.TCNT (MTU7.TCNT) down-counting ( $\text{TCDR} \geq \text{MTU4.TCNT (MTU7.TCNT)} \geq 1$ ). Refer to Figure 24.93.

#### (4) Buffer Transfer

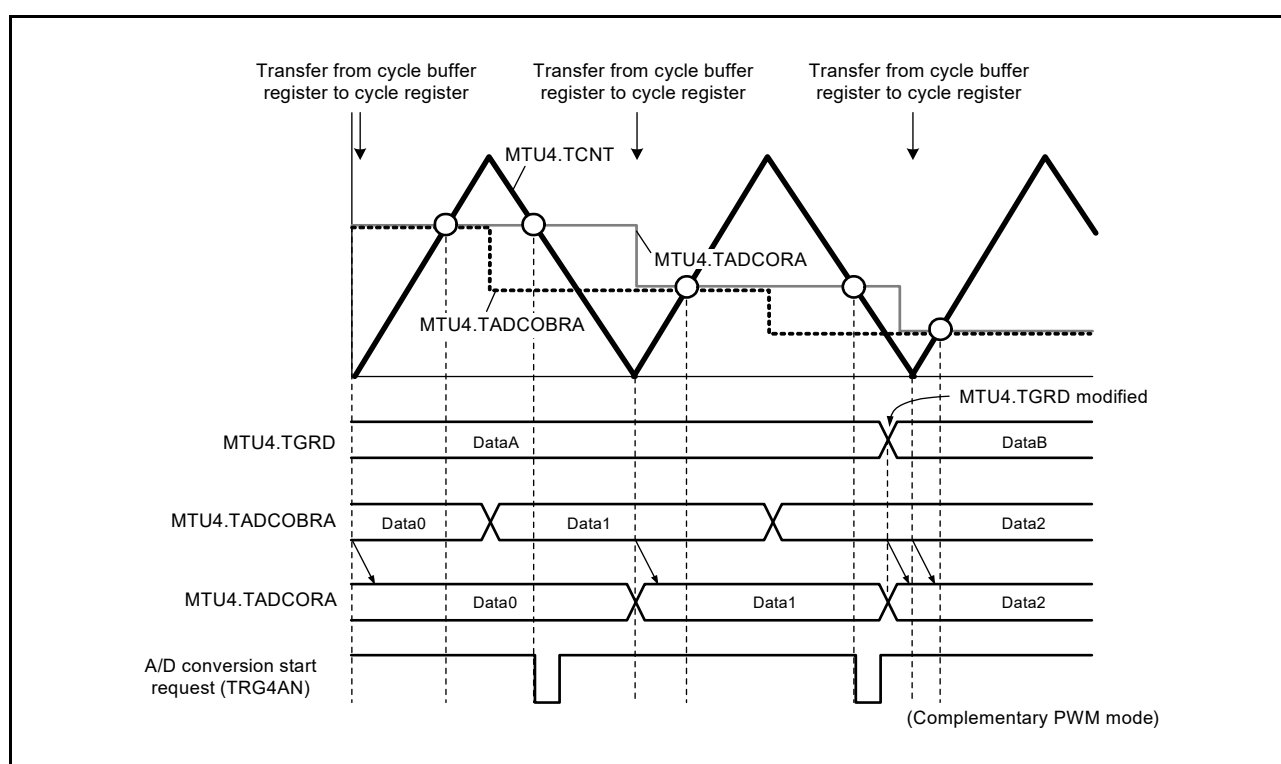
The data in the timer A/D conversion start request cycle set registers (MTU4.TADCORA and MTU4.TADCORB, or MTU7.TADCORA and MTU7.TADCORB) is updated by writing data to the timer A/D conversion start request cycle set buffer registers (MTU4.TADCOBRA and MTU4.TADCOBRB, or MTU7.TADCOBRA and MTU7.TADCOBRB). Data is transferred from the buffer registers to the respective cycle set registers at the timing selected with the BF[1:0] bits in the MTU4.TADCR (MTU7.TADCR) register.

In complementary PWM mode, data is also transferred from the timer A/D conversion start request cycle set buffer registers to the timer A/D conversion start request cycle set registers when MTU4.TGRD (MTU7.TGRD) register is updated.

There are notes on the timing for transferring data when using buffer transfer in complementary PWM mode.

For details, section 24.6.28, Usage Notes on A/D Conversion Start Request Delaying Function in Complementary PWM Mode.

In modes other than complementary PWM mode, set the BF1 bit in the MTU4.TADCR (MTU7.TADCR) register to 0.



**Figure 24.94 Example of A/D Conversion Start Request Signal (TRG4AN) and Buffer Transfer Operation**



### (5) A/D Conversion Start Request Delaying Function Linked with Interrupt Skipping Function 1

In complementary PWM mode, A/D conversion start requests (TRG4AN and TRG4BN (TRG7AN and TRG7BN)) can be issued in coordination with interrupt skipping 1 by making settings in the ITA3AE, ITA4VE, ITB3AE, and ITB4VE (ITA6AE, ITA7VE, ITB6AE, and ITB7VE) bits in the MTU4.TADCR (MTU7.TADCR) register.

Figure 24.95 shows an example of A/D conversion start request signal (TRG4AN (TRG7AN)) operation when TRG4AN (TRG7AN) output is enabled during MTU4.TCNT (MTU7.TCNT) up-counting and down-counting and A/D conversion start requests are linked with interrupt skipping 1.

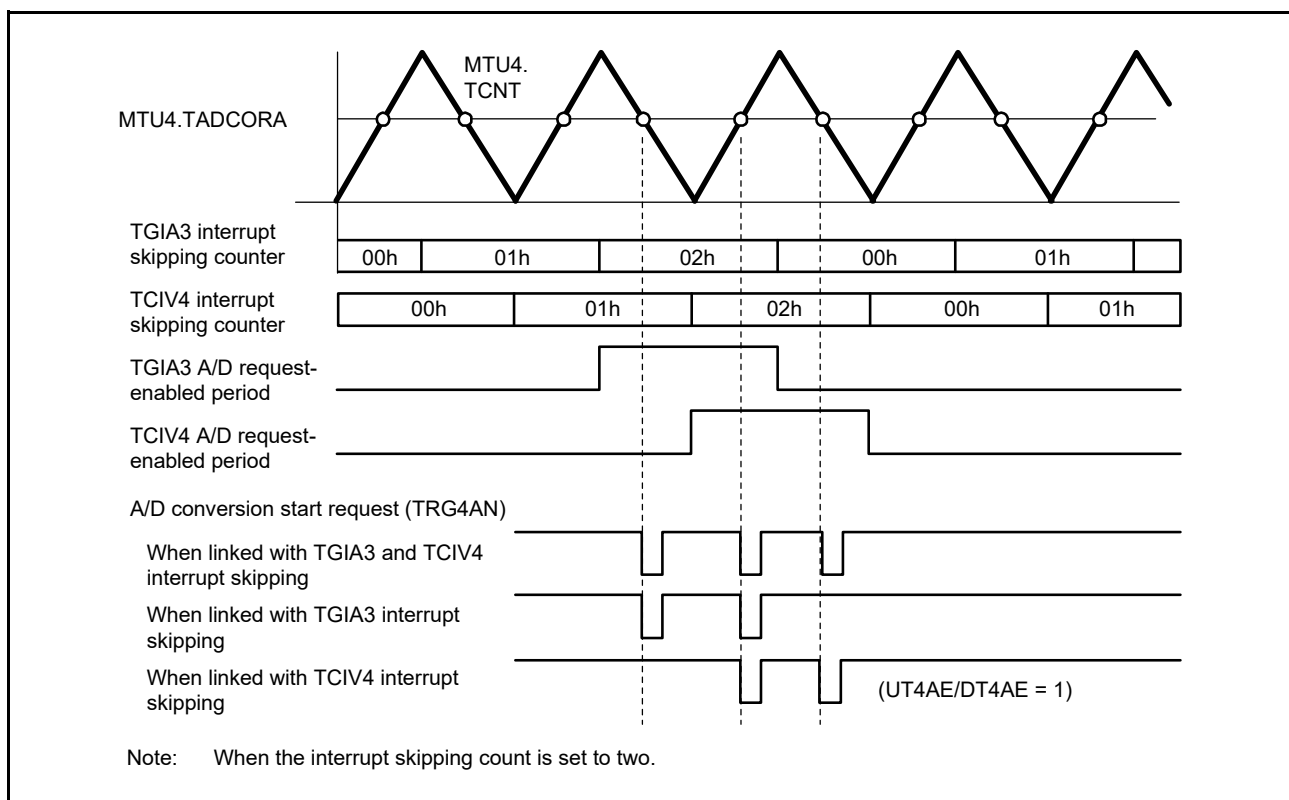
Figure 24.96 shows another example of A/D conversion start request signal (TRG4AN (TRG7AN)) operation when TRG4AN (TRG7AN) output is enabled during MTU4.TCNT (MTU7.TCNT) up-counting and A/D conversion start requests are linked with interrupt skipping 1.

In modes other than complementary PWM mode, do not use the A/D conversion start request delaying function linked with the interrupt skipping function 1.

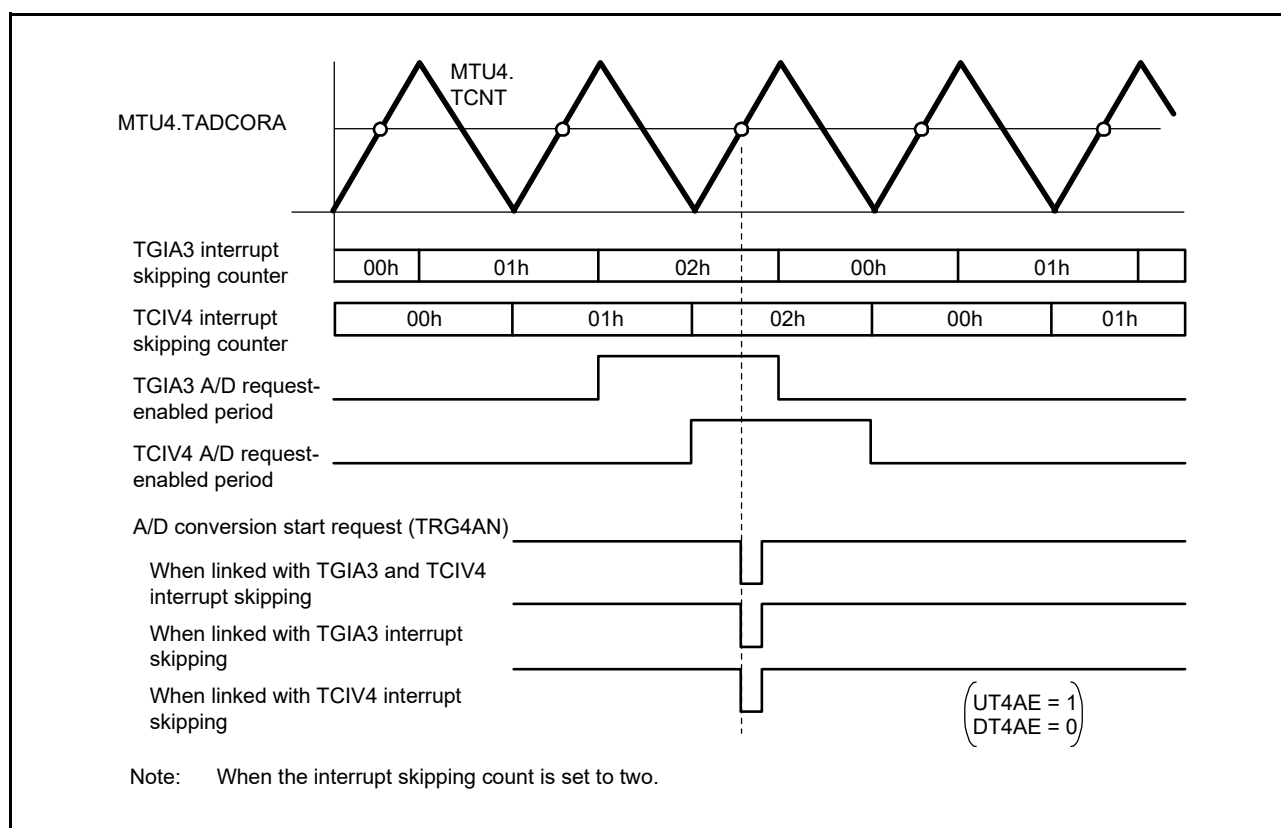
Set the ITA3AE, ITA4VE, ITB3AE, and ITB4VE (ITA6AE, ITA7VE, ITB6AE, and ITB7VE) bits in the MTU4.TADCR (MTU7.TADCR) register to 0.

**Note:** This function should be used in combination with interrupt skipping 1.

When interrupt skipping is disabled (the T3AEN and T4VEN (T6AEN and T7VEN) bits in the timer interrupt skipping set register (TITCR1A (TITCR1B)) are set to 0 or the skipping count set bits (T3ACOR and T4VCOR (T6ACOR and T7VCOR)) in TITCR1A (TITCR1B) are set to 0), make sure that A/D conversion start requests are not linked with interrupt skipping 1 (set the ITA3AE, ITA4VE, ITB3AE, and ITB4VE (ITA6AE, ITA7VE, ITB6AE, and ITB7VE) bits in the timer A/D conversion start request control register (MTU4.TADCR (MTU7.TADCR)) to 0). When this function is used, MTU4.TADCORA and MTU4.TADCORB (MTU7.TADCORA and MTU7.TADCORB) should be set with the value ranging 0002h to the value set in TCDRA minus 2 (value set in TCDRB minus 2).



**Figure 24.95** Example of A/D Conversion Start Request Signal (TRG4AN) Operation Linked with Interrupt Skipping Function 1 (UT4AE and DT4AE = 1)



**Figure 24.96** Example of A/D Conversion Start Request Signal (TRG4AN) Operation Linked with Interrupt Skipping Function 1 (UT4AE = 1, DT4AE = 0)

### (6) A/D Conversion Start Request Delaying Function Linked with Interrupt Skipping Function 2

By setting the TITM bit to 1 in the TITMRA (TITMRB) register, the counter starts down-counting from the value (0 to 7) set in the TRG4COR[2:0] (TRG7COR[2:0]) bits in TITCR2A (TITCR2B) register every time an A/D conversion start trigger (TRG4AN or TRG4BN (TRG7AN or TRG7BN)) is generated. When the counter value reaches 0 and is reloaded, the TRG4AN and TRG4BN (TRG7AN and TRG7BN) interrupts become valid and an A/D conversion start request signal (TRG4ABN (TRG7ABN)) is output.

This function is valid only when the A/D conversion start request delaying function is enabled.

#### (a) Example of Procedure for Setting Interrupt Skipping Function 2

Figure 24.97 shows an example of procedure for setting interrupt skipping function 2.

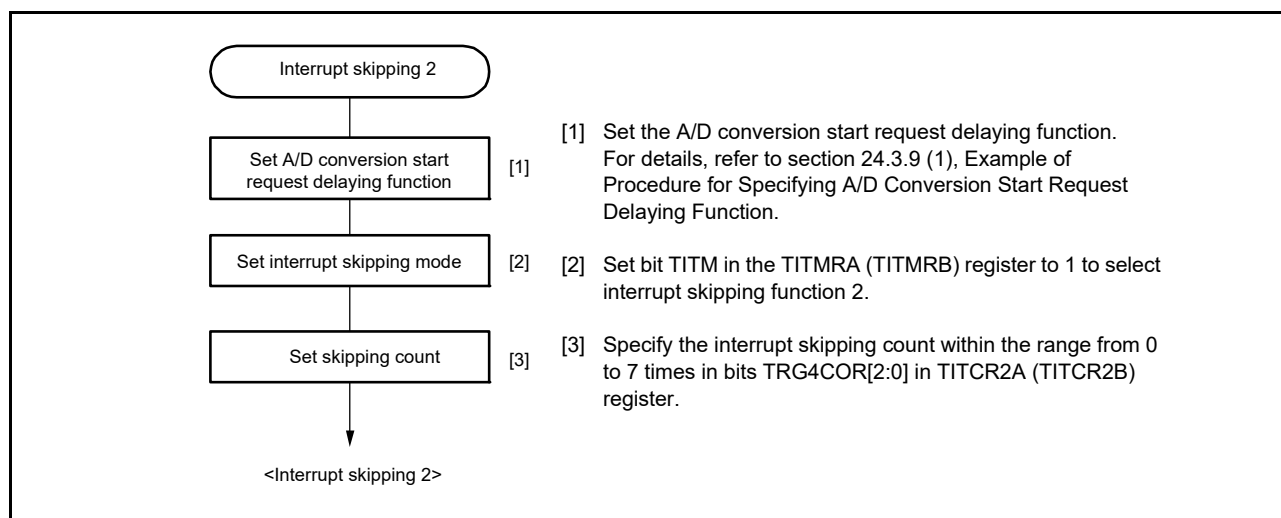


Figure 24.97 Example of Procedure for Setting Interrupt Skipping Function 2

#### (b) Example of Interrupt Skipping Function 2 Operation

Figure 24.98 shows an example of interrupt skipping function 2 operation.

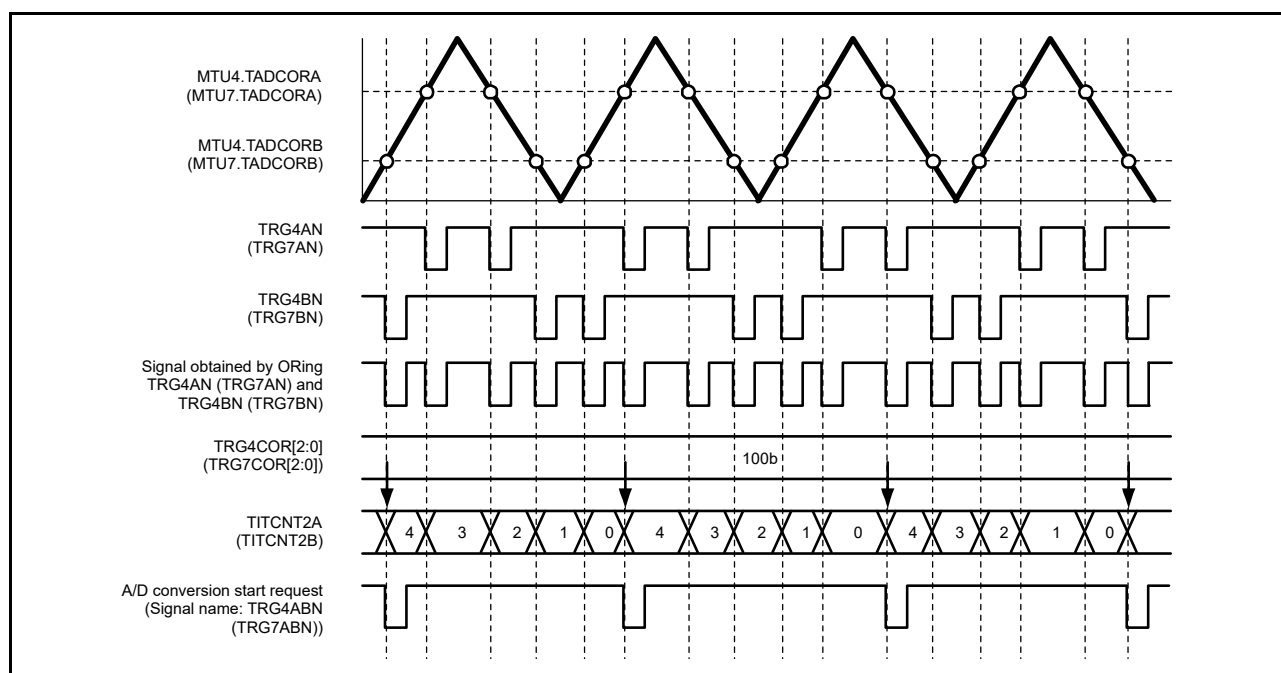


Figure 24.98 Example of Interrupt Skipping Function 2 Operation (Skipping Count is Set to Four)

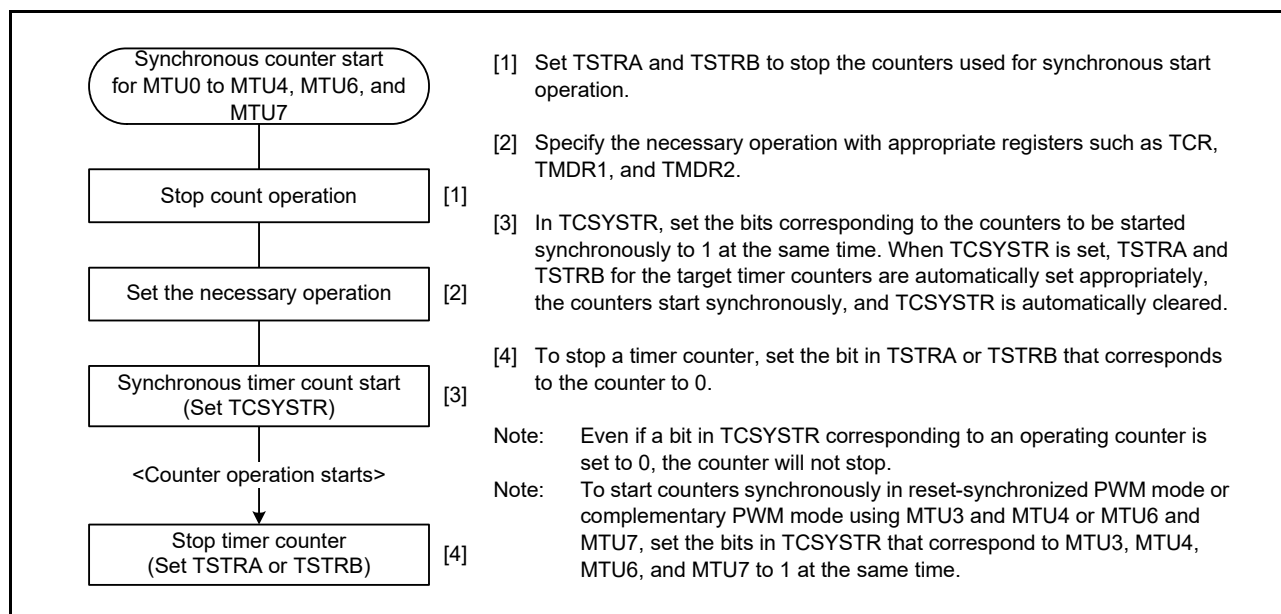
### 24.3.10 Synchronous Operation of MTU0 to MTU4, MTU6, and MTU7

#### (1) Synchronous Counter Start for MTU0 to MTU4, MTU6, and MTU7

The counters in MTU0 to MTU4, MTU6, and MTU7 can be started synchronously by making the TCSYSTR settings.

#### (a) Example of Procedure for Setting Synchronous Counter Start for MTU0 to MTU4, MTU6, and MTU7

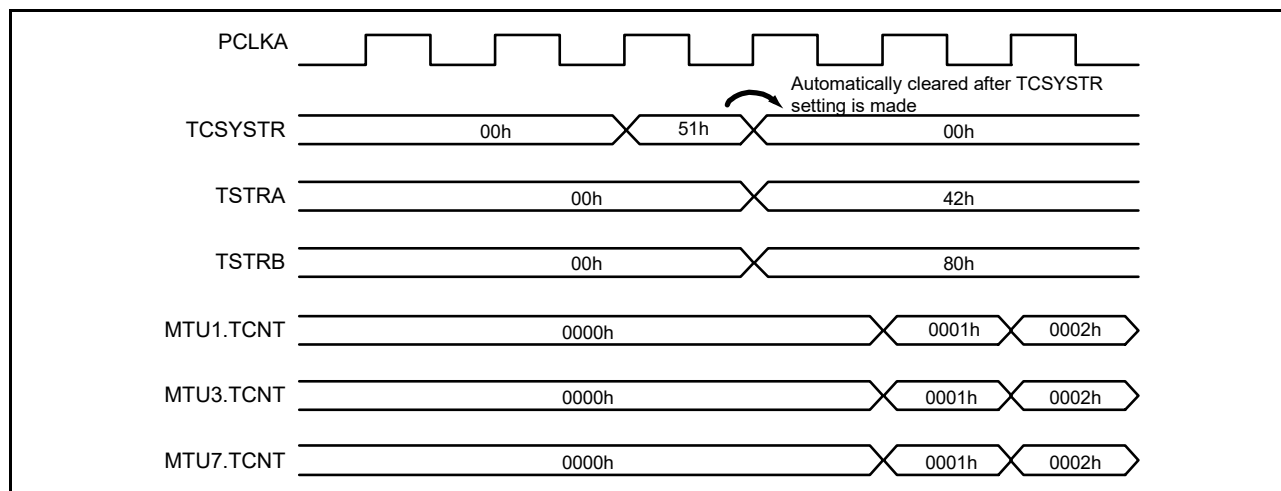
Figure 24.99 shows an example of procedure for setting synchronous counter start for MTU0 to MTU4, MTU6, and MTU7.



**Figure 24.99 Example of Procedure for Setting Synchronous Counter Start for MTU0 to MTU4, MTU6, and MTU7**

#### (b) Examples of Synchronous Counter Start Operation

Figure 24.100 shows an examples of synchronous counter start operation for MTU0 to MTU4, MTU6, and MTU7.



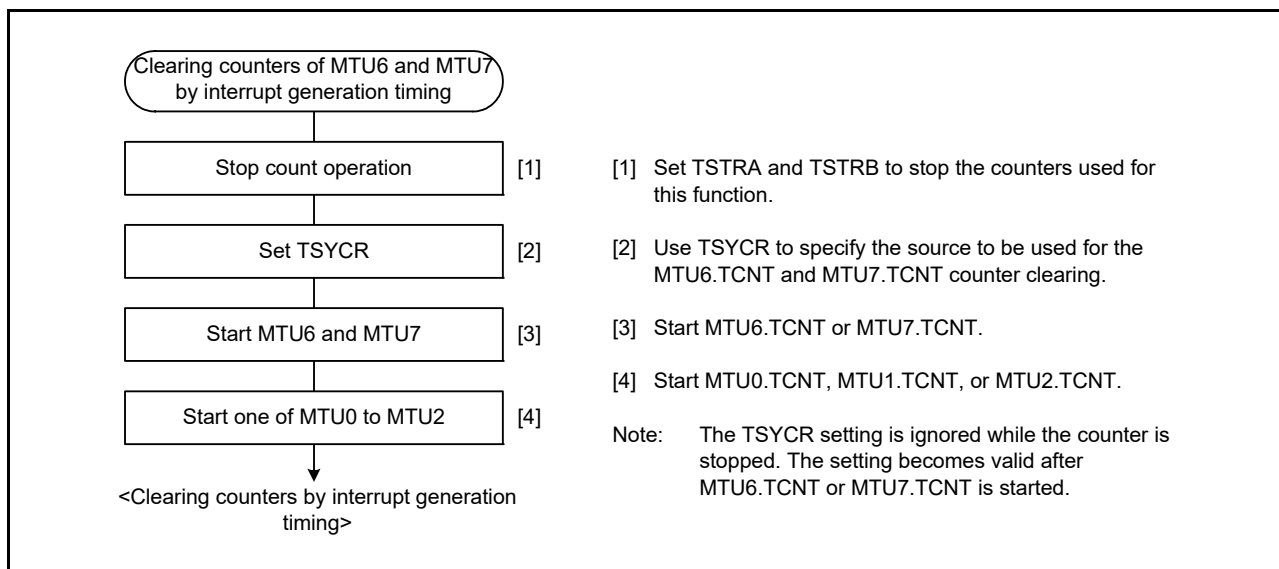
**Figure 24.100 Examples of Synchronous Counter Start Operation for MTU0 to MTU4, MTU6, and MTU7**

## (2) Synchronous Counter Clearing for MTU6 and MTU7

The counters in MTU6 and MTU7 can be cleared by the TGI<sub>mn</sub> interrupt generation timing (m = A to D; n = 0 to 2) through the TSYCR setting.

### (a) Example of Procedure for Specifying Synchronous Counter Clearing for MTU6 and MTU7

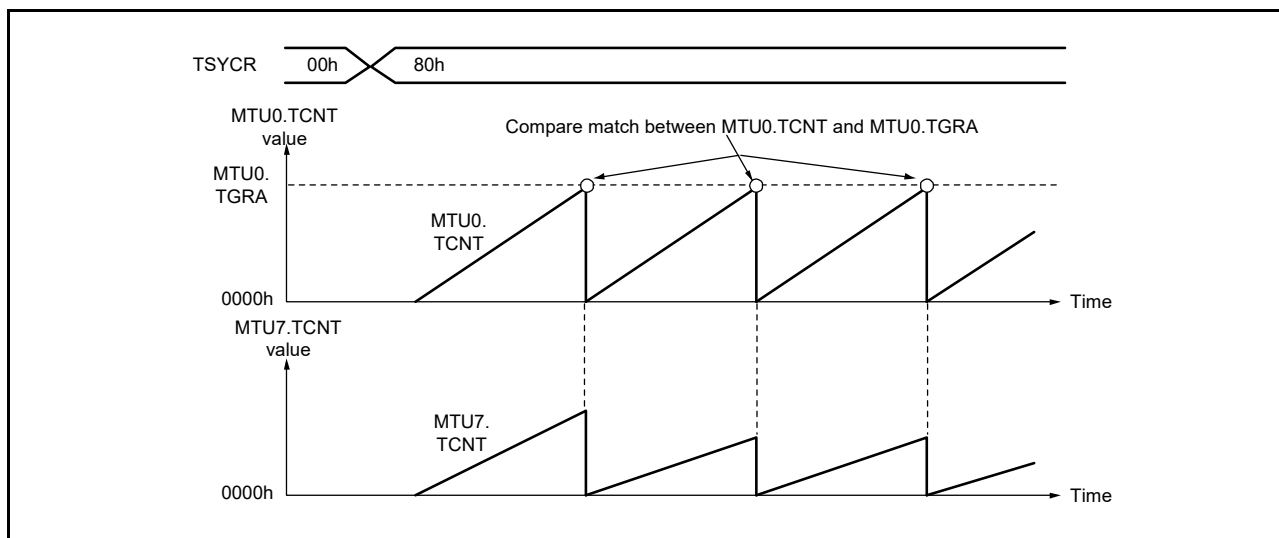
Figure 24.101 shows an example of procedure for specifying synchronous counter clearing for MTU6 and MTU7 by interrupt generation timing.



**Figure 24.101 Example of Procedure for Specifying Synchronous Counter Clearing for MTU6 and MTU7**

### (b) Examples of Synchronous Counter Clearing for MTU6 and MTU7

Figure 24.102 and Figure 24.103 show examples of synchronous counter clearing for MTU6 and MTU7 by interrupt generation timing.



**Figure 24.102 Example of Synchronous Counter Clearing for MTU6 and MTU7 (1)**

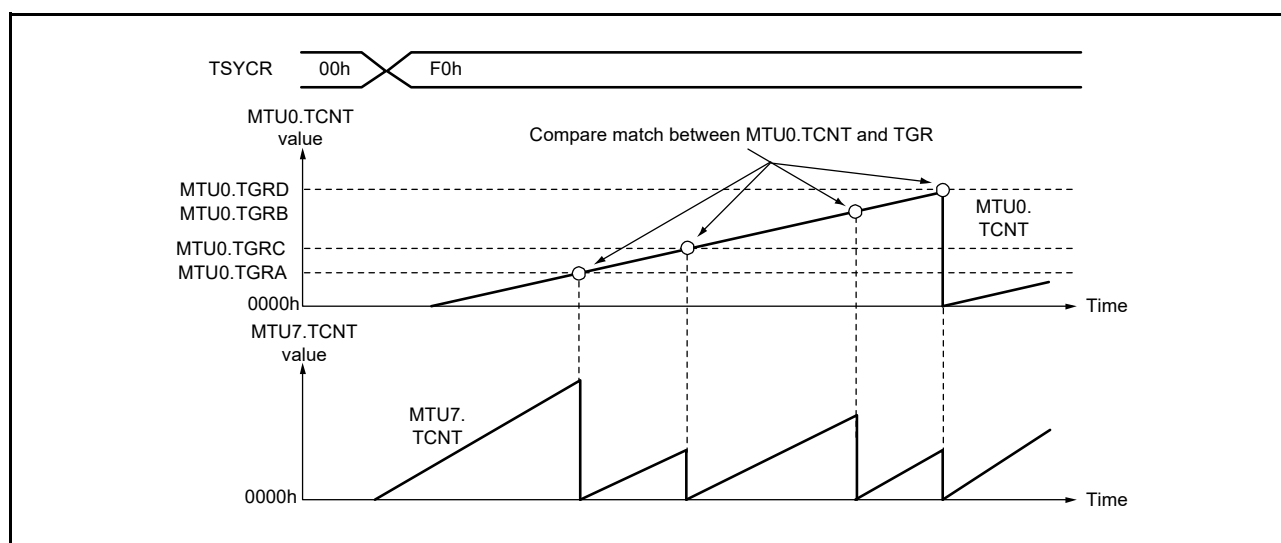


Figure 24.103 Example of Synchronous Counter Clearing for MTU6 and MTU7 (2)

### 24.3.11 External Pulse Width Measurement

The pulse widths of up to three external input lines can be measured in MTU5.

When the IOC[4:0] bits in MTU5.TIORU, MTU5.TIORV, MTU5.TIORW are set for pulse width measurement, the pulse width of the signal input to the MTIC5U, MTIC5V, and MTIC5W pins are measured. TCNTU, TCNTV, and TCNTW count up while the level specified by the IOC[4:0] bits is input.

Figure 24.104 shows an example of setting external pulse width measurement, and Figure 24.105 an example of external pulse width measurement.

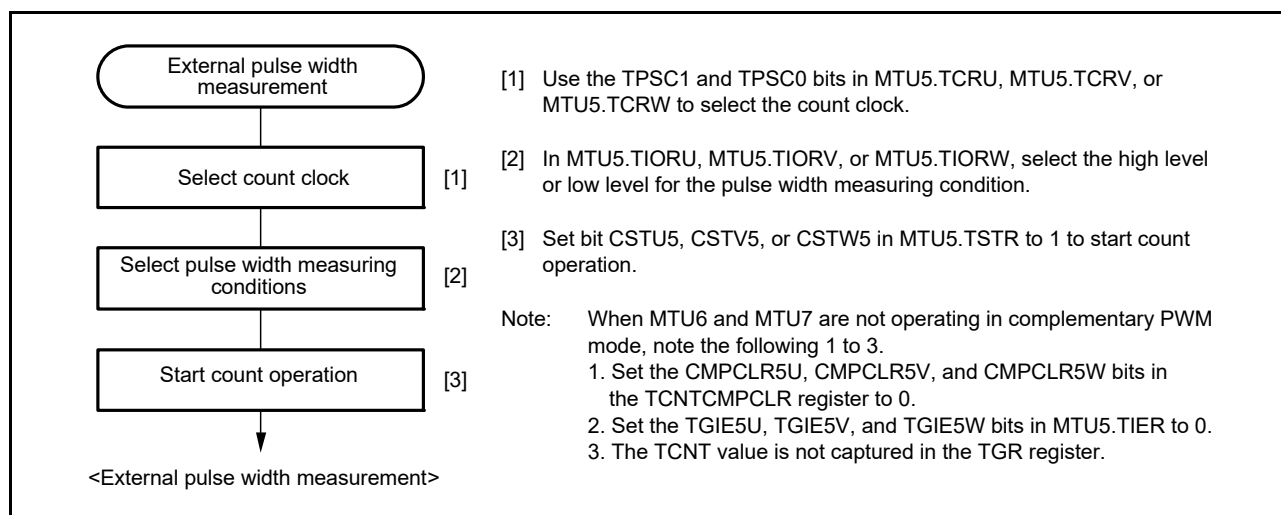


Figure 24.104 Example of External Pulse Width Measurement Setting Procedure

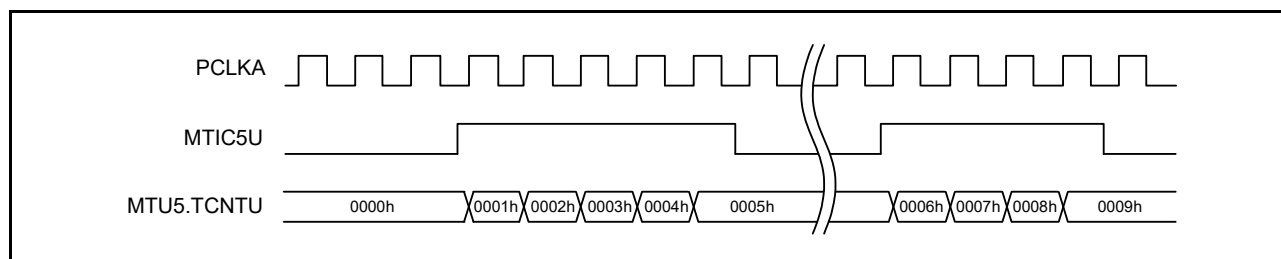


Figure 24.105 Example of External Pulse Width Measurement (Measuring High Pulse Width)

### 24.3.12 Dead Time Compensation

A dead time delay (a propagation delay of the inverter output from the complementary PWM output) can be compensated by combining MTU5 with MTU6 and MTU7. Figure 24.106 shows an example of the motor control circuit compensating a dead time delay by combining MTU5 with MTU6 and MTU7. A dead time for the PWM output waveform during complementary PWM operation using MTU6 and MTU7 can be compensated by adjusting a duty ratio set in a compare register for the PWM output after measuring a delay of the inverter output from the complementary PWM output by an external pulse measurement function for MTU5 (Figure 24.107). Figure 24.108 shows the procedure for setting dead time compensation using MTU5 to MTU7. For details on MTU5 operation at this time, refer to section 24.3.13, TCNTU, TCNTV, and TCNTW Capture at Crest and/or Trough in Complementary PWM Mode.

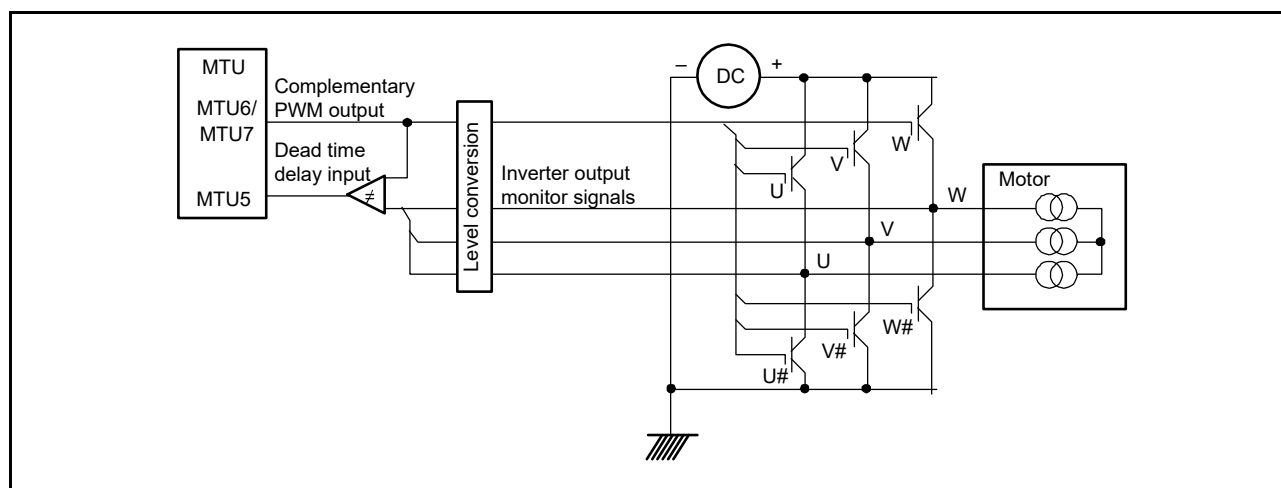


Figure 24.106 Motor Control Circuit Example

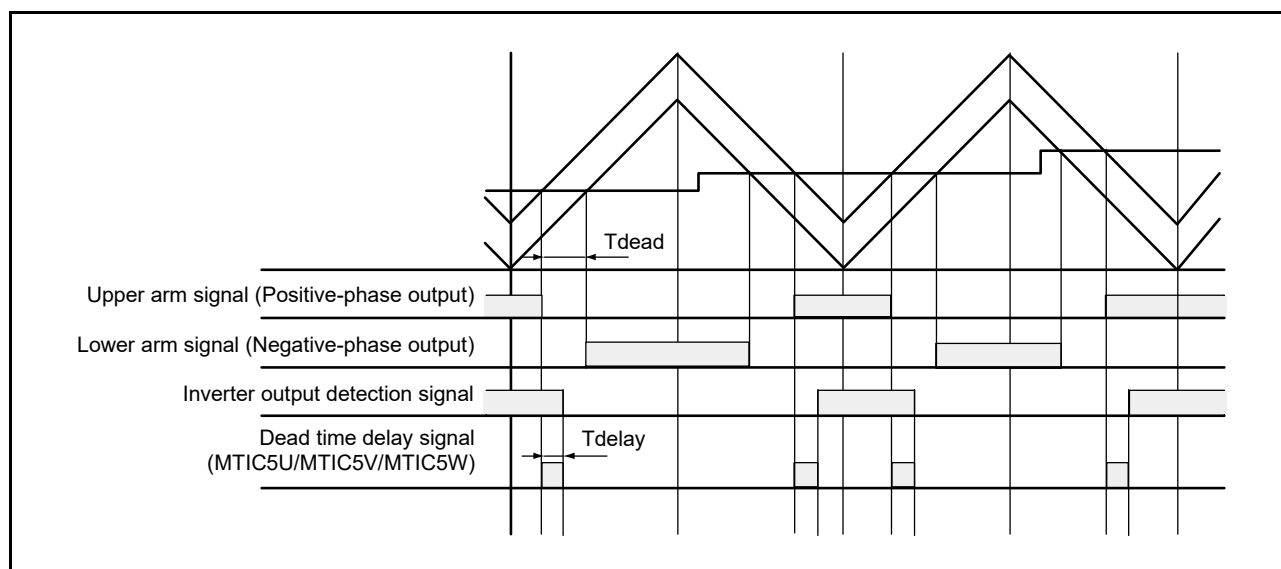
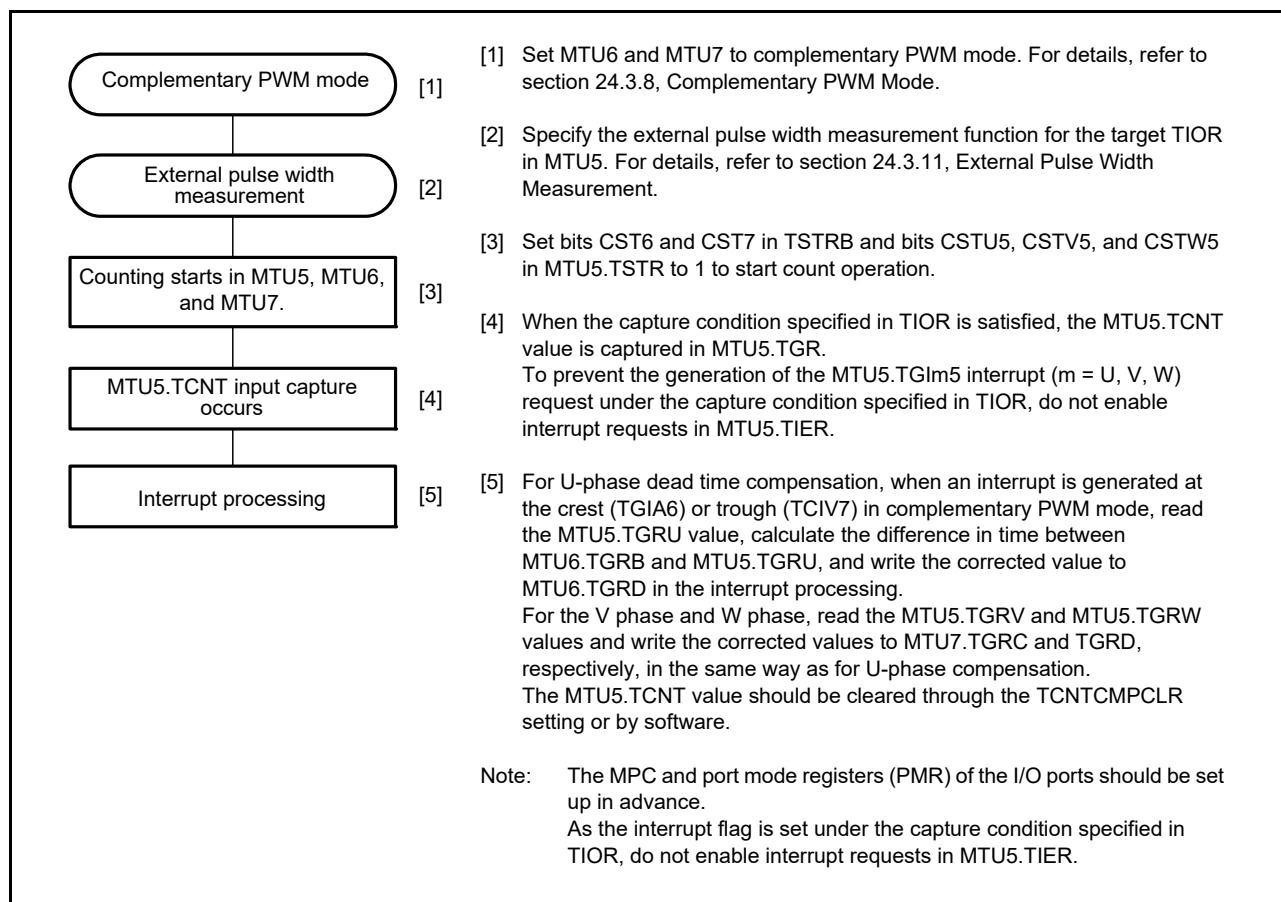


Figure 24.107 Delay in Dead Time in Complementary PWM Operation



## (1) Example of Dead Time Compensation Setting Procedure

Figure 24.108 shows an example of dead time compensation setting procedure by using three counters in MTU5.



**Figure 24.108 Example of Dead Time Compensation Setting Procedure**

### 24.3.13 TCNTU, TCNTV, and TCNTW Capture at Crest and/or Trough in Complementary PWM Mode

The MTU5 external pulse width measurement function allows to transfer the value in TCNTU, TCNTV, and TCNTW to TGRU, TGRV, and TGRW at the crest, or trough, or crest and trough when MTU6 and MTU7 operate in complementary PWM mode. The transfer timing is set in TIORU, TIORV, and TIORW. When the CMPCLR5U, CMPCLR5V, and CMPCLR5W bits in the TCNTCMPCLR register are set to 1, TCNTU, TCNTV, and TCNTW become 0000h at the transfer timing for TGRU, TGRV, and TGRW.

When MTU3 and MTU4 operate in complementary PWM mode, MTU5 cannot operate a capture operation of TCNTU, TCNTV, and TCNTW at the crest, or trough, or crest and trough in complementary PWM mode.

Figure 24.109 shows an operation example in which TCNTU is used as a free-running counter without being cleared, and the value is captured in TGRU at the crest and trough in complementary PWM mode.

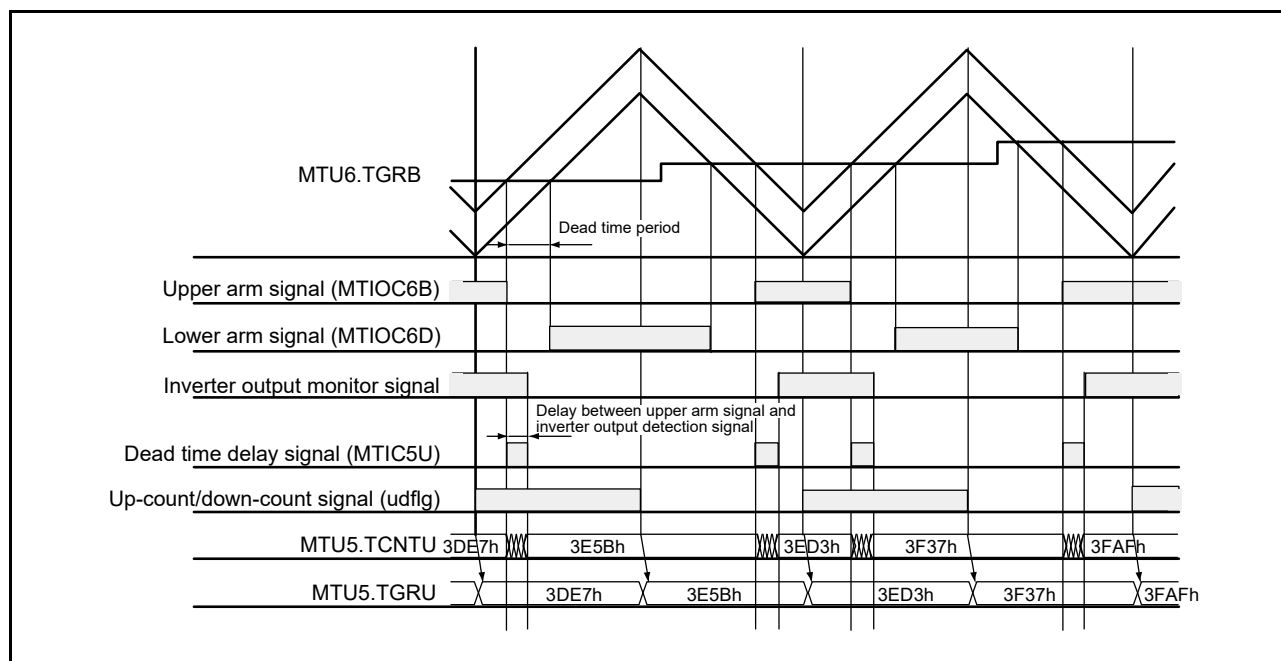


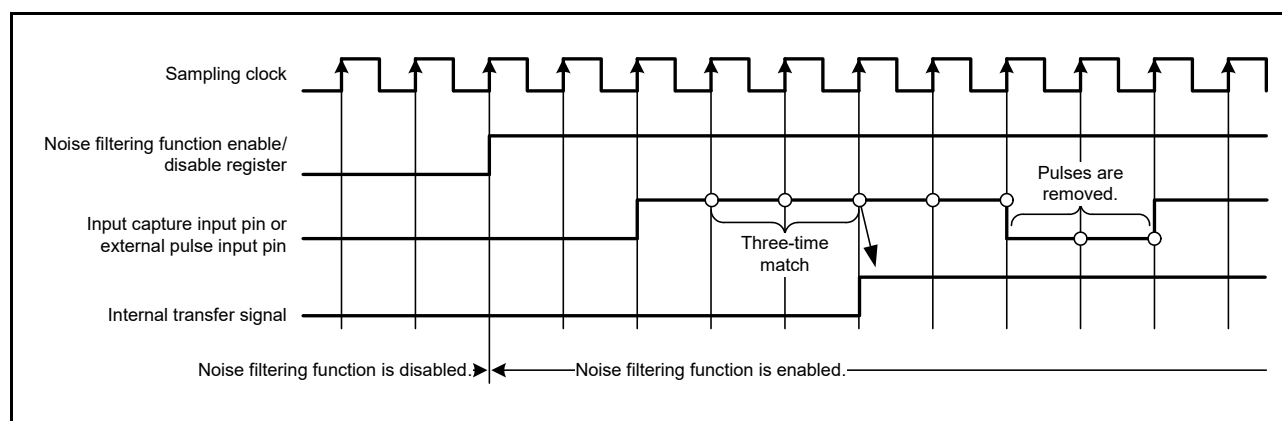
Figure 24.109 TCNTU Capture at Crest and Trough in Complementary PWM Operation

### 24.3.14 Noise Filter Function

The input capture input pins and external pulse input pins have a noise filter function.

Set the NFCRn register (n = 0 to 7, C) to enable or disable the noise filter function and set the sampling clock. The noise filter for each pin can be enabled or disabled individually, and the sampling clock can be set for each channel.

Figure 24.110 shows the timing of noise filtering.



**Figure 24.110 Timing of Noise Filtering**

## 24.4 Interrupt Sources

### 24.4.1 Interrupt Sources and Priorities

There are three kinds of interrupt source; TGR input capture/compare match, TCNT overflow, and TCNT underflow. Each interrupt source has its own enable/disable bit, allowing the generation of interrupt request signals to be enabled or disabled individually.

When an interrupt source is generated, if the corresponding enable/disable bit in TIER is set to 1, an interrupt is requested.

Relative channel priorities can be changed by the interrupt controller; however the priority within a channel is fixed. For details, refer to [section 15, Interrupt Controller \(ICUD\)](#). [Table 24.77](#) lists the MTU interrupt sources.

**Table 24.77 MTU Interrupt Sources**

| Channel | Name  | Interrupt Source                      | DMAC/DTC Activation |
|---------|-------|---------------------------------------|---------------------|
| MTU0    | TGIA0 | MTU0.TGRA input capture/compare match | Possible            |
|         | TGIB0 | MTU0.TGRB input capture/compare match | Possible            |
|         | TGIC0 | MTU0.TGRC input capture/compare match | Possible            |
|         | TGID0 | MTU0.TGRD input capture/compare match | Possible            |
|         | TCIV0 | MTU0.TCNT overflow                    | Not possible        |
|         | TGIE0 | MTU0.TGRE compare match               | Not possible        |
|         | TGIF0 | MTU0.TGRF compare match               | Not possible        |
| MTU1    | TGIA1 | MTU1.TGRA input capture/compare match | Possible            |
|         | TGIB1 | MTU1.TGRB input capture/compare match | Possible            |
|         | TCIV1 | MTU1.TCNT overflow                    | Not possible        |
|         | TCIU1 | MTU1.TCNT underflow                   | Not possible        |
| MTU2    | TGIA2 | MTU2.TGRA input capture/compare match | Possible            |
|         | TGIB2 | MTU2.TGRB input capture/compare match | Possible            |
|         | TCIV2 | MTU2.TCNT overflow                    | Not possible        |
|         | TCIU2 | MTU2.TCNT underflow                   | Not possible        |
| MTU3    | TGIA3 | MTU3.TGRA input capture/compare match | Possible            |
|         | TGIB3 | MTU3.TGRB input capture/compare match | Possible            |
|         | TGIC3 | MTU3.TGRC input capture/compare match | Possible            |
|         | TGID3 | MTU3.TGRD input capture/compare match | Possible            |
|         | TCIV3 | MTU3.TCNT overflow                    | Not possible        |
| MTU4    | TGIA4 | MTU4.TGRA input capture/compare match | Possible            |
|         | TGIB4 | MTU4.TGRB input capture/compare match | Possible            |
|         | TGIC4 | MTU4.TGRC input capture/compare match | Possible            |
|         | TGID4 | MTU4.TGRD input capture/compare match | Possible            |
|         | TCIV4 | MTU4.TCNT overflow/underflow*1        | Possible            |
| MTU5    | TGIU5 | MTU5.TGRU input capture/compare match | Possible            |
|         | TGIV5 | MTU5.TGRV input capture/compare match | Possible            |
|         | TGIW5 | MTU5.TGRW input capture/compare match | Possible            |
| MTU6    | TGIA6 | MTU6.TGRA input capture/compare match | Possible            |
|         | TGIB6 | MTU6.TGRB input capture/compare match | Possible            |
|         | TGIC6 | MTU6.TGRC input capture/compare match | Possible            |
|         | TGID6 | MTU6.TGRD input capture/compare match | Possible            |
|         | TCIV6 | MTU6.TCNT overflow                    | Not possible        |
| MTU7    | TGIA7 | MTU7.TGRA input capture/compare match | Possible            |
|         | TGIB7 | MTU7.TGRB input capture/compare match | Possible            |
|         | TGIC7 | MTU7.TGRC input capture/compare match | Possible            |
|         | TGID7 | MTU7.TGRD input capture/compare match | Possible            |
|         | TCIV7 | MTU7.TCNT overflow/underflow*1        | Possible            |
| MTU8    | TGIA8 | MTU8.TGRA input capture/compare match | Possible            |
|         | TGIB8 | MTU8.TGRB input capture/compare match | Possible            |
|         | TGIC8 | MTU8.TGRC input capture/compare match | Possible            |
|         | TGID8 | MTU8.TGRD input capture/compare match | Possible            |
|         | TCIV8 | MTU8.TCNT overflow                    | Not possible        |

Note: This table shows the initial state immediately after a reset. The relative channel priorities can be changed by the interrupt controller.

Note 1. Underflow is available only in complementary PWM mode.

### (1) Input Capture/Compare Match Interrupt

If the TIER.TGIE bit is set to 1 when a TGR input capture/compare match occurs on a channel, an interrupt is requested. The MTU has 33 input capture/compare match interrupts (six for MTU0, four each for MTU3, MTU4, MTU6, MTU7, and MTU8, two each for MTU1 and MTU2, and three for MTU5).

### (2) Overflow Interrupt

If the TIER.TCIEV bit is set to 1 when a TCNT overflow occurs on a channel, an interrupt is requested. The MTU has eight overflow interrupts (one for each channel except MTU5).

Note that an overflow interrupt is generated also when an underflow of the MTU4.TCNT and MTU7.TCNT occurs while operating in complementary PWM mode.

### (3) Underflow Interrupt

If the TIER.TCIEU bit is set to 1 when a TCNT underflow occurs on a channel, an interrupt is requested. The MTU has two underflow interrupts (one each for MTU1, and MTU2).

## 24.4.2 DTC/DMAC Trigger Sources

### (1) DTC Trigger Sources

The DTC can be triggered by the TGR input capture/compare match interrupt in each channel or the overflow interrupt in MTU4 and MTU7. For details, refer to [section 20, Data Transfer Controller \(DTCb\)](#).

The MTU provides a total of 33 input capture/compare match interrupts and overflow interrupts that can be used as DTC trigger sources: four each for MTU0, MTU3, MTU6, and MTU8, two each for MTU1 and MTU2, five each for MTU4 and MTU7, and three for MTU5.

### (2) DMAC Trigger Sources

The DMAC can be triggered by the TGR input capture/compare match interrupt in each channel and the overflow interrupt in MTU4 and MTU7. For details, refer to [section 18, DMA Controller \(DMACa\)](#).

The MTU provides a total of 33 input capture/compare match interrupts and overflow interrupts that can be used as DMAC trigger sources: four each for MTU0, MTU3, MTU6, and MTU8, two each for MTU1 and MTU2, five each for MTU4 and MTU7, and three for MTU5.

If a DMA transfer is initiated by the MTU, the trigger signal is cleared when the DMAC requests ownership of the internal bus. Accordingly, the state of the internal bus may lead to a wait for the start of DMA transfer.

### (3) EXDMAC Initiation

The EXDMAC can be triggered by MTU interrupts. For details, refer to [section 19, EXDMA Controller \(EXDMACa\)](#).

### 24.4.3 A/D Converter Trigger Sources

The A/D converter can be triggered by one of the following three methods in the MTU. Table 24.78 shows the relationship between interrupt sources and A/D conversion start request signals.

#### (1) A/D Conversion Start by TGRA Input Capture/Compare Match or at Trough of MTU4.TCNT (MTU7.TCNT) in Complementary PWM Mode

The A/D converter can be triggered by the occurrence of a TGRA input capture/compare match in each channel. In addition, if complementary PWM operation is performed while the TTGE2 bit in MTU4.TIER (MTU7.TIER) is set to 1, the A/D converter can be triggered at the trough of MTU4.TCNT (MTU7.TCNT) count (MTU4.TCNT (MTU7.TCNT) = 0000h).

A/D conversion start request TRGAnN is issued to the A/D converter under either of the following conditions (n = 0 to 4, 6, 7).

- When a TGRA input capture/compare match occurs on a channel while the TIER.TTGE bit is set to 1
- When the MTU4.TCNT (MTU7.TCNT) count reaches the trough (MTU4.TCNT (MTU7.TCNT) = 0000h) during complementary PWM operation while the TTGE2 bit in MTU4.TIER (MTU7.TIER) is set to 1

When either condition is satisfied, if A/D conversion start request signal TRGAnN from the MTU is selected as the trigger in the A/D converter, A/D conversion will start.

#### (2) A/D Conversion Start by Compare Match between MTU0.TCNT and MTU0.TGRE

A/D conversion start request TRG0N is issued to the A/D converter when a compare match occurs between MTU0.TCNT and MTU0.TGRE.

When a compare match occurs between MTU0.TCNT and MTU0.TGRE while the TTGE2 bit in MTU0.TIER2 is set to 1, A/D conversion start request TRG0N is issued to the A/D converter. If A/D conversion start request signal TRG0N from the MTU is selected as the trigger in the A/D converter, A/D conversion will start.

#### (3) A/D Conversion Start by A/D Conversion Start Request Delaying Function

The A/D converter can be triggered by generating A/D conversion start request signal TRG4AN or TRG4BN (TRG7AN or TRG7BN) when the MTU4.TCNT (MTU7.TCNT) count matches the MTU4.TADCORA or MTU4.TADCORB (MTU7.TADCORA or MTU7.TADCORB) value if the UT4AE, DT4AE, UT4BE, or DT4BE (UT7AE, DT7AE, UT7BE, or DT7BE) bit in the A/D conversion start request control register (MTU4.TADCR (MTU7.TADCR)) is set to 1. For details, refer to section 24.3.9, A/D Conversion Start Request Delaying Function.

A/D conversion will start when TRG4AN (TRG7AN) is generated if A/D conversion start request signal TRG4AN (TRG7AN) from the MTU is selected as the trigger in the A/D converter, when TRG4BN (TRG7BN) is generated if TRG4BN (TRG7BN) from the MTU is selected as the trigger in the A/D converter, or when TRG4ABN (TRG7ABN) is generated if TRG4ABN (TRG7ABN) from the MTU is selected as the trigger in the A/D converter.

**Table 24.78 Interrupt Sources and A/D Conversion Start Request Signals**

| Target Registers                                          | Interrupt Source                           | A/D Conversion Start Request Signal |
|-----------------------------------------------------------|--------------------------------------------|-------------------------------------|
| MTU0.TGRA and MTU0.TCNT                                   | Input capture/compare match                | TRGA0N                              |
| MTU1.TGRA and MTU1.TCNT                                   |                                            | TRGA1N                              |
| MTU2.TGRA and MTU2.TCNT                                   |                                            | TRGA2N                              |
| MTU3.TGRA and MTU3.TCNT                                   |                                            | TRGA3N                              |
| MTU4.TGRA and MTU4.TCNT*1                                 |                                            | TRGA4N                              |
| MTU4.TCNT                                                 | MTU4.TCNT trough in complementary PWM mode |                                     |
| MTU6.TGRA and MTU6.TCNT                                   | Input capture/compare match                | TRGA6N                              |
| MTU7.TGRA and MTU7.TCNT*1                                 |                                            | TRGA7N                              |
| MTU7.TCNT                                                 | MTU7.TCNT trough in complementary PWM mode |                                     |
| MTU0.TGRE and MTU0.TCNT                                   | Compare match                              | TRG0N                               |
| MTU4.TADCORA and MTU4.TCNT                                |                                            | TRG4AN                              |
| MTU4.TADCORB and MTU4.TCNT                                |                                            | TRG4BN                              |
| MTU7.TADCORA and MTU7.TCNT                                |                                            | TRG7AN                              |
| MTU7.TADCORB and MTU7.TCNT                                |                                            | TRG7BN                              |
| MTU4.TADCORA and MTU4.TCNT,<br>MTU4.TADCORB and MTU4.TCNT |                                            | TRG4ABN                             |
| MTU7.TADCORA and MTU7.TCNT,<br>MTU7.TADCORB and MTU7.TCNT |                                            | TRG7ABN                             |

Note 1. Since PWM waveforms are generated in complementary PWM mode, MTU4.TGRA (MTU7.TGRA) compare match not only with MTU4.TCNT (MTU7.TCNT) but also with MTU3.TCNT (MTU6.TCNT) and TCNTSA (TCNTSB) is detected. Accordingly, when compare match with MTU3.TCNT (MTU6.TCNT) and TCNTSA (TCNTSB) occurs, TRGA4N (TRGA7N) is also generated. When MTU3 and MTU 4 (MTU 6 and MTU7) are made to operate in complementary PWM mode for generating an A/D conversion start request, use the A/D conversion start request by compare match between MTU4.TCNT (MTU7.TCNT) and MTU4.TADCORA/TADCORB (MTU7.TADCORA/TADCORB).



## 24.5 Operation Timing

### 24.5.1 Input/Output Timing

#### (1) TCNT Count Timing

Figure 24.111 and Figure 24.112 show the TCNT count timing in internal clock operation, Figure 24.113 shows the TCNT count timing in external clock operation (normal mode), and Figure 24.114 shows the TCNT count timing in external clock operation (phase counting mode).

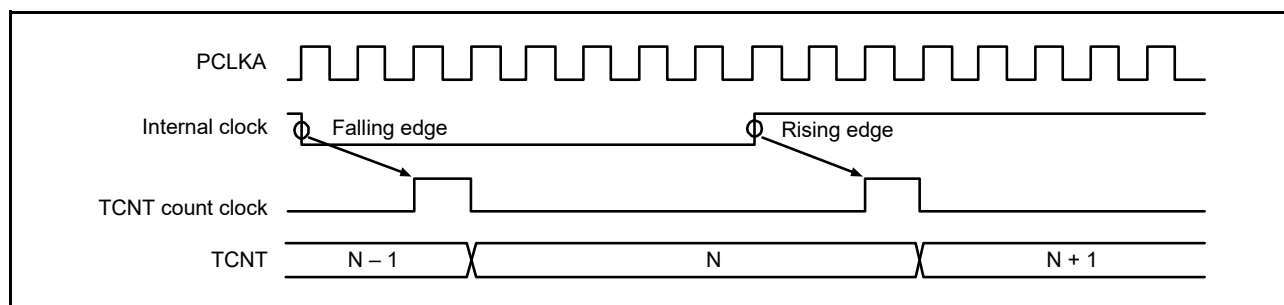


Figure 24.111 Count Timing in Internal Clock Operation (MTU0 to MTU4 and MTU6 to MTU8)

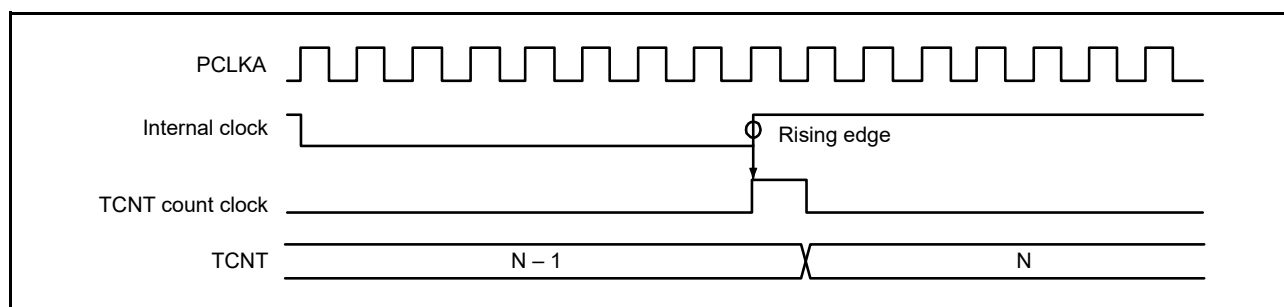


Figure 24.112 Count Timing in Internal Clock Operation (MTU5)

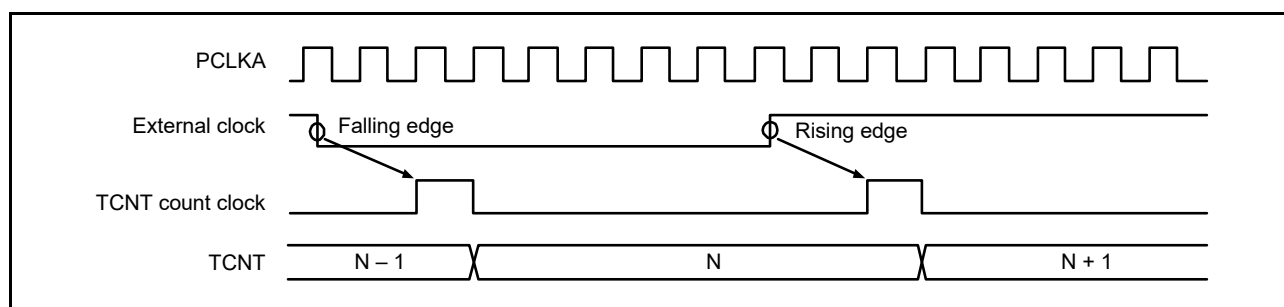


Figure 24.113 Count Timing in External Clock Operation (MTU0 to MTU4 and MTU6 to MTU8)

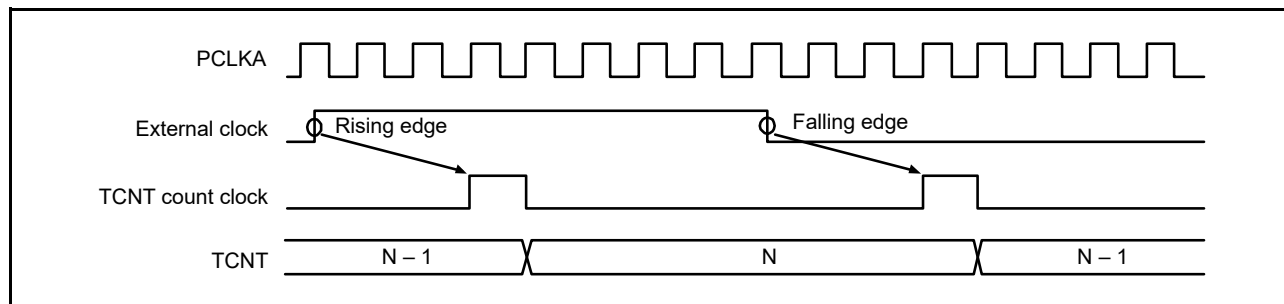
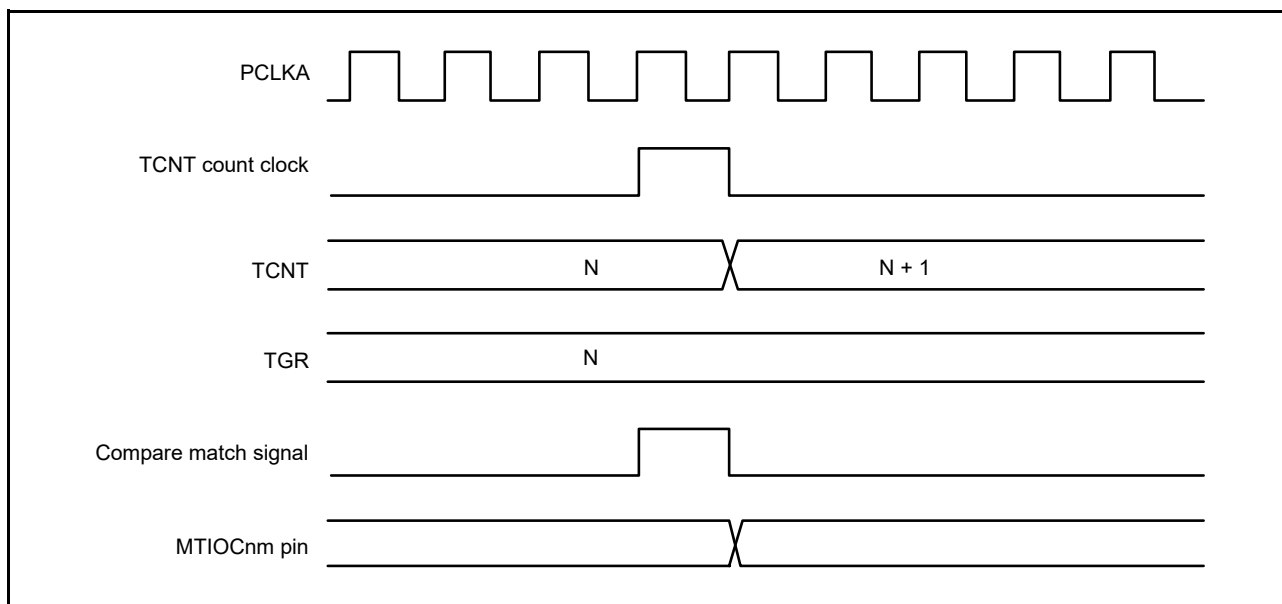


Figure 24.114 Count Timing in External Clock Operation (Phase Counting Mode)

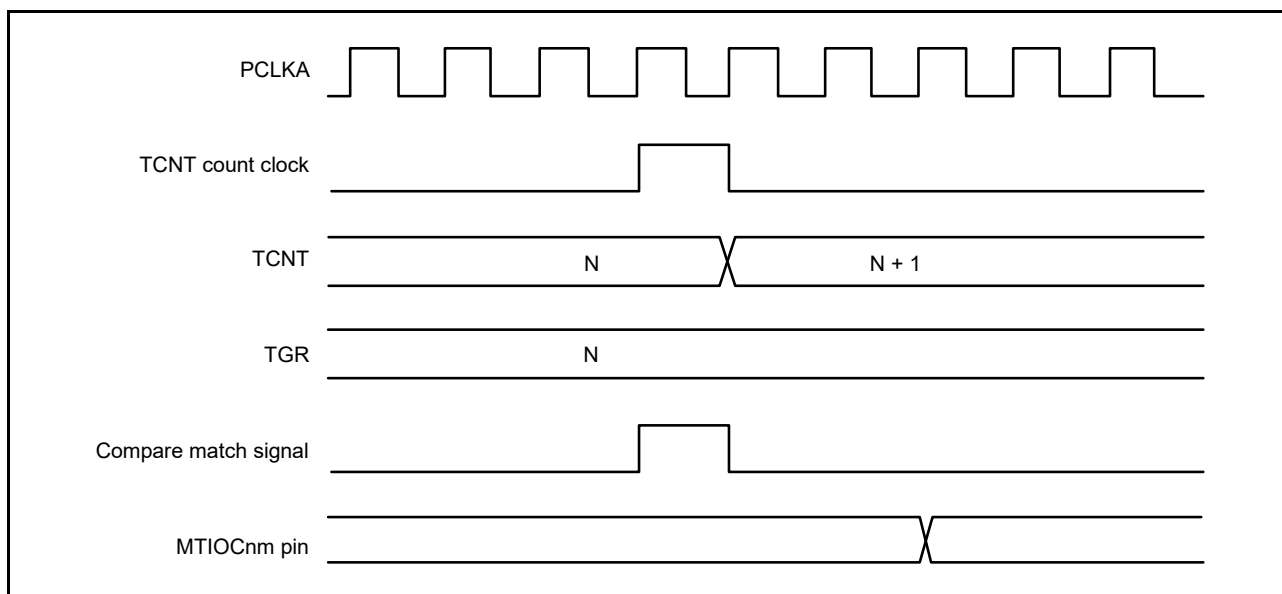
## (2) Output Compare Output Timing

A compare match signal is generated in the final state in which TCNT and TGR match (the point at which the count value matched by TCNT is updated). When a compare match signal is generated, the value set in TIOR is output from MTIOCnm pin ( $n = 0$  to  $4, 6, 7, 8$ ;  $m = A$  to  $D$ ). After a match between TCNT and TGR, the compare match signal is not generated until the TCNT count clock is generated.

Figure 24.115 shows the output compare output timing (normal mode or PWM mode) and Figure 24.116 shows the output compare output timing (complementary PWM mode or reset-synchronized PWM mode).



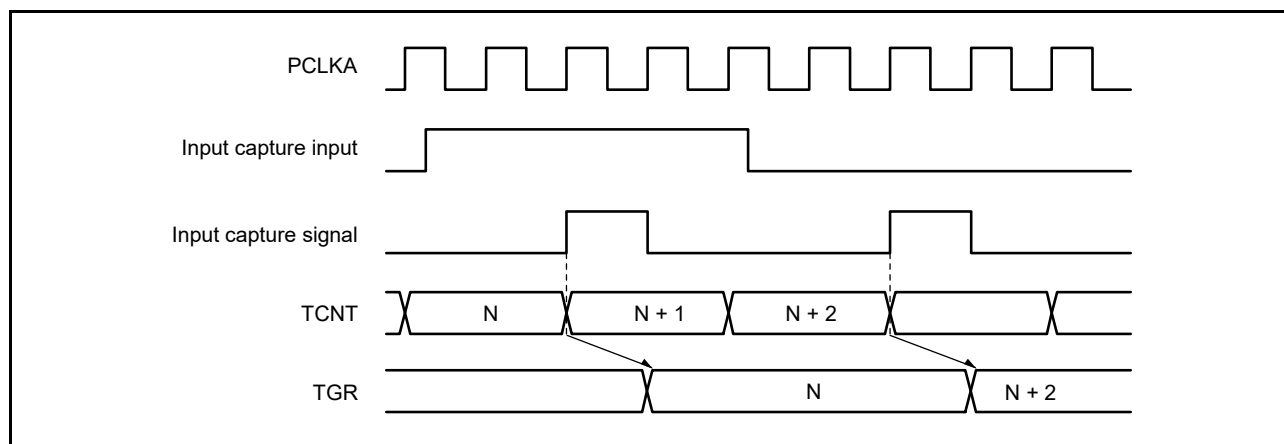
**Figure 24.115 Output Compare Output Timing (Normal Mode or PWM Mode) ( $n = 0$  to  $4, 6, 7, 8$ ;  $m = A$  to  $D$ )**



**Figure 24.116 Output Compare Output Timing (Complementary PWM Mode or Reset-Synchronized PWM Mode) ( $n = 0$  to  $4, 6, 7, 8$ ;  $m = A$  to  $D$ )**

### (3) Input Capture Signal Timing

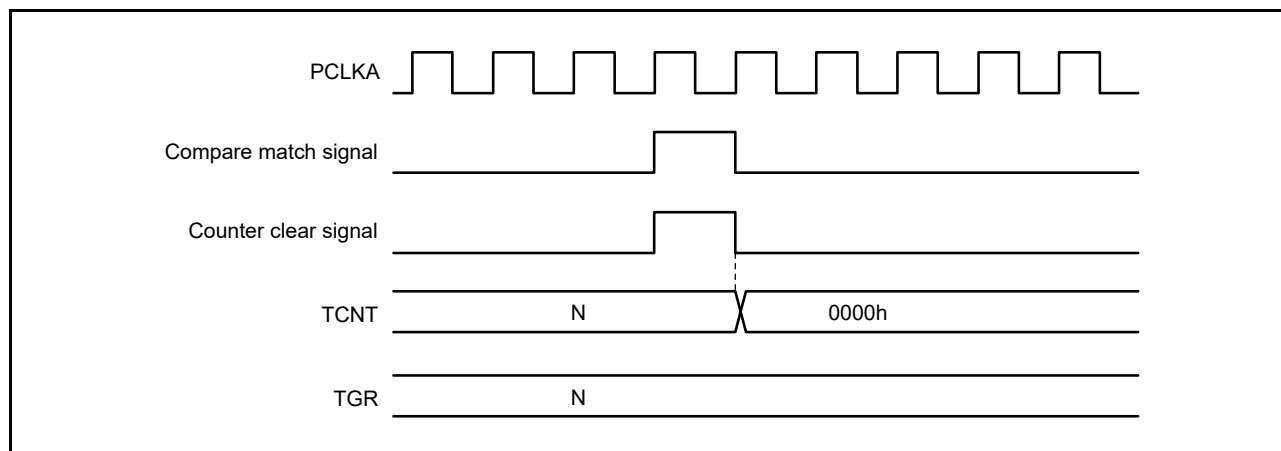
Figure 24.117 shows the input capture signal timing.



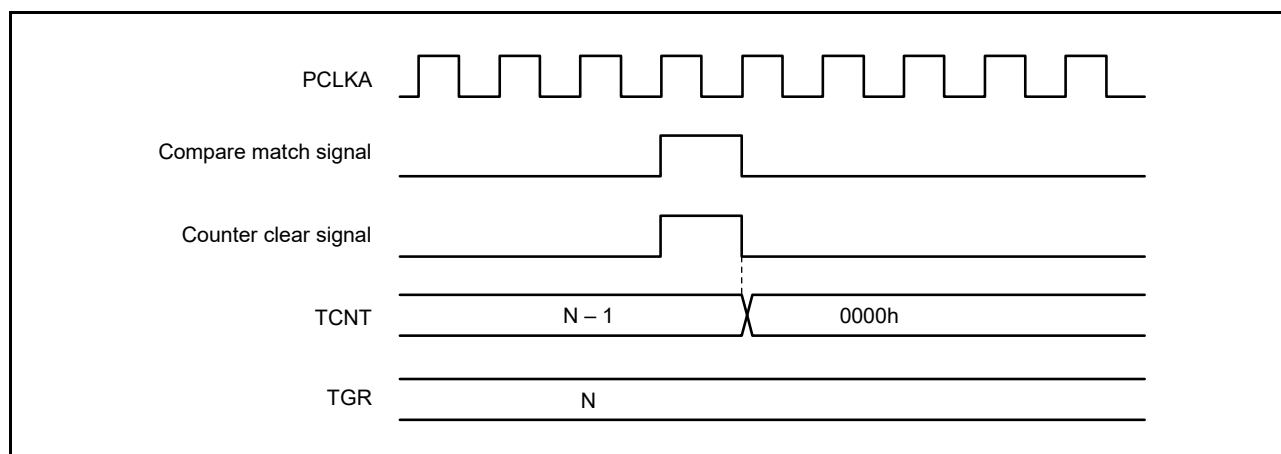
**Figure 24.117 Input Capture Input Signal Timing**

#### (4) Timing for Counter Clearing by Compare Match/Input Capture

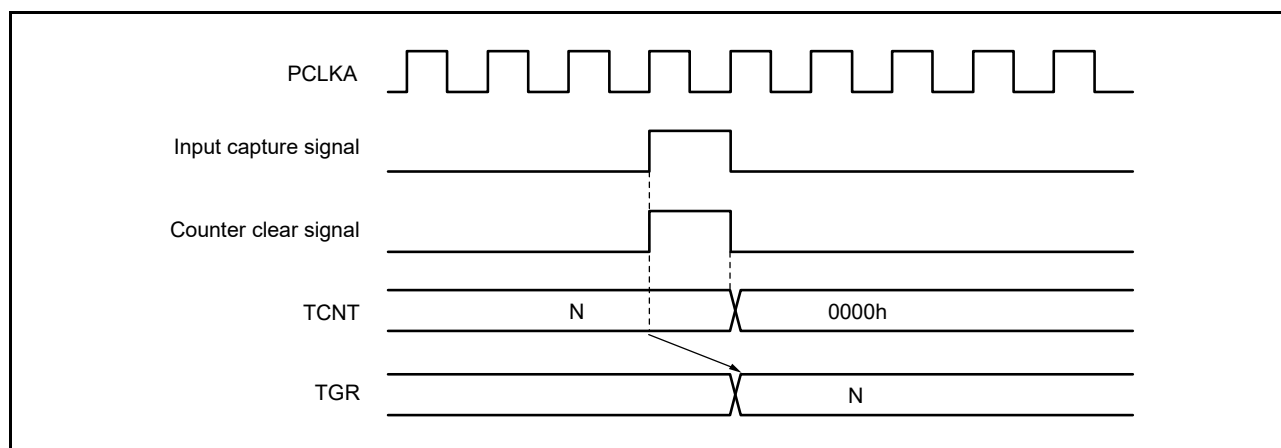
Figure 24.118 and Figure 24.119 show the timing when counter clearing on compare match is specified, and Figure 24.120 shows the timing when counter clearing on input capture is specified.



**Figure 24.118 Counter Clear Timing (Compare Match) (MTU0 to MTU4 and MTU6 to MTU8)**



**Figure 24.119 Counter Clear Timing (Compare Match) (MTU5)**



**Figure 24.120 Counter Clear Timing (Input Capture) (MTU0 to MTU8)**

## (5) Buffer Operation Timing

Figure 24.121 to Figure 24.123 show the timing in buffer operation.

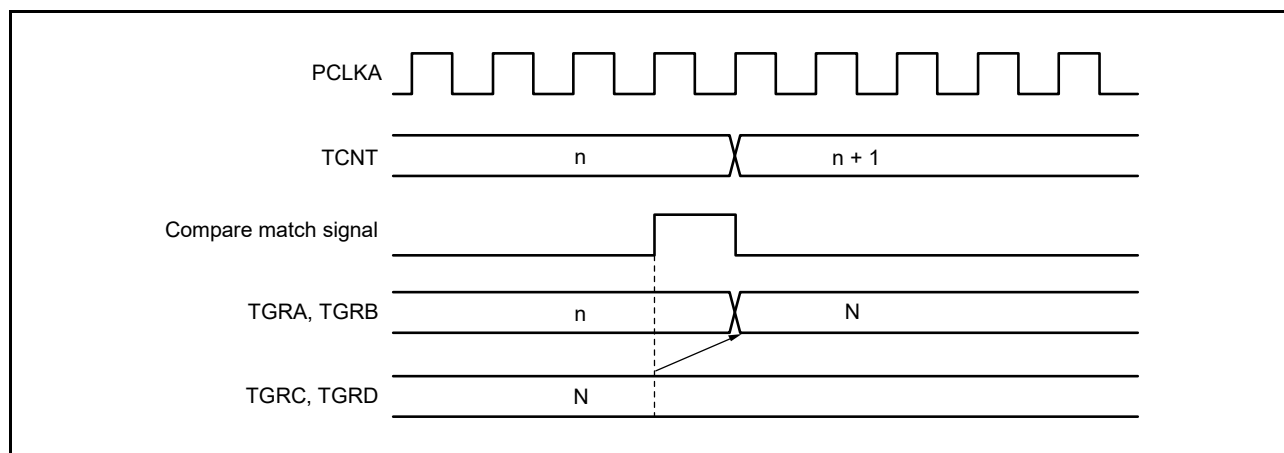


Figure 24.121 Buffer Operation Timing (Compare Match)

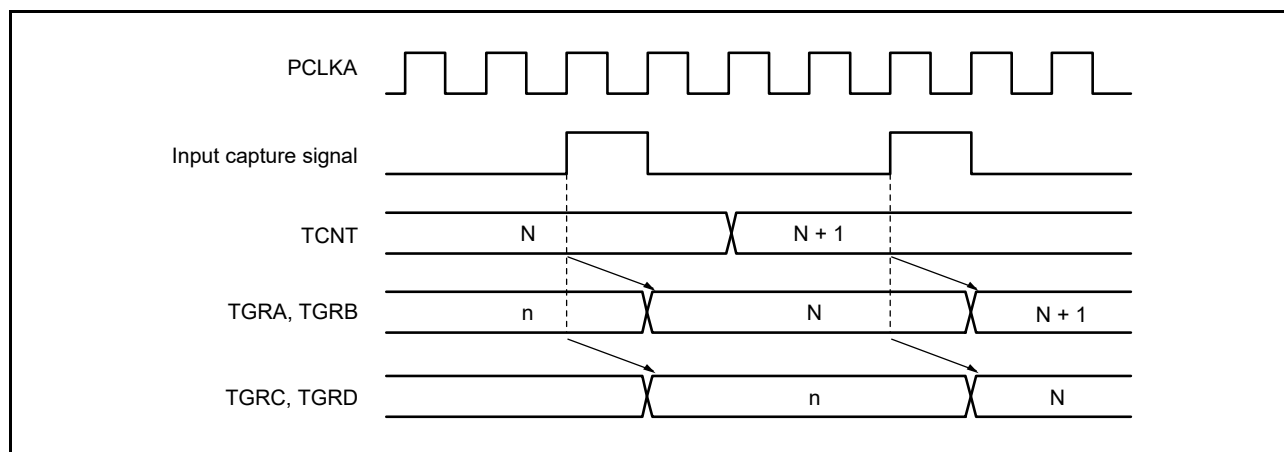


Figure 24.122 Buffer Operation Timing (Input Capture)

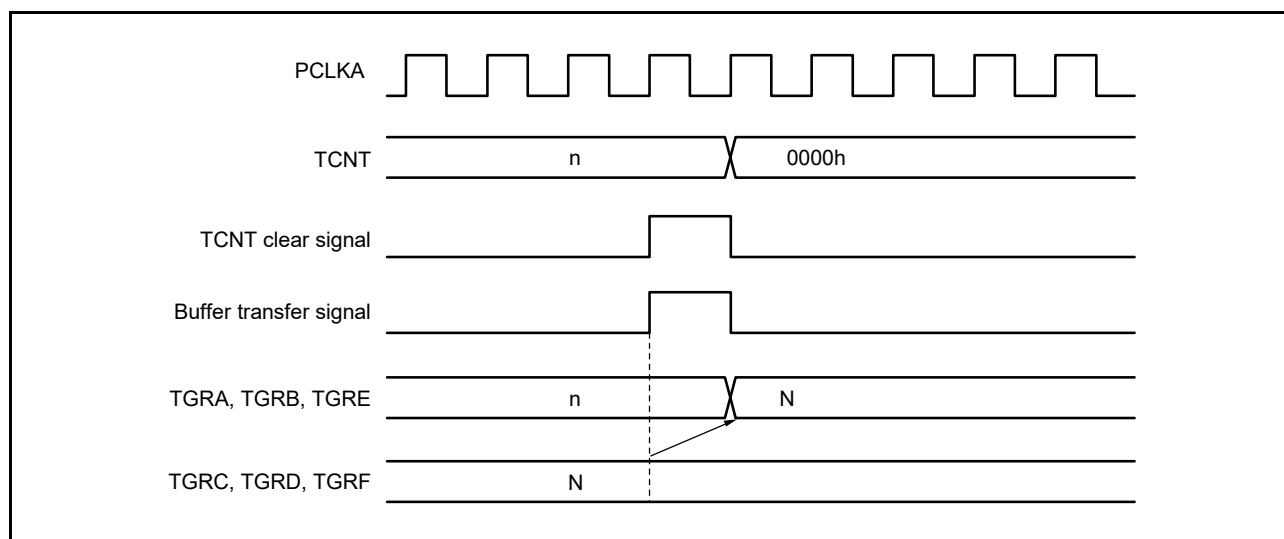


Figure 24.123 Buffer Operation Timing (When TCNT Cleared)

## (6) Buffer Transfer Timing (Complementary PWM Mode)

Figure 24.124 to Figure 24.126 show the buffer transfer timing in complementary PWM mode.

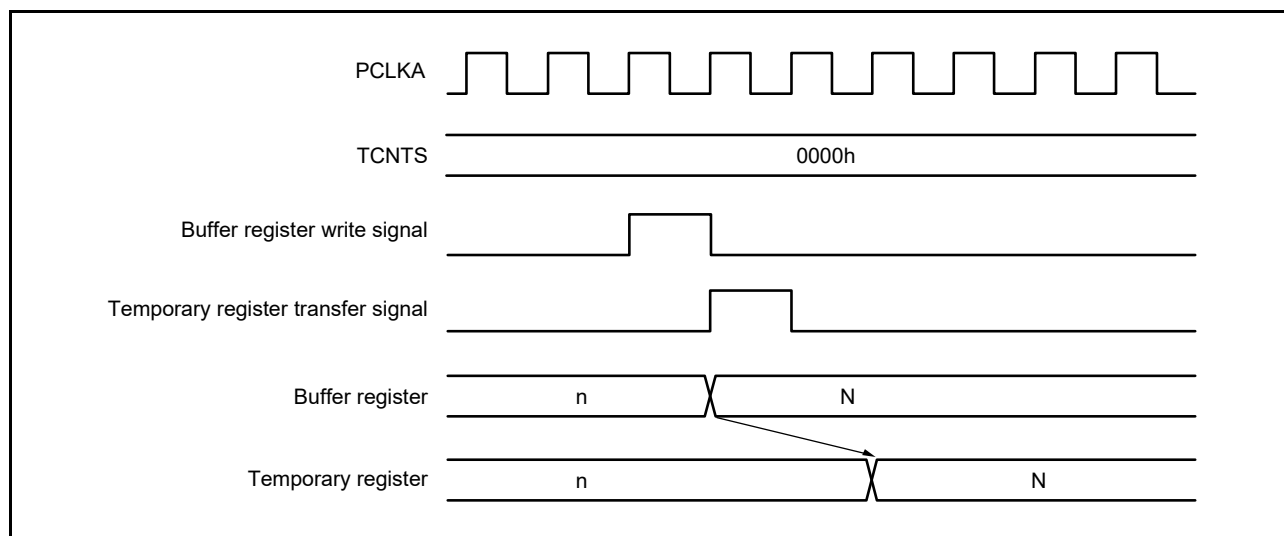


Figure 24.124 Transfer Timing from Buffer Register to Temporary Register (TCNTSA Stopped)

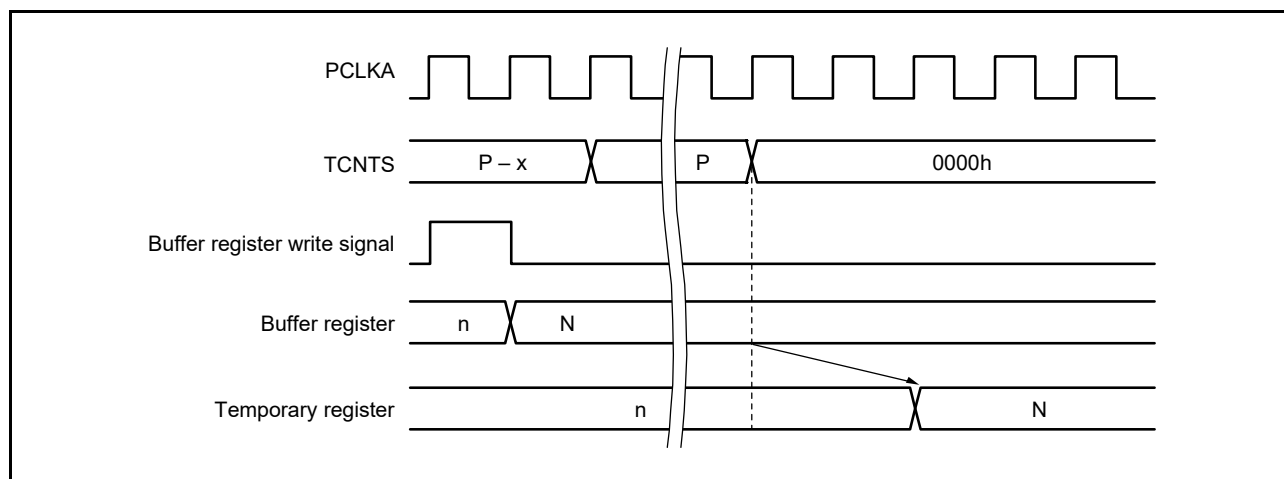


Figure 24.125 Transfer Timing from Buffer Register to Temporary Register (TCNTSA Operating)

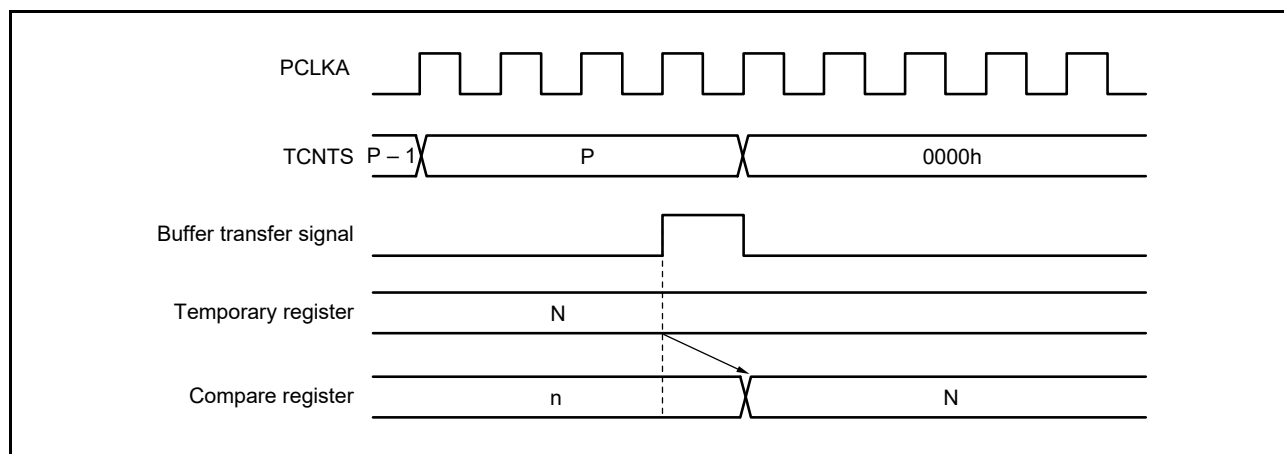


Figure 24.126 Transfer Timing from Temporary Register to Compare Register

## 24.5.2 Interrupt Signal Timing

### (1) TGI Interrupt Timing by Compare Match

Figure 24.127 and Figure 24.128 show the TGI interrupt request signal timing when a compare match occurs.

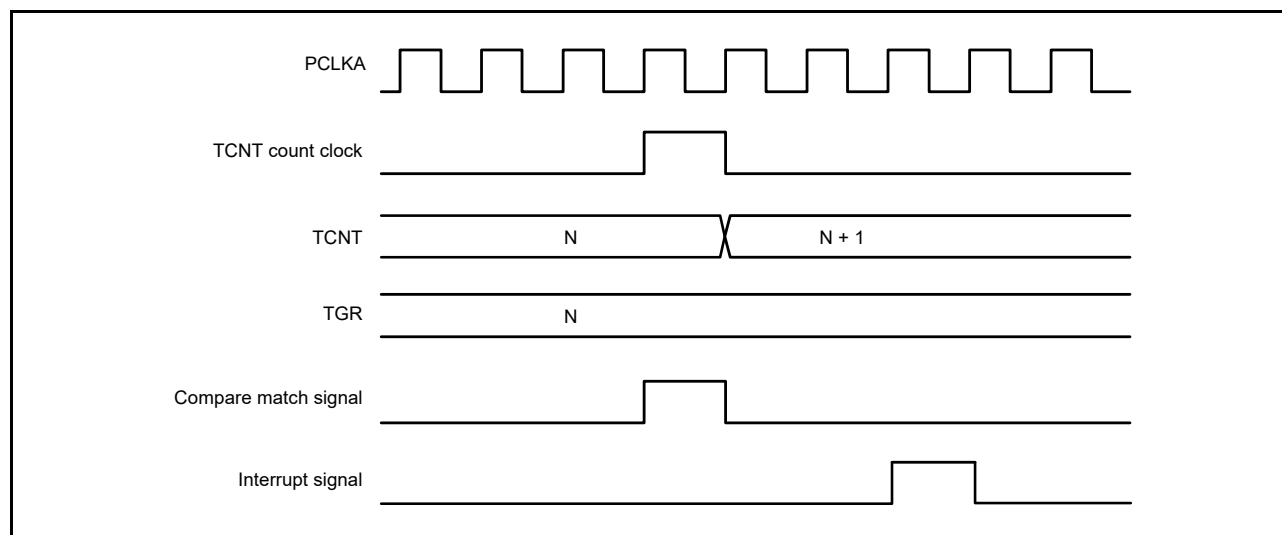


Figure 24.127 TGI Interrupt Timing (Compare Match) (MTU0 to MTU4 and MTU6 to MTU8)

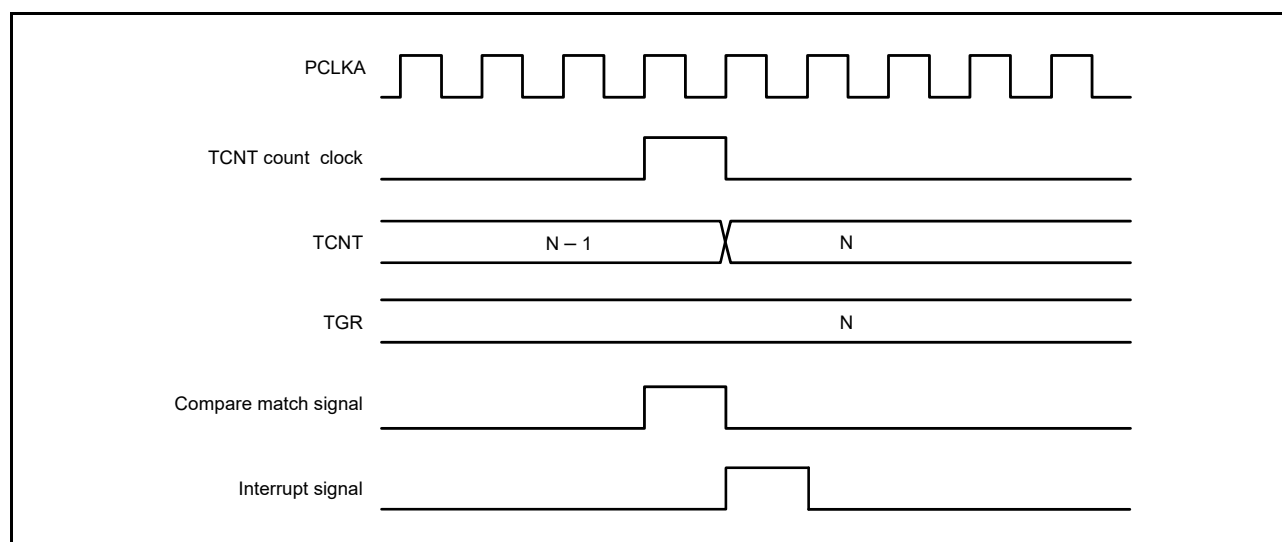


Figure 24.128 TGI Interrupt Timing (Compare Match) (MTU5)

## (2) TGI Interrupt Timing by Input Capture

Figure 24.129 and Figure 24.130 show the TGI interrupt request signal timing when an input capture occurs.

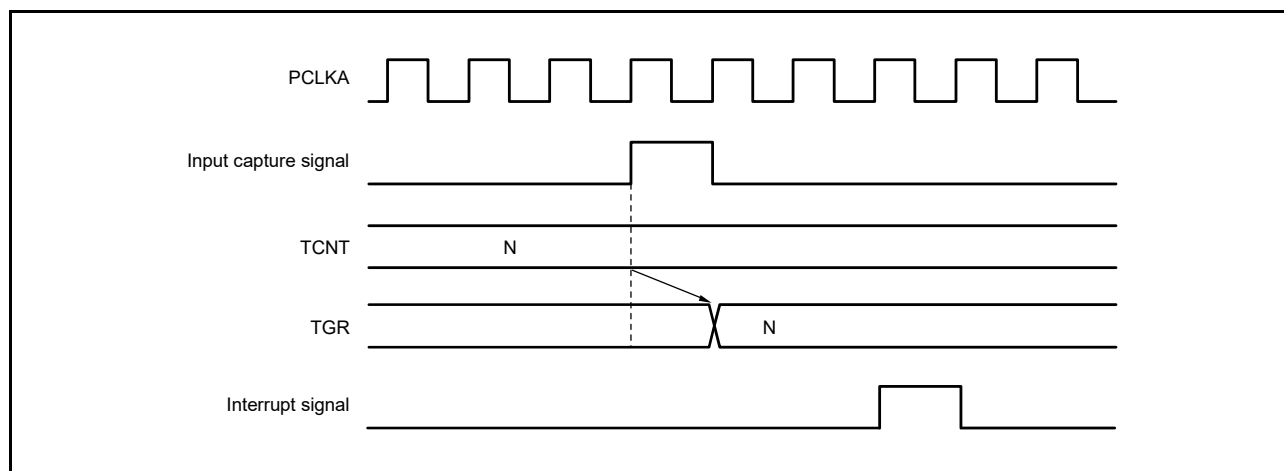


Figure 24.129 TGI Interrupt Timing (Input Capture) (MTU0 to MTU4 and MTU6 to MTU8)

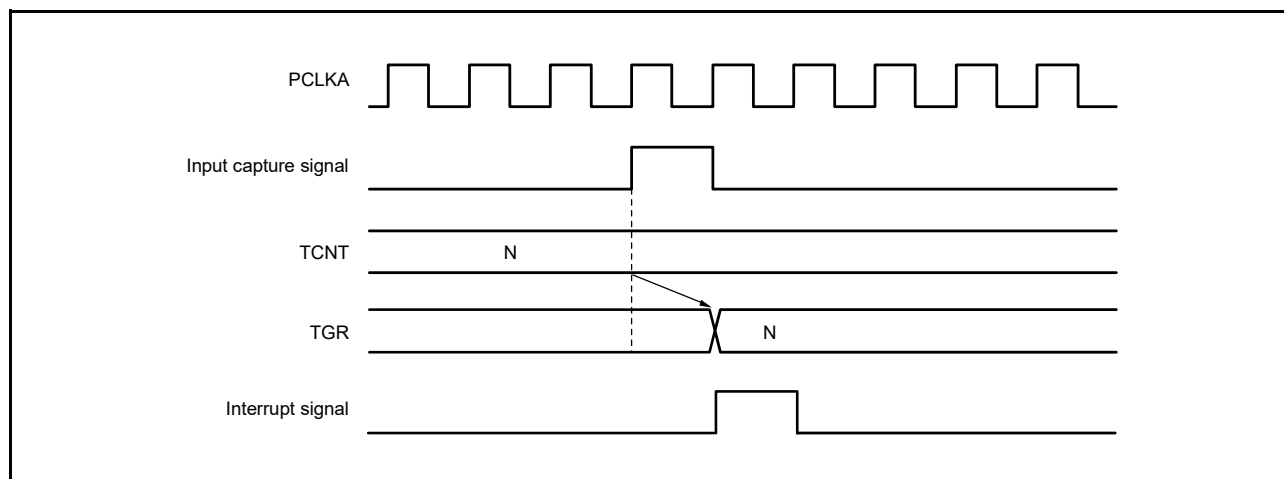


Figure 24.130 TGI Interrupt Timing (Input Capture) (MTU5)



### (3) TCIV and TCIU Interrupt Timing

Figure 24.131 shows the TCIV interrupt request signal timing when an overflow is generated.

Figure 24.132 shows the TCIU interrupt request signal timing when an underflow is generated.

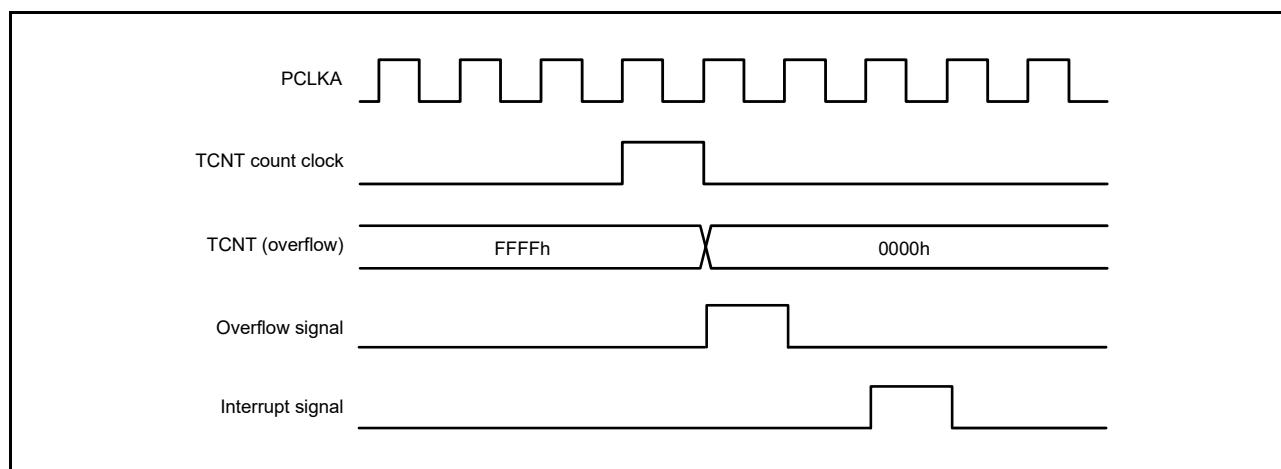


Figure 24.131 TCIV Interrupt Timing

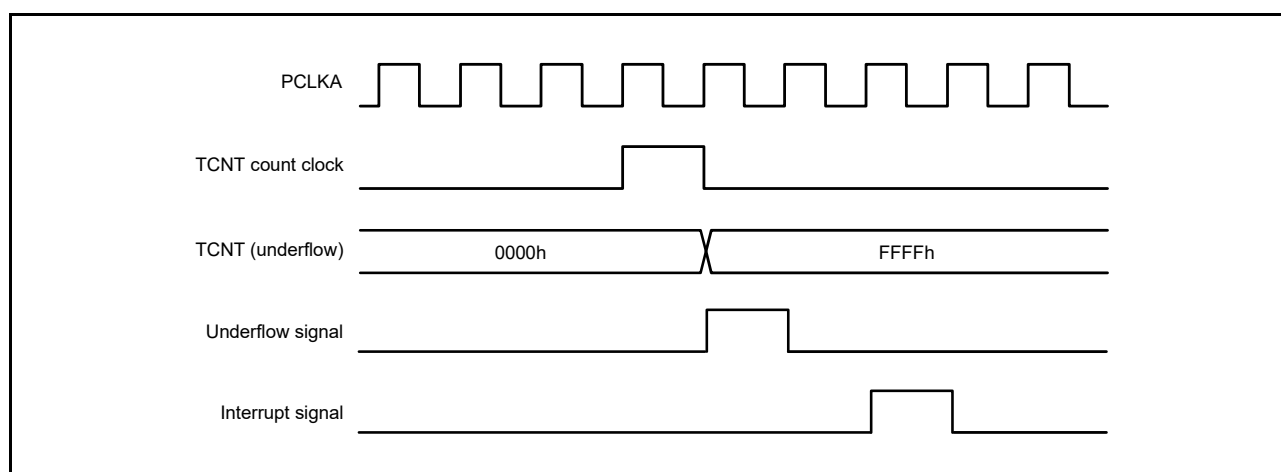


Figure 24.132 TCIU Interrupt Timing

## 24.6 Usage Notes

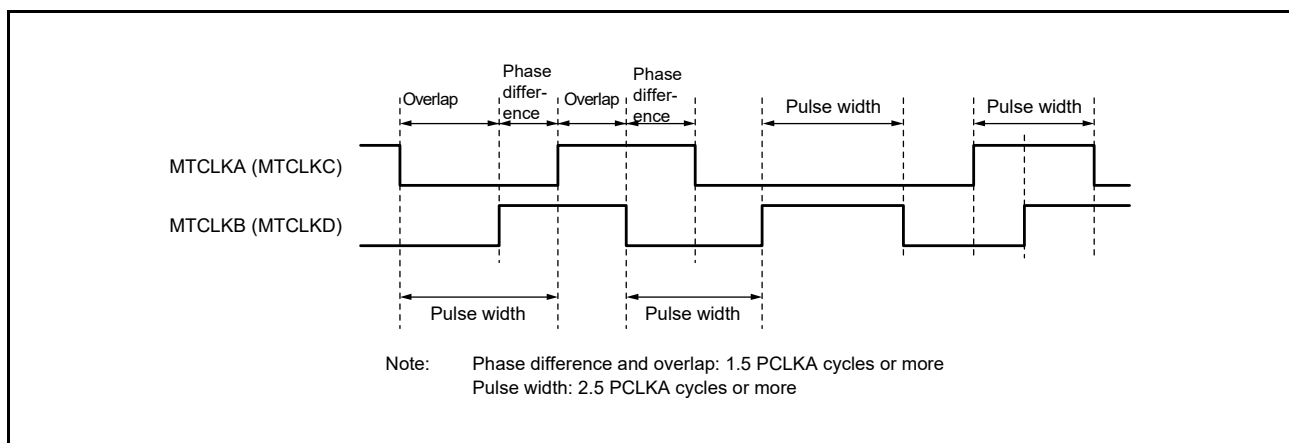
### 24.6.1 Module Stop Function Setting

MTU operation can be disabled or enabled using the module stop control register. MTU operation is stopped with the initial setting. Register access is enabled by releasing the module clock stop state. For details, refer to [section 11, Low Power Consumption](#).

### 24.6.2 Count Clock Restrictions

The count clock source pulse width must be at least 1.5 PCLKA cycles for single-edge detection, and at least 2.5 PCLKA cycles for both-edge detection. The MTU will not operate properly at narrower pulse widths.

In phase counting mode, the phase difference and overlap between the two input clocks must be at least 1.5 PCLKA cycles, and the pulse width must be at least 2.5 PCLKA cycles. Figure 24.133 shows the input clock conditions in phase counting mode.



**Figure 24.133 Phase Difference, Overlap, and Pulse Width in Phase Counting Mode**

### 24.6.3 Note on Period Setting

When counter clearing on compare match is set, TCNT is cleared in the final state in which it matches the TGR value (the point at which TCNT updates the matched count value). Consequently, the actual counter frequency is given by the following formula:

- MTU0 to MTU4 and MTU6 to MTU8

$$f = \frac{\text{CNTCLK}}{N + 1}$$

- MTU5

$$f = \frac{\text{CNTCLK}}{N}$$

f: Counter frequency

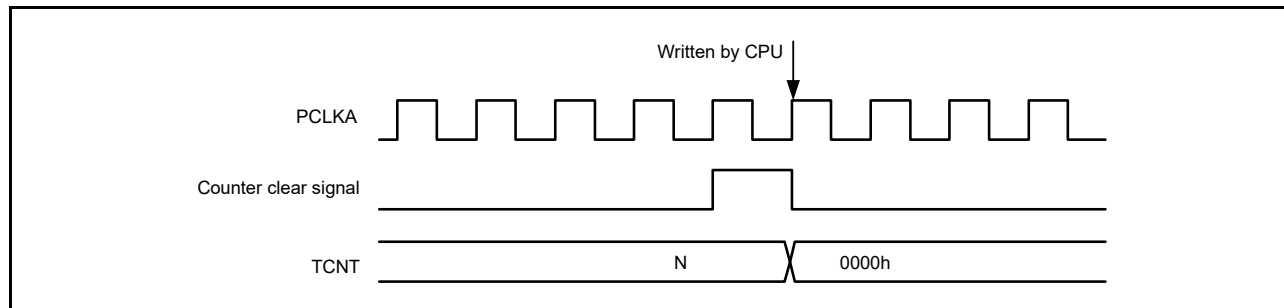
CNTCLK: The count clock frequency set by TCR.TPSC[2:0] and TCR2.TPSC2[2:0]

N: TGR setting

#### 24.6.4 Contention between TCNT Write and Clear Operations

If the counter clear signal is generated in the TCNT write cycle, TCNT clearing takes precedence and TCNT write operation is not performed.

Figure 24.134 shows the timing in this case.

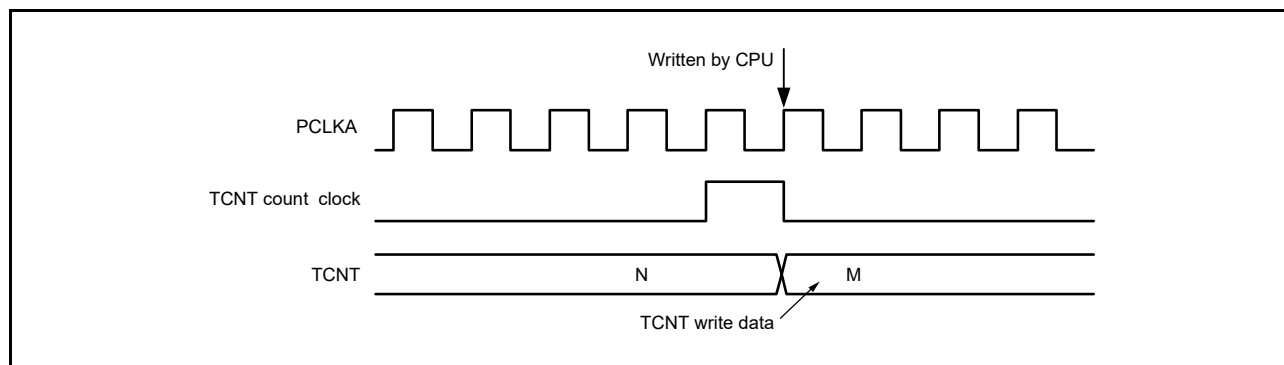


**Figure 24.134 Contention between TCNT Write and Clear Operations**

#### 24.6.5 Contention between TCNT Write and Increment Operations

If incrementing occurs in a TCNT write cycle, TCNT write operation takes precedence and TCNT is not incremented.

Figure 24.135 shows the timing in this case.

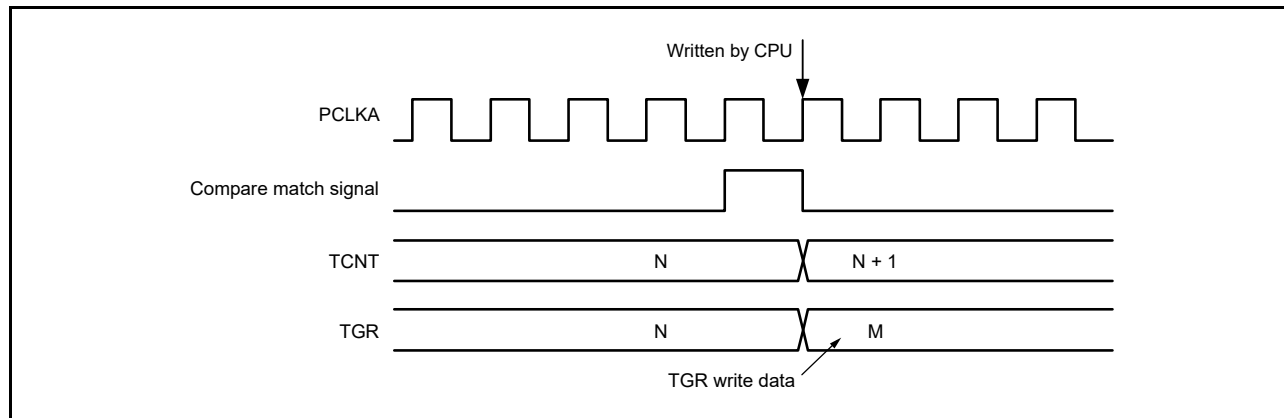


**Figure 24.135 Contention between TCNT Write and Increment Operations**

### 24.6.6 Contention between TGR Write Operation and Compare Match

If a compare match occurs in a TGR write cycle, TGR write operation is executed and the compare match signal is also generated.

Figure 24.136 shows the timing in this case.

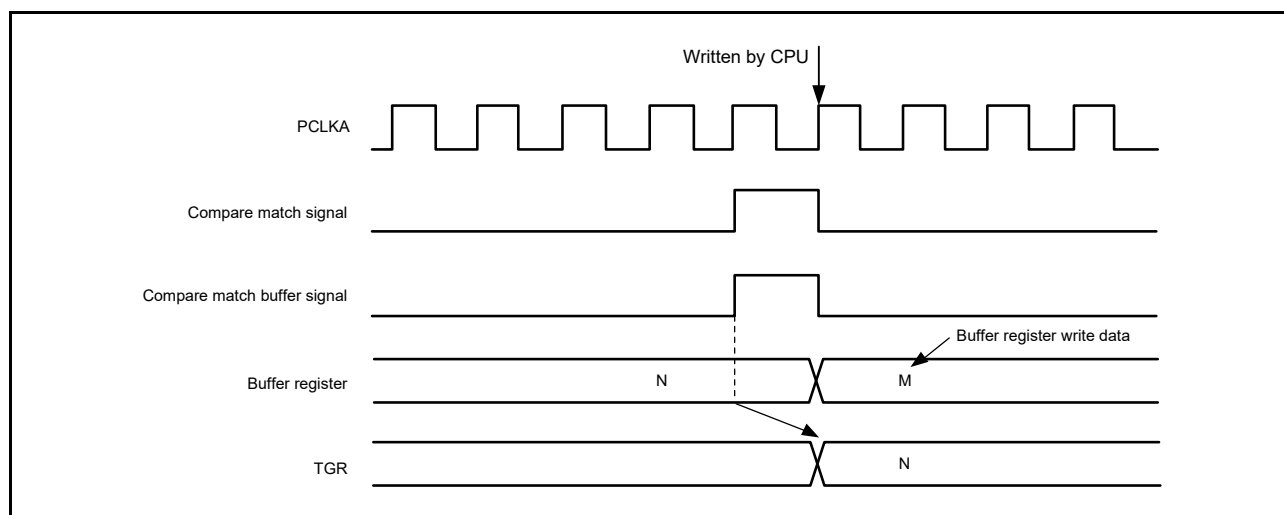


**Figure 24.136 Contention between TGR Write Operation and Compare Match**

### 24.6.7 Contention between Buffer Register Write Operation and Compare Match

If a compare match occurs in the T2 state in a TGR write cycle, the data before write operation is transferred to TGR by the buffer operation.

Figure 24.137 shows the timing in this case.



**Figure 24.137 Contention between Buffer Register Write Operation and Compare Match**

### 24.6.8 Contention between Buffer Register Write and TCNT Clear Operations

When the buffer transfer timing is set at the TCNT clear timing by the timer buffer transfer mode register (TBTM), if TCNT clearing occurs in the TGR write cycle, the data before write operation is transferred to TGR by the buffer operation.

Figure 24.138 shows the timing in this case.

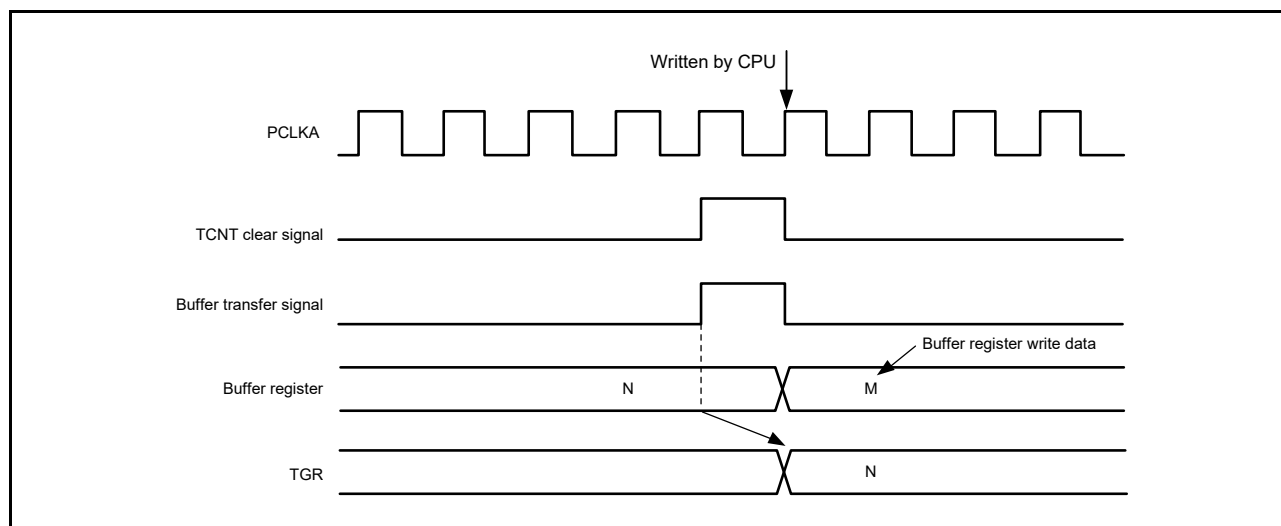


Figure 24.138 Contention between Buffer Register Write and TCNT Clear Operations

### 24.6.9 Contention between TGR Read Operation and Input Capture

If an input capture signal is generated in a TGR read cycle, the data before input capture transfer is read.

Figure 24.139 shows the timing in this case.

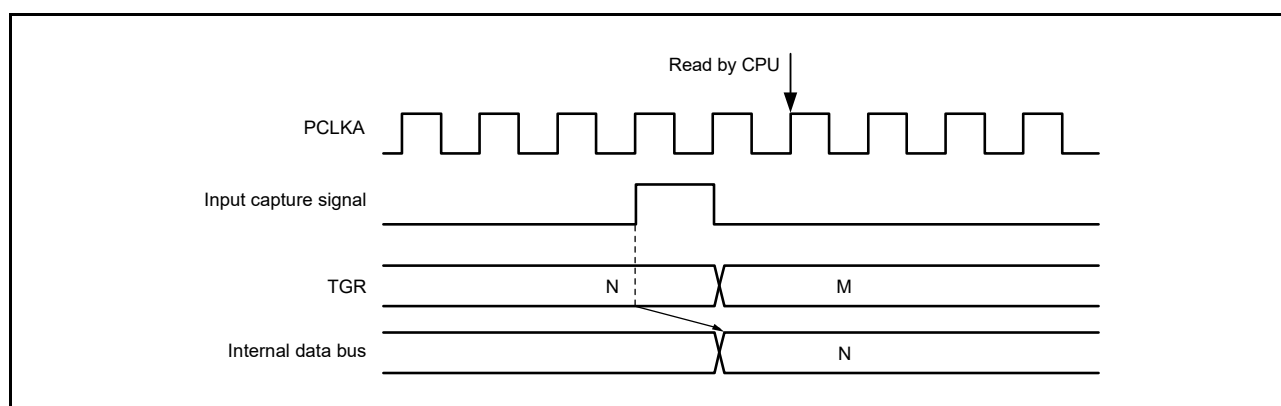


Figure 24.139 Contention between TGR Read Operation and Input Capture (MTU0 to MTU8)

### 24.6.10 Contention between TGR Write Operation and Input Capture

If an input capture signal is generated in the TGR write cycle, the input capture operation takes precedence and the TGR write operation is not performed in MTU0 to MTU4 and MTU6 to MTU8. In MTU5, the TGR write operation is performed and the input capture signal is generated.

Figure 24.140 and Figure 24.141 show the timing in this case.

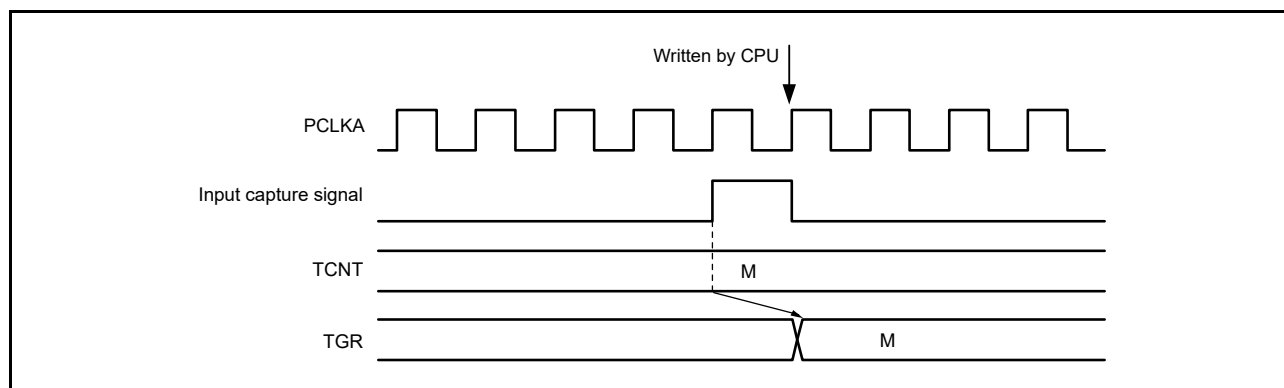


Figure 24.140 Contention between TGR Write Operation and Input Capture (MTU0 to MTU4 and MTU6 to MTU8)

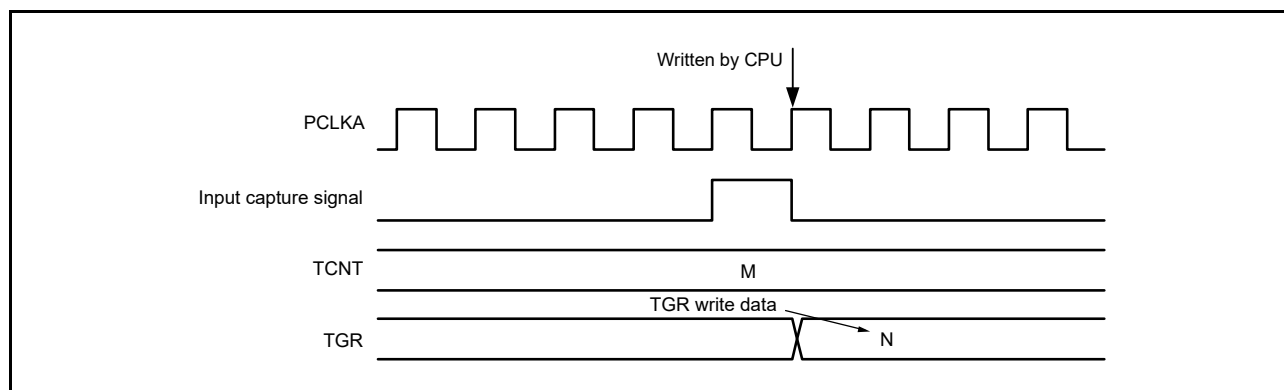


Figure 24.141 Contention between TGR Write Operation and Input Capture (MTU5)

### 24.6.11 Contention between Buffer Register Write Operation and Input Capture

If an input capture signal is generated in the buffer register write cycle, the buffer operation takes precedence and the buffer register write operation is not performed.

Figure 24.142 shows the timing in this case.

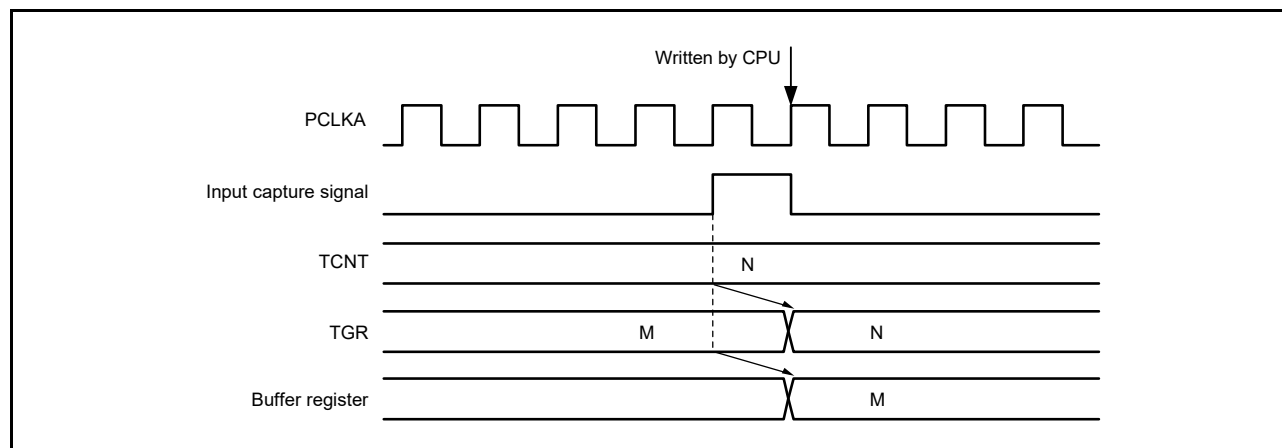


Figure 24.142 Contention between Buffer Register Write Operation and Input Capture

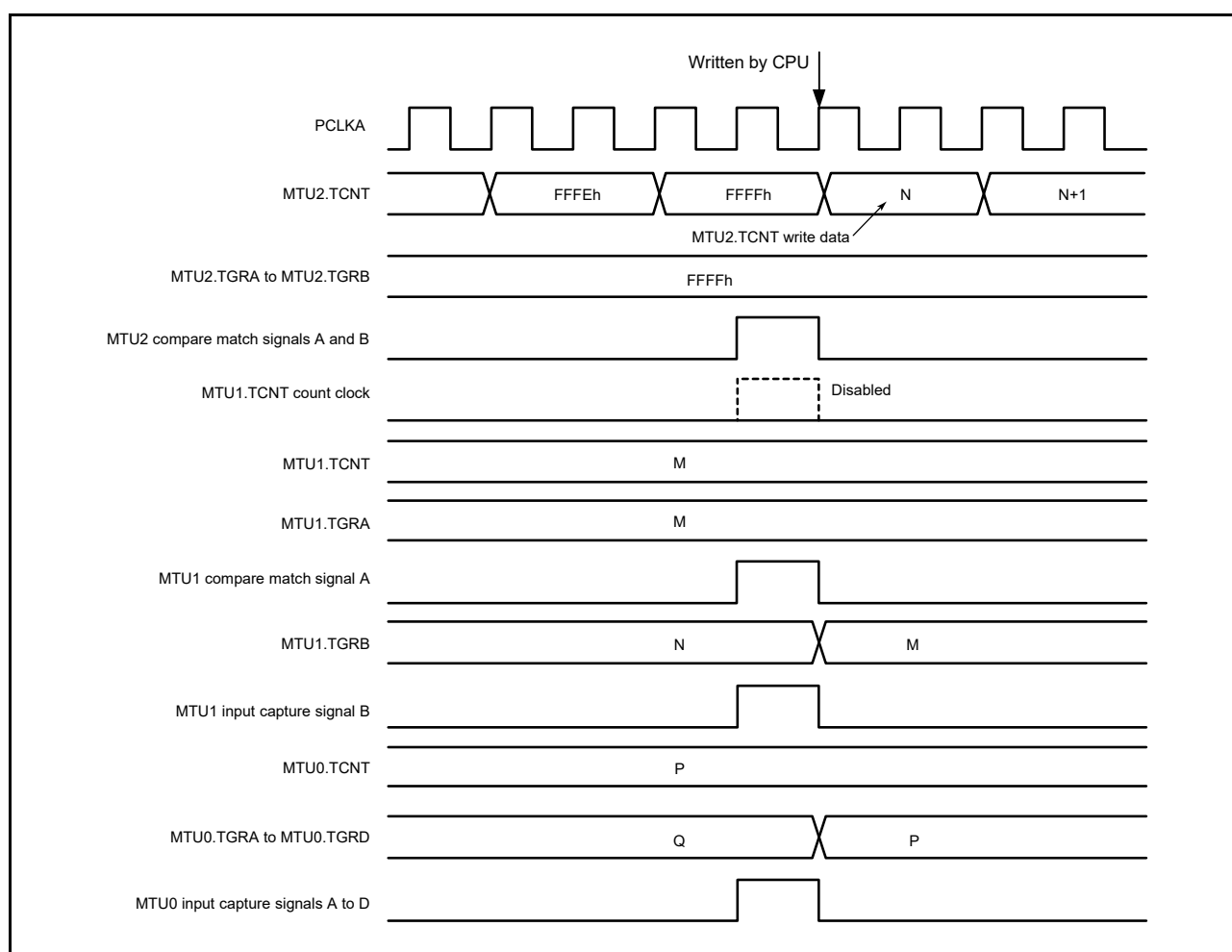
### 24.6.12 Contention between MTU2.TCNT Write Operation and Overflow/Underflow in Cascaded Operation

With timer counters MTU1.TCNT and MTU2.TCNT in a cascade, when a contention occurs between MTU1.TCNT counting (an MTU2.TCNT overflow/underflow) and the MTU2.TCNT write operation, the MTU2.TCNT write operation is performed and the MTU1.TCNT count signal is disabled. In this case, if MTU1.TGRA works as a compare match register and there is a match between the MTU1.TGRA and MTU1.TCNT values, a compare match signal is issued.

Furthermore, when the MTU1.TCNT count clock is selected as the input capture source of MTU0, MTU0.TGRA to MTU0.TGRD work in input capture mode. In addition, when the MTU0.TGRC compare match/input capture is selected as the input capture source of MTU1.TGRB, MTU1.TGRB works in input capture mode.

Figure 24.143 shows the timing in this case.

When setting the TCNT clearing function in cascaded operation, be sure to synchronize MTU1 and MTU2.



**Figure 24.143 Contention between MTU2.TCNT Write Operation and Overflow/Underflow in Cascaded Operation**



### 24.6.13 Counter Value When Count Operation is Stopped in Complementary PWM Mode

When counting operation in MTU3.TCNT and MTU4.TCNT (MTU6.TCNT and MTU7.TCNT) is stopped in complementary PWM mode, the MTU3.TCNT (MTU6.TCNT) value is set to the timer dead time register (TDDRA (TDDRB)) value and MTU4.TCNT (MTU7.TCNT) is set to 0000h.

When operation is restarted in complementary PWM mode, counting begins automatically from the initial setting state. Figure 24.144 shows this operation.

When counting begins in another operating mode, be sure to make initial settings in MTU3.TCNT and MTU4.TCNT (MTU6.TCNT and MTU7.TCNT).

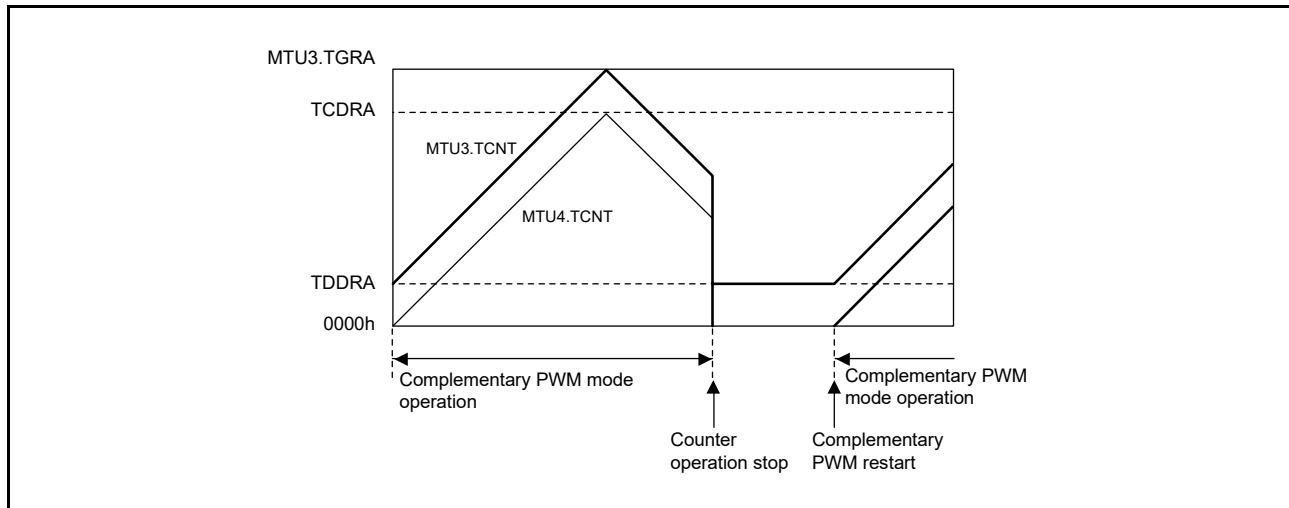


Figure 24.144 Counter Value When Stopped in Complementary PWM Mode

### 24.6.14 Buffer Operation Setting in Complementary PWM Mode

When modifying the PWM period set register (MTU3.TGRA or MTU6.TGRA), timer period data register (TCDRA or TCDRB), and duty set registers (MTU3.TGRB, MTU4.TGRA, and MTU4.TGRB (MTU6.TGRB, MTU7.TGRA, and MTU7.TGRB)) in complementary PWM mode, be sure to use buffer operation. In addition, set the BFA and BFB bits in MTU4.TMDR1 (MTU7.TMDR1) to 0. The MTIOC4C (MTIOC7C) pin cannot output waveforms if the BFA bit in MTU4.TMDR1 (MTU7.TMDR1) is set to 1. Likewise, the MTIOC4D (MTIOC7D) pin cannot output waveforms if the BFB bit in MTU4.TMDR1 (MTU7.TMDR1) is set to 1.

In complementary PWM mode, buffer operation in MTU3 and MTU4 (or MTU6 and MTU7) depends on the settings in bits BFA and BFB of MTU3.TMDR1 (MTU6.TMDR1). When the BFA bit in MTU3.TMDR1 (MTU6.TMDR1) is set to 1, MTU3.TGRC (MTU6.TGRC) functions as a buffer register for MTU3.TGRA (MTU6.TGRA). At the same time, MTU4.TGRC (MTU7.TGRC) functions as a buffer register for MTU4.TGRA (MTU7.TGRA), and TCBRA (TCBRB) functions as a buffer register for TCDRA (TCDRB).

### 24.6.15 Buffer Operation and Compare Match in Reset-Synchronized PWM Mode

When setting buffer operation in reset-synchronized PWM mode, set the BFA and BFB bits in MTU4.TMDR1 (MTU7.TMDR1) to 0. The MTIOC4C (MTIOC7C) pin cannot output waveforms if the BFA bit in MTU4.TMDR1 (MTU7.TMDR1) is set to 1. Likewise, the MTIOC4D (MTIOC7D) pin cannot output waveforms if the BFB bit in MTU4.TMDR1 (MTU7.TMDR1) is set to 1.

In reset-synchronized PWM mode, buffer operation in MTU3 and MTU4 (or MTU 6 and MTU7) depends on the settings in the BFA and BFB bits of MTU3.TMDR1 (MTU6.TMDR1). For example, if the BFA bit in MTU3.TMDR1 (MTU6.TMDR1) is set to 1, MTU3.TGRC (MTU6.TGRC) functions as a buffer register for MTU3.TGRA (MTU6.TGRA). At the same time, MTU4.TGRC (MTU7.TGRC) functions as a buffer register for MTU4.TGRA (MTU7.TGRA).

While the MTU3.TGRC (MTU6.TGRC) and MTU3.TGRD (MTU6.TGRD) are operating as buffer registers, a TGI<sub>mn</sub> interrupt (m = C, D; n = 3, 4 or 6, 7) is not generated.

Figure 24.145 shows an example of MTU3.TGR (MTU6.TGR), MTU4.TGR (MTU7.TGR), MTIOC3 (MTIOC6), and MTIOC4 (MTIOC7) operation with the BFA and BFB bits in MTU3.TMDR1 (MTU6.TMDR1) set to 1 and the BFA and BFB bits in MTU4.TMDR1 (MTU7.TMDR1) set to 0.

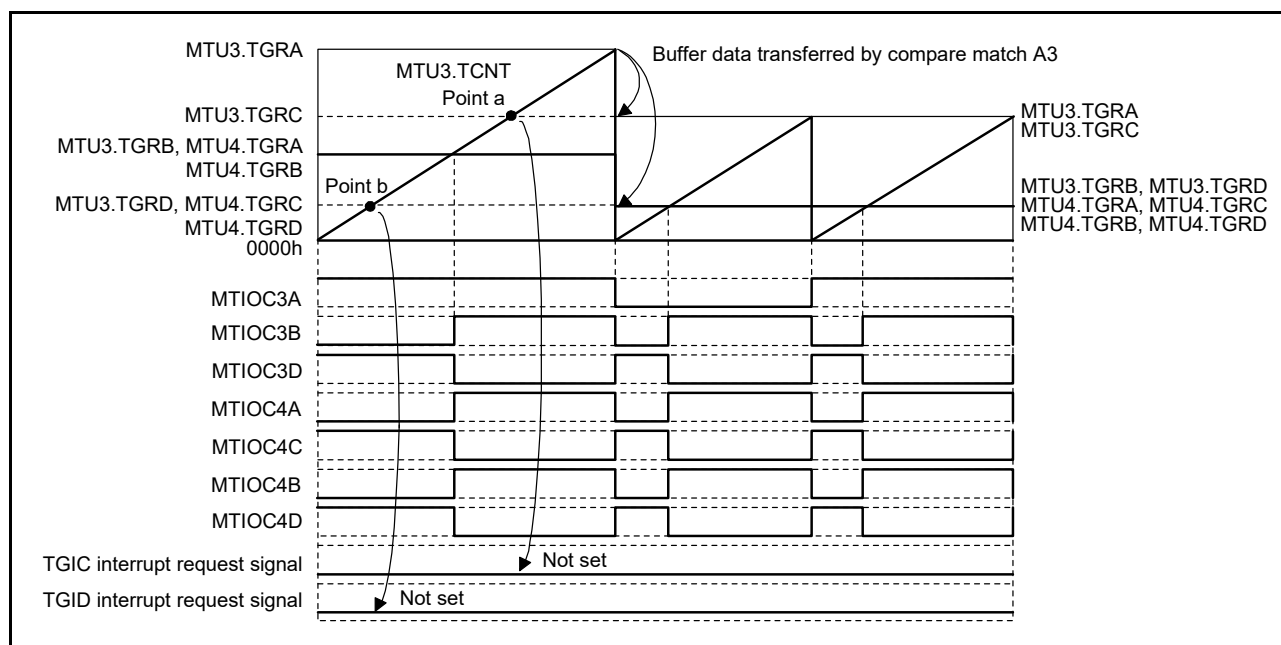


Figure 24.145 Buffer Operation and Compare Match in Reset-Synchronized PWM Mode

### 24.6.16 Overflow in Reset-Synchronized PWM Mode

After reset-synchronized PWM mode is selected, MTU3.TCNT and MTU4.TCNT (MTU6.TCNT and MTU7.TCNT) start counting when the CST3 (CST6) bit of TSTRA (TSTRB) is set to 1. In this state, the MTU4.TCNT (MTU7.TCNT) count clock source and count edge are determined by the MTU3.TCR (MTU6.TCR) setting.

In reset-synchronized PWM mode, with period register MTU3.TGRA (MTU6.TGRA) set to FFFFh and the MTU3.TGRA (MTU6.TGRA) compare match selected as the counter clearing source, MTU3.TCNT and MTU4.TCNT (MTU6.TCNT and MTU7.TCNT) count up to FFFFh, then a compare match occurs with MTU3.TGRA (MTU6.TGRA), and MTU3.TCNT and MTU4.TCNT (MTU6.TCNT and MTU7.TCNT) are both cleared. In this case, a TCIVn interrupt (n = 3, 4 or 6, 7) is not generated.

Figure 24.146 shows an example of operation in reset-synchronized PWM mode with period register MTU3.TGRA (MTU6.TGRA) set to FFFFh and the MTU3.TGRA (MTU6.TGRA) compare match specified for the counter clearing source.

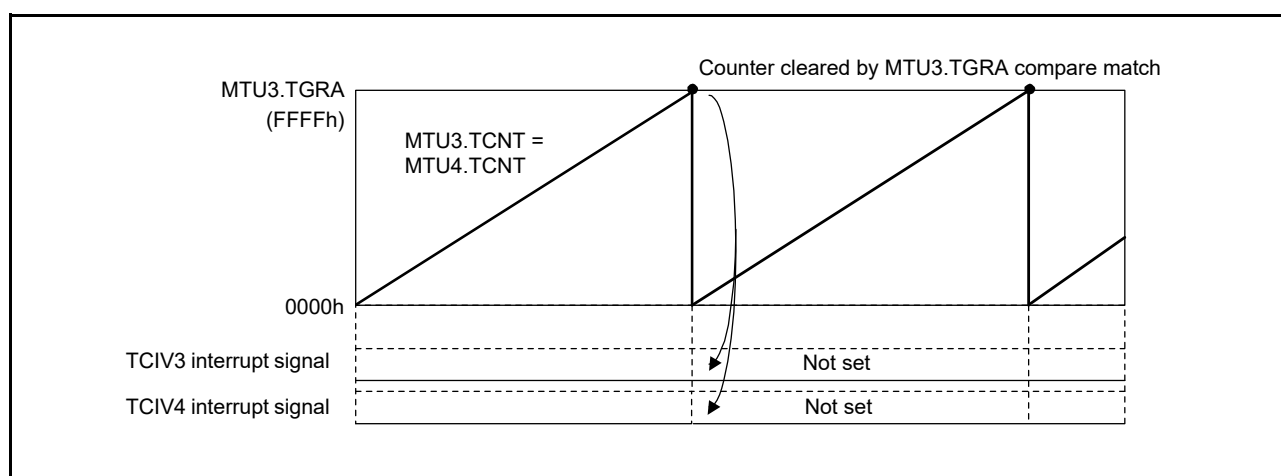


Figure 24.146 Overflow in Reset-Synchronized PWM Mode

### 24.6.17 Contention between Overflow/Underflow and Counter Clearing

If an overflow/underflow and counter clearing occur simultaneously, a TCIV<sub>n</sub> interrupt (n = 0 to 4, 6 to 8) nor a TCIU<sub>n</sub> interrupt (n = 1, 2) is not generated and TCNT clearing takes precedence.

Figure 24.147 shows the operation timing when a TGR compare match is specified as the clearing source and TGR is set to FFFFh.

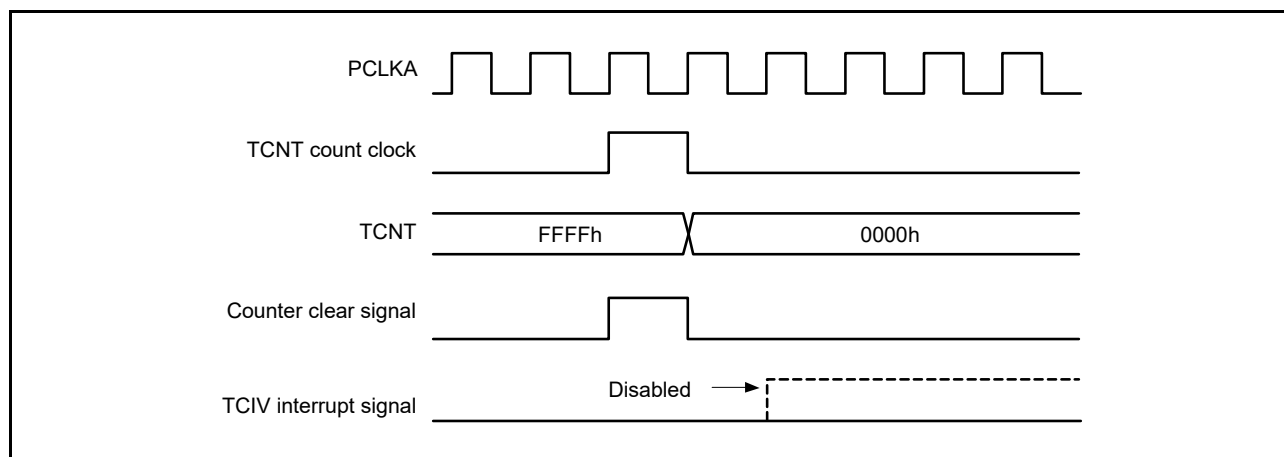


Figure 24.147 Contention between Overflow and Counter Clearing

### 24.6.18 Contention between TCNT Write Operation and Overflow/Underflow

If TCNT counts up or down in a TCNT write cycle and an overflow or an underflow occurs, the TCNT write operation takes precedence. A TCIV<sub>n</sub> interrupt (n = 0 to 4, 6 to 8) nor a TCIU<sub>n</sub> interrupt (n = 1, 2) is not generated.

Figure 24.148 shows the operation timing when there is contention between TCNT write operation and overflow.

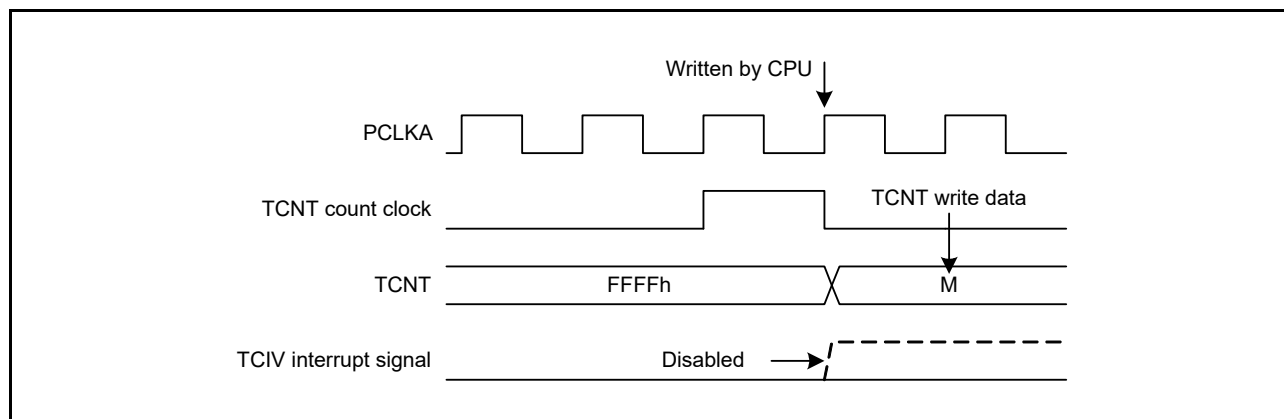


Figure 24.148 Contention between TCNT Write Operation and Overflow

### 24.6.19 Note on Transition from Normal Mode or PWM Mode 1 to Reset-Synchronized PWM Mode

When making a transition from normal mode or PWM mode 1 to reset-synchronized PWM mode in MTU3 and MTU4 (or MTU6 and MTU7), if the counter is stopped while the output pins (MTIOC3B, MTIOC3D, MTIOC4A, MTIOC4C, MTIOC4B, MTIOC4D, MTIOC6B, MTIOC6D, MTIOC7A, MTIOC7C, MTIOC7B, and MTIOC7D) are held at a high level and then operation is started after a transition to reset-synchronized PWM mode, the initial pin output will not be correct.

When making a transition from normal mode to reset-synchronized PWM mode, write 11h to MTU3.TIORH, MTU3.TIORL, MTU4.TIORH, and MTU4.TIORL (MTU6.TIORH, MTU6.TIORL, MTU7.TIORH, and MTU7.TIORL) to initialize the output pin state to a low level, then set the registers to the initial value (00h) before making the mode transition.

When making a transition from PWM mode 1 to reset-synchronized PWM mode, switch to normal mode, initialize the output pin state to a low level, and then set the registers to the initial value (00h) before making the transition to reset-synchronized PWM mode.

### 24.6.20 Output Level in Complementary PWM Mode and Reset-Synchronized PWM Mode

When MTU3 and MTU4 (or MTU6 and MTU7) are in complementary PWM mode or reset-synchronized PWM mode, the PWM waveform output level is determined by the OLSP and OLSN bits in the timer output control register (TOCR1A or TOCR1B). In complementary PWM mode or reset-synchronized PWM mode, TIOR should be set to 00h. The output level in negative phase when the TDERA.TDER (TDERB.TDER) bit is set to 0 in complementary PWM mode (the dead time is not generated) does not depend on the setting of the TOCR1A.OLSN (TOCR1B.OLSN) bit. It is equivalent to the inverted level of positive phase output based on the setting of the TOCR1A.OLSP (TOCR1B.OLSP) bit.

### 24.6.21 Simultaneous Input Capture in MTU1.TCNT and MTU2.TCNT in Cascade Connection

When timer counters 1 and 2 (MTU1.TCNT and MTU2.TCNT) operate as a 32-bit counter in cascade connection, the cascaded counter value cannot be captured successfully in some cases even if input-capture input is simultaneously done to MTIOC1A and MTIOC2A or to MTIOC1B and MTIOC2B. This is because the input timing of MTIOC1A and MTIOC2A or to MTIOC1B and MTIOC2B may not be the same when external input-capture signals input into MTU1.TCNT and MTU2.TCNT are taken in synchronization with the internal clock.

For example, MTU1.TCNT (the counter for upper 16 bits) does not capture the count-up value by an overflow from MTU2.TCNT (the counter for lower 16 bits) but captures the count value before the up-counting. In this case, the values of MTU1.TCNT = FFF1h and MTU2.TCNT = 0000h should be transferred to MTU1.TGRA and MTU2.TGRA or to MTU1.TGRB and MTU2.TGRB, but the values of MTU1.TCNT = FFF0h and MTU2.TCNT = 0000h are erroneously transferred.

The MTU has a function that allows simultaneous capture of MTU1.TCNT and MTU2.TCNT with a single input capture input. This function can be used to read the 32-bit counter such that MTU1.TCNT and MTU2.TCNT are captured at the same time. For details, refer to section 24.2.11, Timer Input Capture Control Register (TICCR).

### 24.6.22 Interrupt Skipping Function 2

When interrupt skipping function 2 is in use and the difference between the values in MTU4.TADCORA and MTU4.TADCORB is small, correct counting of the number skipped may not be possible, in which case requests for A/D conversion will not be generated with the expected timing. The conditions listed below thus apply to these settings. For MTU6 and MTU7, the same conditions apply to the settings of MTU7.TADCORA and MTU7.TADCORB.

- (1) When the number skipped is zero for skipping function 2
  - The difference between the values in MTU4.TADCORA and MTU4.TADCORB must be at least four.
  - The interval of comparison for MTU4.TADCORA must be at least four cycles of PCLKA (the updated value of MTU4.TADCORA is set to the previous value plus or minus at least four).
  - The interval of comparison for MTU4.TADCORB must be at least four cycles of PCLKA (the updated value of MTU4.TADCORB is set to the previous value plus or minus at least four).
- (2) When the number skipped is one or more for skipping function 2
  - The difference between the values in MTU4.TADCORA and MTU4.TADCORB must be at least two.
  - The interval of comparison for MTU4.TADCORB must be at least two cycles of PCLKA (the updated value of MTU4.TADCORB is set to the previous value plus or minus at least two).

### 24.6.23 Notes When Complementary PWM Mode Output Protection Function is Not Used

The complementary PWM mode output protection function is initially enabled. For details, refer to section 25, Port Output Enable 3 (POE3a).

### 24.6.24 Notes Regarding Timer Counter (MTU5.TCNT) and Timer General Register (MTU5.TGR)

Do not set an MTU5.TGR<sub>m</sub> ( $m = U, V, W$ ) bit to the value of the corresponding MTU5.TCNT<sub>m</sub> ( $m = U, V, W$ ) plus one while counting by the MTU5.TCNT<sub>m</sub> ( $m = U, V, W$ ) register is stopped. If an MTU5.TGR<sub>m</sub> ( $m = U, V, W$ ) bit is set to the value of the corresponding MTU5.TCNT<sub>m</sub> ( $m = U, V, W$ ) plus one while counting by the MTU5.TCNT<sub>m</sub> ( $m = U, V, W$ ) is stopped, a compare-match will be generated even though counting is stopped.

In this case, if the compare match enable bit (MTU5.TIER.TGIE5<sub>m</sub> ( $m = U, V, W$ ) bit is set to 1 (enabling interrupts), a compare-match interrupt will also be generated. If the value of the timer compare match clear register is 1 (enabled), the timer is automatically cleared to 0000h when the compare-match is generated, regardless of whether interrupts from the MTU5.TCNT<sub>m</sub> ( $m = U, V, W$ ) are enabled or disabled.

### 24.6.25 Notes to Prevent Malfunctions in Synchronous Clearing for Complementary PWM Mode

If control of the output waveform is enabled (TWCRA.WRE bit = 1 or TWCRB.WRE bit = 1) at the time of synchronous counter clearing in complementary PWM mode, satisfaction of either condition 1 or 2 below has the following effects.

- Dead time on the PWM output pins is shortened (or disappears).
- The active level is output on the negative phase PWM output pins beyond the period for active-level output.

Condition 1: In portion (10) of the initial output inhibition period in Figure 24.149, synchronous clearing occurs within the dead-time period for PWM output.

Condition 2: In portions (10) and (11) of the initial output inhibition period in Figure 24.150, synchronous clearing occurs when any condition from among  $MTU3.TGRB (MTU6.TGRB) \leq TDDRA (TDDRB)$ ,  $MTU4.TGRA (MTU7.TGRA) \leq TDDRA (TDDRB)$ , or  $MTU4.TGRB (MTU7.TGRB) \leq TDDRA (TDDRB)$  is satisfied.

The following method avoids the above phenomena.

Ensure that synchronous clearing proceeds with the value of each comparison register ( $MTU3.TGRB (MTU6.TGRB)$ ,  $MTU4.TGRA (MTU7.TGRA)$ , and  $MTU4.TGRB (MTU7.TGRB)$ ) set to at least double the value of the TDDRA register (TDDRB register).

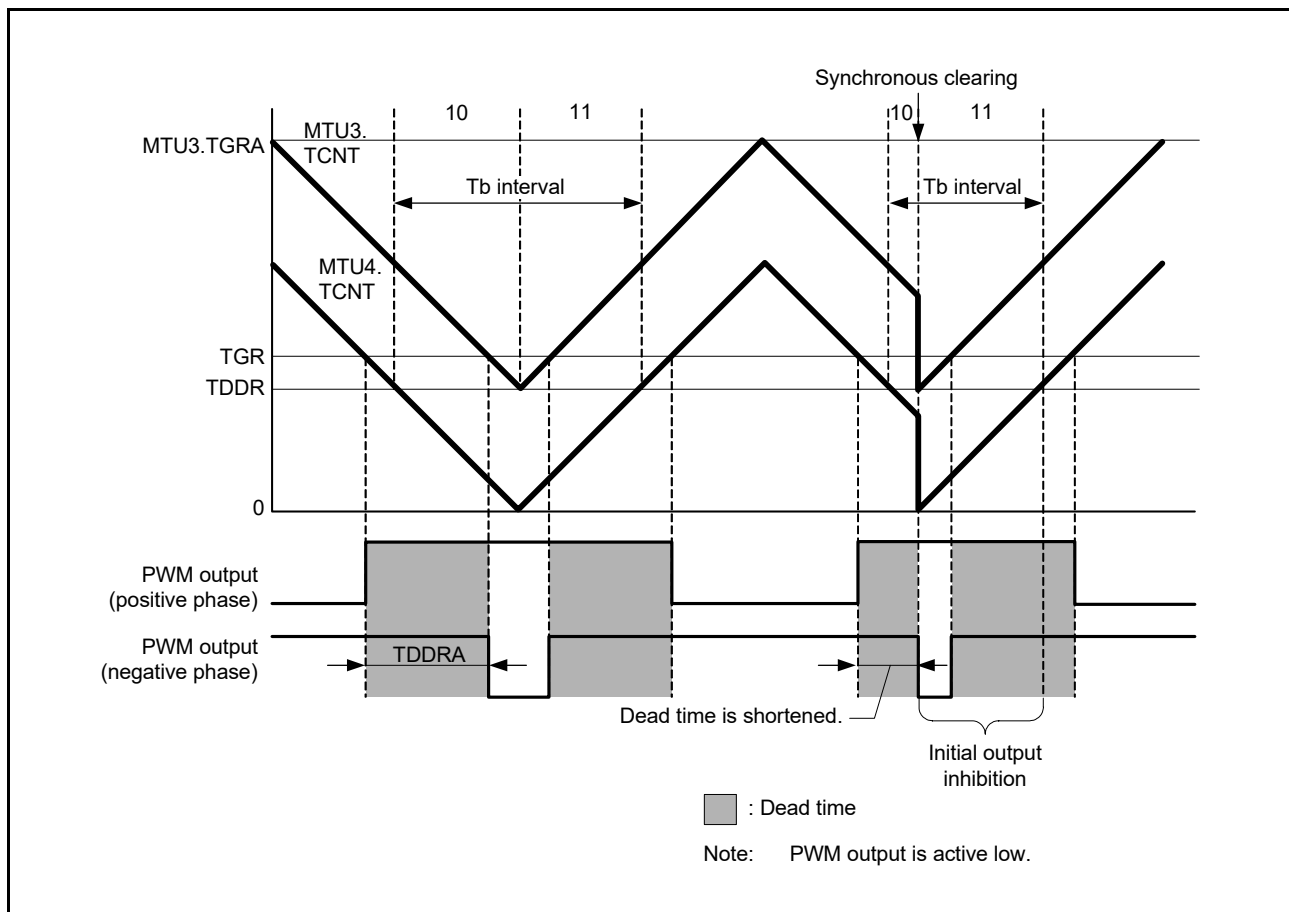
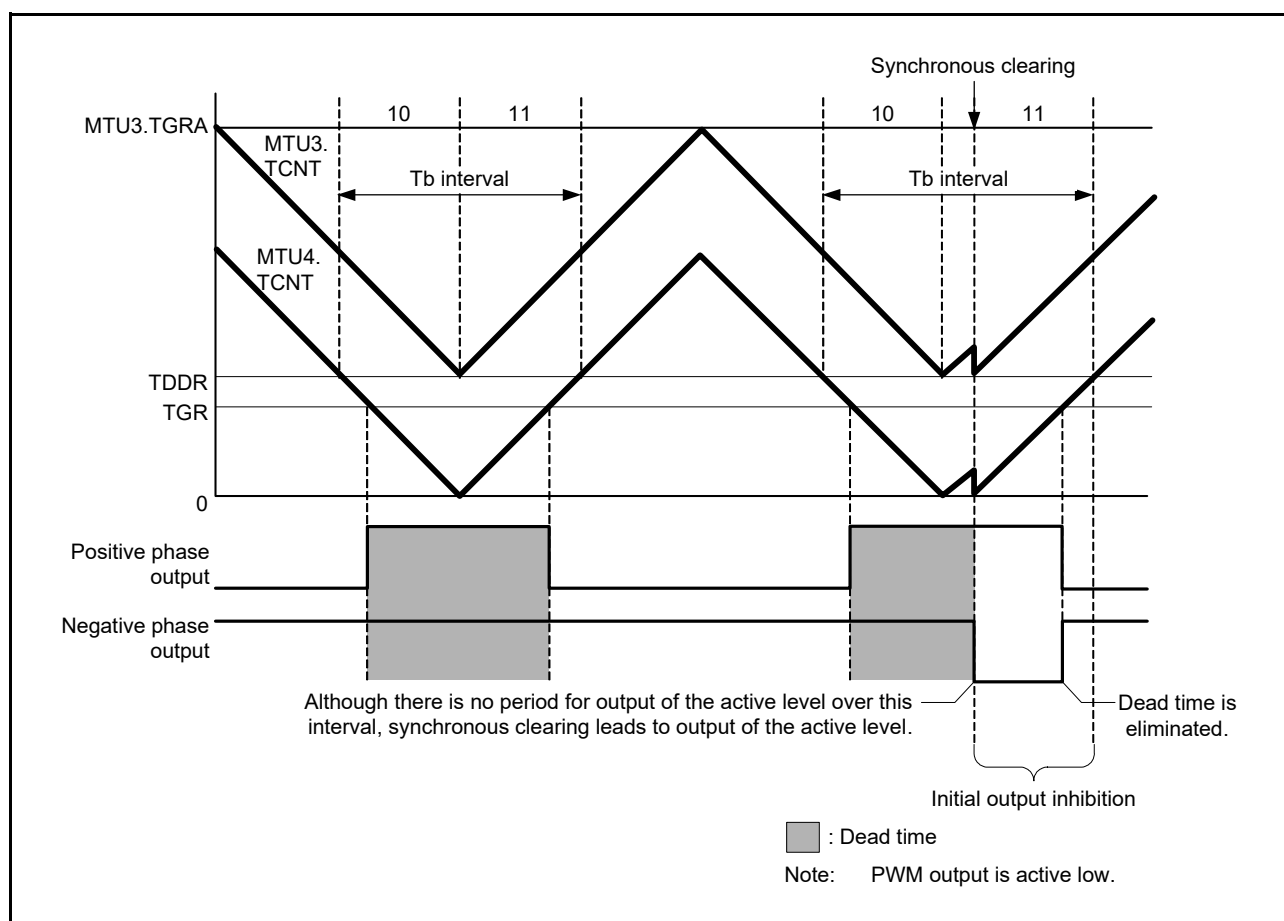


Figure 24.149 Example of Synchronous Clearing (When Condition 1 Applies)



**Figure 24.150 Example of Synchronous Clearing (When Condition 2 Applies)**

#### 24.6.26 Notes on Timer Mode Register Setting for ELC Event Input

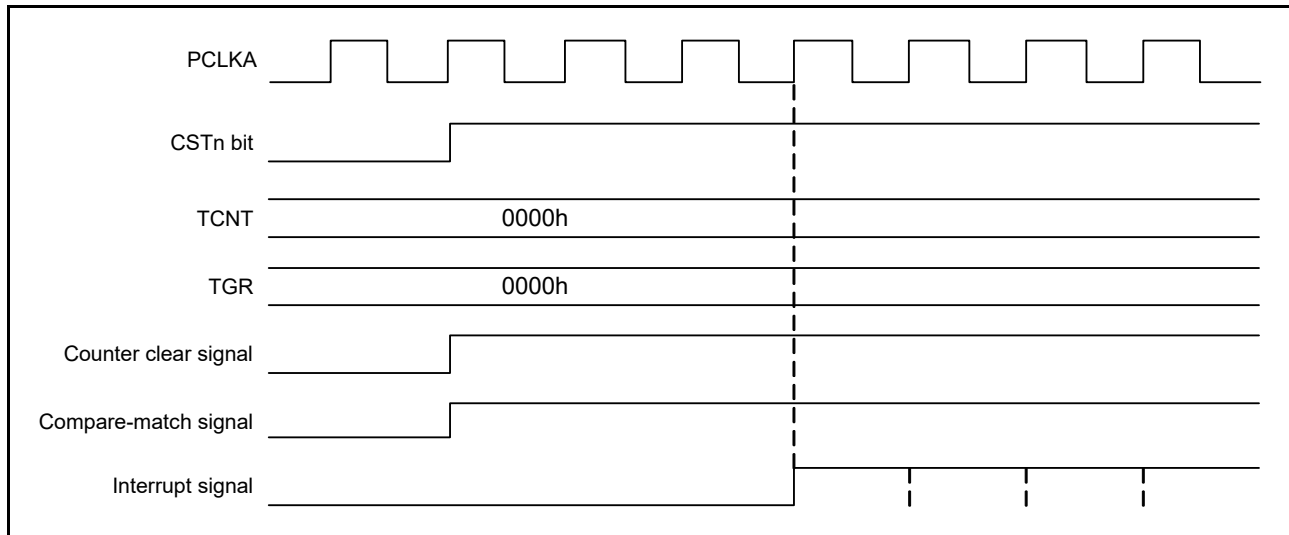
When MTU is used in ELC operation, set the timer mode register (TMDR) of the relevant channel to its initial value (00h).



### 24.6.27 Continuous Output of Interrupt Signal in Response to a Compare Match

When the TGR register is set to 0000h, the PCLKA/1 is set as the count clock, and compare match is set as the trigger for clearing of the count clock, the value of the TCNT counter remains 0000h, and the interrupt signal will be output continuously (i.e. its level will be flat) rather than output over a single cycle. Consequently, interrupts will not be detected in response to second and subsequent compare matches.

Figure 24.151 shows the timing for continuous output of the interrupt signal in response to a compare match.



**Figure 24.151 Continuous Output of Interrupt Signal in Response to a Compare Match**

### 24.6.28 Usage Notes on A/D Conversion Start Request Delaying Function in Complementary PWM Mode

- When data is transferred from a buffer register at the trough of the MTU4.TCNT (MTU7.TCNT) counter while the MTU4.TADCOBRA and MTU4.TADCOBRB (MTU7.TADCOBRA and MTU7.TADCOBRB) registers are set to 0 and the UT4AE and UT4BE (UT7AE and UT7BE) bits in the MTU4.TADCR (MTU7.TADCR) register are set to 1, no A/D conversion start request is issued during up-counting immediately after transfer. Refer to Figure 24.152.
- When data is transferred from a buffer register at the crest of the MTU4.TCNT (MTU7.TCNT) counter while the MTU4.TADCOBRA and MTU4.TADCOBRB (MTU7.TADCOBRA and MTU7.TADCOBRB) registers are set to the same value as the TCDR and the UT4AE and UT4BE (UT7AE and UT7BE) bits in the MTU4.TADCR (MTU7.TADCR) register are set to 1, no A/D conversion start request is issued during down-counting immediately after transfer. Refer to Figure 24.153.
- To issue an A/D conversion start request linked with the interrupt skipping function, set the MTU4.TADCORA and MTU4.TADCORB (MTU7.TADCORA and MTU7.TADCORB) registers so that  $2 \leq \text{MTUn.TADCORA/TADCORB} \leq \text{TCDR} - 2$  is satisfied ( $n = 4, 7$ ).

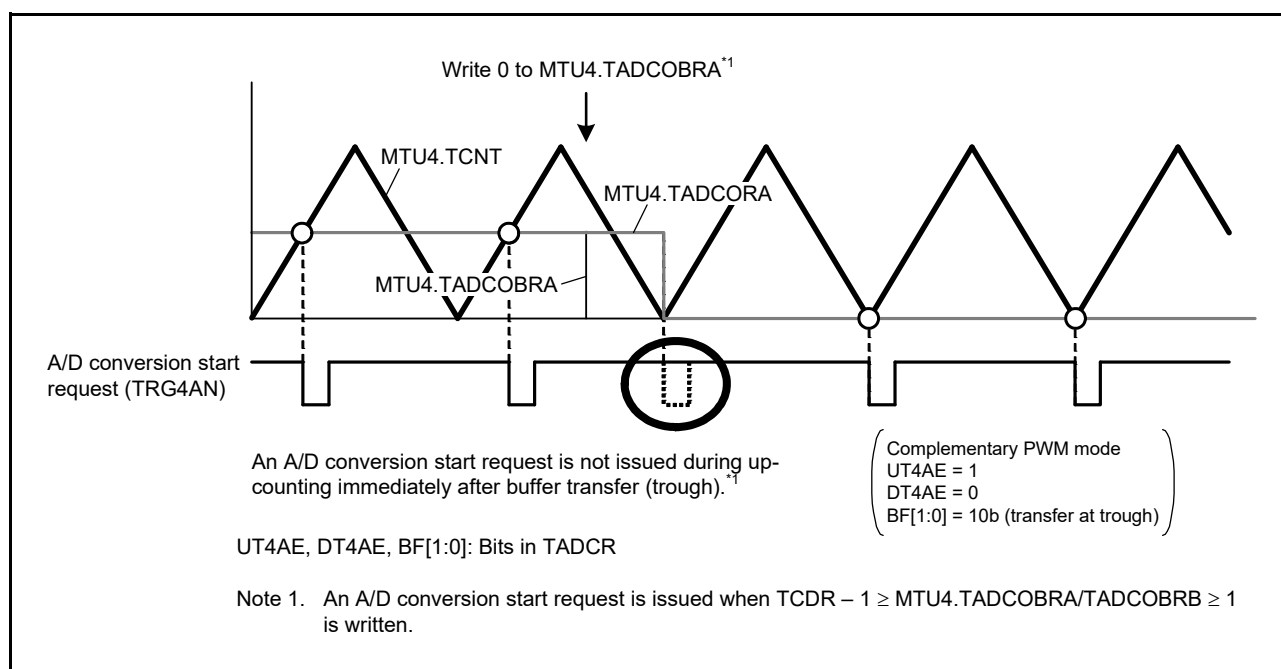


Figure 24.152 A/D Conversion Start Request When 0 is Written to MTU4.TADCOBRA

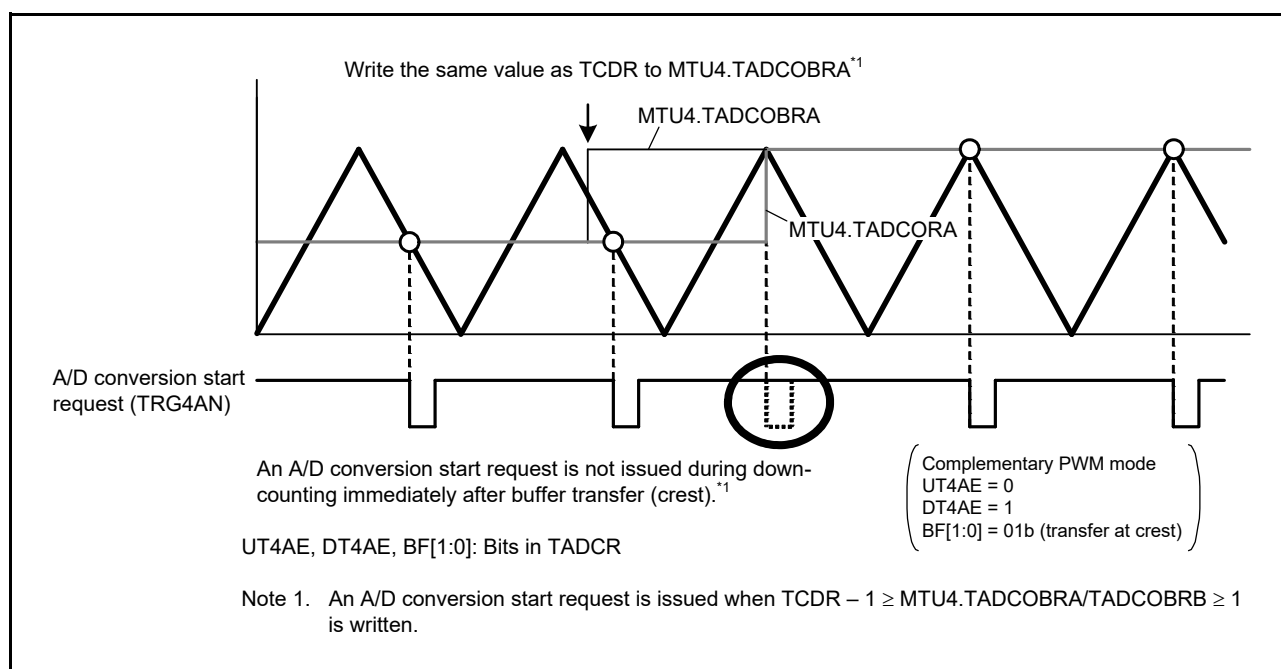


Figure 24.153 A/D Conversion Start Request When the Same Value as TCDR is Written to MTU4.TADCOBRA

## 24.7 MTU Output Pin Initialization

### 24.7.1 Operating Modes

The MTU has the following six operating modes. Waveforms can be output in any of these modes.

- Normal mode (MTU0 to MTU4 and MTU6 to MTU8)
- PWM mode 1 (MTU0 to MTU4, MTU6, and MTU7)
- PWM mode 2 (MTU0 to MTU2)
- Phase counting modes 1 to 5 (MTU1 and MTU2)
- Complementary PWM mode (MTU3, MTU4, MTU6, and MTU7)
- Reset-synchronized PWM mode (MTU3, MTU4, MTU6, and MTU7)

This section describes how to initialize the MTU output pins in each of these modes.

### 24.7.2 Operation in Case of Re-Setting Due to Error during Operation

If an error occurs during MTU operation, MTU output should be cut off by the system. The output can be cut off by allowing non-active level output from the pins by setting the pins as general output ports using the port direction registers (PDR), port output data registers (PODR), and port mode registers (PMR) of the I/O ports. MTU output can be disabled through TIOR settings. Complementary PWM output (MTIOC3B, MTIOC3D, MTIOC4A, MTIOC4B, MTIOC4C, MTIOC4D, MTIOC6B, MTIOC6D, MTIOC7A, MTIOC7B, MTIOC7C, and MTIOC7D) should be specified through TOERA and TOERB settings. For PWM output pins, output can also be cut by hardware, using port output enable 3 (POE3). The pin initialization procedures for re-setting due to an error during operation and the procedures for restarting in a different mode after re-setting are described below.

The MTU has six operating modes, as stated above. There are thus 36 mode transition combinations, but some transitions are not available with certain channel and mode combinations. Available mode transition combinations are shown in Table 24.79.

**Table 24.79 Mode Transition Combinations**

|        | Normal | PWM1 | PWM2          | PCM           | CPWM          | RPWM          |
|--------|--------|------|---------------|---------------|---------------|---------------|
| Normal | (1)    | (2)  | (3)           | (4)           | (5)           | (6)           |
| PWM1   | (7)    | (8)  | (9)           | (10)          | (11)          | (12)          |
| PWM2   | (13)   | (14) | (15)          | (16)          | Not available | Not available |
| PCM    | (17)   | (18) | (19)          | (20)          | Not available | Not available |
| CPWM   | (21)   | (22) | Not available | Not available | (23) (24)     | (25)          |
| RPWM   | (26)   | (27) | Not available | Not available | (28)          | (29)          |

Normal: Normal mode  
 PWM1: PWM mode 1  
 PWM2: PWM mode 2  
 PCM: Phase counting modes 1 to 5  
 CPWM: Complementary PWM mode  
 RPWM: Reset-synchronized PWM mode

### 24.7.3 Overview of Initialization Procedures and Mode Transitions in Case of Error during Operation

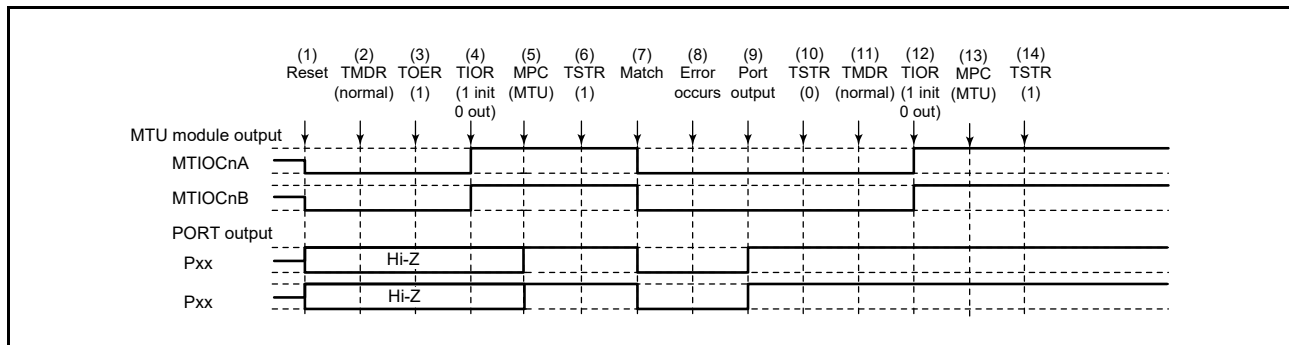
- When making a transition to a mode (Normal, PWM1, PWM2, or PCM) in which the pin output level is selected by the timer I/O control register (TIOR) setting, initialize the pins by means of TIOR setting.
- In PWM mode 1, waveforms are not output to the MTIOCnB and MTIOCnD ( $n = 3, 4, 6, 7$ ) pins. When a pin is configured for MTIOCnB or MTIOCnD, it enters high-impedance state. To output a specified level, set the pin to general output port.
- In PWM mode 2, a waveform is not output to the pin that corresponds to the TGR register used as a period register. When a pin is configured for MTIOCnm ( $n = 0$  to  $2$ ;  $m = A$  to  $D$ ), it enters high-impedance state. To output a specified level, set the pin to general output port.
- In normal mode or PWM mode 2, if TGRC and TGRD operate as buffer registers, waveforms are not output to the corresponding pins (MTIOCnC or MTIOCnD ( $n = 0, 3, 4, 6, 7$ )). When a pin is configured for MTIOCnC or MTIOCnD, it enters high-impedance state. To output a specified level, set the pin to general output port.
- In PWM mode 1, if either TGRC or TGRD operates as a buffer register, waveforms are not output to the corresponding pins (MTIOCnC or MTIOCnD ( $n = 0, 3, 4, 6, 7$ )). When a pin is configured for MTIOCnC or MTIOCnD, it enters high-impedance state. To output a specified level, set the pin to general output port.
- When making a transition to a mode (CPWM or RPWM) in which the pin output level is selected by the timer output control register (TOCR1A, TOCR2A, TOCR1B, or TOCR2B) setting, temporarily disable output in MTU3 and MTU4 (or MTU6 and MTU7) with the timer output master enable register (TOERA or TOERB). At this time, when a pin is configured for MTIOCnm ( $n = 3, 4, 6, 7$ ;  $m = A$  to  $D$ ), it enters high-impedance state. To output a specified level, set the pin to general output port. Switch to normal mode, perform initialization with TIOR, restore TIOR to its initial value, then operate the MTU in accordance with the mode setting procedure (TOCR1A setting, TOCR2A setting, TMDR1 setting, and TOERA setting (TOCR1B setting, TOCR2B setting, TMDR1 setting, and TOERB setting)).

Note: Channel number is substituted for “n” indicated in this section.

Pin initialization procedures are described below for the numbered combinations in Table 24.79. The active level is assumed to be low.

#### (1) Operation When Error Occurs in Normal Mode and Operation is Restarted in Normal Mode

Figure 24.154 shows a case in which an error occurs in normal mode and operation is restarted in normal mode after re-setting.

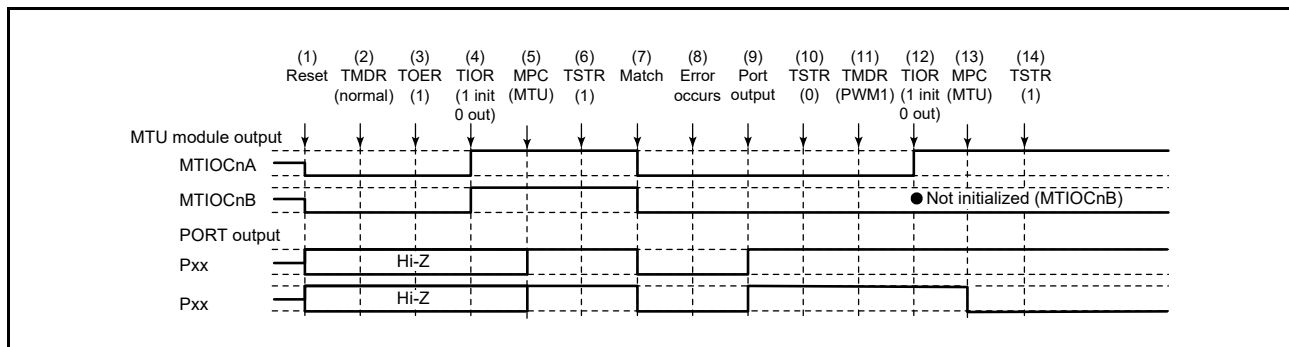


**Figure 24.154 Error Occurrence in Normal Mode, Recovery in Normal Mode**

- (1) After a reset, the MTU output goes low and the ports enter high-impedance state.
- (2) After a reset, the TMDR1 setting is for normal mode.
- (3) For MTU3 and MTU4 (MTU6 and MTU7), enable output with the TOERA (TOERB) register before initializing the pins with the TIOR register.
- (4) Initialize the pins with the TIOR register. (In the example, the initial output is a high level, and a low level is output on compare match occurrence.)
- (5) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.
- (6) Start count operation by setting the TSTRA (TSTRB) register.
- (7) Output goes low on compare match occurrence.
- (8) An error occurs.
- (9) Allow non-active level output by setting the pins as general output ports using the port direction registers (PDR), port output data registers (PODR), and port mode registers (PMR) of the I/O ports.
- (10) Stop count operation by setting the TSTRA (TSTRB) register.
- (11) This step is not necessary when restarting in normal mode.
- (12) Initialize the pins with the TIOR register.
- (13) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.
- (14) Restart operation by setting the TSTRA (TSTRB) register.

## (2) Operation When Error Occurs in Normal Mode and Operation is Restarted in PWM Mode 1

Figure 24.155 shows a case in which an error occurs in normal mode and operation is restarted in PWM mode 1 after re-setting.



**Figure 24.155 Error Occurrence in Normal Mode, Recovery in PWM Mode 1**

(1) to (10) are the same as in Figure 24.154.

(11) Set PWM mode 1.

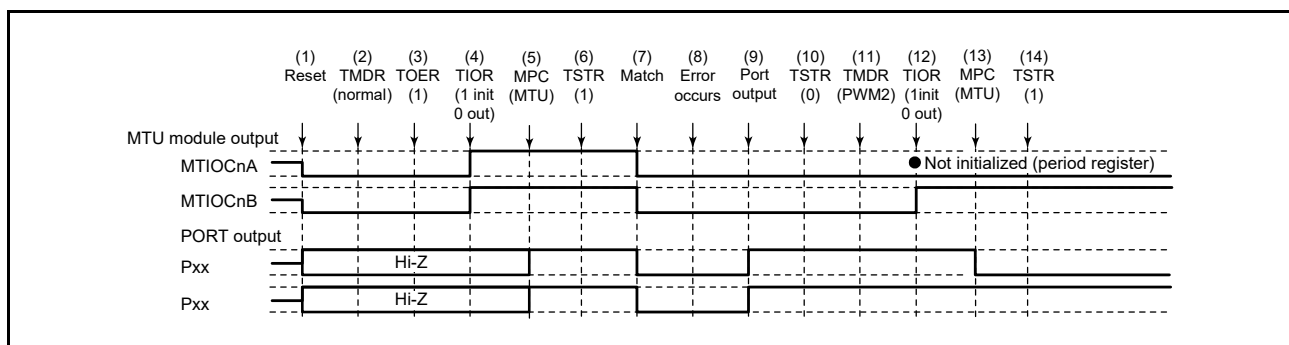
(12) Initialize the pins with the TIOR register. (In PWM mode 1, waveforms are not output to the MTIOCnB (MTIOCnD) pins. To output a specified level, make necessary settings for general output ports in the port direction registers (PDR) and port output data registers (PODR) of the I/O ports.)

(13) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(14) Restart operation by setting the TSTR (TSTRB) register.

## (3) Operation When Error Occurs in Normal Mode and Operation is Restarted in PWM mode 2

Figure 24.156 shows a case in which an error occurs in normal mode and operation is restarted in PWM mode 2 after re-setting.



**Figure 24.156 Error Occurrence in Normal Mode, Recovery in PWM Mode 2**

(1) to (10) are the same as in Figure 24.154.

(11) Set PWM mode 2.

(12) Initialize the pins with the TIOR register. (In PWM mode 2, a waveform is not output to the pin that corresponds to the TGR register used as a period register. To output a specified level, make necessary settings for general output ports in the port direction registers (PDR) and port output data registers (PODR) of the I/O ports.)

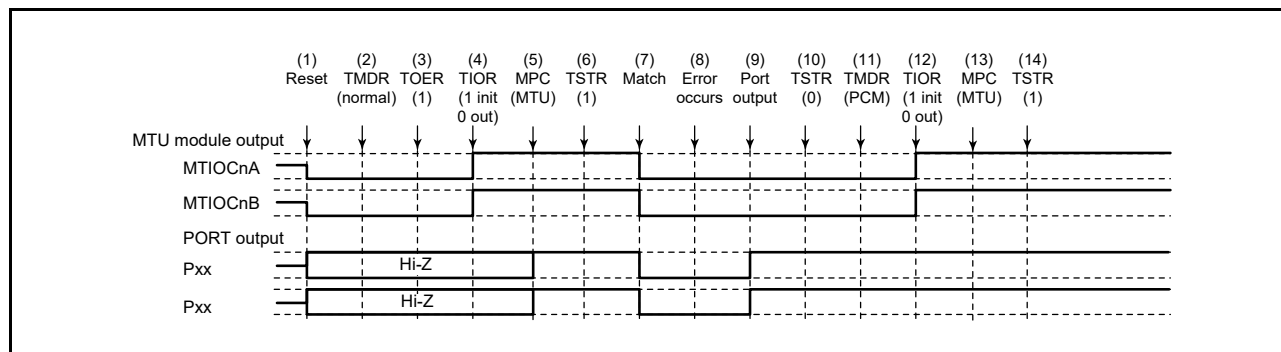
(13) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(14) Restart operation by setting the TSTR (TSTRB) register.

**Note:** PWM mode 2 can only be selected for MTU0 to MTU2, and therefore the TOERA register setting is not necessary.

#### (4) Operation When Error Occurs in Normal Mode and Operation is Restarted in Phase Counting Mode

Figure 24.157 shows a case in which an error occurs in normal mode and operation is restarted in phase counting mode after re-setting.



**Figure 24.157 Error Occurrence in Normal Mode, Recovery in Phase Counting Mode**

(1) to (10) are the same as in Figure 24.154.

(11) Set the phase counting mode.

(12) Initialize the pins with the TIOR register.

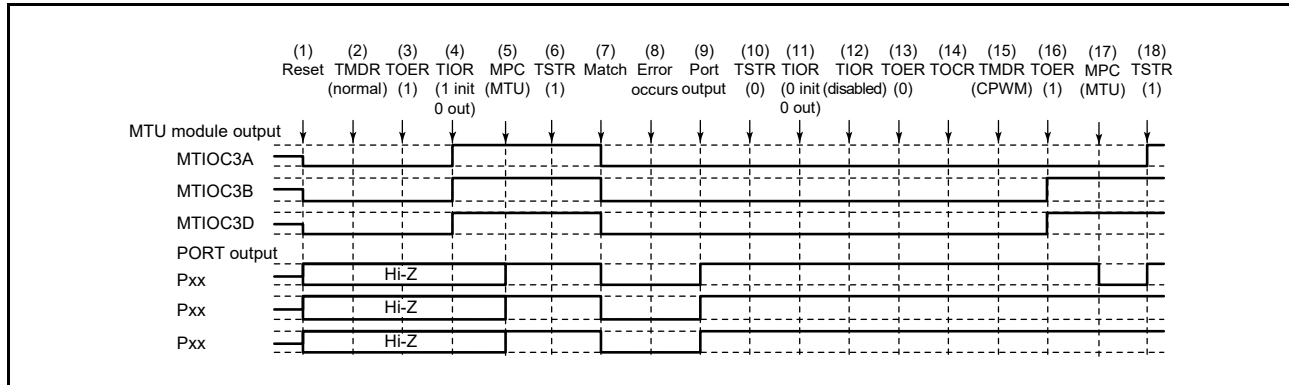
(13) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(14) Restart operation by setting the TSTR register.

**Note:** The phase counting mode can only be selected for MTU1 and MTU2, and therefore the TOERA register setting is not necessary.

### (5) Operation When Error Occurs in Normal Mode and Operation is Restarted in Complementary PWM Mode

Figure 24.158 shows a case in which an error occurs in normal mode and operation is restarted in complementary PWM mode after re-setting.



**Figure 24.158 Error Occurrence in Normal Mode, Recovery in Complementary PWM Mode**

(1) to (10) are the same as in Figure 24.154.

(11) Initialize the normal mode waveform generation block with the TIOR register.

(12) Disable operation of the normal mode waveform generation block with the TIOR register.

(13) Disable output in MTU3 and MTU4 (MTU6 and MTU7) with the TOERA (TOERB) register.

(14) Select the complementary PWM output level and enable or disable cyclic output with registers TOCR1A and TOCR2A (TOCR1B and TOCR2B).

(15) Set complementary PWM mode.

(16) Enable output in MTU3 and MTU4 (MTU6 and MTU7) with the TOERA (TOERB) register.

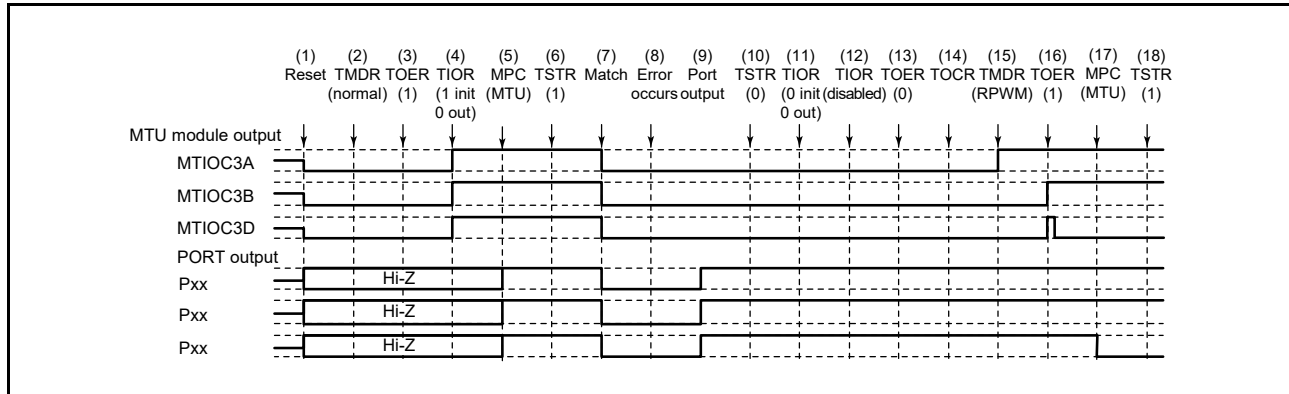
(17) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(18) Restart operation by setting the TSTR (TSTRB) register.



### (6) Operation When Error Occurs in Normal Mode and Operation is Restarted in Reset-Synchronized PWM Mode

Figure 24.159 shows a case in which an error occurs in normal mode and operation is restarted in reset-synchronized PWM mode after re-setting.



**Figure 24.159 Error Occurrence in Normal Mode, Recovery in Reset-Synchronized PWM Mode**

(1) to (13) are the same as in Figure 24.158.

(14) Select the reset-synchronized PWM output level and enable or disable cyclic output with registers TOCR1A and TOCR2A (TOCR1B and TOCR2B).

(15) Set reset-synchronized PWM mode.

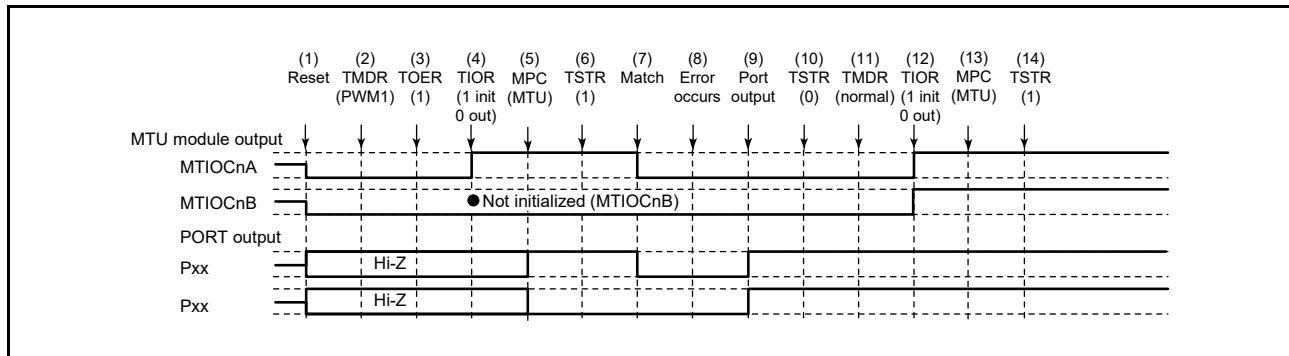
(16) Enable output in MTU3 and MTU4 (MTU6 and MTU7) with the TOERA (TOERB) register.

(17) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(18) Restart operation by setting the TSTRA (TSTRB) register.

## (7) Operation When Error Occurs in PWM Mode 1 and Operation is Restarted in Normal Mode

Figure 24.160 shows a case in which an error occurs in PWM mode 1 and operation is restarted in normal mode after re-setting.

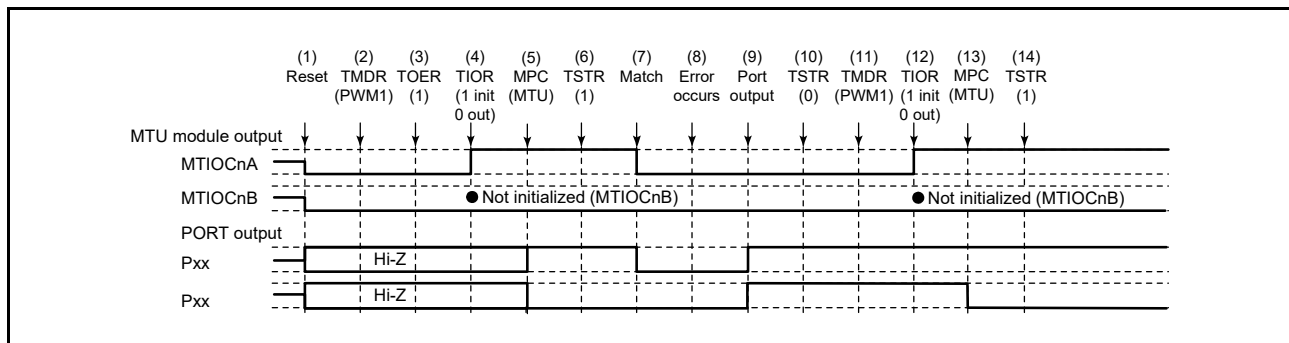


**Figure 24.160 Error Occurrence in PWM Mode 1, Recovery in Normal Mode**

- (1) After a reset, the MTU output goes low and the ports enter high-impedance state.
- (2) Set PWM mode 1.
- (3) For MTU3 and MTU4 (MTU6 and MTU7), enable output with the TOERA (TOERB) register before initializing the pins with the TIOR register.
- (4) Initialize the pins with the TIOR register. (In the example, the initial output is a high level, and a low level is output on compare match occurrence. In PWM mode 1, the MTIOcNB side is not initialized.)
- (5) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.
- (6) Start count operation by setting the TSTRA (TSTRB) register.
- (7) Output goes low on compare match occurrence.
- (8) An error occurs.
- (9) Allow non-active level output by setting the pins as general output ports using the port direction registers (PDR), port output data registers (PODR), and port mode registers (PMR) of the I/O ports.
- (10) Stop count operation by setting the TSTRA (TSTRB) register.
- (11) Set normal mode.
- (12) Initialize the pins with the TIOR register.
- (13) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.
- (14) Restart operation by setting the TSTRA (TSTRB) register.

### (8) Operation When Error Occurs in PWM Mode 1 and Operation is Restarted in PWM mode 1

Figure 24.161 shows a case in which an error occurs in PWM mode 1 and operation is restarted in PWM mode 1 after re-setting.



**Figure 24.161 Error Occurrence in PWM Mode 1, Recovery in PWM Mode 1**

(1) to (10) are the same as in Figure 24.160.

(11) This step is not necessary when restarting in PWM mode 1.

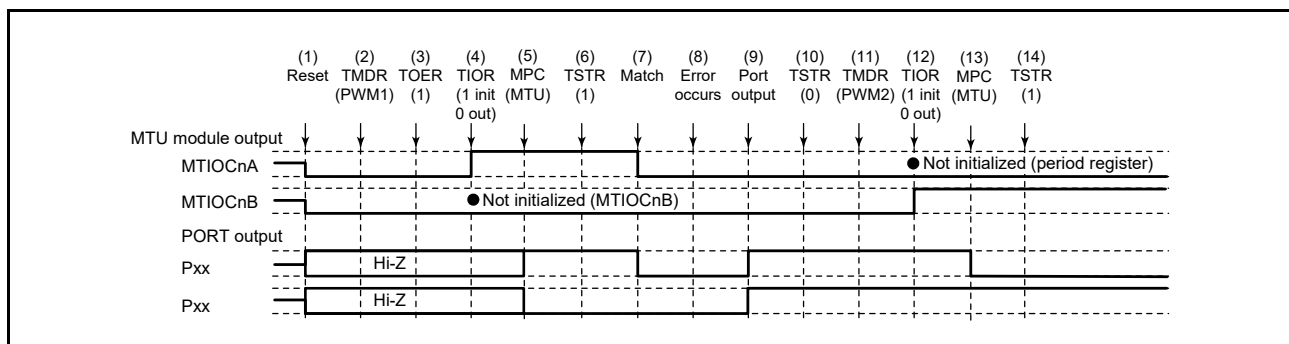
(12) Initialize the pins with the TIOR register. (In PWM mode 1, waveforms are not output to the MTIOCnB (MTIOCnD) pins. To output a specified level, make necessary settings for general output ports in the port direction registers (PDR) and port output data registers (PODR) of the I/O ports.)

(13) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(14) Restart operation by setting the TSTR (TSTRB) register.

### (9) Operation When Error Occurs in PWM Mode 1 and Operation is Restarted in PWM mode 2

Figure 24.162 shows a case in which an error occurs in PWM mode 1 and operation is restarted in PWM mode 2 after re-setting.



**Figure 24.162 Error Occurrence in PWM Mode 1, Recovery in PWM Mode 2**

(1) to (10) are the same as in Figure 24.160.

(11) Set PWM mode 2.

(12) Initialize the pins with the TIOR register. (In PWM mode 2, a waveform is not output to the pin that corresponds to the TGR register used as a period register. To output a specified level, make necessary settings for general output ports in the port direction registers (PDR) and port output data registers (PODR) of the I/O ports.)

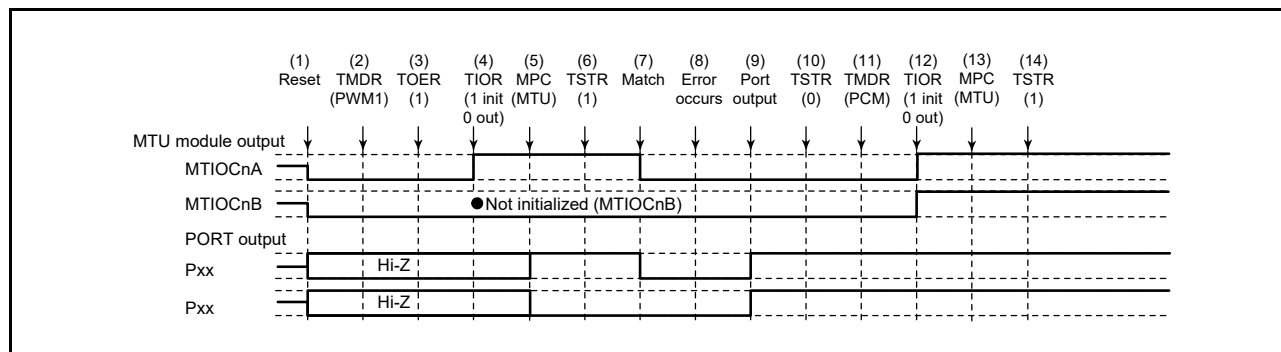
(13) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(14) Restart operation by setting the TSTR (TSTRB) register.

**Note:** PWM mode 2 can only be selected for MTU0 to MTU2, and therefore the TOERA register setting is not necessary.

### (10) Operation When Error Occurs in PWM Mode 1 and Operation is Restarted in Phase Counting Mode

Figure 24.163 shows a case in which an error occurs in PWM mode 1 and operation is restarted in phase counting mode after re-setting.



**Figure 24.163 Error Occurrence in PWM Mode 1, Recovery in Phase Counting Mode**

(1) to (10) are the same as in Figure 24.160.

(11) Set the phase counting mode.

(12) Initialize the pins with the TIOR register.

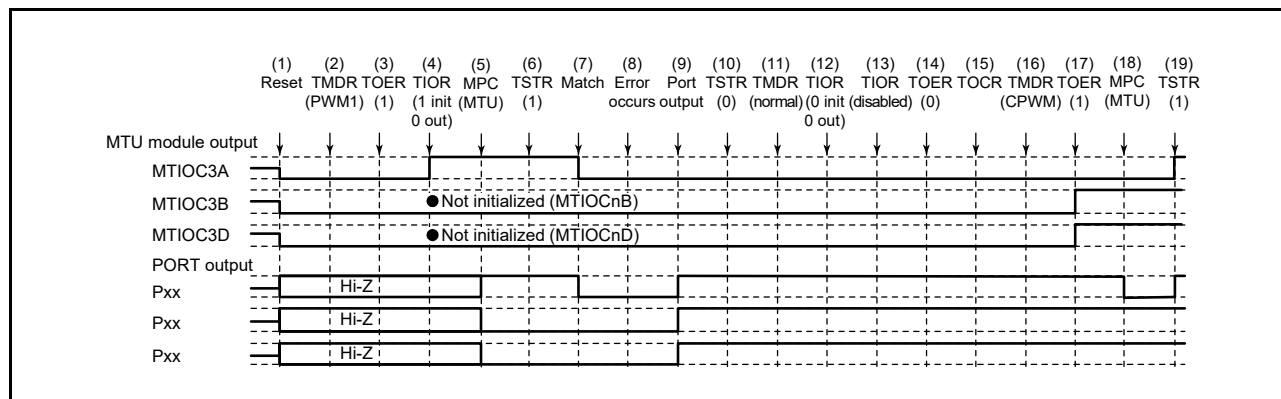
(13) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(14) Restart operation by setting the TSTR register.

**Note:** The phase counting mode can only be selected for MTU1 and MTU2, and therefore the TOERA register setting is not necessary.

### (11) Operation When Error Occurs in PWM Mode 1 and Operation is Restarted in Complementary PWM Mode

Figure 24.164 shows a case in which an error occurs in PWM mode 1 and operation is restarted in complementary PWM mode after re-setting.



**Figure 24.164 Error Occurrence in PWM Mode 1, Recovery in Complementary PWM Mode**

(1) to (10) are the same as in Figure 24.160.

(11) Set normal mode to initialize the normal mode waveform generation block.

(12) Initialize the PWM mode 1 waveform generation block with the TIOR register.

(13) Disable operation of the PWM mode 1 waveform generation block with the TIOR register.

(14) Disable output in MTU3 and MTU4 (MTU6 and MTU7) with the TOERA (TOERB) register.

(15) Select the complementary PWM output level and enable or disable cyclic output with registers TOCR1A and TOCR2A (TSTRB).

(16) Set complementary PWM mode.

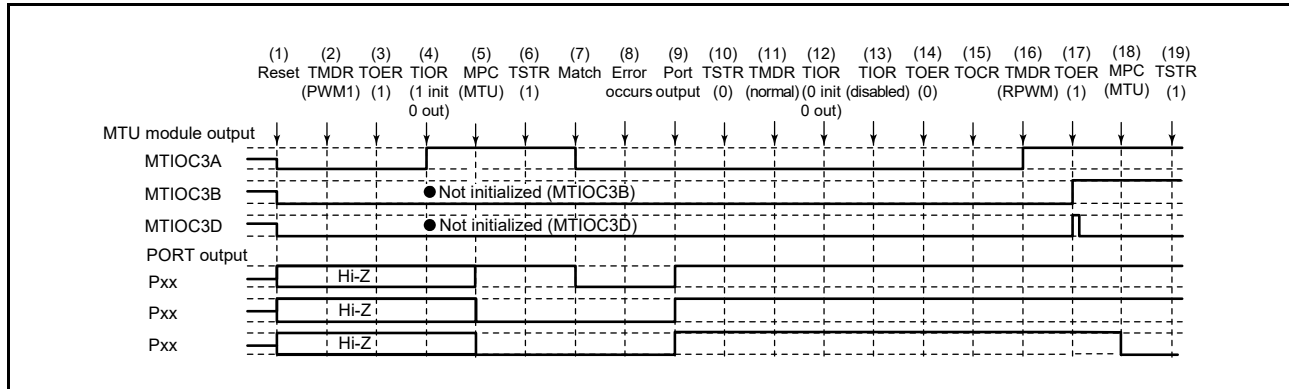
(17) Enable output in MTU3 and MTU4 (MTU6 and MTU7) with the TOERA (TOERB) register.

(18) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(19) Restart operation by setting the TSTR (TSTRB) register.

### (12) Operation When Error Occurs in PWM Mode 1 and Operation is Restarted in Reset-Synchronized PWM Mode

Figure 24.165 shows a case in which an error occurs in PWM mode 1 and operation is restarted in reset-synchronized PWM mode after re-setting.



**Figure 24.165 Error Occurrence in PWM Mode 1, Recovery in Reset-Synchronized PWM Mode**

(1) to (14) are the same as in Figure 24.164.

(15) Select the reset-synchronized PWM output level and enable or disable cyclic output with registers TOCR1A and TOCR2A (TOCR1B and TOCR2B).

(16) Set reset-synchronized PWM mode.

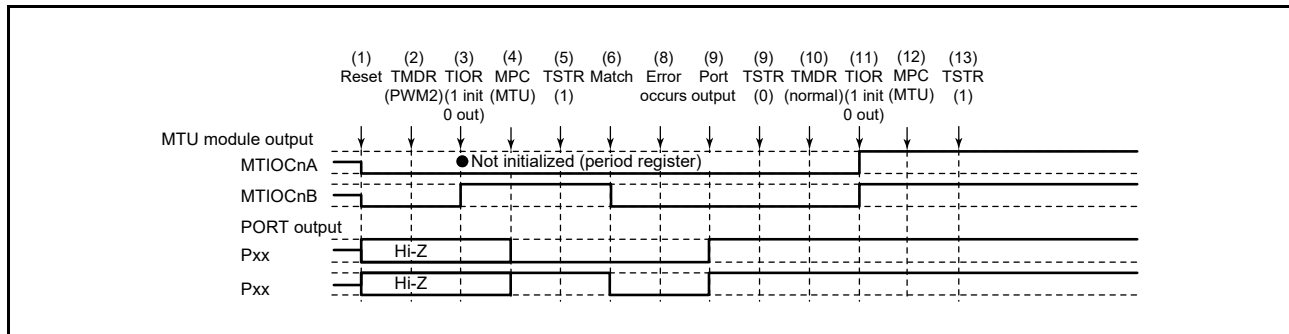
(17) Enable output in MTU3 and MTU4 (MTU6 and MTU7) with the TOERA (TOERB) register.

(18) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(19) Restart operation by setting the TSTR (TSTRB) register.

## (13) Operation When Error Occurs in PWM Mode 2 and Operation is Restarted in Normal Mode

Figure 24.166 shows a case in which an error occurs in PWM mode 2 and operation is restarted in normal mode after re-setting.

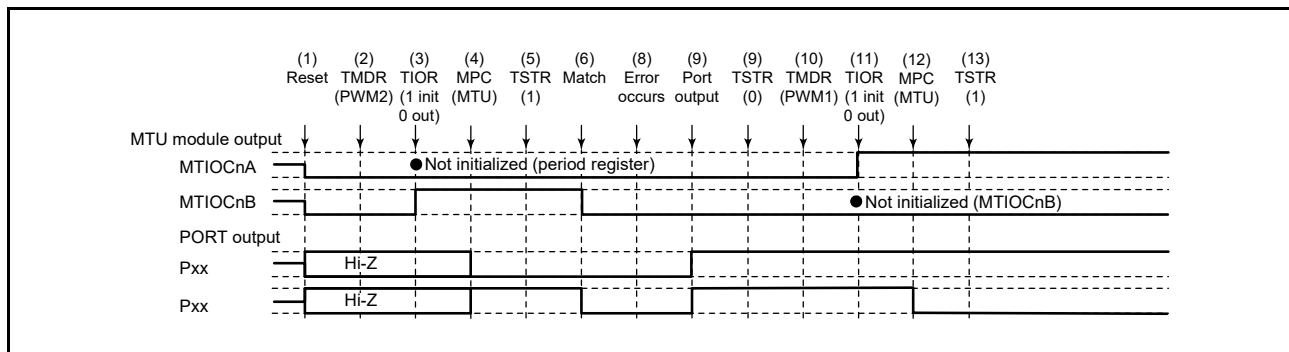


**Figure 24.166 Error Occurrence in PWM Mode 2, Recovery in Normal Mode**

- (1) After a reset, the MTU output goes low and the ports enter high-impedance state.
- (2) Set PWM mode 2.
- (3) Initialize the pins with the TIOR register. (In the example, the initial output is a high level, and a low level is output on compare match occurrence. In PWM mode 2, the pin that corresponds to the TGR register used as a period register is not initialized. In the example, the MTUUn.TGRA register is used as a period register.)
- (4) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.
- (5) Start count operation by setting the TSTR register.
- (6) Output goes low on compare match occurrence.
- (7) An error occurs.
- (8) Allow non-active level output by setting the pins as general output ports using the port direction registers (PDR), port output data registers (PODR), and port mode registers (PMR) of the I/O ports.
- (9) Stop count operation by setting the TSTR register.
- (10) Set normal mode.
- (11) Initialize the pins with the TIOR register.
- (12) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.
- (13) Restart operation by setting the TSTR register.

## (14) Operation When Error Occurs in PWM Mode 2 and Operation is Restarted in PWM Mode 1

Figure 24.167 shows a case in which an error occurs in PWM mode 2 and operation is restarted in PWM mode 1 after re-setting.



**Figure 24.167 Error Occurrence in PWM Mode 2, Recovery in PWM Mode 1**

(1) to (9) are the same as in Figure 24.166.

(10) Set PWM mode 1.

(11) Initialize the pins with the TIOR register. (In PWM mode 1, waveforms are not output to the MTIOcNB (MTIOcND) pins. To output a specified level, make necessary settings for general output ports in the port direction registers (PDR) and port output data registers (PODR) of the I/O ports.)

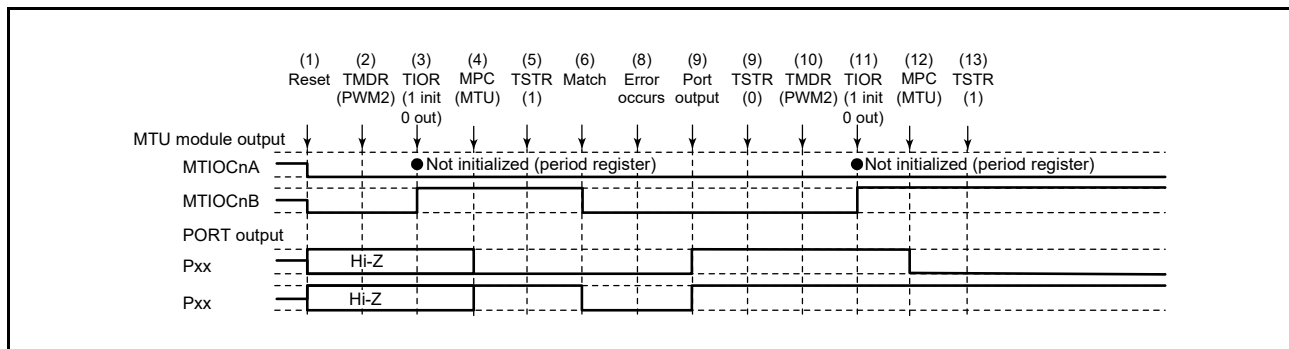
(12) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(13) Restart operation by setting the TSTRA register.



## (15) Operation When Error Occurs in PWM Mode 2 and Operation is Restarted in PWM Mode 2

Figure 24.168 shows a case in which an error occurs in PWM mode 2 and operation is restarted in PWM mode 2 after re-setting.



**Figure 24.168 Error Occurrence in PWM Mode 2, Recovery in PWM Mode 2**

(1) to (9) are the same as in Figure 24.166.

(10) This step is not necessary when restarting in PWM mode 2.

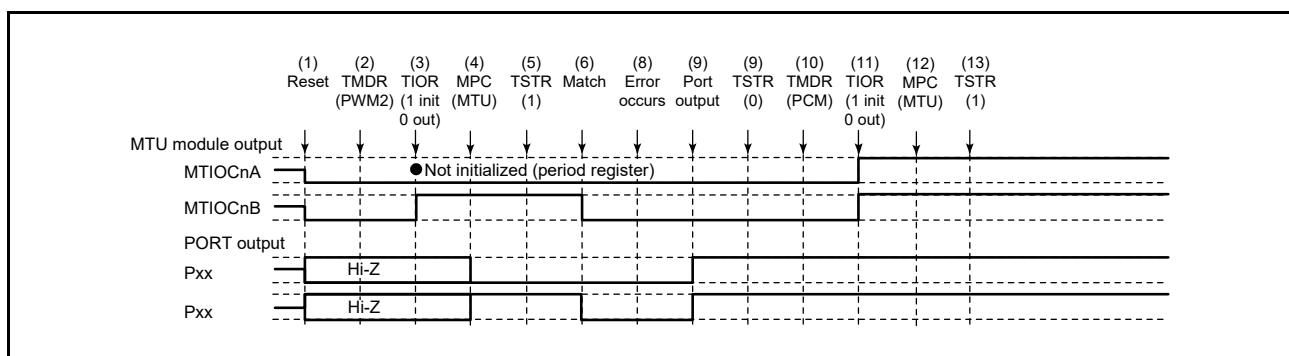
(11) Initialize the pins with the TIOR register. (In PWM mode 2, a waveform is not output to the pin that corresponds to the TGR register used as a period register. To output a specified level, make necessary settings for general output ports in the port direction registers (PDR) and port output data registers (PODR) of the I/O ports.)

(12) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(13) Restart operation by setting the TSTR register.

## (16) Operation When Error Occurs in PWM Mode 2 and Operation is Restarted in Phase Counting Mode

Figure 24.169 shows a case in which an error occurs in PWM mode 2 and operation is restarted in phase counting mode after re-setting.



**Figure 24.169 Error Occurrence in PWM Mode 2, Recovery in Phase Counting Mode**

(1) to (9) are the same as in Figure 24.166.

(10) Set the phase counting mode.

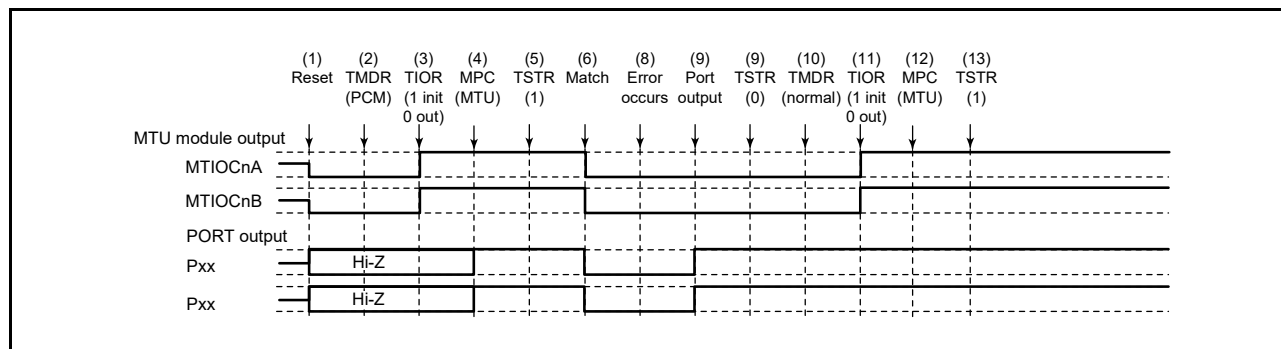
(11) Initialize the pins with the TIOR register.

(12) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(13) Restart operation by setting the TSTR register.

### (17) Operation When Error Occurs in Phase Counting Mode and Operation is Restarted in Normal Mode

Figure 24.170 shows a case in which an error occurs in phase counting mode and operation is restarted in normal mode after re-setting.

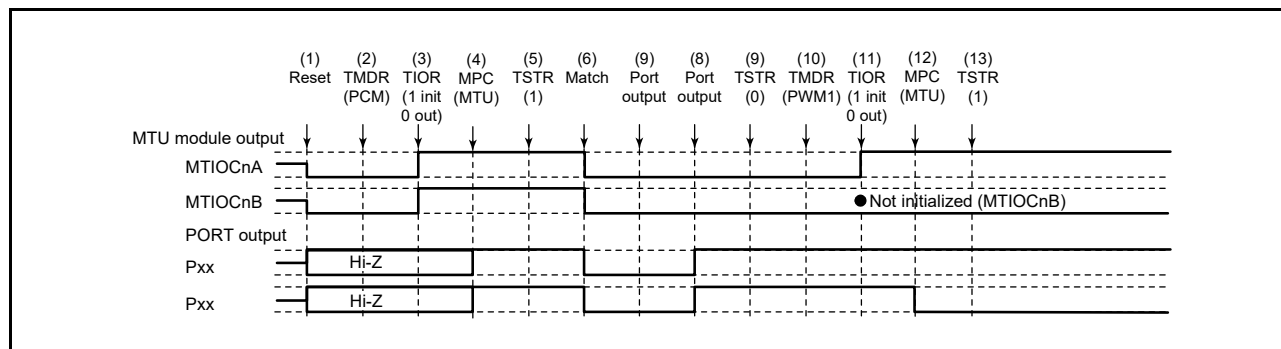


**Figure 24.170 Error Occurrence in Phase Counting Mode, Recovery in Normal Mode**

- (1) After a reset, the MTU output goes low and the ports enter high-impedance state.
- (2) Set phase counting mode.
- (3) Initialize the pins with the TIOR register. (In the example, the initial output is a high level, and a low level is output on compare match occurrence.)
- (4) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.
- (5) Start count operation by setting the TSTR register.
- (6) Output goes low on compare match occurrence.
- (7) An error occurs.
- (8) Allow non-active level output by setting the pins as general output ports using the port direction registers (PDR), port output data registers (PODR), and port mode registers (PMR) of the I/O ports.
- (9) Stop count operation by setting the TSTR register.
- (10) Set normal mode.
- (11) Initialize the pins with the TIOR register.
- (12) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.
- (13) Restart operation by setting the TSTR register.

### (18) Operation When Error Occurs in Phase Counting Mode and Operation is Restarted in PWM Mode 1

Figure 24.171 shows a case in which an error occurs in phase counting mode and operation is restarted in PWM mode 1 after re-setting.



**Figure 24.171 Error Occurrence in Phase Counting Mode, Recovery in PWM Mode 1**

(1) to (9) are the same as in Figure 24.170.

(10) Set PWM mode 1.

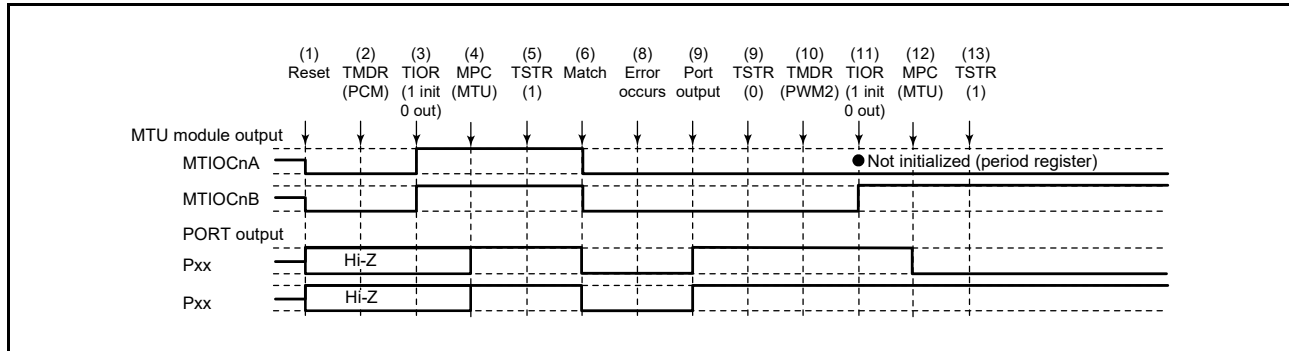
(11) Initialize the pins with the TIOR register. (In PWM mode 1, waveforms are not output to the MTIOCnB (MTIOCnD) pins. To output a specified level, make necessary settings for general output ports in the port direction registers (PDR) and port output data registers (PODR) of the I/O ports.)

(12) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(13) Restart operation by setting the TSTR register.

### (19) Operation When Error Occurs in Phase Counting Mode and Operation is Restarted in PWM Mode 2

Figure 24.172 shows a case in which an error occurs in phase counting mode and operation is restarted in PWM mode 2 after re-setting.



**Figure 24.172 Error Occurrence in Phase Counting Mode, Recovery in PWM Mode 2**

(1) to (9) are the same as in Figure 24.170.

(10) Set PWM mode 2.

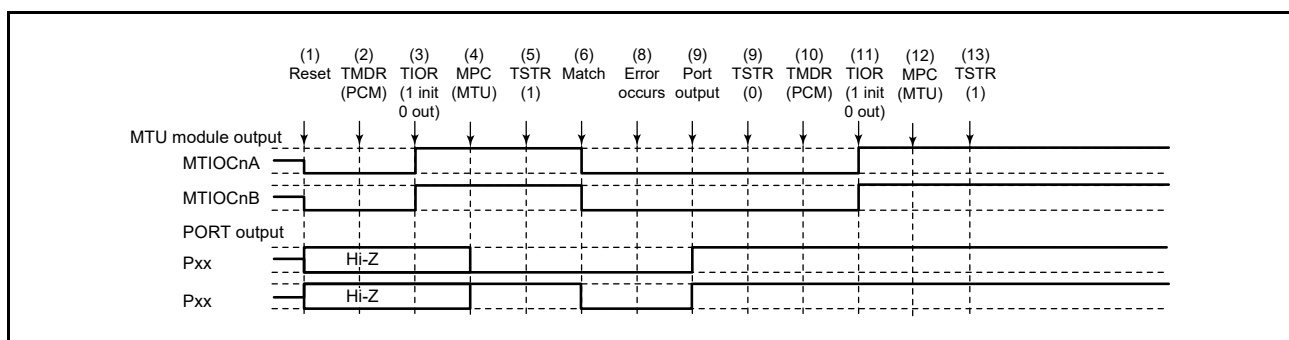
(11) Initialize the pins with the TIOR register. (In PWM mode 2, a waveform is not output to the pin that corresponds to the TGR register used as a period register. To output a specified level, make necessary settings for general output ports in the port direction registers (PDR) and port output data registers (PODR) of the I/O ports.)

(12) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(13) Restart operation by setting the TSTR register.

### (20) Operation When Error Occurs in Phase Counting Mode and Operation is Restarted in Phase Counting Mode

Figure 24.173 shows a case in which an error occurs in phase counting mode and operation is restarted in phase counting mode after re-setting.



**Figure 24.173 Error Occurrence in Phase Counting Mode, Recovery in Phase Counting Mode**

(1) to (9) are the same as in Figure 24.170.

(10) This step is not necessary when restarting in phase counting mode.

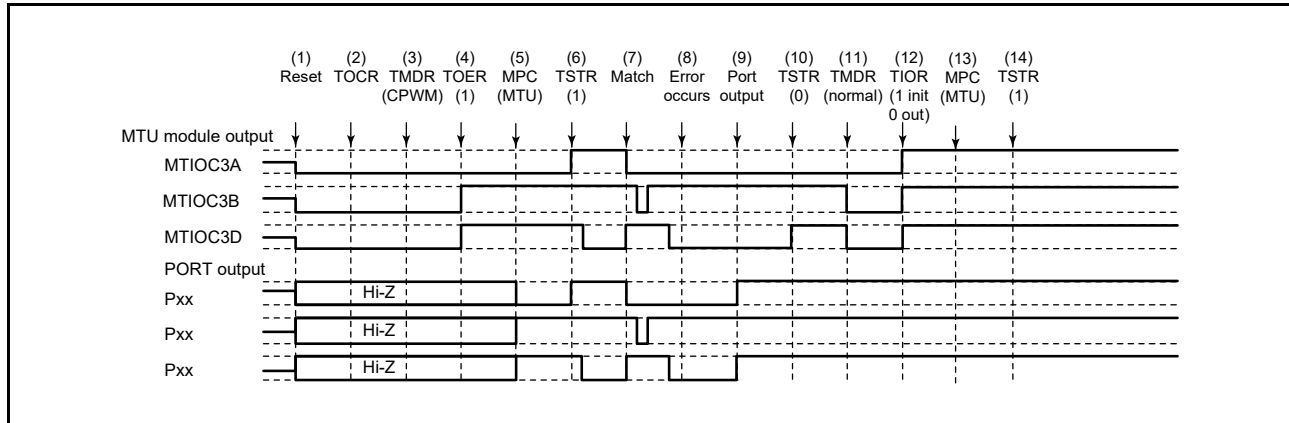
(11) Initialize the pins with the TIOR register.

(12) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(13) Restart operation by setting the TSTR register.

### (21) Operation When Error Occurs in Complementary PWM Mode and Operation is Restarted in Normal Mode

Figure 24.174 shows a case in which an error occurs in complementary PWM mode and operation is restarted in normal mode after re-setting.

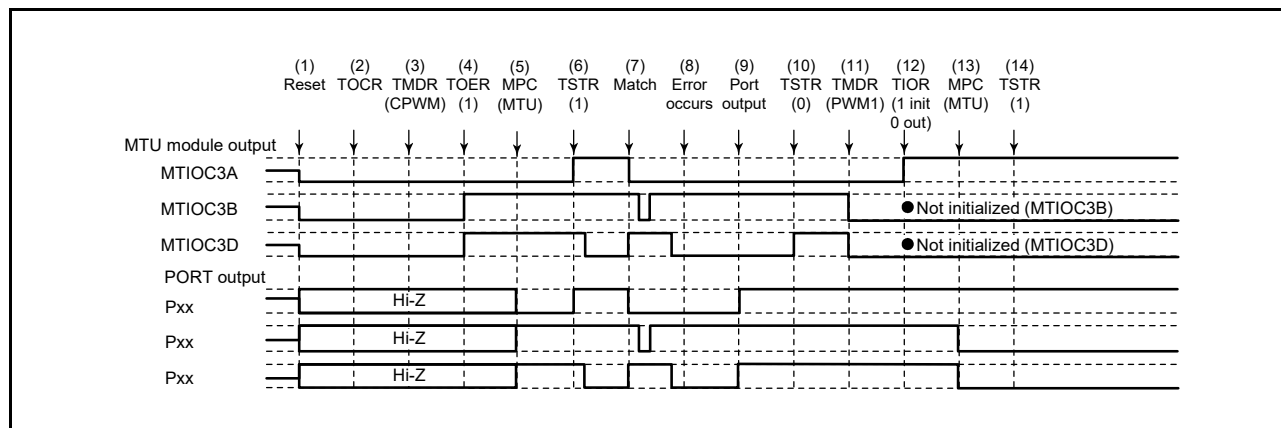


**Figure 24.174 Error Occurrence in Complementary PWM Mode, Recovery in Normal Mode**

- (1) After a reset, the MTU3 output goes low and the ports enter high-impedance state.
- (2) Select the complementary PWM output level and enable or disable cyclic output with registers TOCR1A and TOCR2A (TOCR1B and TOCR2B).
- (3) Set complementary PWM mode.
- (4) Enable output in MTU3 and MTU4 (MTU6 and MTU7) with the TOERA (TOERB) register.
- (5) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.
- (6) Start count operation by setting the TSTRA (TSTRB) register.
- (7) The complementary PWM waveform is output on compare match occurrence.
- (8) An error occurs.
- (9) Allow non-active level output by setting the pins as general output ports using the port direction registers (PDR), port output data registers (PODR), and port mode registers (PMR) of the I/O ports.
- (10) Stop count operation by setting the TSTRA (TSTRB) register. (MTU output becomes the initial complementary PWM output value).
- (11) Set normal mode (MTU output goes low).
- (12) Initialize the pins with the TIOR register.
- (13) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.
- (14) Restart operation by setting the TSTRA (TSTRB) register.

## (22) Operation When Error Occurs in Complementary PWM Mode and Operation is Restarted in PWM Mode 1

Figure 24.175 shows a case in which an error occurs in complementary PWM mode and operation is restarted in PWM mode 1 after re-setting.



**Figure 24.175 Error Occurrence in Complementary PWM Mode, Recovery in PWM Mode 1**

(1) to (10) are the same as in Figure 24.174.

(11) Set PWM mode 1 (MTU output goes low).

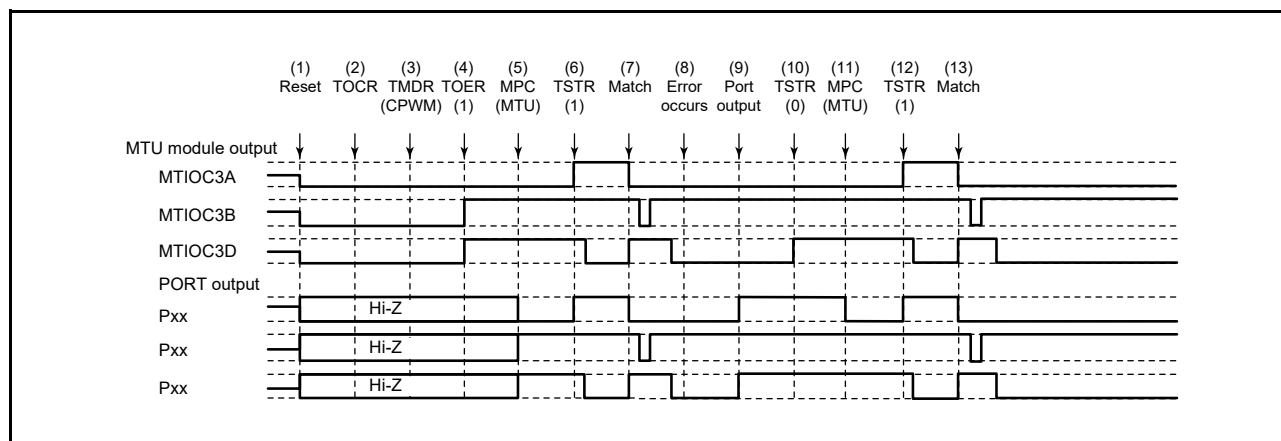
(12) Initialize the pins with the TIOR register. (In PWM mode 1, waveforms are not output to the MTIOCnB (MTIOCnD) pins. To output a specified level, make necessary settings for general output ports in the port direction registers (PDR) and port output data registers (PODR) of the I/O ports.)

(13) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(14) Restart operation by setting the TSTR (TSTRB) register.

### (23) Operation When Error Occurs in Complementary PWM Mode and Operation is Restarted in Complementary PWM Mode

Figure 24.176 shows a case in which an error occurs in complementary PWM mode and operation is restarted in complementary PWM mode after re-setting (when operation is restarted using the period and duty settings at the time of stopping the counter).



**Figure 24.176 Error Occurrence in Complementary PWM Mode, Recovery in Complementary PWM Mode (1)**

(1) to (10) are the same as in Figure 24.174.

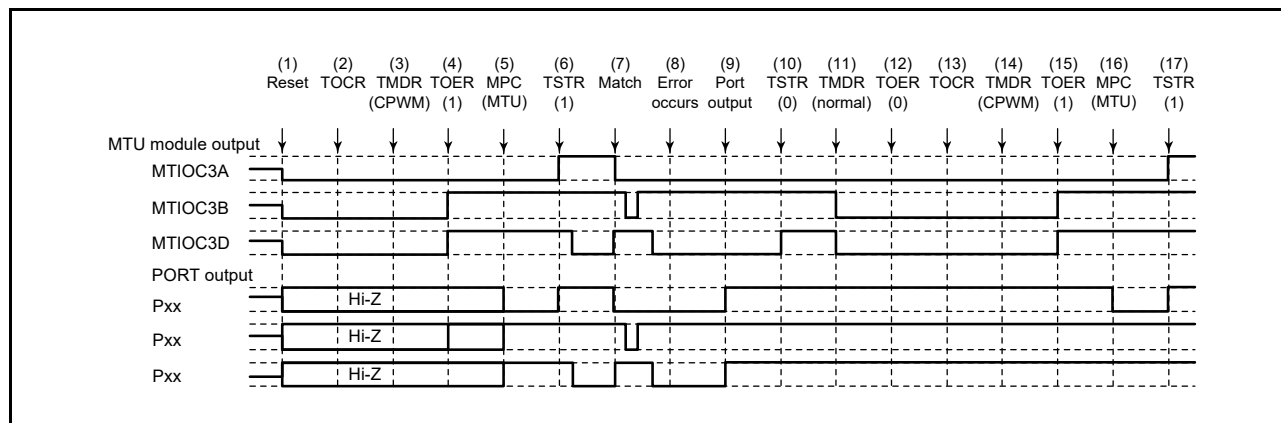
(11) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(12) Restart operation by setting the TSTR (TSTRB) register.

(13) The complementary PWM waveform is output on compare match occurrence.

### (24) Operation When Error Occurs in Complementary PWM Mode and Operation is Restarted in Complementary PWM Mode with New Settings

Figure 24.177 shows a case in which an error occurs in complementary PWM mode and operation is restarted in complementary PWM mode after re-setting (operation is restarted using new period and duty ratio settings).



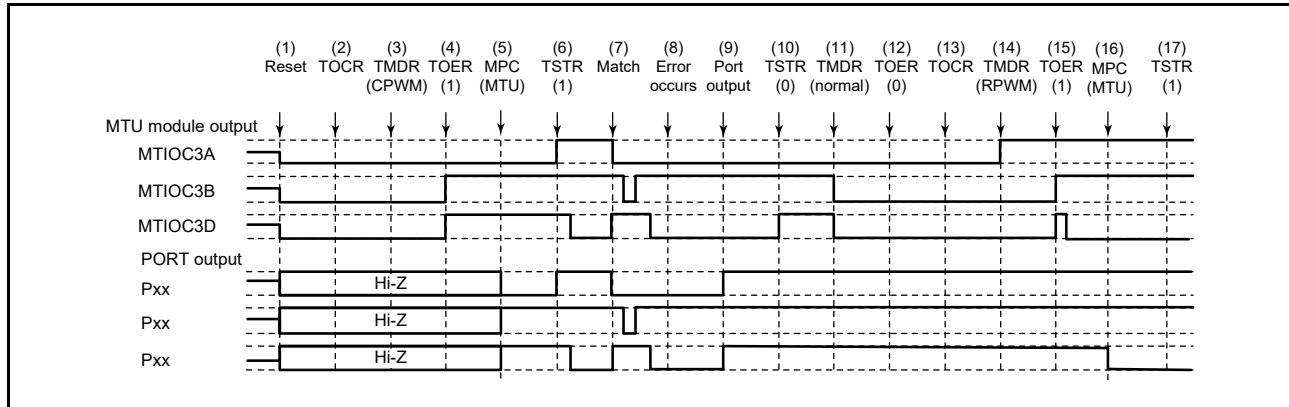
**Figure 24.177 Error Occurrence in Complementary PWM Mode, Recovery in Complementary PWM Mode (2)**

- (1) to (10) are the same as in Figure 24.174.
- (11) Set normal mode and make new settings (MTU output goes low).
- (12) Disable output in MTU3 and MTU4 (MTU6 and MTU7) with the TOERA (TOERB) register.
- (13) Select the complementary PWM output level and enable or disable cyclic output with registers TOCR1A and TOCR2A (TOCR1B and TOCR2B).
- (14) Set complementary PWM mode.
- (15) Enable output in MTU3 and MTU4 (MTU6 and MTU7) with the TOERA (TOERB) register.
- (16) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.
- (17) Restart operation by setting the TSTR (TSTRB) register.



### (25) Operation When Error Occurs in Complementary PWM Mode and Operation is Restarted in Reset-Synchronized PWM Mode

Figure 24.178 shows a case in which an error occurs in complementary PWM mode and operation is restarted in reset-synchronized PWM mode after re-setting.

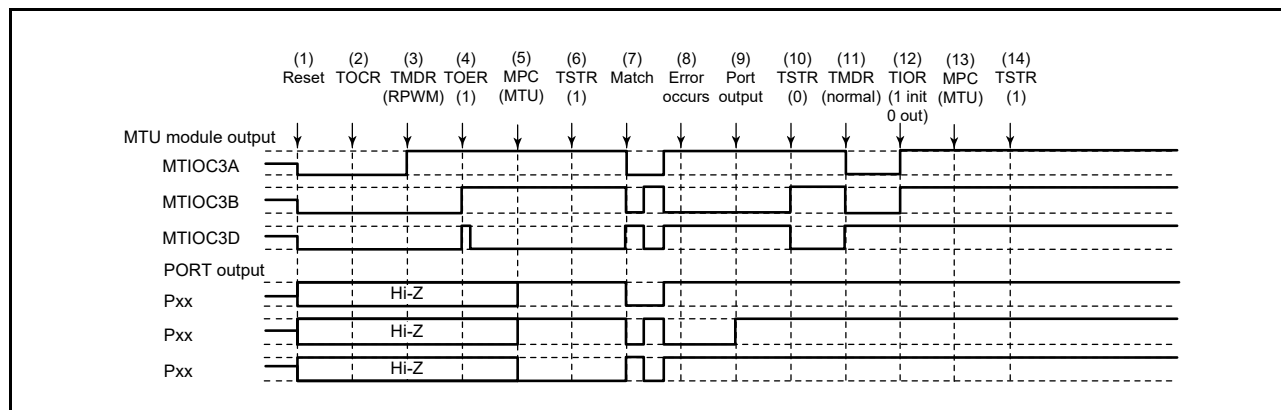


**Figure 24.178 Error Occurrence in Complementary PWM Mode, Recovery in Reset-Synchronized PWM Mode**

- (1) to (10) are the same as in Figure 24.174.
- (11) Set normal mode (MTU output goes low).
- (12) Disable output in MTU3 and MTU4 (MTU6 and MTU7) with the TOERA (TOERB) register.
- (13) Select the reset-synchronized PWM output level and enable or disable cyclic output with registers TOCR1A and TOCR2A (TOCR1B and TOCR2B).
- (14) Set reset-synchronized PWM mode.
- (15) Enable output in MTU3 and MTU4 (MTU6 and MTU7) with the TOERA (TOERB) register.
- (16) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.
- (17) Restart operation by setting the TSTR (TSTRB) register.

## (26) Operation When Error Occurs in Reset-Synchronized PWM Mode and Operation is Restarted in Normal Mode

Figure 24.179 shows a case in which an error occurs in reset-synchronized PWM mode and operation is restarted in normal mode after re-setting.

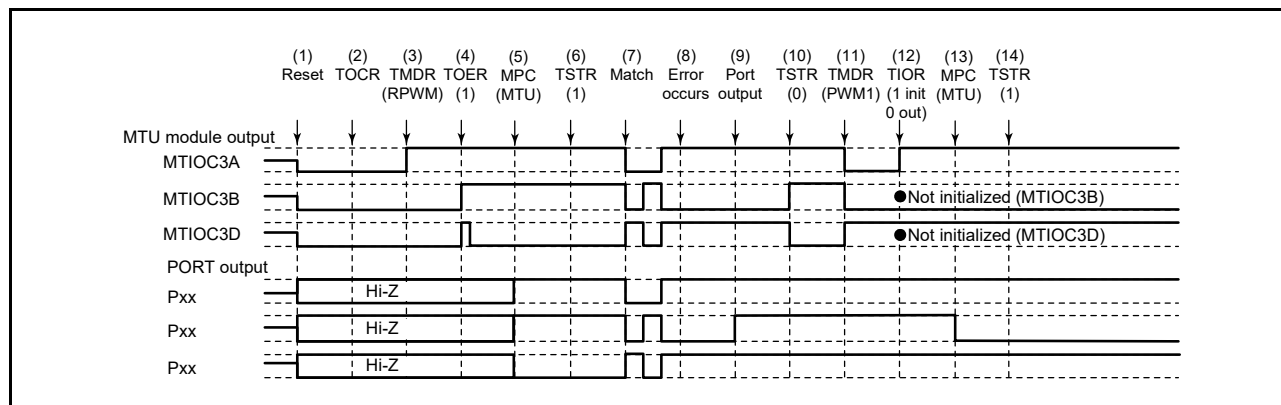


**Figure 24.179 Error Occurrence in Reset-Synchronized PWM Mode, Recovery in Normal Mode**

- (1) After a reset, the MTU output goes low and the ports enter high-impedance state.
- (2) Select the reset-synchronized PWM output level and enable or disable cyclic output with registers TOCR1A and TOCR2A (TOCR1B and TOCR2B).
- (3) Set reset-synchronized PWM mode.
- (4) Enable output in MTU3 and MTU4 (MTU6 and MTU7) with the TOERA (TOERB) register.
- (5) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.
- (6) Start count operation by setting the TSTRA (TSTRB) register.
- (7) The reset-synchronized PWM waveform is output on compare match occurrence.
- (8) An error occurs.
- (9) Allow non-active level output by setting the pins as general output ports using the port direction registers (PDR), port output data registers (PODR), and port mode registers (PMR) of the I/O ports.
- (10) Stop count operation by setting the TSTRA (TSTRB) register. (MTU output becomes the initial reset-synchronized PWM output value.)
- (11) Set normal mode (positive-phase MTU output goes low, and negative-phase output goes high).
- (12) Initialize the pins with the TIOR register.
- (13) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.
- (14) Restart operation by setting the TSTRA (TSTRB) register.

### (27) Operation When Error Occurs in Reset-Synchronized PWM Mode and Operation is Restarted in PWM Mode 1

Figure 24.180 shows a case in which an error occurs in reset-synchronized PWM mode and operation is restarted in PWM mode 1 after re-setting.



**Figure 24.180 Error Occurrence in Reset-Synchronized PWM Mode, Recovery in PWM Mode 1**

(1) to (10) are the same as in Figure 24.179.

(11) Set PWM mode 1 (positive-phase MTU output goes low, and negative-phase output goes high).

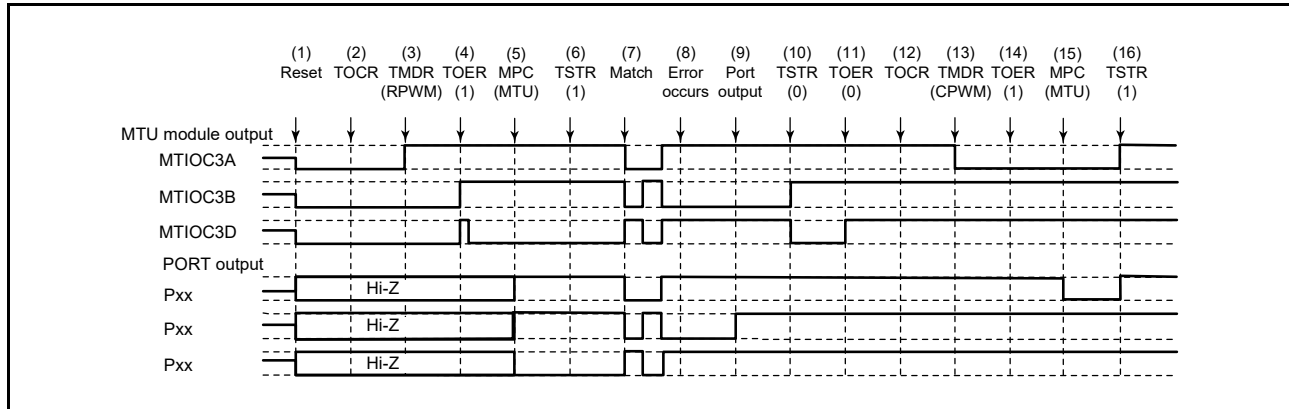
(12) Initialize the pins with the TIOR register. (In PWM mode 1, waveforms are not output to the MTIOCnB (MTIOCnD) pins. To output a specified level, make necessary settings for general output ports in the port direction registers (PDR) and port output data registers (PODR) of the I/O ports.)

(13) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(14) Restart operation by setting the TSTRA (TSTRB) register.

### (28) Operation When Error Occurs in Reset-Synchronized PWM Mode and Operation is Restarted in Complementary PWM Mode

Figure 24.181 shows a case in which an error occurs in reset-synchronized PWM mode and operation is restarted in complementary PWM mode after re-setting.



**Figure 24.181 Error Occurrence in Reset-Synchronized PWM Mode, Recovery in Complementary PWM Mode**

(1) to (10) are the same as in Figure 24.179.

(11) Disable output in MTU3 and MTU4 (MTU6 and MTU7) with the TOERA (TOERB) register.

(12) Select the complementary PWM output level and enable or disable cyclic output with registers TOCR1A and TOCR2A (TOCR1B and TOCR2B).

(13) Set complementary PWM mode (MTU cyclic output pin goes low).

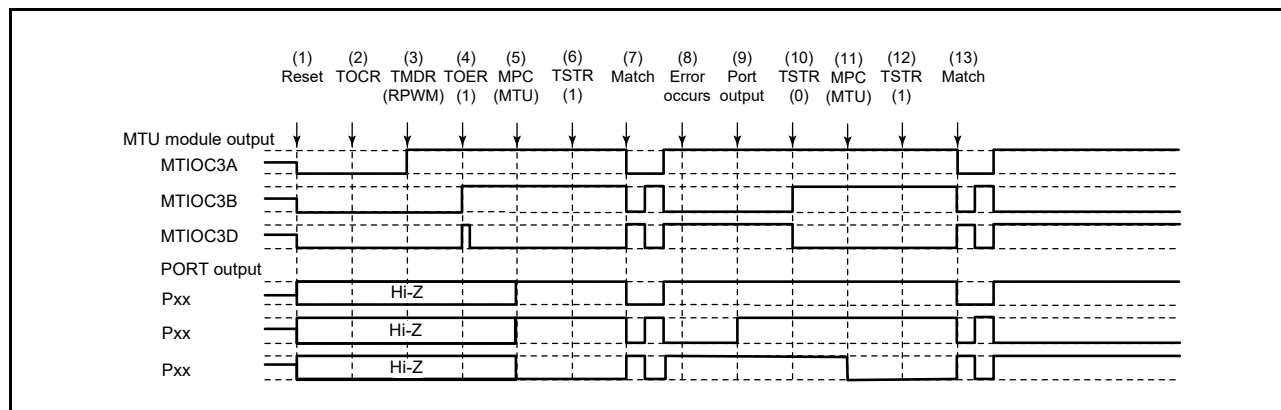
(14) Enable output in MTU3 and MTU4 (MTU6 and MTU7) with the TOERA (TOERB) register.

(15) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(16) Restart operation by setting the TSTR (TSTRB) register.

### (29) Operation When Error Occurs in Reset-Synchronized PWM Mode and Operation is Restarted in Reset-Synchronized PWM Mode

Figure 24.182 shows a case in which an error occurs in reset-synchronized PWM mode and operation is restarted in reset-synchronized PWM mode after re-setting.



**Figure 24.182 Error Occurrence in Reset-Synchronized PWM Mode, Recovery in Reset-Synchronized PWM Mode**

(1) to (10) are the same as in Figure 24.179.

(11) Set MTU output using the MPC and port mode registers (PMR) corresponding to the I/O ports.

(12) Restart operation by setting the TSTR (TSTRB) register.

(13) The reset-synchronized PWM waveform is output on compare match occurrence.

## 24.8 Operations Linked by the ELC

### 24.8.1 Event Signal Output to the ELC

The MTU is capable of operation linked with another module set in advance when its interrupt request signal is used as an event signal by the event link controller (ELC).

The MTU outputs the event signal regardless of the setting of the corresponding interrupt request enable bit.

### 24.8.2 MTU Operations in Response to Receiving Event Signals from the ELC

The MTU can perform the following operations in response to the event set in advance in the ELSRn register of the event link controller (ELC).

#### (1) Start Counting Operation

Select “counting is started” as the operation of the MTU by setting the ELOPA or ELOPB register in the ELC. The ELOPA register controls the operation of MTU0 and MTU3 and the ELOPB register controls the operation of MTU4. When the event specified in the ELSRn register occurs, the CSTn bit in the TSTRA register shown in Table 24.80 is set to 1, and the MTU counter starts.

However, when the specified event is generated while the CSTn bit in the TSTRA register has already been set to 1, the event has no effect. Table 24.80 lists the TSTRA register bits used for each channel.

**Table 24.80 Counter Start Bit Set by the ELC**

| Channel No. | Counter Start Bit |
|-------------|-------------------|
| MTU0        | TSTRA.CST0 bit    |
| MTU3        | TSTRA.CST3 bit    |
| MTU4        | TSTRA.CST4 bit    |

#### (2) Input Capture Operation

Select “input capture” as the operation of the MTU by setting the ELOPA or ELOPB register in the ELC. The ELOPA register controls the operation of MTU0 and MTU3 and the ELOPB register controls the operation of MTU4. When the event specified in the ELSRn register occurs, the value of the TCNT register is captured in the TGR register. When using input capture in response to an event, the corresponding bit of the TIOR register in the MTU should be set for input capture and the CSTn bit of TSTRA register should be set to 1 to start counting by the counter.

In this case, the TIOCnA pin (input capture pin) input has no effect.

Table 24.81 lists the timer general register and I/O control bit used for each channel in input capture operations in response to the ELC.

**Table 24.81 Timer General Register and I/O Control Bit Used in the Input Capture Operation**

| Channel No. | Timer General Register | I/O Control Bit          |
|-------------|------------------------|--------------------------|
| MTU0        | MTU0.TGRA              | MTU0.TIORH.IOA[3:0] bits |
| MTU3        | MTU3.TGRA              | MTU3.TIORH.IOA[3:0] bits |
| MTU4        | MTU4.TGRA              | MTU4.TIORH.IOA[3:0] bits |

#### (3) Restart Counting (Clear Counter) Operation

Select “counting is restarted” as the operation of the MTU by setting the ELOPA or ELOPB register in the ELC. The ELOPA register controls the operation of MTU0 and MTU3 and the ELOPB register controls the operation of MTU4. When the event specified in the ELSRn register occurs, the TCNT register is cleared. If the corresponding CSTn bit in the TSTRA register is set to 1, counting continues. For the CSTn bits in the TSTRA register, refer to Table 24.80.

### 24.8.3 Usage Notes on MTU Operation by Event Signal Reception from the ELC

The following notes on usage apply when the MTU is used in event link operation.

#### (1) Start Counting

If the event specified in the ELSRn register occurs during a cycle of writing to a CSTn bit in the TSTRA register, writing to the CSTn bit in the TSTRA register does not proceed because setting of the bit to 1 due to the event takes priority.

#### (2) Restart Counting (Clear Counter)

If the event specified in the ELSRn register occurs during a cycle of writing to the TCNT counter, writing to the TCNT counter does not proceed because clearing of the counter due to the event takes priority.

In addition, for MTU3 and MTU4 in complementary PWM mode, do not use the counter restarting by the ELC.